GEOMETRY · COMMON CORE HIGH SCHOOL MATH

Question Collection

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

105 question sets • 3160 questions

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Precise definition of a circle

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 3	Use the center, radius, and diameter of a circle	**7.G.B.4**
BACKWARD 4	Find the distance from a center to points on a grid	**8.G.B.8**
FORWARD 1	Test definitions of circle vocabulary with counterexamples	**HSG-CO.A.1**
FORWARD 2	Write precise definitions of circle parts from the definition of a circ…	**HSG-CO.A.1**

SET 1 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of a circle. Base it on the notions of point, plane, and distance (you may use ordinary words such as "set", "all", and "given"). Name the center and the radius in your definition.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q1/5

Kai writes: "A circle is a round curve with a center."

Explain what makes this imprecise, and rewrite it as a precise definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q2/5

Lee writes: "The circle with center O and radius 4 is all the points of the plane that are within 4 units of O."

What figure does Lee's sentence describe? Rewrite it so that it defines the circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q3/5

Mia writes: "A circle is all the points that are 5 cm from point O."

What must she add to make the definition precise? Explain why.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q4/5

Ana writes: "A circle is the set of points in a plane at a distance from a point O in that plane."

Which condition is missing? Rewrite the definition so that it is precise.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSG-CO.A.1

BACKWARD 3

Use the center, radius, and diameter of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 7 cm. Point P is in the plane of the circle, 7 cm from O. Is P on the circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q1/5

A circle has a diameter of 18 in. What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q2/5

A circle has center C and radius 4. Point D is in the same plane, 3 units from C. Where is D?

A. Inside the circle

B. Outside the circle

C. On the circle

D. There is not enough information

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q3/5

True or false?

Every radius of a circle has the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q4/5

True or false?

A point 10 units from the center of a circle of radius 5 lies on the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSG-CO.A.1

BACKWARD 4

Find the distance from a center to points on a grid • Grade 8 • Geometry • 8.G.B.8

Find the distance from O(0, 0) to P(3, 4).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B4-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q1/5

Find the distance from C(1, 1) to Q(1, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B4-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q2/5

Which point is 5 units from O(0, 0)?

A. (5, 5)

B. (2, 3)

C. (4, 4)

D. (−4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B4-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q3/5

True or false?

The points (5, 0) and (0, −5) are the same distance from the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B4-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q4/5

True or false?

The point (3, 3) is 3 units from the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B4-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 1 • BACKWARD 4 • Q5/5

SET 1 • HSG-CO.A.1

FORWARD 1

Test definitions of circle vocabulary with counterexamples • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Proposed definition: "A circle is a set of points that are all the same distance from a point O."

Give a set that fits this wording but is NOT a circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q1/5

Proposed definition: "A radius of a circle is a segment from the center."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q2/5

Which set fits the description "all points 2 units from point O" but is NOT a circle?

A. A circle of radius 2 centered at O in a plane

B. A sphere of radius 2 centered at O

C. A disk of radius 2

D. A segment of length 2 from O

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q3/5

Proposed definition: "A chord is a segment inside a circle."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q4/5

Proposed definition: "A diameter is a segment that passes through the center of a circle."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSG-CO.A.1

FORWARD 2

Write precise definitions of circle parts from the definition of a circle • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of a chord of a circle.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q1/5

Write a precise definition of a diameter of a circle with center O.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q2/5

Segment PQ is a diameter of a circle with center O. Write a precise definition of a semicircle with endpoints P and Q.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q3/5

Which is a precise definition of the interior of a circle with center O and radius r?

A. All points of the plane whose distance from O is at most r

B. All points of the circle

C. All points whose distance from O is less than r

D. All points of the plane of the circle whose distance from O is less than r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q4/5

Two circles in the same plane have the same center. Write a precise definition of this pair ("concentric circles") and explain why two different concentric circles never meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q5/5

Precise definition of an angle

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	See an angle as two rays with a common endpoint, measured by…	**4.MD.C.5.a**
BACKWARD 2	Identify rays and their endpoints	**4.G.A.1**
FORWARD 1	Define angle relationships from the definition of an angle	**HSG-CO.A.1**
FORWARD 2	Prove angle facts from the angle definitions	**HSG-CO.C.9**

SET 97 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of an angle. Name its vertex and its sides in your definition.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q1/5

Sam writes: "An angle is the space between two lines."

Explain two things that are wrong, and rewrite it as a precise definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q2/5

Jo writes: "An angle is two rays."

Give a pair of rays that fits Jo's wording but is not an angle, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q3/5

A 1° angle is defined using distance around a circular arc. Complete the definition, and explain why the measure of an angle does not depend on the radius of the circle used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q4/5

Rays QP and QR have the common endpoint Q, and ray RS starts at R. Using the definition, explain which pair of rays forms an angle and name the angle, and which pair does not.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • MAIN • Q5/5

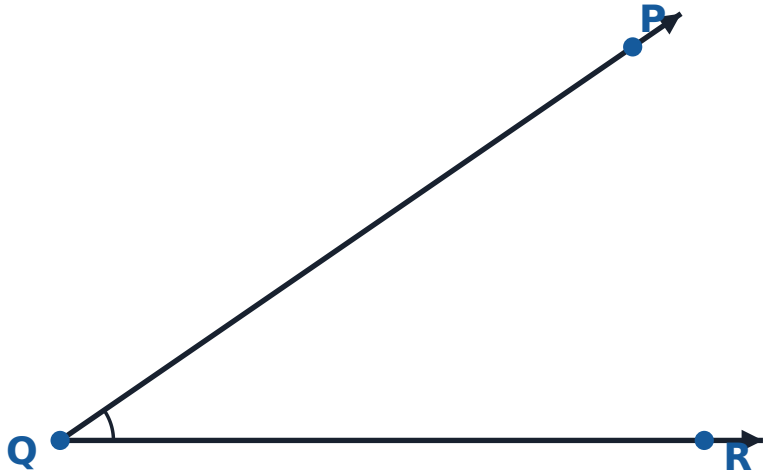
SET 97 • HSG-CO.A.1

BACKWARD 1

See an angle as two rays with a common endpoint, measured by an arc • Grade 4 • Measurement & Data • 4.MD.C.5.a

Name the vertex and the two sides of ∠PQR.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q1/5

An angle has its vertex at the center of a circle and cuts off $\frac{1}{4}$ of the circle. What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q2/5

An angle measures 60°. What fraction of a circle centered at its vertex does it cut off?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q3/5

True or false?

The measure of an angle depends on how long its sides are drawn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q4/5

True or false?

A one-degree angle turns through $\frac{1}{360}$ of a full circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 97 • BACKWARD 1 • Q5/5

SET 97 • HSG-CO.A.1

BACKWARD 2

Identify rays and their endpoints • Grade 4 • Geometry • 4.G.A.1

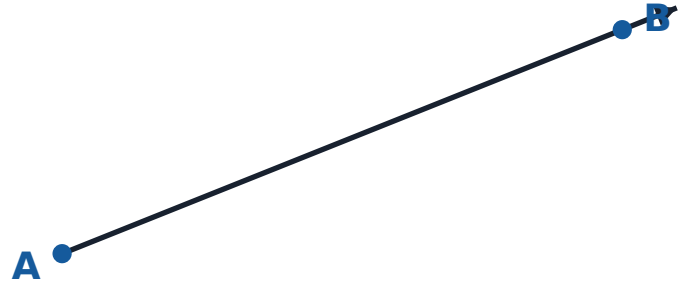
What is shown in the figure?

A. Segment AB

B. Ray BA

C. Line AB

D. Ray AB



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q1/5

How many endpoints does a ray have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q2/5

Which figure goes on without end in exactly one direction?

A. A line segment

B. A point

C. A ray

D. A line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q3/5

True or false?

Ray AB and ray BA are the same ray.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q4/5

True or false?

Rays QA and QB share the endpoint Q.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 97 • BACKWARD 2 • Q5/5

SET 97 • HSG-CO.A.1

FORWARD 1

Define angle relationships from the definition of an angle • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of adjacent angles.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q1/5

Write a precise definition of vertical angles.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q2/5

Write a precise definition of a linear pair of angles.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q3/5

Proposed definition: "Two angles are vertical angles if they have the same vertex and are not adjacent."

Give a counterexample and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q4/5

Which is a precise definition of the bisector of ∠ABC?

A. Any line through B

B. A ray from B that lies outside ∠ABC

C. Ray BD, where D is in the interior of ∠ABC and m∠ABD = m∠DBC

D. The segment that joins A and C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 97 • FORWARD 1 • Q5/5

SET 97 • HSG-CO.A.1

FORWARD 2

Prove angle facts from the angle definitions • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

$\angle 1$ and $\angle 2$ form a linear pair, and $\angle 2$ and $\angle 3$ form a linear pair, where $\angle 1$ and $\angle 3$ are vertical angles. Prove that $m\angle 1 = m\angle 3$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q1/5

Ray BD bisects ∠ABC and m∠ABD = (2x + 10)°, m∠DBC = (4x − 20)°. Find m∠ABC.

m∠ABC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q2/5

∠PQR and ∠RQS form a linear pair, and m∠PQR is 4 times m∠RQS. Find both angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q3/5

Rays OA and OB are opposite rays, and ray OC is not on line AB. Ray OD bisects ∠AOC and ray OE bisects ∠COB. Prove that m∠DOE = 90°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q4/5

True or false?

If two angles are adjacent and supplementary, their noncommon sides are opposite rays.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S97-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 97 • FORWARD 2 • Q5/5

SET 98 • HSG-CO.A.1 • CONGRUENCE: DEFINITIONS AND TRANSFORMATIONS

Precise definition of a line segment

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Identify points, lines, rays, and segments	**4.G.A.1**
BACKWARD 2	Find distances between points on a number line	**6.NS.C.8**
FORWARD 1	Define figures built from segments	**HSG-CO.A.1**
FORWARD 2	Test segment-based definitions with counterexamples	**HSG-CO.A.1**

SET 98 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Points A and B lie on line ℓ. Which is a precise definition of segment AB?

A. Point A and every point of line ℓ on the same side of A as B

B. Every point of the plane that is closer to A than to B

C. Every point of line ℓ

D. Points A and B together with every point C of line ℓ for which AC + CB = AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q1/5

Write a precise definition of segment AB using the word "between".

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q2/5

Ravi writes: "A segment is part of a line."

Give a part of a line that is not a segment, and rewrite the definition precisely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q3/5

Points A, B, and C are on one line with AC = 3, CB = 5, and AB = 8. Is C on segment AB? Use the definition to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q4/5

Points A, B, and D are on one line with AD = 3, DB = 11, and AB = 8. Is D on segment AB? Use the definition to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • MAIN • Q5/5

SET 98 • HSG-CO.A.1

BACKWARD 1

Identify points, lines, rays, and segments • Grade 4 • Geometry • 4.G.A.1

How many endpoints does a line segment have? How many does a ray have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q1/5

True or false?

A line segment has exactly two endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q2/5

Which figure has two endpoints?

A. A ray

B. A line

C. An angle

D. A segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q3/5

True or false?

A line has two endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q4/5

Which name describes the part of line AB from A to B, including A and B?

A. Ray BA

B. Line AB

C. Segment AB

D. Ray AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 98 • BACKWARD 1 • Q5/5

SET 98 • HSG-CO.A.1

BACKWARD 2

Find distances between points on a number line • Grade 6 • The Number System • 6.NS.C.8

On a number line, A is at 2 and B is at 9. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q1/5

On a number line, A is at −3 and C is at 4. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q2/5

On a number line, A is at 1, C is at 4, B is at 10. Which is true?

A. AB < AC

B. AC + CB > AB

C. AC + CB = AB

D. AC = CB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q3/5

True or false?

On a number line, the point at 5 is between the points at 2 and 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q4/5

True or false?

On a number line, the point at 11 is between the points at 2 and 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 98 • BACKWARD 2 • Q5/5

SET 98 • HSG-CO.A.1

FORWARD 1

Define figures built from segments • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of ray AB. Use the word "between".

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q1/5

Write a precise definition of the midpoint of segment AB.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q2/5

Write a precise definition of a triangle using segments.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q3/5

Write a precise definition of congruent segments.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q4/5

Write a precise definition of a segment bisector.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 1 • Q5/5

SET 98 • HSG-CO.A.1

FORWARD 2

Test segment-based definitions with counterexamples • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Proposed definition: "Point M is the midpoint of segment AB if AM = MB."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q1/5

Proposed definition: "A triangle is three segments."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q2/5

Proposed definition: "C is between A and B if AC + CB = AB."

Is this enough when A, B, and C are any three points of a plane? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q3/5

True or false?

"The midpoint of AB is a point M with AM = MB" is precise enough, because every such point lies on segment AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q4/5

Proposed definition: "A segment bisector is a line that cuts a segment into two parts."

Give a counterexample and repair it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S98-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 98 • FORWARD 2 • Q5/5

Precise definition of perpendicular lines

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Recognize perpendicular lines and right angles	**4.G.A.1**
BACKWARD 2	Find the other angles when two lines cross	**7.G.B.5**
FORWARD 1	Prove consequences of the definition of perpendicular lines	**HSG-CO.C.9**
FORWARD 2	Reason about perpendicular lines with a third line, in a plane and…	**HSG-CO.C.9**

SET 2 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Ben writes: "Perpendicular lines are lines that make a T."

Explain what is imprecise, and write a precise definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q1/5

Write a precise definition of perpendicular lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q2/5

Lines a and b intersect at P. Two angles at P form a linear pair and have equal measures.

Use the definition to decide whether a and b are perpendicular. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q3/5

Line m passes through (0, 0) and (2, 1). Line n passes through (0, 0) and (−1, 2). Use the definition (intersect and form a right angle) to decide whether m and n are perpendicular. Show how you know the angle at (0, 0) is a right angle.

Answer: _____

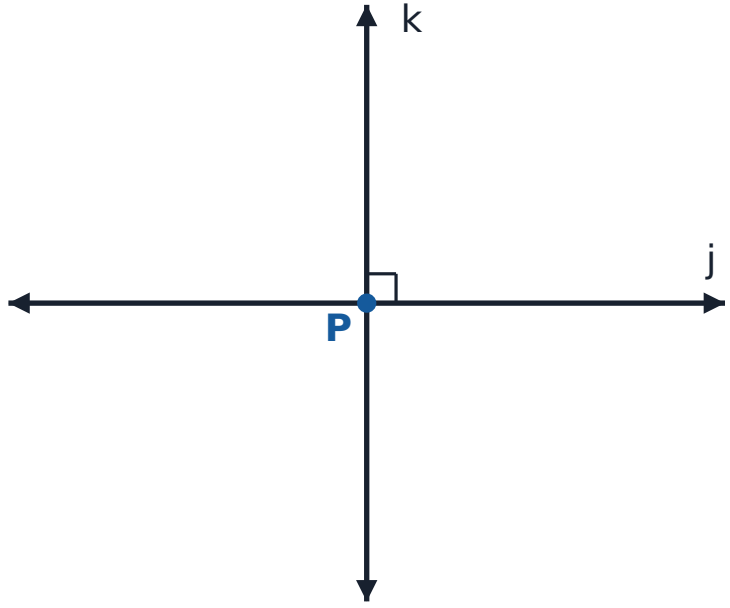
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q4/5

Lines j and k meet at point P, and one of the angles they form is marked as a right angle.

Are j and k perpendicular? Use the definition to explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q5/5

SET 2 • HSG-CO.A.1

BACKWARD 1

Recognize perpendicular lines and right angles • Grade 4 • Geometry • 4.G.A.1

True or false?

Lines that meet at a 90° angle are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q1/5

True or false?

Any two lines that cross are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q2/5

Two perpendicular lines cross. How many right angles do they form?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q3/5

What is true of two sides of a rectangle that share a vertex?

A. They are perpendicular

B. They are parallel

C. They never meet

D. They form a 60° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q4/5

Which pair of streets is most like perpendicular lines?

A. Two straight streets that never meet

B. Two streets that curve around a park

C. Two streets that meet at a sharp point

D. Two streets that cross at a square corner

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

BACKWARD 2

Find the other angles when two lines cross • Grade 7 • Geometry • 7.G.B.5

Two lines cross. One of the angles measures 90°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q1/5

Two lines cross. One of the angles measures 70°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q2/5

True or false?

Vertical angles formed by two crossing lines are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q3/5

Two angles form a linear pair. One measures 90°. What does the other measure?

A. 180°

B. 45°

C. 90°

D. 270°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q4/5

True or false?

Two lines cross and one angle measures 100°. The angle next to it also measures 100°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSG-CO.A.1

FORWARD 1

Prove consequences of the definition of perpendicular lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are perpendicular and meet at point P. Prove that all four angles at P are right angles.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q1/5

Two lines meet at point P and form a linear pair of congruent angles. Prove that the lines are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q2/5

Lines a and b intersect, and one of the angles they form measures (3x + 15)°. For what value of x are a and b perpendicular? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q3/5

Two intersecting lines form a pair of vertical angles that are supplementary. Prove that the lines are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q4/5

Lines p and q meet at O. Ana says: "Two adjacent angles at O add to 180°, so p ⊥ q." Explain why her reason does not prove perpendicularity, and state a fact that does.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

FORWARD 2

Reason about perpendicular lines with a third line, in a plane and in space • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

In a plane, lines a and b are both perpendicular to line t. Prove that a and b are parallel.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q1/5

True or false?

In three-dimensional space, two lines that are each perpendicular to the same line must be parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q2/5

In a plane, line a is parallel to line b, and line t is perpendicular to line a. What is true about t and b?

A. t and b do not meet

B. t is perpendicular to b

C. t is parallel to b

D. The angle between t and b cannot be known

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q3/5

In a coordinate plane, line t is the x-axis. Line a is perpendicular to t at (0, 0), and line b is perpendicular to t and passes through (5, 2). Write an equation of line b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q4/5

True or false?

In a plane, through a point on a line there is exactly one line perpendicular to that line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q5/5

Precise definition of parallel lines

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Recognize parallel lines	**4.G.A.1**
BACKWARD 2	Decide whether two lines in the coordinate plane intersect	**8.EE.C.8**
BACKWARD 3	Visualize lines and planes in a box	**7.G.A.3**
FORWARD 1	Prove that lines are parallel from angle facts	**HSG-CO.C.9**
FORWARD 2	Use the parallel postulate	**HSG-CO.C.9**

SET 99 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of parallel lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q1/5

Lines ℓ and m do not intersect. Which additional fact makes them parallel?

A. They lie in the same plane

B. They both look horizontal in a drawing

C. They are drawn the same length

D. They are both perpendicular to one segment in space

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q2/5

Eli writes: "Parallel lines are lines that never meet."

Give an example of two lines that fit Eli's sentence but are not parallel, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q3/5

Box ABCD-EFGH has top face ABCD directly above bottom face EFGH (A above E, B above F, C above G, D above H). Using the definition, decide whether line AB is parallel to line HG, and whether line AB is parallel to line FG. Explain each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q4/5

Line p is $y = 2x + 1$ and line q is $y = 2x - 3$. Use the definition to decide whether p and q are parallel. Show that they do not intersect.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 99 • MAIN • Q5/5

SET 99 • HSG-CO.A.1

BACKWARD 1

Recognize parallel lines • Grade 4 • Geometry • 4.G.A.1

True or false?

Two lines in a plane that never meet are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q1/5

What is true of the opposite sides of a rectangle?

A. They are parallel

B. They are perpendicular

C. They meet at one point

D. They are neither parallel nor perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q2/5

True or false?

Any two lines that look slanted the same amount in a drawing are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q3/5

Which pair of streets is most like parallel lines?

A. Two streets that meet at a corner

B. Two streets that cross

C. A street and a bridge above it going another way

D. Two straight streets on flat ground that never meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q4/5

How many times do two parallel lines intersect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 99 • BACKWARD 1 • Q5/5

SET 99 • HSG-CO.A.1

BACKWARD 2

Decide whether two lines in the coordinate plane intersect • Grade 8 • Expressions & Equations • 8.EE.C.8

Do the lines $y = x + 2$ and $y = x - 1$ intersect? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q1/5

Do the lines $y = 2x$ and $y = -x + 6$ intersect? If so, where?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q2/5

Which line never meets $y = 3x + 1$?

A. $y = x + 1$

B. $y = 3$

C. $y = -3x + 1$

D. $y = 3x - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q3/5

True or false?

Two different lines with the same slope never intersect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q4/5

True or false?

The lines $y = 2x + 5$ and $y = 5$ never meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8 • Set 99 • BACKWARD 2 • Q5/5

SET 99 • HSG-CO.A.1

BACKWARD 3

Visualize lines and planes in a box • Grade 7 • Geometry • 7.G.A.3

True or false?

Any two edges of a box that never meet are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B3-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q1/5

In a box, the top front edge and the bottom back edge (running the same direction) are

A. perpendicular

B. skew

C. parallel

D. the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q2/5

How many faces of a box are there?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q3/5

Which pair of box edges are skew?

A. A top front edge and the bottom front edge below it

B. A top front edge and a bottom side edge that runs front to back

C. Two top edges that meet at a corner

D. Two vertical edges

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q4/5

True or false?

Two vertical edges of a box are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 99 • BACKWARD 3 • Q5/5

SET 99 • HSG-CO.A.1

FORWARD 1

Prove that lines are parallel from angle facts • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by transversal t, and a pair of alternate interior angles are congruent. Explain why ℓ and m cannot meet (so they are parallel).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q1/5

Lines ℓ and m are cut by a transversal. A pair of corresponding angles measure $(5x - 10)°$ and $(3x + 20)°$. For what x are ℓ and m parallel?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q2/5

Same-side interior angles formed by lines a, b, and a transversal measure 115° and 65°. Are a and b parallel? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q3/5

In a plane, line a ∥ line b and line b ∥ line c, where a and c are different lines. Prove that a ∥ c.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q4/5

Which fact proves that two coplanar lines cut by a transversal are parallel?

A. A linear pair adds to 180°

B. Two angles are acute

C. A pair of alternate interior angles are congruent

D. A pair of vertical angles are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 1 • Q5/5

SET 99 • HSG-CO.A.1

FORWARD 2

Use the parallel postulate • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

In a plane, through a point not on a line there is exactly one line parallel to the given line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q1/5

Point P is not on line ℓ. Line m passes through P and is parallel to ℓ. Line n also passes through P and is parallel to ℓ. What can you conclude about m and n? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q2/5

In the coordinate plane, find an equation of the line through (2, 5) parallel to $y = -3x + 1$, and explain why it is the only one.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q3/5

Which statement is the parallel postulate?

A. Vertical angles are congruent

B. All right angles are congruent

C. Through any two points there is exactly one line

D. Through a point not on a line there is exactly one parallel to the line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q4/5

Why does the parallel postulate say "a point not on the line"?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S99-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 99 • FORWARD 2 • Q5/5

Transformations as functions: inputs, outputs, and preimages

MAIN	Geometry main question	**HSG-CO.A.2**
BACKWARD 1	Evaluate a rule for a given input	**8.F.A.1**
BACKWARD 2	Apply coordinate rules for translations, reflections, and rotations	**8.G.A.3**
BACKWARD 3	Read and locate ordered pairs in all four quadrants	**6.NS.C.6.c**
FORWARD 1	Compose two transformations and write one rule	**HSG-CO.A.2**
FORWARD 2	Find the inverse of a transformation	**HSG-CO.A.2**

SET 3 • HSG-CO.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

The transformation T is defined by T(x, y) = (x + 4, y − 3). Find T(2, 5).

T(2, 5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q1/5

The transformation R is defined by $R(x, y) = (-y, x)$. Find $R(3, -1)$.

$R(3, -1) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q2/5

The transformation F takes each point (x, y) to (x, −y). The output point is F(P) = (4, 7). Find the input point P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q3/5

Every point of the plane is moved 5 units right and 2 units up. Which function describes this transformation?

A. $T(x, y) = (x - 5, y - 2)$

B. $T(x, y) = (5x, 2y)$

C. $T(x, y) = (x + 2, y + 5)$

D. $T(x, y) = (x + 5, y + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q4/5

The transformation D is defined by D(x, y) = (2x, 2y). Find every input point P whose output is itself (D(P) = P). Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 1

Evaluate a rule for a given input • Grade 8 • Functions • 8.F.A.1

f(x) = 3x − 2. Find f(4).

f(4) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q1/5

A rule takes a number, doubles it, and adds 1. What is the output for the input 6?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q2/5

g(x) = x + 5. Which input gives the output 12?

Input: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

A function assigns exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

The rule "the input 3 gives the outputs 4 and 5" describes a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 2

Apply coordinate rules for translations, reflections, and rotations • Grade 8 • Geometry • 8.G.A.3

Translate the point (2, −1) 3 units left and 4 units up. Where does it go?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q1/5

Reflect the point (5, 2) across the y-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q2/5

Rotate the point (4, 1) by 180° about the origin.

A. (1, 4)

B. (−1, 4)

C. (4, −1)

D. (−4, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q3/5

True or false?

Reflecting the point (3, 6) across the x-axis gives (3, −6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q4/5

True or false?

Rotating the point (0, 5) by 90° counterclockwise about the origin gives (5, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

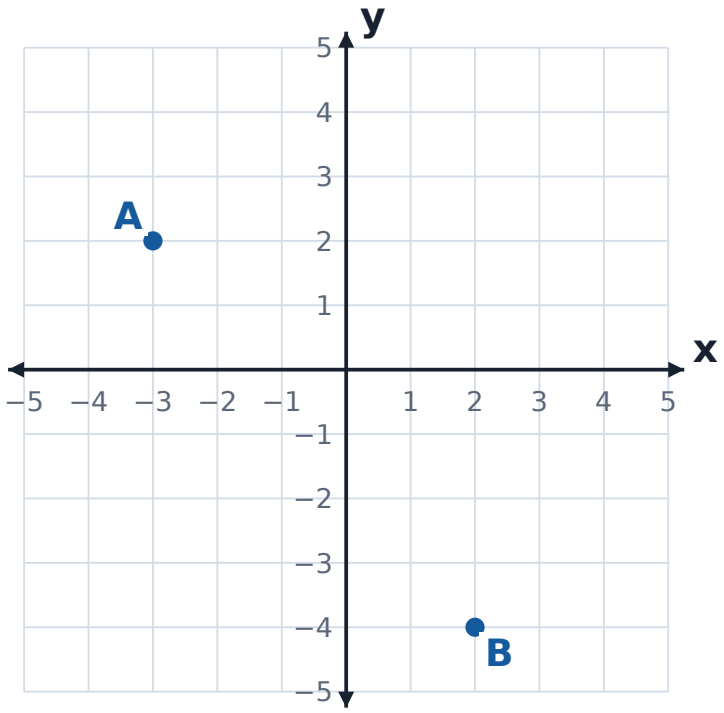
Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 3

Read and locate ordered pairs in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?



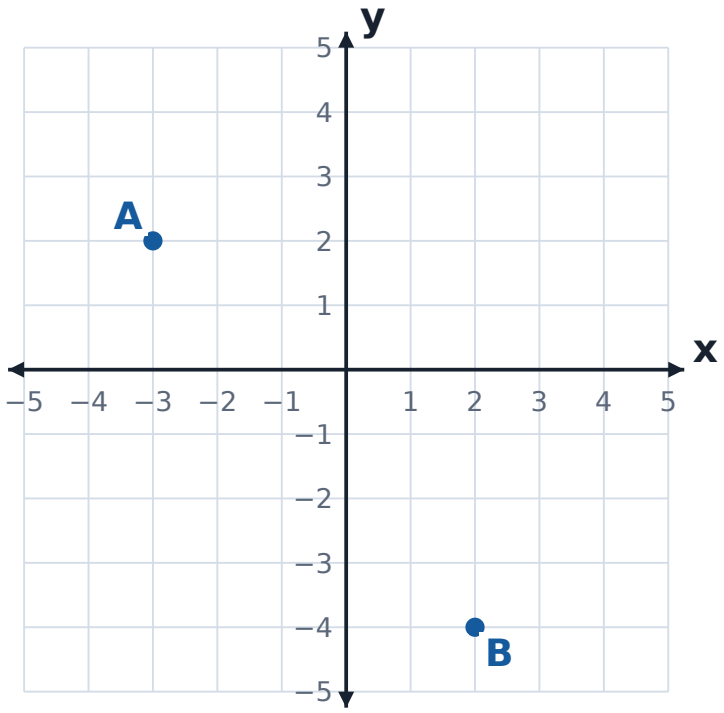
A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q1/5

What are the coordinates of point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q2/5

Which point is in Quadrant III?

A. $(-2, -5)$

B. $(-2, 5)$

C. $(2, 5)$

D. $(2, -5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q3/5

True or false?

The point (0, −3) lies on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q4/5

True or false?

The point $(-2, 5)$ is 2 units to the right of the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSG-CO.A.2

FORWARD 1

Compose two transformations and write one rule • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Find $R(T(1, 4))$.

$R(T(1, 4)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q1/5

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Write a single rule for $R(T(x, y))$.

$R(T(x, y)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q2/5

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Find $T(R(1, 4))$.

$T(R(1, 4)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q3/5

True or false?

For any two transformations, doing them in either order gives the same result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q4/5

$T(x, y) = (x + 3, y - 1)$ and $S(x, y) = (x - 1, y + 5)$. Which rule is $S(T(x, y))$?

A. $(x + 2, y + 4)$

B. $(x - 3, y - 5)$

C. $(x + 2, y - 4)$

D. $(x + 4, y - 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

FORWARD 2

Find the inverse of a transformation • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Find a rule for the inverse of $T(x, y) = (x - 6, y + 2)$.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q1/5

$R(x, y) = (-y, x)$ is a 90° counterclockwise rotation about the origin. Find a rule for its inverse.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q2/5

True or false?

Every rotation is its own inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q3/5

D(x, y) = (3x, 3y). Which rule undoes D?

A. $(x - 3, y - 3)$

B. $(3y, 3x)$

C. $(-3x, -3y)$

D. $(\frac{x}{3}, \frac{y}{3})$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q4/5

True or false?

The transformation P(x, y) = (x, 0), which moves every point straight up or down to the x-axis, has an inverse transformation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q5/5

Compare transformations that preserve distance and angle with those that do not

MAIN	Geometry main question	**HSG-CO.A.2**
BACKWARD 1	Find the distance between two points	**8.G.B.8**
BACKWARD 2	Know that rigid motions keep lengths and angle measures	**8.G.A.1**
FORWARD 1	Classify transformations as rigid motions, similarity transformatio…	**HSG-SRT.A.2**
FORWARD 2	Prove with variables that a transformation preserves or scales ev…	**HSG-SRT.A.2**

SET 4 • HSG-CO.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Which transformation does NOT preserve distance?

A. $R(x, y) = (-x, y)$

B. $T(x, y) = (x + 4, y - 1)$

C. $H(x, y) = (2x, y)$

D. $Q(x, y) = (-y, x)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q1/5

H(x, y) = (2x, y). Points A(0, 0) and B(3, 4) are 5 units apart.

Find H(A), H(B), and the distance between them. Does H preserve distance?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q2/5

The translation $T(x, y) = (x - 2, y + 5)$ maps $A(1, 1)$ and $B(4, 5)$ to A' and B'. Show that $A'B' = AB$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q3/5

$V(x, y) = (x, 3y)$. Does V preserve the measure of every angle? Use the right angle at $O(0, 0)$ between the rays through $(1, 1)$ and $(1, -1)$ to decide.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q4/5

S(x, y) = (3x, 3y). Does S preserve distance? Does S preserve angle measure? Explain each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q5/5

SET 4 • HSG-CO.A.2

BACKWARD 1

Find the distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q1/5

Find the distance between $(-2, 3)$ and $(4, 3)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q2/5

Find the distance between (0, 0) and (5, 12).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q3/5

True or false?

The distance from (0, 0) to (1, 1) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q4/5

Find the distance between (2, −1) and (5, 3).

A. 25

B. $\sqrt{7}$

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

BACKWARD 2

Know that rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

True or false?

A rotation keeps every segment the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q1/5

True or false?

A reflection can change the measure of an angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q2/5

A triangle is translated. What changes?

A. Its angle measures

B. Its side lengths

C. The positions of its vertices

D. Its area

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q3/5

A segment 7 cm long is reflected across a line. How long is its image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q4/5

A 35° angle is rotated 90° about a point. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSG-CO.A.2

FORWARD 1

Classify transformations as rigid motions, similarity transformations, or neither • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Which transformation preserves angle measure but not distance?

A. $(x, y) \rightarrow (4x, y)$

B. $(x, y) \rightarrow (x + 4, y)$

C. $(x, y) \rightarrow (x, y + 4)$

D. $(x, y) \rightarrow (4x, 4y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q1/5

Classify $(x, y) \to (-2x, -2y)$ as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q2/5

Classify $(x, y) \to (x + y, y)$ as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q3/5

True or false?

Every similarity transformation is a rigid motion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q4/5

True or false?

A transformation that multiplies every distance by the same number k > 0 always changes angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

FORWARD 2

Prove with variables that a transformation preserves or scales every distance • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Prove that $T(x, y) = (x + a, y + b)$ preserves the distance between ANY two points (x_1, y_1) and (x_2, y_2).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q1/5

Prove that D(x, y) = (3x, 3y) multiplies the distance between ANY two points by 3.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q2/5

$H(x, y) = (2x, y)$. The distance between the images of (x_1, y_1) and (x_2, y_2) is $\sqrt{(4(x_2 - x_1)^2 + (y_2 - y_1)^2)}$.

For which pairs of points is the distance unchanged?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q3/5

True or false?

One example in which a distance stays the same is enough to prove that a transformation preserves every distance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q4/5

True or false?

If a transformation preserves the distance between every pair of points, it also preserves every angle measure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q5/5

Rotations and reflections that carry a rectangle, parallelogram, or trapezoid onto itself

MAIN	Geometry main question	**HSG-CO.A.3**
BACKWARD 1	Identify lines of symmetry of figures	**4.G.A.3**
BACKWARD 2	Know the side and angle properties of parallelograms, rectangles,…	**5.G.B.3**
BACKWARD 3	Rotate points by 90° and 180° about the origin	**8.G.A.3**
FORWARD 1	Use a symmetry to deduce congruent parts of the figure	**HSG-CO.B.6**
FORWARD 2	Combine symmetries and identify a figure from its symmetries	**HSG-CO.A.5**

SET 5 • HSG-CO.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Rectangle ABCD is not a square. Describe every rotation (with angle greater than 0° and at most 360°) and every reflection that carries the rectangle onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q1/5

Parallelogram PQRS has no right angles, and its sides are not all the same length. Describe every rotation (angle greater than 0° and at most 360°, about any point) and every reflection that carries it onto itself.

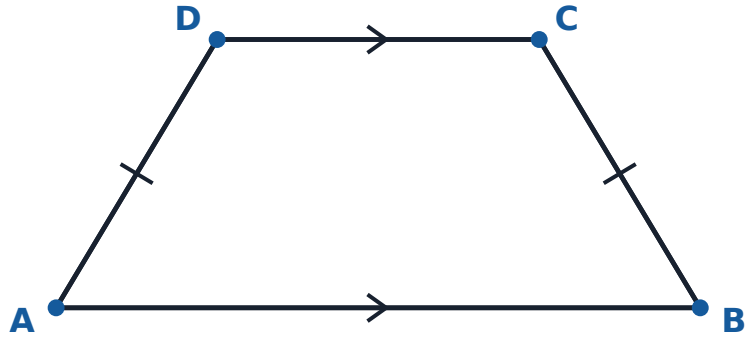
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q2/5

Trapezoid ABCD has exactly one pair of parallel sides, the bases AB and DC (AB ≠ DC), and congruent legs AD = BC. Describe every rotation (angle greater than 0° and at most 360°, about any point) and every reflection that carries it onto itself.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q3/5

Rectangle ABCD has A(1, 1), B(7, 1), C(7, 4), D(1, 4). Give the center of every rotation that carries it onto itself (angle greater than 0° and at most 360°), and the equation of every line of reflection that carries it onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q4/5

Trapezoid EFGH has exactly one pair of parallel sides, EF ∥ HG, with EF = 4, HG = 8, and legs FG = 5 and EH = 3. Which rotations (angle greater than 0° and at most 360°, about any point) and which reflections carry it onto itself?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 1

Identify lines of symmetry of figures • Grade 4 • Geometry • 4.G.A.3

How many lines of symmetry does a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q1/5

How many lines of symmetry does a rectangle that is not a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q2/5

True or false?

Each diagonal of a rectangle that is not a square is a line of symmetry of the rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q3/5

Which figure has exactly one line of symmetry?

A. An isosceles triangle that is not equilateral

B. A square

C. A circle

D. A scalene triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q4/5

True or false?

An isosceles trapezoid has a line of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 2

Know the side and angle properties of parallelograms, rectangles, and trapezoids • Grade 5 • Geometry • 5.G.B.3

True or false?

In every parallelogram, the diagonals have equal lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q1/5

True or false?

Every rectangle is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q2/5

Which property do all rectangles have?

A. Diagonals that are perpendicular

B. Four sides of equal length

C. Four right angles

D. Exactly one pair of parallel sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q3/5

A parallelogram has one angle of 70°. Find its other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q4/5

An isosceles trapezoid has a 65° angle at one end of its longer base. What is the angle at the other end of that base?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 3

Rotate points by 90° and 180° about the origin • Grade 8 • Geometry • 8.G.A.3

Rotate the point (3, 1) by 180° about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q1/5

Rotate the point (2, 0) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q2/5

True or false?

A 180° rotation about the origin sends (x, y) to (−x, −y).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q3/5

Rotate the point $(-4, 2)$ by 90° counterclockwise about the origin.

A. $(4, -2)$

B. $(-4, -2)$

C. $(2, 4)$

D. $(-2, -4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q4/5

True or false?

A rotation of 360° about a point moves every point to a new position.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSG-CO.A.3

FORWARD 1

Use a symmetry to deduce congruent parts of the figure • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Isosceles trapezoid ABCD (AB ∥ DC) is carried onto itself by the reflection across the line through the midpoints of AB and DC. Which vertex does A go to? Use this to name a pair of congruent angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q1/5

A 180° rotation about point O carries parallelogram PQRS onto itself, sending P to R and Q to S. Name two pairs of congruent sides that follow.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q2/5

True or false?

Because a reflection carries an isosceles trapezoid onto itself, its two diagonals are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q3/5

A reflection carries rectangle ABCD onto itself and swaps vertices A and B (and C and D). Which equality follows?

A. AB = BC

B. AC = AB

C. AD = AB

D. AD = BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q4/5

True or false?

If a 180° rotation about a point carries a quadrilateral onto itself, the quadrilateral must be a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

FORWARD 2

Combine symmetries and identify a figure from its symmetries • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Rectangle ABCD (not a square) has center O. Reflect it across the line through the midpoints of AB and CD, then across the line through the midpoints of BC and DA. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q1/5

A quadrilateral is carried onto itself by exactly two transformations: the 360° rotation and one reflection across a line through two of its vertices. Which could it be?

A. A parallelogram that is not a rectangle or a rhombus

B. A square

C. A rectangle that is not a square

D. A kite that is not a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q2/5

True or false?

A quadrilateral that is carried onto itself by a 180° rotation about a point must be a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q3/5

A quadrilateral is carried onto itself by a 180° rotation but by no reflection. What is the most specific name for it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q4/5

True or false?

Doing the same reflection twice gives the same result as the 360° rotation (the identity).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q5/5

Rotations and reflections that carry a regular polygon onto itself

MAIN	Geometry main question	**HSG-CO.A.3**
BACKWARD 1	Find the angle of a fraction of a full turn	**4.MD.C.5.b**
BACKWARD 2	Recognize regular polygons	**5.G.B.3**
BACKWARD 3	Find lines of symmetry of regular polygons	**4.G.A.3**
FORWARD 1	Work backward from a symmetry to the regular polygon	**HSG-CO.A.3**
FORWARD 2	Combine symmetries of a regular polygon	**HSG-CO.A.5**

SET 6 • HSG-CO.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Describe every rotation about its center, with angle greater than 0° and at most 360°, that carries a regular pentagon onto itself.

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q1/5

How many reflections carry a regular hexagon onto itself? Describe their lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q2/5

Describe every rotation about its center, with angle greater than 0° and at most 360°, that carries a regular octagon onto itself.

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q3/5

How many reflections carry a regular nonagon (9 sides) onto itself? Describe their lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q4/5

Describe every reflection that carries an equilateral triangle onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 1

Find the angle of a fraction of a full turn • Grade 4 • Measurement & Data • 4.MD.C.5.b

How many degrees are in $\frac{1}{5}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q1/5

How many degrees are in $\frac{1}{8}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q2/5

How many degrees are in $\frac{3}{6}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{12}$ of a full turn is 30°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q4/5

True or false?

$\frac{1}{9}$ of a full turn is 36°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 2

Recognize regular polygons • Grade 5 • Geometry • 5.G.B.3

Which is a regular polygon?

A. A right triangle

B. A rhombus that is not a square

C. A rectangle that is not a square

D. An equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q1/5

True or false?

A polygon with all sides equal must be regular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q2/5

How many sides does a regular decagon have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q3/5

What is the name of the regular polygon with 4 sides?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q4/5

True or false?

A regular hexagon has 6 equal angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 3

Find lines of symmetry of regular polygons • Grade 4 • Geometry • 4.G.A.3

How many lines of symmetry does an equilateral triangle have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q1/5

How many lines of symmetry does a regular pentagon have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q2/5

True or false?

Every line of symmetry of a regular polygon passes through its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q3/5

A line of symmetry of a square passes through

A. two adjacent vertices

B. exactly one vertex and no midpoint

C. no point of the square

D. two opposite vertices, or the midpoints of two opposite sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q4/5

True or false?

A regular hexagon has exactly 3 lines of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSG-CO.A.3

FORWARD 1

Work backward from a symmetry to the regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

The smallest positive rotation that carries a regular polygon onto itself is 24°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q1/5

A regular polygon has 20 lines of symmetry. What is the smallest positive rotation that carries it onto itself?

Rotation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A regular polygon can have 50° as its smallest positive rotation onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q3/5

Which can be the smallest positive rotation that carries a regular polygon onto itself?

A. 35°

B. 70°

C. 80°

D. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q4/5

Counting the 360° rotation, how many rotations and reflections in all carry a regular 12-gon onto itself?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSG-CO.A.3

FORWARD 2

Combine symmetries of a regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Regular hexagon ABCDEF is labeled counterclockwise. It is rotated 120° counterclockwise about its center. Which vertex position does A move to?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q1/5

Regular hexagon ABCDEF is labeled counterclockwise. Rotate it 60° counterclockwise about its center, then reflect it across line AD (through the original positions of A and D). Which original vertex position does vertex B end at?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q2/5

True or false?

For a regular pentagon, a reflection followed by a different reflection that carries it onto itself is a rotation that carries it onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q3/5

Square ABCD is labeled counterclockwise. Reflect it across line AC, then rotate it 90° counterclockwise about its center. Which original vertex position does vertex B end at?

A. The position of C

B. The position of A

C. The position of D

D. The position of B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q4/5

Two successive 72° rotations of a regular pentagon about its center equal which single transformation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q5/5

Definition of a reflection

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Find the midpoint of a horizontal or vertical segment	**6.NS.C.8**
BACKWARD 2	Recognize perpendicular lines on a grid	**4.G.A.1**
BACKWARD 3	Reflect points with coordinate rules	**8.G.A.3**
FORWARD 1	Find lines of reflection and images in slanted, non-axis cases	**HSG-CO.A.4**
FORWARD 2	Prove from the definition that reflections keep lengths	**HSG-CO.A.4**

SET 7 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Complete a precise definition. A reflection across line ℓ maps each point P not on ℓ to the point P′ such that _____. Each point on ℓ maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q1/5

Noa writes: "A reflection across ℓ flips each point over ℓ."

Explain what is imprecise, and write a precise definition.

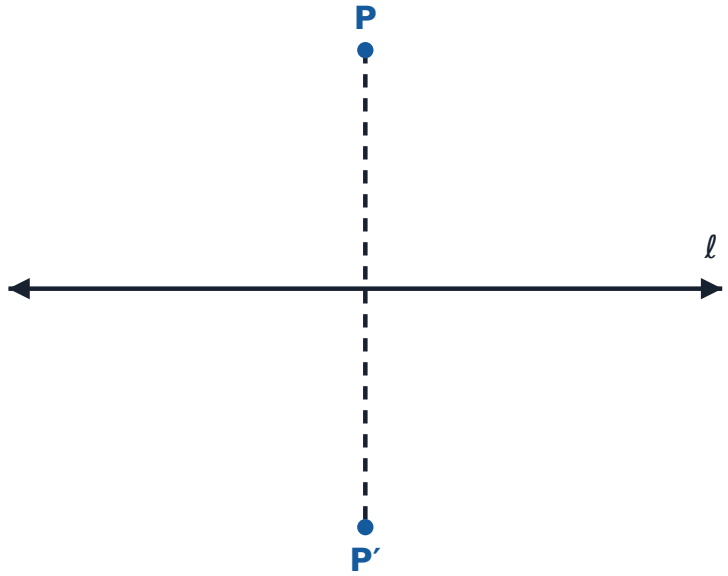
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q2/5

Point P′ is the image of point P under a reflection across line ℓ, and P is not on ℓ. Which must be true?



A. ℓ is perpendicular to PP′ and passes through its midpoint

B. ℓ passes through P

C. PP′ is parallel to ℓ

D. P and P′ are on the same side of ℓ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q3/5

Line ℓ is y = 1. Use the definition to check that the reflection across ℓ maps P(2, 5) to P′(2, −3): test both conditions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q4/5

A reflection maps P(1, 2) to P′(5, 2). Use the definition to find the line of reflection.

Line: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 1

Find the midpoint of a horizontal or vertical segment • Grade 6 • The Number System • 6.NS.C.8

Find the midpoint of the segment from (2, 4) to (8, 4).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from (1, 1) to (1, 9).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q2/5

On a number line, what number is halfway between −3 and 7?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q3/5

True or false?

The midpoint of the segment from (−4, 0) to (6, 0) is (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q4/5

True or false?

The midpoint of a segment is closer to one endpoint than to the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 2

Recognize perpendicular lines on a grid • Grade 4 • Geometry • 4.G.A.1

True or false?

A horizontal line and a vertical line are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q1/5

True or false?

Any two segments of the same length lie on parallel lines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q2/5

Which pair of segments lie on perpendicular lines?

A. From (0, 0) to (4, 0) and from (2, −1) to (2, 3)

B. From (1, 1) to (2, 3) and from (1, 1) to (3, 2)

C. From (0, 0) to (3, 1) and from (0, 2) to (3, 3)

D. From (0, 0) to (3, 1) and from (0, 0) to (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q3/5

A segment goes from (2, 1) to (2, 6). Describe a segment through (5, 3) whose line is perpendicular to the line containing the first segment.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q4/5

Which describes two lines that are both perpendicular to the x-axis?

A. They are parallel or the same line

B. They are perpendicular to each other

C. They meet at the origin

D. They always meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 3

Reflect points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Reflect the point (4, −3) across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q1/5

Reflect the point $(-1, 5)$ across the y-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q2/5

Reflect the point (2, 7) across the line y = x.

A. (7, 2)

B. (−7, −2)

C. (−2, 7)

D. (2, −7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q3/5

True or false?

Reflecting the point $(-6, 1)$ across the y-axis gives $(6, 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q4/5

True or false?

Reflecting (3, 3) across the x-axis gives (−3, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSG-CO.A.4

FORWARD 1

Find lines of reflection and images in slanted, non-axis cases • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A reflection maps P(0, 0) to P′(4, 2). Find an equation of the line of reflection.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q1/5

A reflection maps A(−2, 3) to A′(4, 3). Where does it map B(0, −1)?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q2/5

True or false?

A reflection maps P to P′ and Q to Q′, where P and Q are not on the line of reflection. Then PP′ is parallel to QQ′ (or on the same line).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q3/5

A reflection maps A(1, 3) to A′(3, 1). Is the line of reflection y = x? Use the definition to decide.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q4/5

A reflection maps P(−1, 4) to P′(5, 4). Which point is on the line of reflection?

A. (−1, 0)

B. (5, 0)

C. (2, −3)

D. (4, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

FORWARD 2

Prove from the definition that reflections keep lengths • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Points P and Q are on the same side of line ℓ, and line PQ is not perpendicular to ℓ. A reflection across ℓ maps them to P′ and Q′. Let PP′ meet ℓ at M and QQ′ meet ℓ at N. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q1/5

Point P is on line ℓ and point Q is not. The reflection across ℓ maps Q to Q′. Prove that PQ = PQ′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q2/5

Using coordinates with ℓ as the x-axis, the reflection is $(x, y) \to (x, -y)$. Prove that it keeps the distance between any two points (x_1, y_1) and (x_2, y_2).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q3/5

True or false?

Under a reflection, a point not on the line of reflection and its image are on the same side of the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q4/5

In a proof that a reflection across ℓ keeps the length of segment PQ, which fact is used first?

A. ℓ is the perpendicular bisector of PP′ and of QQ′

B. ℓ bisects $\angle PQP'$

C. ℓ is parallel to PQ

D. ℓ passes through P

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q5/5

Definition of a translation

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Recognize parallel segments on a grid	**4.G.A.1**
BACKWARD 2	Translate points with coordinate rules	**8.G.A.3**
FORWARD 1	Use the definition with points and their images	**HSG-CO.A.4**
FORWARD 2	Prove from the definition that translations keep lengths	**HSG-CO.A.4**

SET 100 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Points A and B are different. Write a precise definition of the translation that maps A to B.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S100-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q1/5

Tia writes: "A translation slides a figure."

Explain what is imprecise, and write a precise definition of the translation that maps A to B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q2/5

The translation that maps A(1, 2) to B(4, 6) maps P(−2, 0) to P′. Use the definition (same length, parallel, same direction) to find P′, and check the length.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q3/5

A translation maps A to A′ and B to B′. Explain, using the definition, why AA′ and BB′ are parallel (or on one line) and have the same length.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q4/5

Is $(x, y) \to (x + 3, y - 4)$ a translation? Use the definition to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • MAIN • Q5/5

SET 100 • HSG-CO.A.4

BACKWARD 1

Recognize parallel segments on a grid • Grade 4 • Geometry • 4.G.A.1

True or false?

Two segments that each go 3 right and 1 up lie on parallel lines (or on one line).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q1/5

True or false?

Any two segments of the same length are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q2/5

Which pair of segments is parallel?

A. From (0, 0) to (3, 1) and from (0, 2) to (3, 3)

B. From (0, 0) to (4, 0) and from (0, 0) to (0, 4)

C. From (1, 1) to (2, 3) and from (1, 1) to (3, 2)

D. From (0, 0) to (3, 1) and from (0, 0) to (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q3/5

A segment goes from (1, 1) to (4, 3). Give the other endpoint of a parallel segment of the same length that starts at (0, 5).

Endpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q4/5

Which describes two lines that are both perpendicular to the x-axis?

A. They are parallel or the same line

B. They are perpendicular to each other

C. They meet at the origin

D. They always meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 100 • BACKWARD 1 • Q5/5

SET 100 • HSG-CO.A.4

BACKWARD 2

Translate points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Translate the point $(-1, 5)$ 6 units right and 2 units down.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q1/5

Translate the point $(3, -2)$ 4 units left.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q2/5

A translation moves (0, 0) to (2, −3). Where does it move (1, 1)?

A. (3, 4)

B. (2, −3)

C. (−1, 4)

D. (3, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q3/5

True or false?

Translating (3, 3) by 2 units left gives (1, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q4/5

True or false?

Translating (3, 3) by 2 units left gives (3, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 100 • BACKWARD 2 • Q5/5

SET 100 • HSG-CO.A.4

FORWARD 1

Use the definition with points and their images • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A translation maps $A(2, -1)$ to $A'(-3, 4)$. Where does it map $B(0, 6)$?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q1/5

A translation maps P to P′(4, 1) and moves every point 2 left and 3 up. Find P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q2/5

True or false?

Under a translation, all the segments that join points to their images are parallel (or on one line) and have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q3/5

A translation maps triangle ABC with A(0, 0), B(2, 0), C(0, 3) so that C goes to (5, 7). Find the images of A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q4/5

A translation maps (1, 2) to (4, 0). Which point maps to (0, 0)?

A. (−4, 0)

B. (−3, 2)

C. (3, −2)

D. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 1 • Q5/5

SET 100 • HSG-CO.A.4

FORWARD 2

Prove from the definition that translations keep lengths • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A translation maps P to P′ and Q to Q′, where Q is not on line PP′. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q1/5

True or false?

A translation maps P to P′ and Q to Q′. Then PP′Q′Q is always a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q2/5

Prove that $T(x, y) = (x + a, y + b)$ maps every nonvertical line $y = mx + k$ to a line with the same slope m (so the image is parallel to the line or is the line itself).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q3/5

A translation maps P to P′ and Q to Q′, where Q IS on line PP′. Prove that PQ = P′Q′ in this case too.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q4/5

In the proof that a translation keeps the length of PQ (Q not on line PP′), which figure is used?

A. Rectangle PQRS

B. A circle centered at P

C. Parallelogram PP′Q′Q

D. Triangle PQQ′ only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S100-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 100 • FORWARD 2 • Q5/5

Definition of a rotation

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Describe turns in degrees, clockwise and counterclockwise	**4.MD.C.5.b**
BACKWARD 2	Points on a circle are the same distance from its center	**7.G.B.4**
BACKWARD 3	Rotate points 90°, 180°, and 270° about the origin	**8.G.A.3**
FORWARD 1	Rotate about a center that is not the origin	**HSG-CO.A.4**
FORWARD 2	Find the center and angle of a rotation from points and their imag..	**HSG-CO.A.4**

SET 8 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Complete a precise definition. A counterclockwise rotation about point O through angle θ maps each point P ≠ O to the point P′ such that _____ and _____. Point O maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q1/5

A rotation about point O maps point P to P′ (P ≠ P′). Which must be true?

A. O is the midpoint of PP′

B. O is on the perpendicular bisector of PP′

C. OP′ = 2 · OP

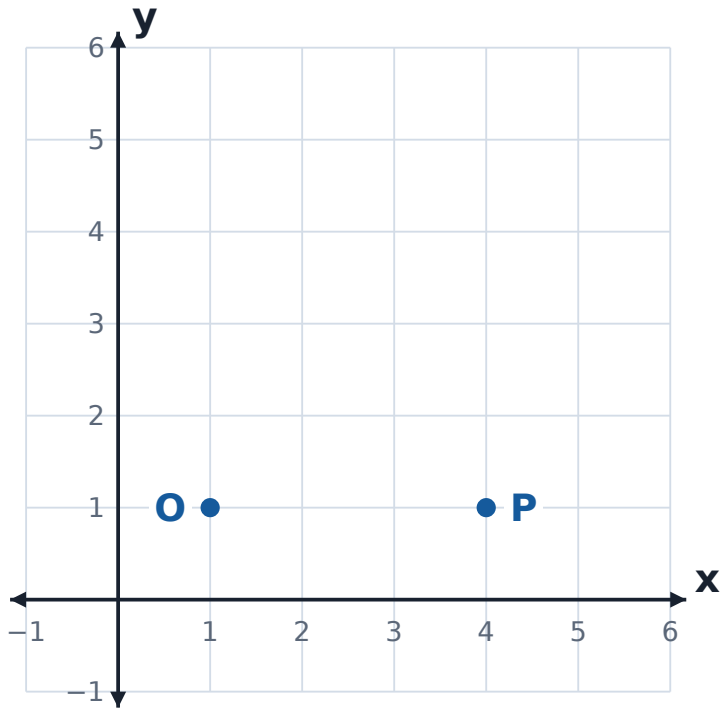
D. PP′ is perpendicular to OP

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q2/5

A rotation of 90° counterclockwise about O(1, 1) maps P(4, 1) to P′. Use the definition to find P′.



P′ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q3/5

A rotation about point O through 70° maps a point P ≠ O to P′. Use the definition to explain why P′ lies on the circle with center O through P, and why O is the only point that the rotation leaves where it is.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q4/5

A rotation maps P(2, 0) to P′(0, 2). Explain why its center must lie on the line y = x.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 1

Describe turns in degrees, clockwise and counterclockwise • Grade 4 • Measurement & Data • 4.MD.C.5.b

A ray points right. It turns 90° counterclockwise. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q1/5

A ray points up. It turns 180°. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q2/5

True or false?

A clockwise turn of 270° ends in the same direction as a counterclockwise turn of 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q3/5

A ray points up. It turns 90° clockwise. Which way does it point now?

A. Down

B. Left

C. Right

D. Up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q4/5

How many degrees are in $\frac{3}{4}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 2

Points on a circle are the same distance from its center • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 6. Point P is on the circle. What is OP?

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q1/5

True or false?

If OA = OB, then A and B lie on one circle with center O.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q2/5

Which point is on the circle with center (0, 0) and radius 5?

A. (4, 4)

B. (3, 4)

C. (2, 3)

D. (5, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q3/5

A circle has center O and diameter 10 cm. How far is every point of the circle from O?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q4/5

True or false?

Two points the same distance from O are always the same point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 3

Rotate points 90°, 180°, and 270° about the origin • Grade 8 • Geometry • 8.G.A.3

Rotate (5, 2) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q1/5

Rotate $(-3, 4)$ by $180°$ about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q2/5

Rotate (1, 6) by 90° clockwise about the origin.

A. (−6, 1)

B. (1, −6)

C. (−1, −6)

D. (6, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q3/5

True or false?

Rotating (2, 3) by 270° counterclockwise about the origin gives (3, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q4/5

True or false?

Rotating (4, 0) by 90° counterclockwise about the origin gives (0, −4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSG-CO.A.4

FORWARD 1

Rotate about a center that is not the origin • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Rotate P(5, 4) by 90° clockwise about C(2, 3).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q1/5

Rotate P(1, 5) by 180° about C(−1, 2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q2/5

Rotate P(3, −2) by 90° counterclockwise about C(0, −2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q3/5

Rotate P(1, 4) by 270° counterclockwise about C(1, 1).

A. (4, 4)

B. (4, 1)

C. (1, −2)

D. (−2, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q4/5

True or false?

Rotating P(4, 0) by 90° counterclockwise about C(2, 0) gives the same image as rotating P by 90° counterclockwise about the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

FORWARD 2

Find the center and angle of a rotation from points and their images • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A rotation maps B(1, 0) to B′(0, 1) and A(3, 1) to A′(−1, 3). The center is where the perpendicular bisectors of BB′ and AA′ meet. Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q1/5

A rotation maps P(4, 1) to P′(2, 3) and Q(4, 3) to Q′(0, 3). Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q2/5

True or false?

The center of a nonzero rotation moves to a new position under the rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q3/5

A rotation about C maps P to P′. $CP = 5$ and $PP' = 5\sqrt{2}$. Which could be the angle of rotation?

A. 60°

B. 90°

C. 45°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q4/5

A rotation about the origin maps (5, 0) to (3, 4). What is the counterclockwise angle of rotation, to the nearest degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q5/5

Draw the image of a figure under a rotation, reflection, or translation

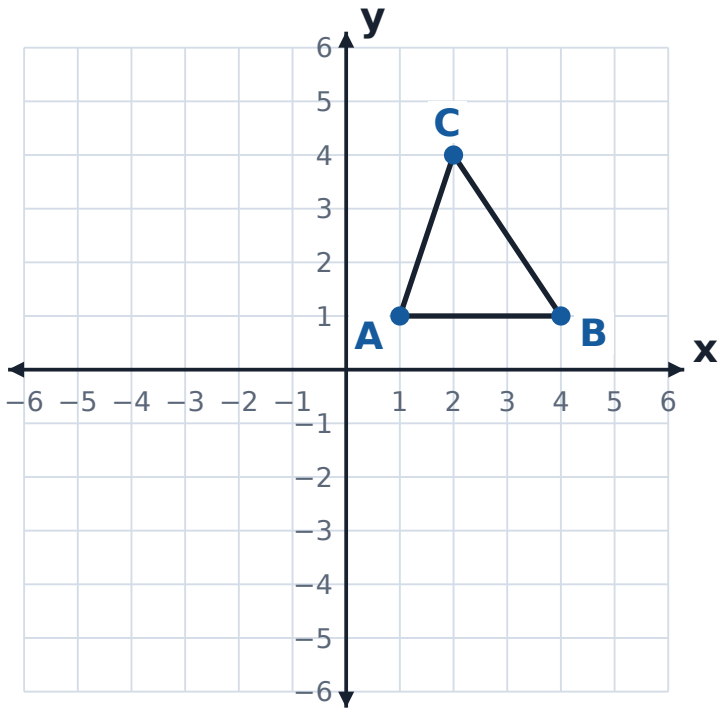
MAIN	Geometry main question	**HSG-CO.A.5**
BACKWARD 1	Plot and read points in all four quadrants	**6.NS.C.6.c**
BACKWARD 2	Find image points with coordinate rules	**8.G.A.3**
BACKWARD 3	Count the distance from a point to a horizontal or vertical line	**6.NS.C.8**
FORWARD 1	Draw the image under a sequence of two transformations	**HSG-CO.A.5**
FORWARD 2	Draw rotations and reflections off the grid with compass, protract…	**HSG-CO.A.5**

SET 9 • HSG-CO.A.5

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Draw the image of triangle ABC under the translation 5 units left and 2 units down. Label the image A′B′C′.

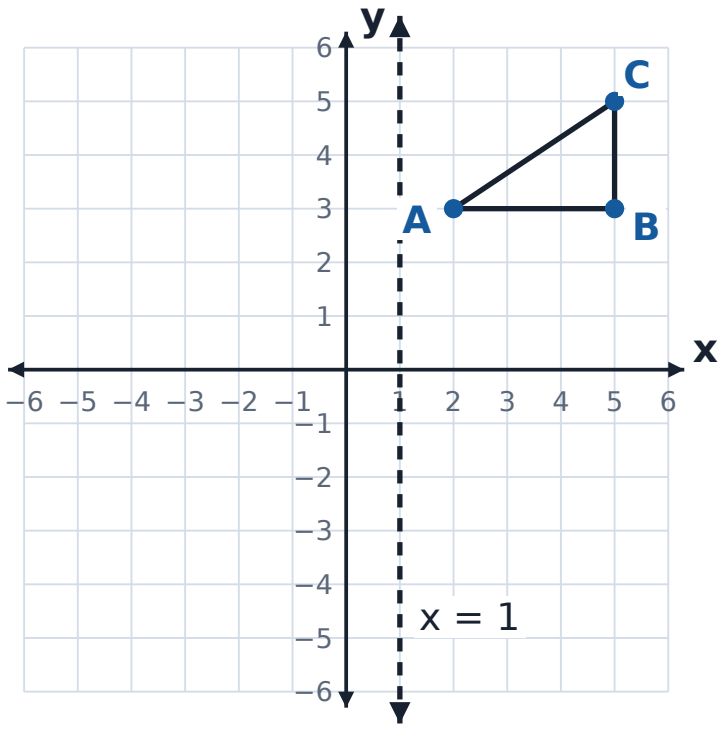


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q1/5

Draw the image of triangle ABC under the reflection across the line x = 1. Label the image A′B′C′.

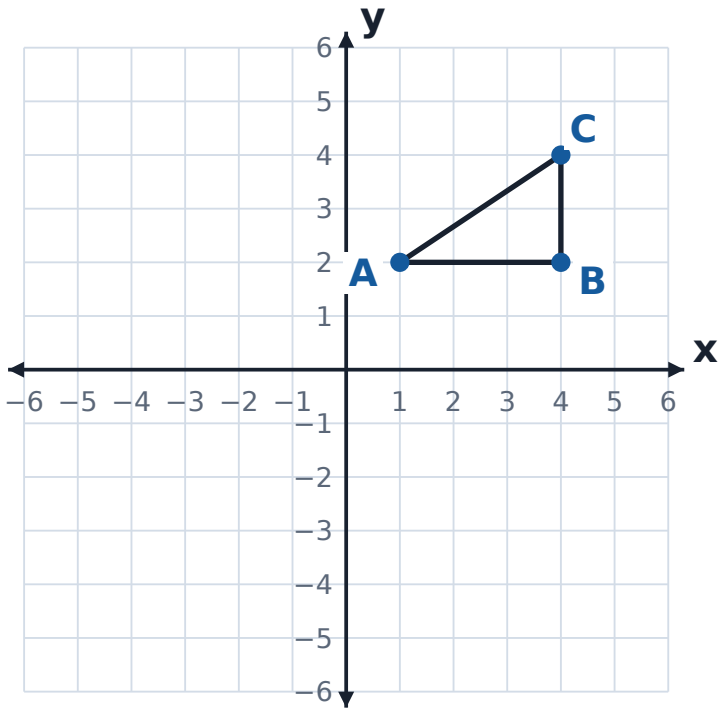


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q2/5

Draw the image of triangle ABC under the rotation of 90° counterclockwise about the origin. Label the image A′B′C′.

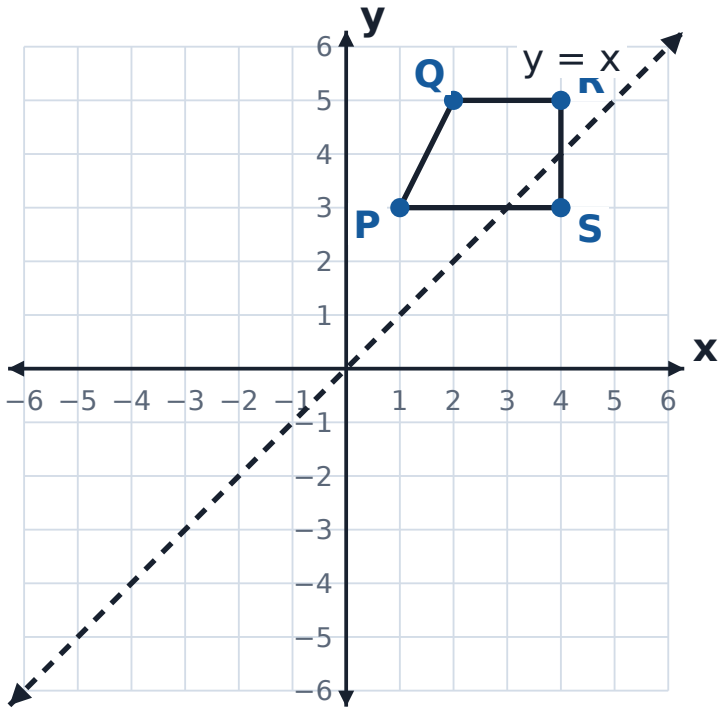


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q3/5

Draw the image of quadrilateral PQRS under the reflection across the line $y = x$. Label the image $P'Q'R'S'$.

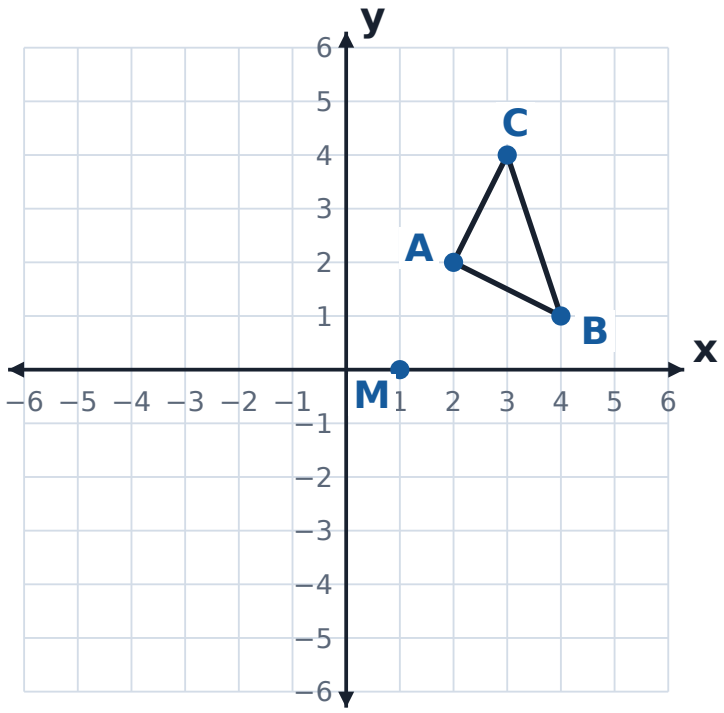


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q4/5

Draw the image of triangle ABC under the rotation of 180° about the point M(1, 0). Label the image A′B′C′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

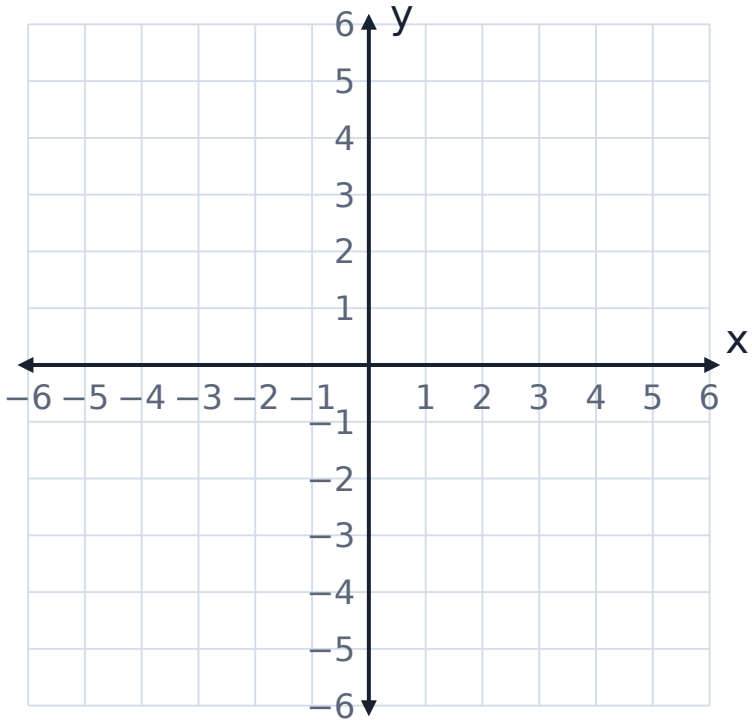
High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 1

Plot and read points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

Plot and label the points A(−3, 2), B(0, −4), and C(5, 1).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q1/5

A point is 4 units left of the y-axis and 1 unit below the x-axis. What are its coordinates?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q2/5

Which point is 2 units above the x-axis and on the y-axis?

A. (2, 2)

B. (0, −2)

C. (0, 2)

D. (2, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q3/5

True or false?

The point (3, −2) is in Quadrant IV.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q4/5

True or false?

To plot (2, 5), start at the origin and move 2 units up and 5 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 2

Find image points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Find the image of $(3, -2)$ under a translation 4 units left and 1 unit up.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q1/5

Find the image of $(-2, 5)$ under a reflection across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q2/5

Find the image of (3, 1) under a 90° clockwise rotation about the origin.

A. (−3, −1)

B. (3, −1)

C. (1, −3)

D. (−1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q3/5

True or false?

The image of $(4, -1)$ under a reflection across the y-axis is $(-4, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q4/5

True or false?

The image of (2, 2) under a 180° rotation about the origin is (2, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 3

Count the distance from a point to a horizontal or vertical line • Grade 6 • The Number System • 6.NS.C.8

How far is the point (5, 3) from the vertical line x = 1?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q1/5

How far is the point $(-2, -1)$ from the horizontal line $y = 3$?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q2/5

Which point is 3 units from the line x = 2, on its left?

A. (−3, 4)

B. (−1, 4)

C. (2, 1)

D. (5, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q3/5

True or false?

The point (0, 4) is 4 units from the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q4/5

True or false?

The point (6, 2) is 6 units from the line $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

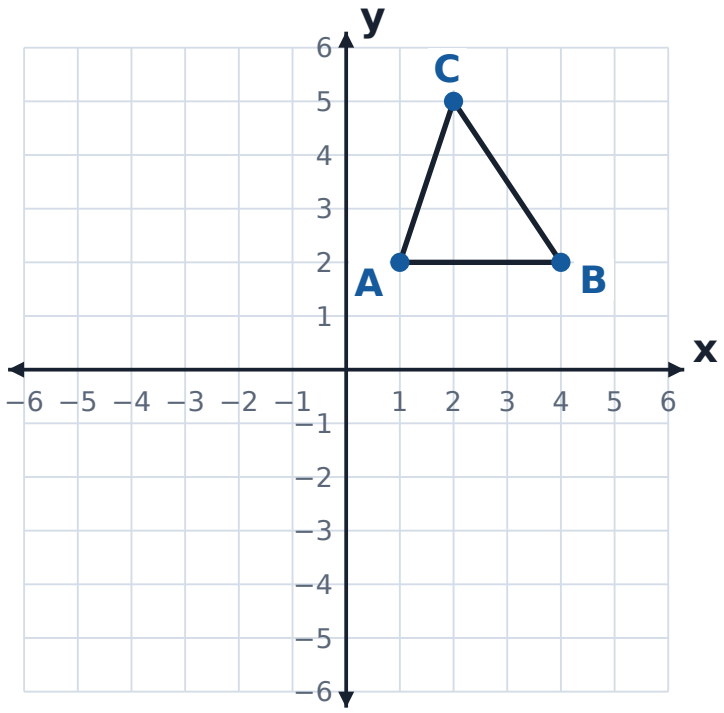
Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSG-CO.A.5

FORWARD 1

Draw the image under a sequence of two transformations • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Reflect triangle ABC across the y-axis, then translate the result 3 units down. Draw and label the final image A″B″C″.

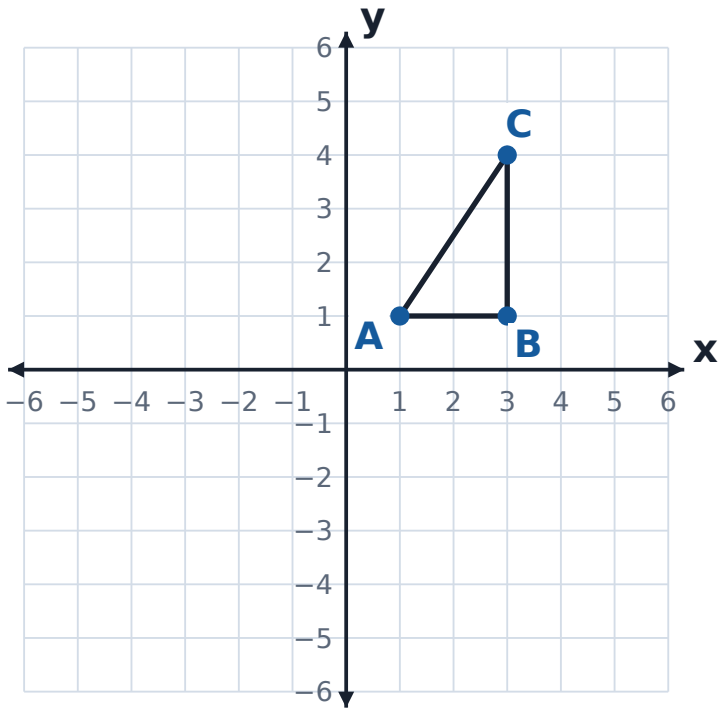


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q1/5

Rotate triangle ABC by 90° counterclockwise about the origin, then reflect the result across the x-axis. Draw and label the final image A″B″C″.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q2/5

Triangle ABC has A(2, 1). It is translated 2 units right and then rotated 180° about the origin. Where does A end up?

A″ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q3/5

True or false?

Reflecting a figure across the x-axis and then across the y-axis gives the same image as reflecting it across the line y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q4/5

Point P(1, 3) is translated 2 units left and then reflected across the line y = x. Where does it end up?

A. (3, 1)

B. (−1, 3)

C. (1, −3)

D. (3, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

FORWARD 2

Draw rotations and reflections off the grid with compass, protractor, and straightedge • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

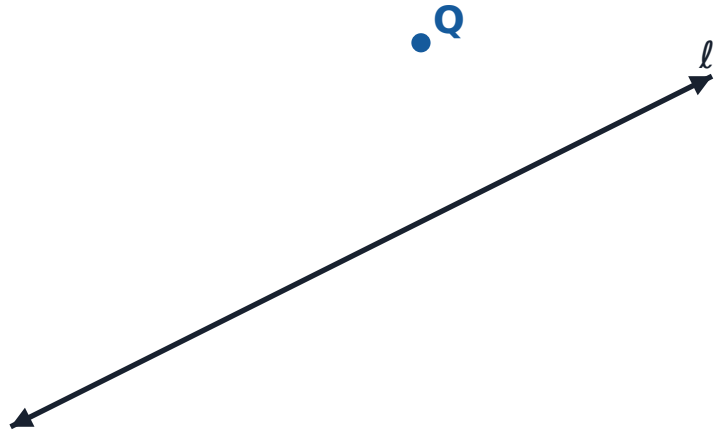
Use a compass and protractor. Rotate point P 60° counterclockwise about point O. Mark and label the image P′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q1/5

Use a compass and straightedge. Reflect point Q across line ℓ. Mark and label the image Q′.

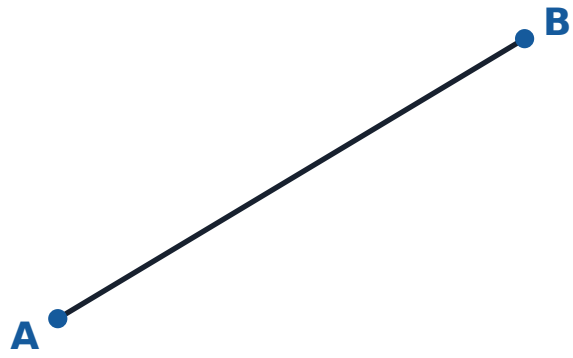


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q2/5

Use a compass and protractor. Rotate segment AB 90° clockwise about point A. Draw and label the image AB′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q3/5

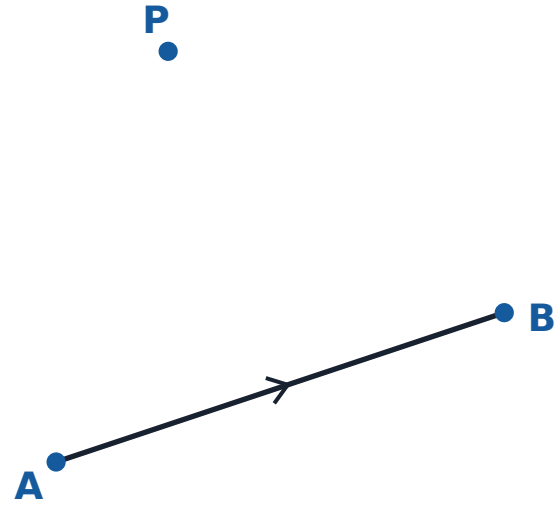
Use a compass and straightedge. Rotate point P 180° about point O. Mark and label the image P′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q4/5

Use a compass and straightedge. The translation maps A to B. Draw the image P′ of point P (P is not on line AB).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q5/5

Specify a sequence of transformations that carries one figure onto another

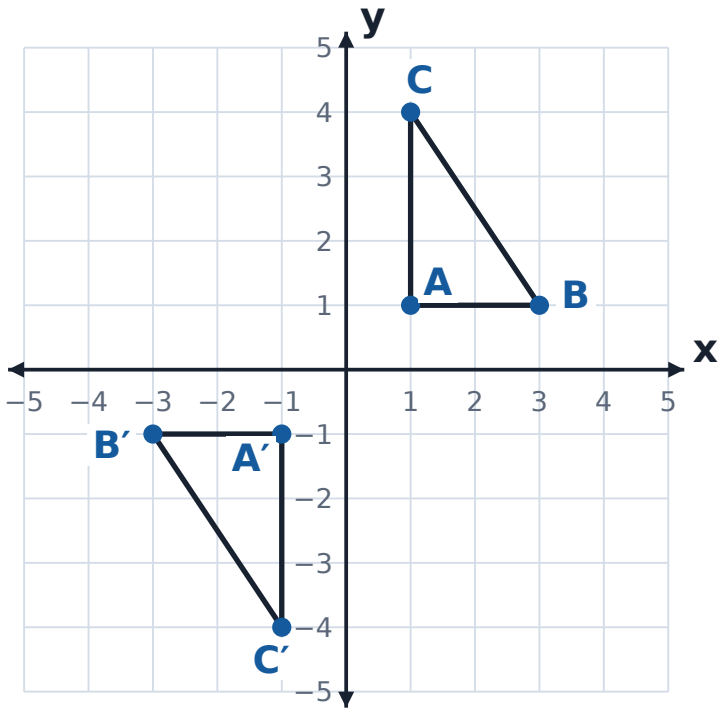
MAIN	Geometry main question	**HSG-CO.A.5**
BACKWARD 1	Understand congruence as the result of one rigid motion	**8.G.A.2**
BACKWARD 2	Find images with coordinate rules for sequences	**8.G.A.3**
BACKWARD 3	Read the orientation (clockwise or counterclockwise order) of a tr…	**8.G.A.1**
FORWARD 1	Use the fewest transformations and justify	**HSG-CO.A.5**
FORWARD 2	Specify a similarity sequence: a dilation and rigid motions	**HSG-SRT.A.2**

SET 10 • HSG-CO.A.5

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′ (A to A′, B to B′, C to C′).



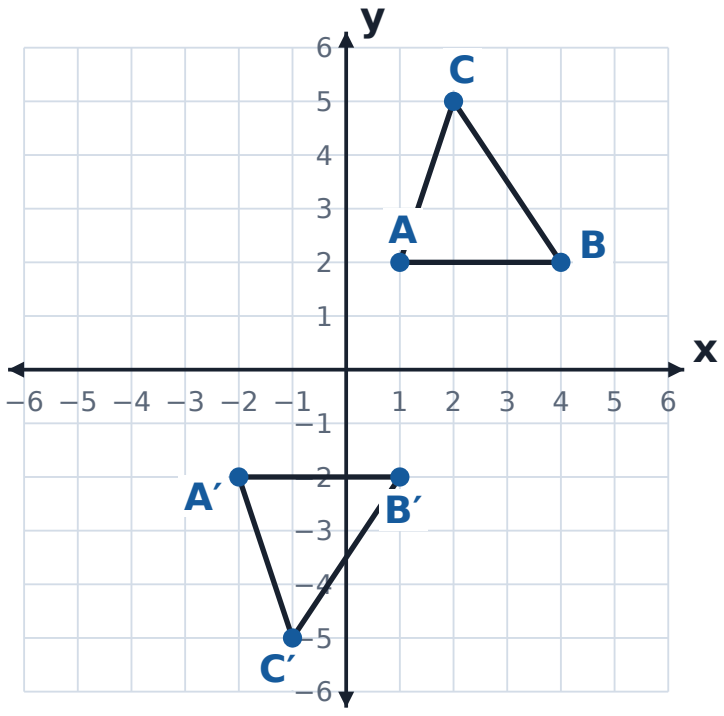
Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q1/5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′.



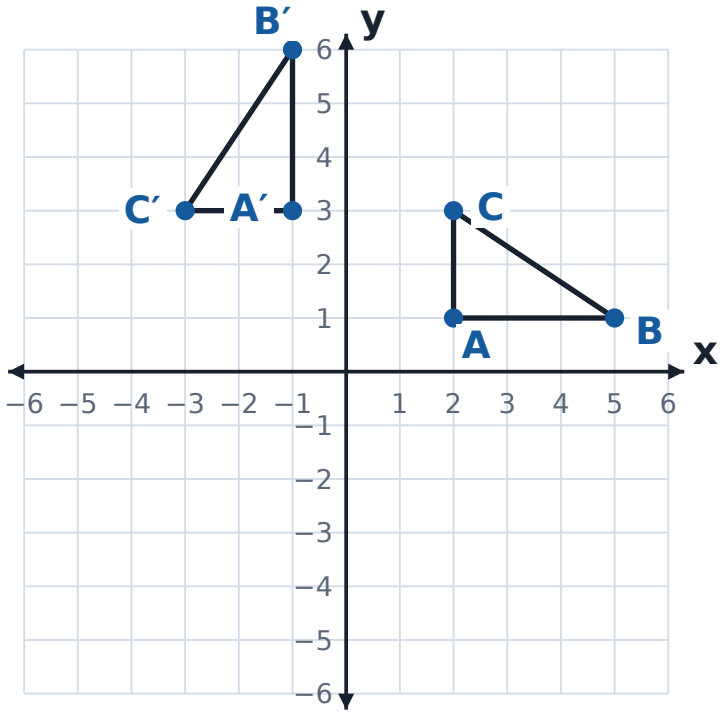
Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q2/5

On the grid, triangle A′B′C′ is congruent to triangle ABC. Describe a sequence of rigid transformations that carries ABC onto A′B′C′, with A → A′, B → B′, C → C′.



Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q3/5

Right triangle ABC has A(0, 0), B(2, 0), C(0, 3). Triangle A′B′C′ has A′(4, 0), B′(4, 2), C′(1, 0). Which sequence carries ABC onto A′B′C′?

A. Reflect across the y-axis, then translate 4 units right

B. Rotate 90° counterclockwise about the origin, then translate 4 units right

C. Rotate 90° clockwise about the origin, then translate 4 units right

D. Translate 4 units right, then rotate 90° counterclockwise about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q4/5

Triangle PQR has P(1, 1), Q(4, 1), R(1, 3). Triangle P′Q′R′ has P′(1, −1), Q′(4, −1), R′(1, −3). Describe a sequence of rigid transformations that carries PQR onto P′Q′R′, and explain why it must include a reflection.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 1

Understand congruence as the result of one rigid motion • Grade 8 • Geometry • 8.G.A.2

Triangle 2 is the image of triangle 1 after one rigid motion. Every point moved 3 units right. Which motion was it?

A. A reflection

B. A translation

C. A rotation

D. A dilation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q1/5

A triangle and its image face opposite ways, as in a mirror. Which kind of rigid motion was most likely used?

Motion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q2/5

True or false?

If one figure is the image of another under a sequence of rigid motions, the figures are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q3/5

True or false?

A figure and its image under a rigid motion can have different areas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q4/5

Segment AB goes from (0, 0) to (2, 0). Its image goes from (0, 0) to (0, 2). Which single rigid motion could do this?

A. A translation 2 units up

B. A reflection across the x-axis

C. A 90° counterclockwise rotation about the origin

D. A 180° rotation about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 2

Find images with coordinate rules for sequences • Grade 8 • Geometry • 8.G.A.3

Point (2, −3) is reflected across the y-axis. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q1/5

Point $(-1, 4)$ is rotated 90° counterclockwise about the origin. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q2/5

Point (5, 1) is translated 2 units left and 3 units down. Find its image.

A. (3, −2)

B. (7, 4)

C. (7, −2)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q3/5

True or false?

Point (0, 3) rotated 180° about the origin is (0, −3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q4/5

True or false?

Point (4, 1) reflected across the line $y = x$ is $(4, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 3

Read the orientation (clockwise or counterclockwise order) of a triangle • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has A(0, 0), B(4, 0), C(0, 3). Going A → B → C, is the order clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q1/5

True or false?

A reflection reverses the orientation of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q2/5

True or false?

A rotation reverses the orientation of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q3/5

Triangle PQR is clockwise. After two reflections, its image is

A. counterclockwise

B. either, depending on the lines

C. clockwise

D. not a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q4/5

Triangle ABC is counterclockwise. It is translated and then reflected. Is the final image clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSG-CO.A.5

FORWARD 1

Use the fewest transformations and justify • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Two reflections are done one after the other, across parallel lines that are 3 units apart. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q1/5

Two reflections are done one after the other, across lines that meet at point O and form a 40° angle. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q2/5

Triangle ABC is clockwise and triangle A′B′C′ is congruent to it and counterclockwise. Explain why no single rotation or translation carries ABC onto A′B′C′.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q3/5

True or false?

Every rigid motion of the plane can be written as a sequence of at most three reflections.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q4/5

Triangle A′B′C′ is the image of triangle ABC under some rigid motion, and both are counterclockwise. What is the fewest number of reflections that can always carry ABC onto A′B′C′?

A. 2

B. 4

C. 3

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

FORWARD 2

Specify a similarity sequence: a dilation and rigid motions • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has A(0, 0), B(2, 0), C(0, 1). Triangle A′B′C′ has A′(0, 0), B′(0, 6), C′(−3, 0). Describe a sequence (a dilation and a rigid motion) that carries ABC onto A′B′C′.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q1/5

Square S has vertices (0, 0), (1, 0), (1, 1), (0, 1). Square T has vertices (2, 2), (4, 2), (4, 4), (2, 4). Describe a similarity sequence carrying S onto T.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q2/5

True or false?

Any two figures that a sequence of rigid motions and a dilation carries one onto the other are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q3/5

Segment AB is from (0, 0) to (3, 0). Segment CD is from (1, 1) to (1, 7). Which sequence carries AB onto CD (A to C, B to D)?

A. Rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

B. Dilate by 2 about the origin, then translate 1 right and 1 up

C. Dilate by 3 about the origin, rotate 90° counterclockwise, then translate 1 right and 1 up

D. Dilate by 2 about the origin, rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q4/5

Triangle PQR has sides 3, 4, 5. Triangle XYZ has sides 6, 8, 10. What scale factor must the dilation in a similarity sequence from PQR to XYZ have?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q5/5

Predict the effect of a rigid motion on a figure

MAIN	Geometry main question	**HSG-CO.B.6**
BACKWARD 1	Know what rigid motions keep: lines, lengths, angles, and parallel l…	**8.G.A.1**
BACKWARD 2	Match the vertices and sides of a figure with its image	**8.G.A.2**
FORWARD 1	Predict the effect of a sequence of rigid motions	**HSG-CO.B.6**
FORWARD 2	Use a rigid motion to conclude that parts of a figure are congruent	**HSG-CO.B.7**

SET 11 • HSG-CO.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC has AB = 7, BC = 5, and m∠B = 48°. It is reflected across a line ℓ. What are A′B′ and m∠B′?

A. 14 and 48°

B. 7 and 132°

C. 7 and 48°

D. They depend on where ℓ is

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q1/5

Quadrilateral PQRS has PQ ∥ SR. It is rotated 90° about a point O. What is true about P′Q′ and S′R′? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q2/5

Point A is 4 cm from line m. Triangle ABC is reflected across m. How far is A′ from m, and on which side of m is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q3/5

Triangle ABC is labeled counterclockwise (A → B → C). It is translated 5 units right and then reflected across a line m. Is the final image A″B″C″ labeled clockwise or counterclockwise? Explain each step.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q4/5

Segment AB lies on line ℓ. A 180° rotation about a point O on ℓ maps AB to A′B′. Where does A′B′ lie? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q5/5

SET 11 • HSG-CO.B.6

BACKWARD 1

Know what rigid motions keep: lines, lengths, angles, and parallel lines • Grade 8 • Geometry • 8.G.A.1

True or false?

Under a rotation, the image of a line is a line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q1/5

True or false?

A reflection can turn two parallel lines into two lines that meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q2/5

A 52° angle is translated 6 units to the right. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q3/5

A 9 cm segment is rotated 30° about one of its endpoints. How long is its image?

A. 30 cm

B. 4.5 cm

C. 39 cm

D. 9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q4/5

Name two properties of a figure that a reflection does NOT change.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

BACKWARD 2

Match the vertices and sides of a figure with its image • Grade 8 • Geometry • 8.G.A.2

Triangle ABC is rotated to triangle A′B′C′. Which side of the image corresponds to side BC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q1/5

Quadrilateral PQRS is reflected to P′Q′R′S′. Which angle corresponds to ∠Q?

A. ∠R′

B. ∠P′

C. ∠Q′

D. ∠S′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q2/5

True or false?

If triangle ABC is translated to triangle A′B′C′, then AB = A′B′.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q3/5

Triangle DEF is reflected to D′E′F′. DE = 6 and EF = 9. Find D′E′ and E′F′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q4/5

True or false?

After a rotation, side A′B′ of the image always lies on top of side AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSG-CO.B.6

FORWARD 1

Predict the effect of a sequence of rigid motions • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC is reflected across line ℓ and then across line m, where $\ell \parallel$ m. Is the final image clockwise or counterclockwise if ABC is counterclockwise? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q1/5

A figure is rotated 70° counterclockwise about point O and then rotated 50° clockwise about the same point O. Which single rigid motion has the same effect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q2/5

True or false?

A translation that is not the identity has exactly one fixed point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q3/5

Which rigid motion keeps exactly one point fixed?

A. A translation 3 units right

B. A reflection across line ℓ

C. A rotation of 90° about point O

D. The identity

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q4/5

Line ℓ is reflected across line m, and ℓ is perpendicular to m. What is the image of ℓ? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

FORWARD 2

Use a rigid motion to conclude that parts of a figure are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rotation about O maps triangle OAB onto triangle OCD, with A → C and B → D. OA = 5 and AB = 7. Find OC and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q1/5

A reflection across line ℓ maps triangle PQR onto triangle PSR (Q → S, and P and R are on ℓ). $m\angle PQR = 64°$. Find $m\angle PSR$.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q2/5

True or false?

If a rigid motion maps segment AB onto segment CD, then AB could be longer than CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q3/5

A translation maps triangle KLM onto triangle K′L′M′. Which conclusion is NOT justified?

A. KL ∥ K′L′ or KL is on line K′L′

B. ∠L ≅ ∠L′

C. KL is on the same line as K′L′

D. KL = K′L′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q4/5

Explain why the image of an isosceles triangle under any rigid motion is also isosceles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q5/5

Decide whether two figures are congruent using rigid motions

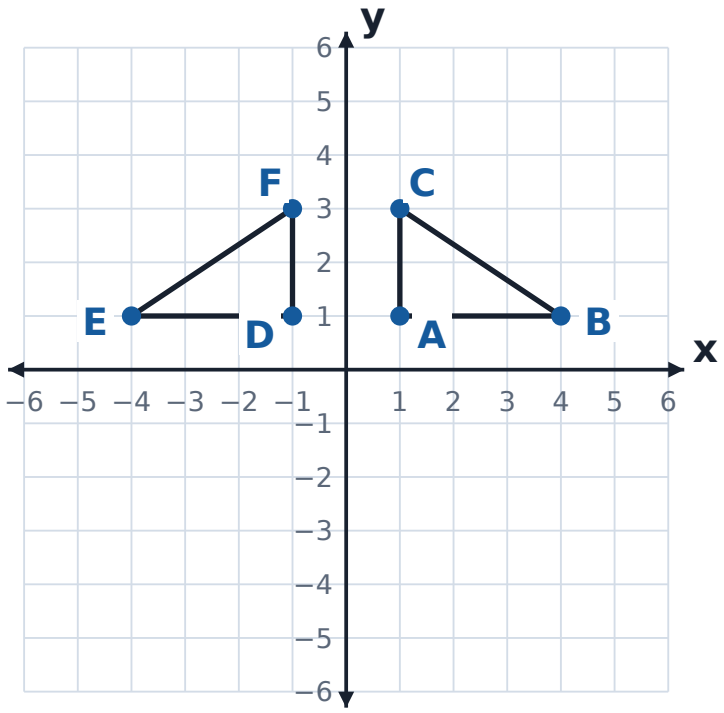
MAIN	Geometry main question	**HSG-CO.B.6**
BACKWARD 1	Recognize congruent figures as images under rigid motions	**8.G.A.2**
BACKWARD 2	Find lengths on a grid with the Pythagorean Theorem	**8.G.B.8**
FORWARD 1	Decide congruence for figures given by measurements, including…	**HSG-CO.B.6**
FORWARD 2	Use an established congruence to find unknown parts	**HSG-CO.B.7**

SET 12 • HSG-CO.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Is triangle ABC congruent to triangle DEF? If so, give a rigid motion that carries ABC onto DEF. If not, explain why.



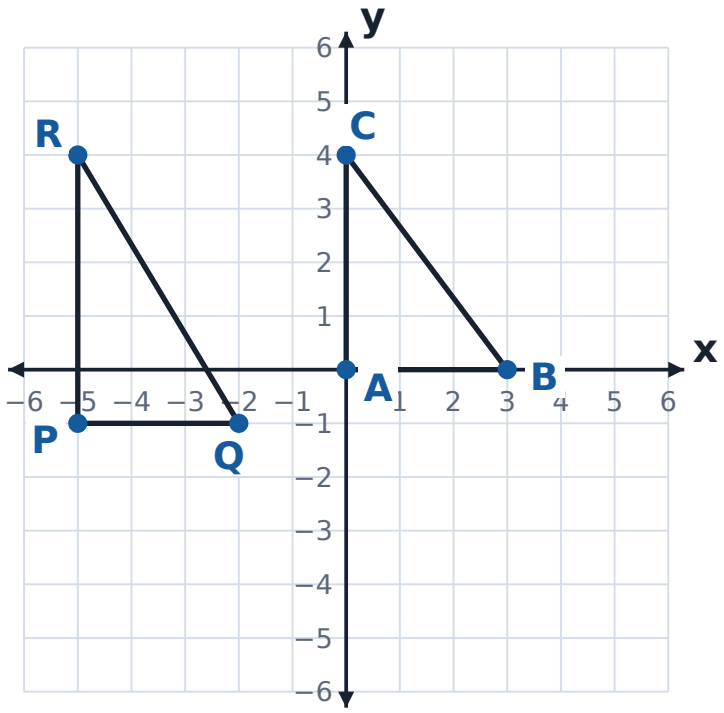
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q1/5

Is triangle ABC congruent to triangle PQR? If so, give a rigid motion that carries one onto the other. If not, explain why.



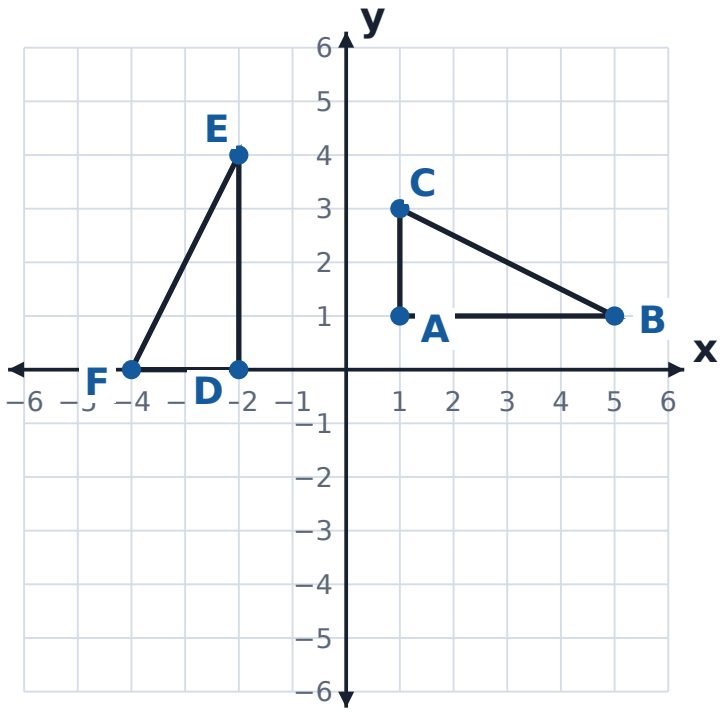
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q2/5

Is triangle ABC congruent to triangle DEF? If so, give a rigid motion (or sequence) that carries ABC onto DEF. If not, explain why.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q3/5

Quadrilateral Q has sides 3, 4, 3, 4 and four right angles. Quadrilateral R has sides 3, 4, 3, 4 and two 70° angles. Is Q congruent to R? Explain using rigid motions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q4/5

Rectangle R is 4 cm wide and 6 cm tall. Rectangle S is 6 cm wide and 4 cm tall. Are they congruent? Give a rigid motion or explain why not.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q5/5

SET 12 • HSG-CO.B.6

BACKWARD 1

Recognize congruent figures as images under rigid motions • Grade 8 • Geometry • 8.G.A.2

True or false?

If one figure is a translation of another, the figures are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q1/5

True or false?

Two figures with the same shape but different sizes are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q2/5

Which pair of figures must be congruent?

A. A square and a larger square

B. A triangle and its reflection

C. A circle of radius 2 and a circle of radius 3

D. A triangle and a dilation of it by 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q3/5

A triangle has sides 5, 6, and 7. It is rotated 90°. What are the side lengths of its image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q4/5

Is a figure congruent to its image after a reflection followed by a translation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

BACKWARD 2

Find lengths on a grid with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.8

Find the length of the segment from (1, 2) to (5, 2).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q1/5

Find the length of the segment from (0, 0) to (3, 4).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q2/5

Find the length of the segment from $(-2, 1)$ to $(1, 5)$.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q3/5

True or false?

The segments from (0, 0) to (2, 1) and from (0, 0) to (1, 2) have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q4/5

True or false?

The segment from (0, 0) to (3, 3) has length 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSG-CO.B.6

FORWARD 1

Decide congruence for figures given by measurements, including curved figures • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Circle P has radius 3 and circle Q has radius 3. Explain, using a rigid motion, why they are congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q1/5

Two quadrilaterals each have sides 5, 5, 5, 5. One is a square and the other has a 60° angle. Are they congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q2/5

True or false?

Two triangles with the same perimeter must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q3/5

Sector P has radius 4 and central angle 50°. Sector Q has radius 4 and central angle 50°, drawn in another place and turned. Are they congruent? Explain with a rigid motion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q4/5

Angle 1 measures 35° and angle 2 measures 35°. Describe a rigid motion that carries angle 1 onto angle 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

FORWARD 2

Use an established congruence to find unknown parts • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries quadrilateral ABCD onto quadrilateral EFGH with A → E, B → F, C → G, D → H. AB = 6, ∠C = 110°, and GH = 4. Find EF, ∠G, and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q1/5

Triangle RST is congruent to triangle XYZ by a rigid motion with R → X, S → Y, T → Z. RS = 2x + 3 and XY = 15. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q2/5

True or false?

If a rigid motion carries triangle ABC onto triangle DEF with A → D, B → E, C → F, then ∠B ≅ ∠F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q3/5

A reflection carries pentagon PQRST onto pentagon P′Q′R′S′T′. Which pair must be congruent?

A. ∠S and ∠S′

B. PQ and RS

C. QR and S′T′

D. ∠S and ∠T′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q4/5

Triangle ABC is congruent to triangle CDA by a rigid motion with A → C, B → D, C → A. m∠BAC = 3y° and m∠DCA = 42°. Find y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q5/5

Congruent triangles: a rigid motion exactly when all corresponding parts are congruent

MAIN	Geometry main question	**HSG-CO.B.7**
BACKWARD 1	Read a congruence statement to match corresponding parts	**8.G.A.2**
BACKWARD 2	Rigid motions keep lengths and angle measures	**8.G.A.1**
BACKWARD 3	Move a point onto another and a ray onto another with rigid motio…	**8.G.A.3**
FORWARD 1	Find which three corresponding parts are enough	**HSG-CO.B.8**
FORWARD 2	Use congruent triangles to prove a relationship in a figure	**HSG-SRT.B.5**

SET 13 • HSG-CO.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries $\triangle ABC$ onto $\triangle DEF$ with $A \to D$, $B \to E$, $C \to F$. Explain why $AB = DE$, $BC = EF$, $CA = FD$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $\angle C \cong \angle F$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q1/5

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $BC = EF$, $CA = FD$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $\angle C \cong \angle F$. Explain how to build a rigid motion that carries $\triangle ABC$ onto $\triangle DEF$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q2/5

The rotation of 90° counterclockwise about the origin carries $\triangle ABC$, with A(1, 0), B(4, 0), C(1, 2), onto $\triangle A'B'C'$. Find A′, B′, C′, check that all six pairs of corresponding parts are congruent, and explain why the definition of congruence guarantees this for every rigid motion.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q3/5

△ABC has A(0, 0), B(3, 0), C(0, 4). △DEF has D(5, 1), E(5, 4), F(1, 1). Show that all six pairs of corresponding parts (A ↔ D, B ↔ E, C ↔ F) are congruent, and then give a rigid motion that carries △ABC onto △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q4/5

Explain why the converse direction needs the sides as well as the angles: give two triangles with all three pairs of angles congruent for which no rigid motion carries one onto the other, and say which part of the definition fails.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 1

Read a congruence statement to match corresponding parts • Grade 8 • Geometry • 8.G.A.2

△ABC ≅ △XYZ. Which angle corresponds to ∠B?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

△ABC ≅ △XYZ. Which side corresponds to AC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

△PQR ≅ △LMN. Which pair corresponds?

A. QR and MN

B. QR and LN

C. PQ and MN

D. PR and LM

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

True or false?

$\triangle DEF \cong \triangle GHI$ means that $\angle F \cong \angle I$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

True or false?

$\triangle DEF \cong \triangle GHI$ means that $DE = HI$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 2

Rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

True or false?

A rotation keeps the measure of every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q1/5

Segment PQ is 12 units long. It is reflected and then translated. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q2/5

A 71° angle is reflected across a line. Its image measures

A. 289°

B. 109°

C. 71°

D. 19°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q3/5

True or false?

A translation can change the length of a segment if it moves the segment far enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q4/5

Which three basic rigid motions keep lengths and angle measures?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 3

Move a point onto another and a ray onto another with rigid motions • Grade 8 • Geometry • 8.G.A.3

Which translation moves (2, 3) onto (5, −1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q1/5

A ray starts at the origin and points along the positive x-axis. Which rotation about the origin makes it point along the positive y-axis?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q2/5

After a rotation, point C is below line DE but should be above it, and D and E are on the line. Which motion moves C across without moving D or E?

A. A rotation about D

B. A translation along DE

C. A rotation about the midpoint of DE

D. A reflection across line DE

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q3/5

True or false?

Reflecting across a line keeps every point of that line fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q4/5

True or false?

Translating a triangle can keep exactly one of its vertices fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q5/5

SET 13 • HSG-CO.B.7

FORWARD 1

Find which three corresponding parts are enough • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Must two triangles be congruent if all three pairs of corresponding sides are congruent? Use the result of the full six-part test.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q1/5

Give an example of two triangles with two pairs of congruent sides and a pair of congruent angles that are NOT congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q2/5

True or false?

Knowing that all three pairs of corresponding angles are congruent is enough to prove two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q3/5

Which set of corresponding parts is NOT always enough for congruence?

A. Three sides

B. Two sides and an angle not between them

C. Two sides and the angle between them

D. Two angles and the side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q4/5

Two triangles have AB = DE, ∠A ≅ ∠D, and ∠B ≅ ∠E. Must ∠C ≅ ∠F? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

FORWARD 2

Use congruent triangles to prove a relationship in a figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB = CB and AD = CD. Prove that diagonal BD bisects ∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q1/5

Segments AC and BD bisect each other at E. Prove that AB = CD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q2/5

In $\triangle PQR$ and $\triangle STU$, $PQ = ST$, $QR = TU$, and $PR = SU$. The triangles are congruent. What must be true of $\angle R$ and $\angle U$? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q3/5

True or false?

Once two triangles are proved congruent, only the parts used in the proof are known to be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q4/5

To prove that point M is the midpoint of segment XY using congruent triangles, which pair of parts must you show is congruent?

A. ∠M and ∠X

B. ∠X and ∠Y

C. XM and YM

D. XY and XM

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q5/5

Explain SAS congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Identify the included angle and included side	**7.G.A.2**
BACKWARD 2	Rigid motions map rays, lengths, and angles to equal ones	**8.G.A.1**
BACKWARD 3	Choose a translation, rotation, or reflection to move a point or ray…	**8.G.A.2**
FORWARD 1	Prove triangles congruent by SAS inside a larger figure	**HSG-SRT.B.5**
FORWARD 2	Decide whether given information is enough for SAS, and name a…	**HSG-SRT.B.5**

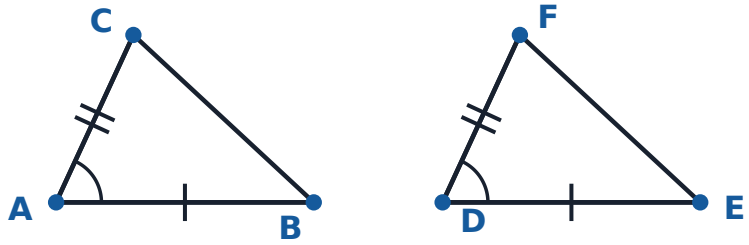
SET 14 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $AC = DF$, and $\angle A \cong \angle D$. Explain, using rigid motions, why $\triangle ABC \cong \triangle DEF$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q1/5

In $\triangle JKL$ and $\triangle MNO$, $JK = MN$, $JL = MO$, and $\angle J \cong \angle M$. $\triangle MNO$ is a mirror image of $\triangle JKL$ (its vertices run the opposite way around). Explain, using rigid motions, why $\triangle JKL \cong \triangle MNO$, and say at which step a reflection is needed.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q2/5

△ABC has A(0, 0), B(4, 0), C(1, 3). △DEF has D(2, −1), E(2, 3), F(−1, 0). AB = DE = 4, AC = DF = √10, and ∠A ≅ ∠D. Name a translation and a rotation that carry A to D and ray AB onto ray DE, and explain, using only the SAS facts, why C must then land on F.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q3/5

In quadrilateral ABCD, AB = AD and diagonal AC bisects ∠BAD. Explain, using one reflection, why △ABC ≅ △ADC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q4/5

Segments AC and BD bisect each other at E (A, E, C on one line and B, E, D on another). Explain, using a 180° rotation about E, why $\triangle AEB \cong \triangle CED$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 1

Identify the included angle and included side • Grade 7 • Geometry • 7.G.A.2

In △ABC, which angle is included between sides AB and AC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q1/5

In △PQR, which side is included between ∠P and ∠Q?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q2/5

In △XYZ, which angle is NOT included between sides XY and YZ?

A. Both ∠X and ∠Y

B. ∠X

C. ∠Y

D. None of the angles is excluded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q3/5

True or false?

Two sides of 5 cm and 7 cm with a 40° angle between them determine one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q4/5

True or false?

Two sides of 5 cm and 7 cm with a 40° angle that is not between them always determine one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 2

Rigid motions map rays, lengths, and angles to equal ones • Grade 8 • Geometry • 8.G.A.1

True or false?

Under a rigid motion, the image of a ray is a ray.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q1/5

A 30° angle is rotated about its vertex. What is the measure of the image angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q2/5

A point is 5 units from D along a ray. After a rotation about D, the image point is

A. closer to D

B. not on the image ray

C. 5 units from D on the image ray

D. farther from D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q3/5

True or false?

A reflection across line DE can move point D.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q4/5

Segment AB is translated so that A lands on D. AB = 9. How far is the image of B from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 3

Choose a translation, rotation, or reflection to move a point or ray onto a target • Grade 8 • Geometry • 8.G.A.2

Which kind of rigid motion moves point A onto point D without turning anything?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q1/5

A and D are already together. Which kind of rigid motion turns ray AB onto ray DE without moving D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q2/5

D and E are fixed, and point C is on the wrong side of line DE. Which motion fixes D and E and moves C across the line?

A. A translation along DE

B. A rotation of 90° about D

C. A rotation of 180° about E

D. A reflection across line DE

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q3/5

True or false?

A rotation about point D keeps D fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q4/5

True or false?

A translation that moves A onto D also keeps A fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSG-CO.B.8

FORWARD 1

Prove triangles congruent by SAS inside a larger figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB ∥ DC and AB = DC. Diagonal AC is drawn. Prove that △ABC ≅ △CDA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q1/5

Point M is the midpoint of segment AB, and line ℓ is perpendicular to AB at M. Point P is on ℓ, and P ≠ M. Prove that △PMA ≅ △PMB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q2/5

Segments AC and BD cross at E, with AE = CE and BE = DE. Name the congruent triangles and the criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q3/5

True or false?

In an SAS proof, a side that the two triangles share can be one of the pairs of congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q4/5

Segments PQ and RS bisect each other at T. Which reason gives the included angles for SAS?

A. The angles are both right angles

B. Vertical angles are congruent

C. Base angles are congruent

D. Corresponding angles are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

FORWARD 2

Decide whether given information is enough for SAS, and name a missing condition • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In △ABC and △DEF, AB = DE and BC = EF. Name ONE more pair of congruent parts that would prove congruence by SAS.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q1/5

In $\triangle PQR$ and $\triangle XYZ$, $PQ = XY$, $PR = XZ$, and $\angle Q \cong \angle Y$. Is this enough to prove the triangles congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q2/5

True or false?

In $\triangle ABC$ and $\triangle ADC$ (sharing AC), $AB = AD$ and $\angle BCA \cong \angle DCA$. These facts prove $\triangle ABC \cong \triangle ADC$ by SAS.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q3/5

Kites ABCD and EFGH have AB = EF, BC = FG, and ∠B ≅ ∠F. Which triangles are congruent by SAS?

A. △BCD and △FGH

B. △ABC and △EFG

C. △ABD and △EFH

D. △ACD and △EGH

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q4/5

Two triangles have sides 7 and 9 with a 50° angle between them. A student says they need not be congruent because the third sides were not measured. Is the student right? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q5/5

Explain ASA congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Two angles and the included side determine a triangle	**7.G.A.2**
BACKWARD 2	Two different lines meet in at most one point	**8.EE.C.8.a**
BACKWARD 3	Rigid motions keep rays and angle measures	**8.G.A.1**
FORWARD 1	Explain AAS from ASA and the triangle angle sum	**HSG-CO.C.10**
FORWARD 2	Prove congruence with ASA or AAS in overlapping triangles	**HSG-SRT.B.5**

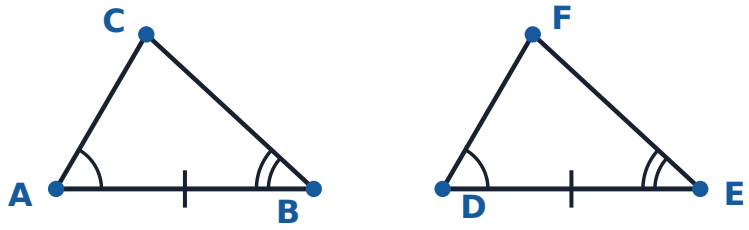
SET 15 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle ABC$ and $\triangle DEF$, $\angle A \cong \angle D$, $AB = DE$, and $\angle B \cong \angle E$. Explain, using rigid motions, why $\triangle ABC \cong \triangle DEF$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q1/5

In △PQR and △STU, ∠P ≅ ∠S, PQ = ST, and ∠Q ≅ ∠T. △STU is a mirror image of △PQR (its vertices run the opposite way around). Explain, using rigid motions, why △PQR ≅ △STU, including where the reflection is used and why R lands on U.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q2/5

△ABC has A(0, 0), B(4, 0), C(1, 3). △DEF has D(5, 5), E(1, 5), F(4, 2). ∠A ≅ ∠D, AB = DE, and ∠B ≅ ∠E. The 180° rotation about (2.5, 2.5) sends A to D and B to E. Explain, using only the ASA facts (not the coordinates of C), why it must send C to F.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q3/5

In $\triangle ABC$, $\angle B \cong \angle C$. Let ℓ be the perpendicular bisector of BC. Explain, using the reflection across ℓ and the ASA argument, why the reflection carries $\triangle ABC$ onto $\triangle ACB$, and conclude that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q4/5

Segments AD and BC cross at E, which is the midpoint of AD, and AB ∥ CD (B and C on opposite sides of AD). Explain, using a 180° rotation about E, why △ABE ≅ △DCE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 1

Two angles and the included side determine a triangle • Grade 7 • Geometry • 7.G.A.2

True or false?

Two angles of 40° and 60° with a 5 cm side between them determine exactly one triangle (up to position).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q1/5

True or false?

Any two angles with a side between them make a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q2/5

A triangle has angles of 50° and 70° at the ends of a 6 cm side. What is the third angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q3/5

Which set of conditions determines exactly one triangle?

A. Angles 30° and 80° with a 4 cm side between them

B. One angle of 45° and nothing else

C. Angles 30°, 60°, and 90°

D. Angles 100° and 90° with a 3 cm side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q4/5

Two angles of a triangle are 95° and 90°. Can the triangle exist? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 2

Two different lines meet in at most one point • Grade 8 • Expressions & Equations • 8.EE.C.8.a

How many points do the lines $y = x + 1$ and $y = -x + 3$ have in common?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q1/5

True or false?

Two different lines can meet in exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q2/5

Lines $y = 2x + 1$ and $y = 2x - 4$

A. never meet

B. are the same line

C. meet at one point

D. meet at two points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q3/5

True or false?

If two lines share two points, they are the same line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q4/5

Where do the lines $x = 2$ and $y = -3$ meet?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 3

Rigid motions keep rays and angle measures • Grade 8 • Geometry • 8.G.A.1

A 64° angle is translated and then reflected. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q1/5

True or false?

A rigid motion maps an angle to an angle with the same measure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q2/5

A ray from E makes a 35° angle with ray ED above line DE. How many such rays are there?

A. Exactly one

B. Two

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q3/5

True or false?

After a rotation, an angle can become larger if its sides are longer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q4/5

Segment AB = 4 cm is rotated so that A lands on D. Where is B, relative to D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSG-CO.B.8

FORWARD 1

Explain AAS from ASA and the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In $\triangle ABC$ and $\triangle DEF$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $BC = EF$ (the side is NOT between the angles). Prove $\triangle ABC \cong \triangle DEF$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q1/5

Two triangles have angles 40° and 75° and a 9 cm side opposite the 75° angle. Must they be congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q2/5

In the argument above, how does AAS follow from ASA? Name the theorem used to get the third pair of angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q3/5

In $\triangle PQR$ and $\triangle XYZ$, $\angle P \cong \angle X$, $\angle Q \cong \angle Y$, and $PR = XZ$. Which criterion proves congruence?

A. ASA

B. AAS

C. SAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q4/5

Which corresponding side would make AAS apply if $\angle A \cong \angle D$ and $\angle C \cong \angle F$?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

FORWARD 2

Prove congruence with ASA or AAS in overlapping triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Segments AD and BC cross at E. AB ∥ CD and E is the midpoint of AD. Prove that △ABE ≅ △DCE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q1/5

In $\triangle ABC$, the bisector of $\angle A$ is perpendicular to BC at D. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q2/5

In overlapping triangles ABE and ACD, $\angle A$ is shared, $AB = AC$, and $\angle B \cong \angle C$. Which criterion proves $\triangle ABE \cong \triangle ACD$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q3/5

True or false?

An angle shared by two overlapping triangles cannot be used in a congruence proof, because it is the same angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q4/5

In △JKL and △JML, ∠KJL ≅ ∠MJL and ∠JLK ≅ ∠JLM. Which side completes ASA?

A. JL = JL

B. JK = JM

C. KL = ML

D. JK = KL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q5/5

Explain SSS congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Three side lengths determine a triangle (triangle inequality)	**7.G.A.2**
BACKWARD 2	Find the points at given distances from two points (two circles)	**7.G.B.4**
BACKWARD 3	Rigid motions keep every length	**8.G.A.1**
FORWARD 1	Explain the Hypotenuse-Leg criterion for right triangles	**HSG-SRT.B.5**
FORWARD 2	Decide which criteria guarantee congruence, with counterexampl…	**HSG-SRT.B.5**

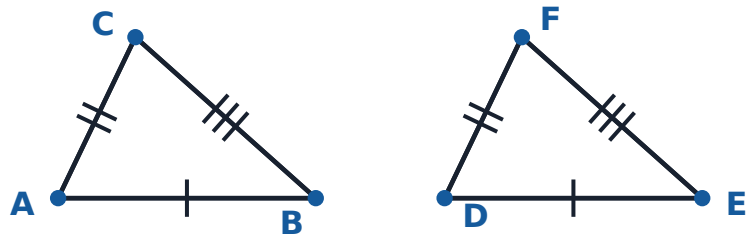
SET 16 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $BC = EF$, and $CA = FD$. Explain, using rigid motions, why $\triangle ABC \cong \triangle DEF$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q1/5

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $BC = EF$, and $CA = FD$. A translation and a rotation have moved A onto D and B onto E, and the image C' of C is on the opposite side of line DE from F, with $DC' = DF$ and $EC' = EF$. Explain why the reflection across line DE carries C' onto F. Justify each step without using SSS (it is what is being explained).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q2/5

△ABC has A(0, 0), B(5, 0), C(1, 3). △DEF has D(1, 1), E(1, 6), F(−2, 2). Show that the three pairs of corresponding sides are congruent, and explain, using rigid motions, why the triangles are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q3/5

Quadrilateral ABCD has AB = AD and CB = CD (a kite). Explain, using the reflection across line AC, why $\triangle ABC \cong \triangle ADC$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q4/5

A triangle has sides 4, 5, and 6. Its 6-unit side is placed on segment DE (DE = 6). Explain why the third vertex can be in exactly two places, one on each side of line DE, and why the reflection across DE carries one place onto the other.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 1

Three side lengths determine a triangle (triangle inequality) • Grade 7 • Geometry • 7.G.A.2

True or false?

Side lengths 3, 4, and 8 can form a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q1/5

True or false?

Side lengths 5, 6, and 10 form exactly one triangle (up to position).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q2/5

Which three lengths can form a triangle?

A. 5, 5, 11

B. 4, 5, 7

C. 2, 3, 6

D. 1, 1, 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q3/5

Two sides of a triangle are 6 and 10. Give the range of possible lengths for the third side.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q4/5

Can two different-shaped triangles both have sides 7, 8, and 9? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 2

Find the points at given distances from two points (two circles) • Grade 7 • Geometry • 7.G.B.4

Point P is 5 units from A and 5 units from B, and A and B are 6 units apart. How many such points P are there in the plane?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q1/5

True or false?

In a given plane containing A, all points 3 cm from A form a circle with center A and radius 3 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q2/5

Circles of radius 4 centered at A and B meet at two points. AB = 6. Where are the two points?

A. Both on line AB

B. One on each side of line AB

C. At A and B

D. Both on the same side of AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q3/5

True or false?

Two circles always meet in exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q4/5

Points A and B are 10 apart. Is there a point 3 units from A and 4 units from B? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 3

Rigid motions keep every length • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has sides 3, 5, and 7. After a rotation and a reflection, what are the side lengths of the image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q1/5

True or false?

Under a rigid motion, the distance from the image of A to the image of C equals AC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q2/5

Which can a rigid motion change?

A. The perimeter

B. The length of a side

C. The area

D. The position of a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q3/5

True or false?

A reflection doubles the distance from a point to the line of reflection.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q4/5

A point C is 6 units from D. After a reflection across a line through D, how far is the image of C from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSG-CO.B.8

FORWARD 1

Explain the Hypotenuse-Leg criterion for right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Right triangles ABC and DEF have right angles at C and F, hypotenuses AB = DE, and legs AC = DF. Prove that $\triangle ABC \cong \triangle DEF$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q1/5

Two right triangles each have hypotenuse 13 and one leg 5. Are they congruent? What is the other leg?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q2/5

True or false?

Hypotenuse-Leg is a special case of SSA that does guarantee congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q3/5

Which information proves two right triangles congruent?

A. One pair of acute angles is equal

B. Their right angles are equal

C. Their hypotenuses are equal and one pair of legs is equal

D. Their hypotenuses are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q4/5

In right triangles PQR and XYZ (right angles at R and Z), PQ = XY = 10 and QR = YZ = 6. Find PR and XZ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

FORWARD 2

Decide which criteria guarantee congruence, with counterexamples • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Explain with a counterexample why AAA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q1/5

Explain with a counterexample why SSA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q2/5

Which list contains only valid triangle congruence criteria?

A. SSS, SAS, ASA, AAS

B. SSS, SSA, ASA, AAS

C. SSA, AAA, SAS, HL

D. SAS, AAA, ASA, SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q3/5

True or false?

If two triangles have the same three side lengths, they have the same three angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q4/5

True or false?

Two triangles with two pairs of congruent sides and equal areas must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q5/5

Prove that vertical angles are congruent

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Angles in a linear pair add to 180°	**7.G.B.5**
BACKWARD 2	Use properties of equality as reasons	**HSA-REI.A.1**
BACKWARD 3	Identify vertical angles and linear pairs in a figure	**7.G.B.5**
FORWARD 1	Use the vertical angle theorem in multistep proofs	**HSG-CO.C.9**
FORWARD 2	Prove statements about the bisectors of angles formed by two lin…	**HSG-CO.C.9**

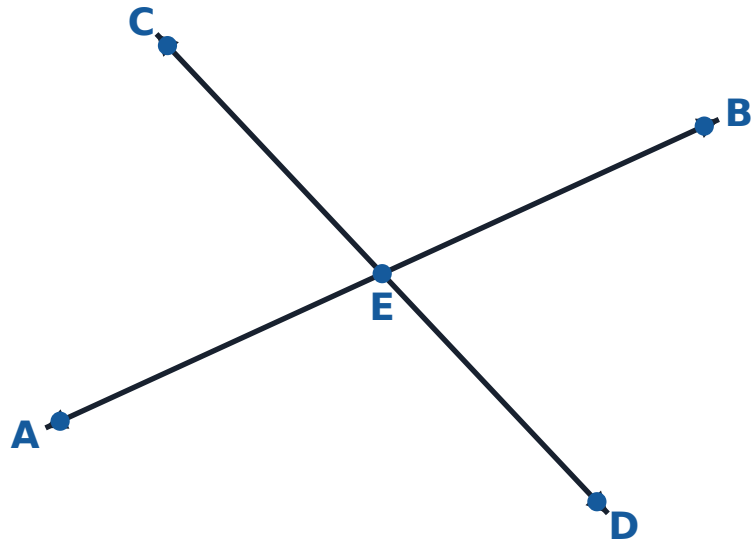
SET 17 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines AB and CD intersect at E. Prove that ∠AEC ≅ ∠BED.

Work space



Conclusion: _____

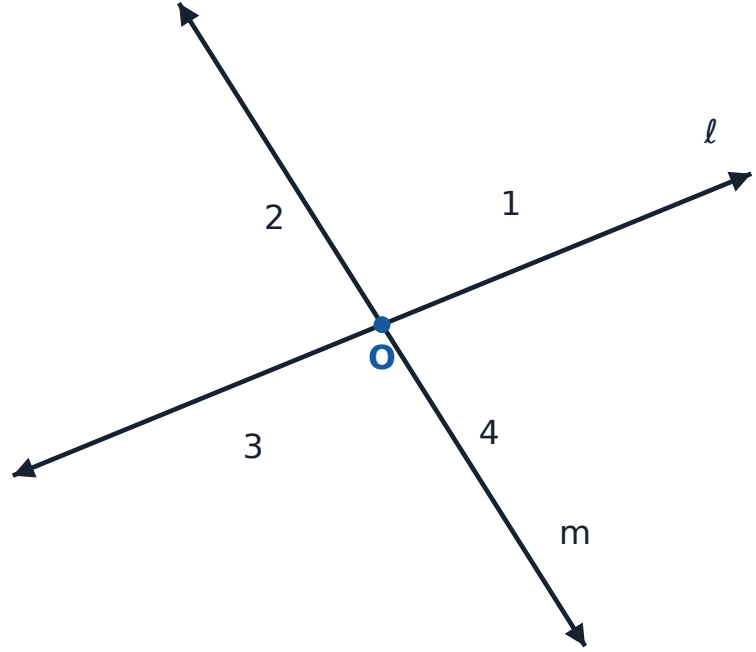
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q1/5

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, and $\angle 4$ in order around O. Prove that $\angle 2 \cong \angle 4$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q2/5

Lines PQ and RS intersect at T (P, T, Q on one line and R, T, S on the other). Prove that ∠PTR ≅ ∠QTS.

Work space

Conclusion: _____

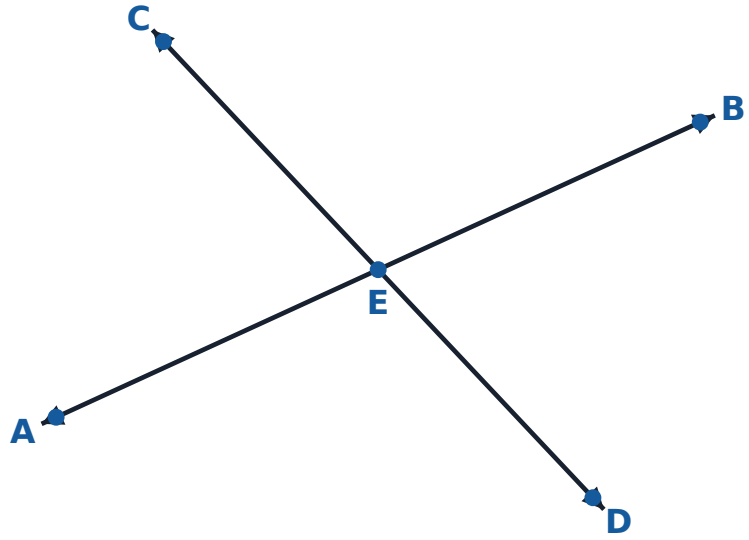
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q3/5

Lines AB and CD intersect at E. Explain, using a rigid motion, why ∠AEC ≅ ∠BED.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q4/5

Two lines intersect, forming vertical angles $\angle 1$ and $\angle 3$, and $\angle 2$ is adjacent to both. Suppose $m\angle 1 = x°$. Prove that $m\angle 3 = x°$ for every value of x, giving a reason for each step.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 1

Angles in a linear pair add to 180° • Grade 7 • Geometry • 7.G.B.5

Two angles form a linear pair. One measures 128°. Find the other.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q1/5

∠PQR and ∠RQS form a linear pair, and m∠PQR = 3 · m∠RQS. Find m∠RQS.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q2/5

True or false?

The two angles of a linear pair are always supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q3/5

Points A, B, C are on a line with B between A and C, and ray BD is drawn. Which angles form a linear pair?

A. ∠DBC and ∠ABC

B. ∠BAD and ∠BCD

C. ∠ABD and ∠DBC

D. ∠ABD and ∠ABC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q4/5

True or false?

Two angles that add to 180° must form a linear pair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 2

Use properties of equality as reasons • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

From $m\angle 1 + 40° = m\angle 2 + 40°$, which property gives $m\angle 1 = m\angle 2$?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q1/5

If $x + y = 180$ and $y + z = 180$, which property lets you write $x + y = y + z$?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q2/5

Which property justifies: if a = b, then b = a?

A. Reflexive property

B. Symmetric property

C. Distributive property

D. Transitive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q3/5

True or false?

The reflexive property says that any quantity equals itself, such as $m\angle A = m\angle A$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q4/5

True or false?

From $2x = 2y$ you may conclude $x = y$ by the subtraction property of equality.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

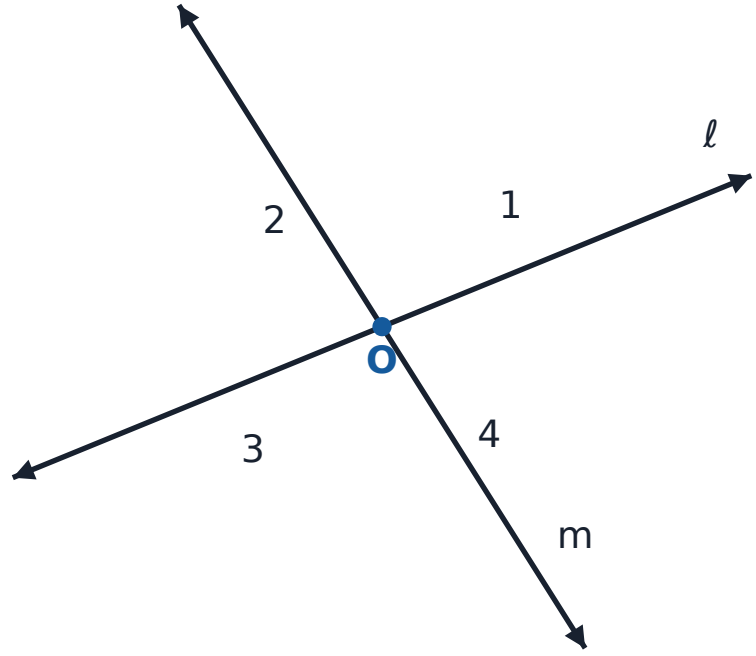
High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 3

Identify vertical angles and linear pairs in a figure • Grade 7 • Geometry • 7.G.B.5

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, $\angle 4$ in order. Which angle is vertical to $\angle 1$?



A. $\angle 3$

B. $\angle 1$

C. $\angle 4$

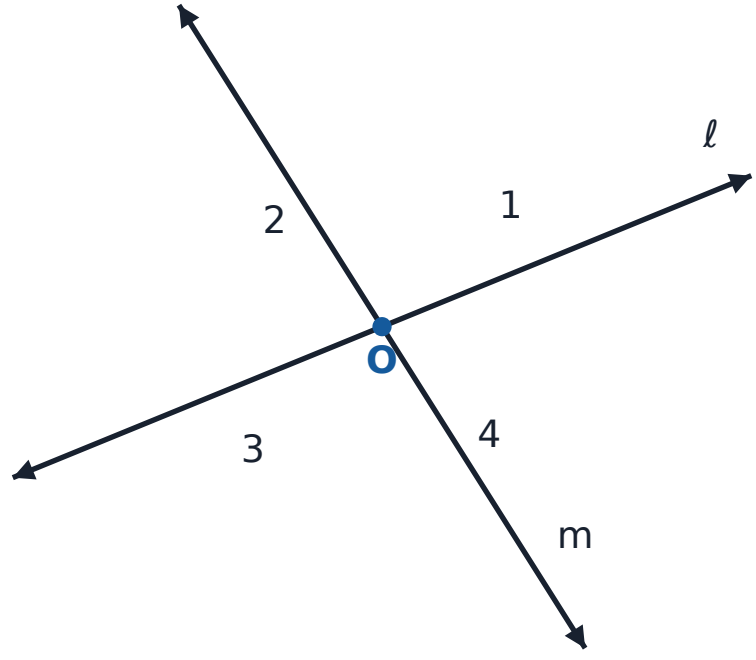
D. $\angle 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q1/5

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, $\angle 4$ in order. Name the two angles that form linear pairs with $\angle 2$.



Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q2/5

True or false?

Vertical angles share a side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q3/5

True or false?

When two lines cross, they form two pairs of vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q4/5

Two lines cross. One angle is 63°. Find its vertical angle and the angles next to it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • HSG-CO.C.9

FORWARD 1

Use the vertical angle theorem in multistep proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines AB and CD intersect at E, and ray EF bisects ∠AEC. Ray EG is the ray opposite ray EF. Prove that ray EG bisects ∠BED.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

Three lines meet at point O, forming six angles. Two of the angles that are next to each other are 50° and 70°. Prove that the six angles are 50°, 70°, 60°, 50°, 70°, 60° in order.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

Two lines meet. One angle is (4x + 10)° and its vertical angle is (6x − 30)°. Find x and all four angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

True or false?

If one angle formed by two crossing lines is a right angle, all four angles are right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

Lines AB and CD meet at E with m∠AEC = 2x and m∠CEB = 3x. What is m∠BED?

A. 72°

B. 144°

C. 36°

D. 108°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

FORWARD 2

Prove statements about the bisectors of angles formed by two lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

∠ABD and ∠DBC form a linear pair. Ray BE bisects ∠ABD and ray BF bisects ∠DBC. Prove that ∠EBF is a right angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q1/5

Is the converse of the vertical angle theorem true: "If two angles are congruent, they are vertical angles"? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q2/5

Two lines meet at O. Prove that the bisectors of a pair of vertical angles lie on one line.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q3/5

True or false?

The bisectors of a pair of vertical angles are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q4/5

Two lines meet, and the bisectors of two adjacent angles are drawn. What angle do the bisectors form?

A. 45°

B. It depends on the lines

C. 180°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q5/5

SET 18 • HSG-CO.C.9 • CONGRUENCE: PROVING THEOREMS

Prove the angle theorems for parallel lines cut by a transversal

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Name and use angles formed by parallel lines and a transversal	**8.G.A.5**
BACKWARD 2	Vertical angles and linear pairs as reasons	**HSG-CO.C.9**
BACKWARD 3	The parallel postulate and the definition of parallel lines	**HSG-CO.A.1**
FORWARD 1	Prove the converse: congruent alternate interior angles make line...	**HSG-CO.C.9**
FORWARD 2	Use parallel-line theorems to prove the triangle angle sum	**HSG-CO.C.10**

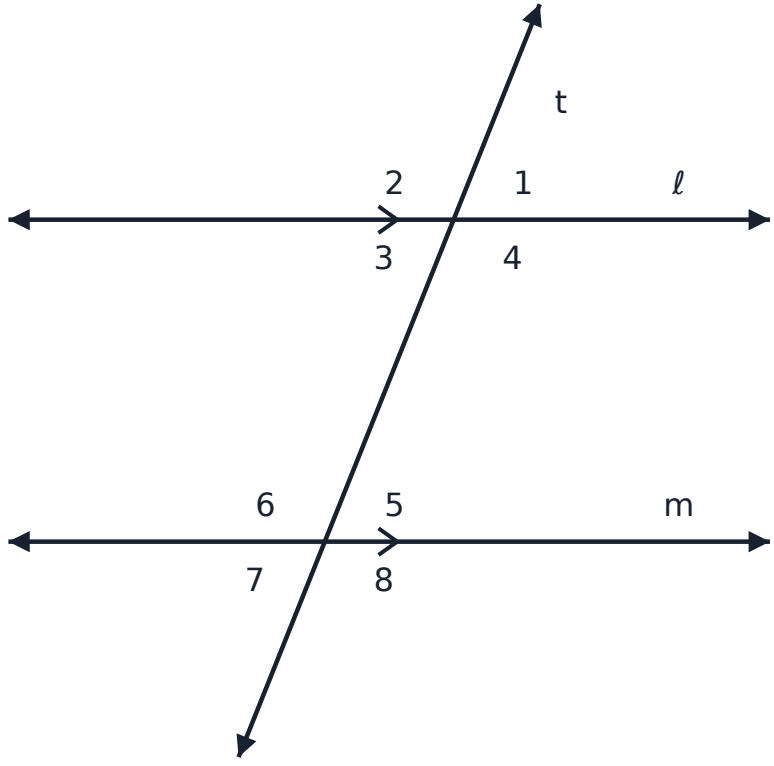
SET 18 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are parallel and are cut by transversal t, crossing ℓ at P and m at Q. Explain, using a translation, why corresponding angles $\angle 1$ and $\angle 5$ are congruent. (This argument uses only the parallel postulate and rigid motions, so the later items may use its result.)

Work space



Conclusion: _____

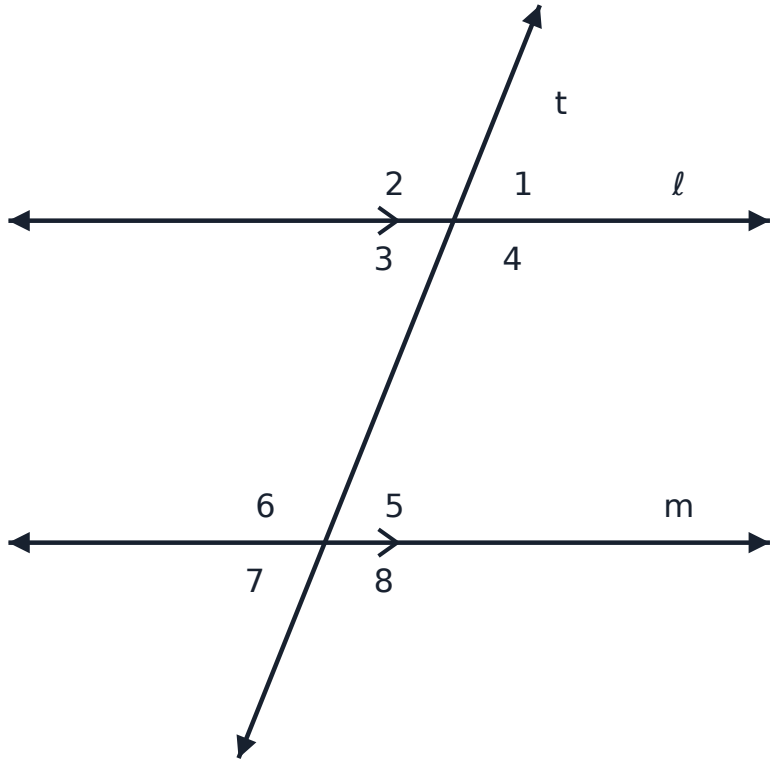
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q1/5

Lines ℓ and m are parallel and are cut by transversal t. Use the corresponding angles theorem (proved with a translation) to prove that alternate interior angles $\angle 3$ and $\angle 5$ are congruent.

Work space



Conclusion: _____

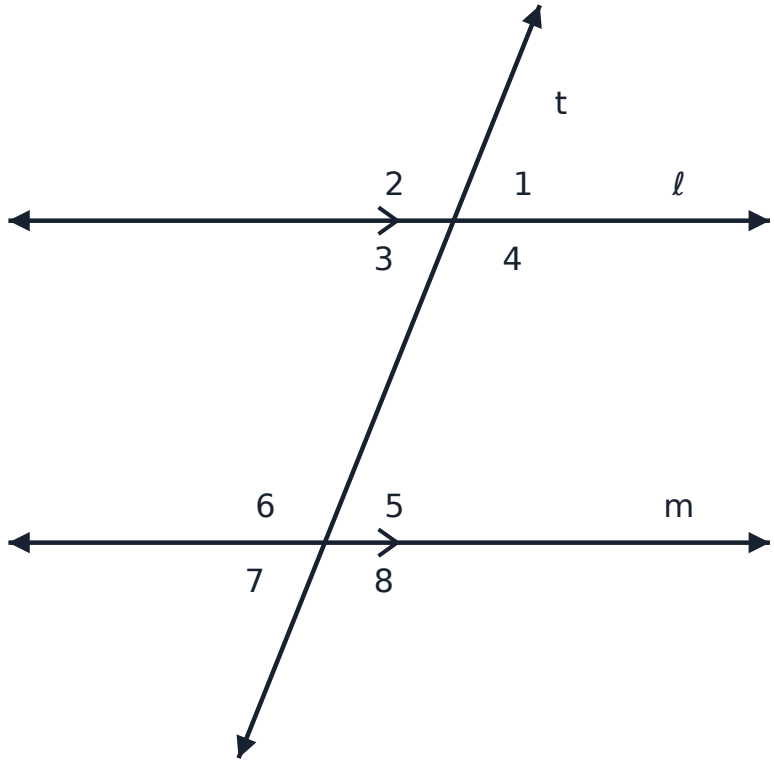
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q2/5

Lines ℓ and m are parallel and are cut by transversal t. Use the corresponding angles theorem to prove that alternate exterior angles $\angle 2$ and $\angle 8$ are congruent.

Work space



Conclusion: _____

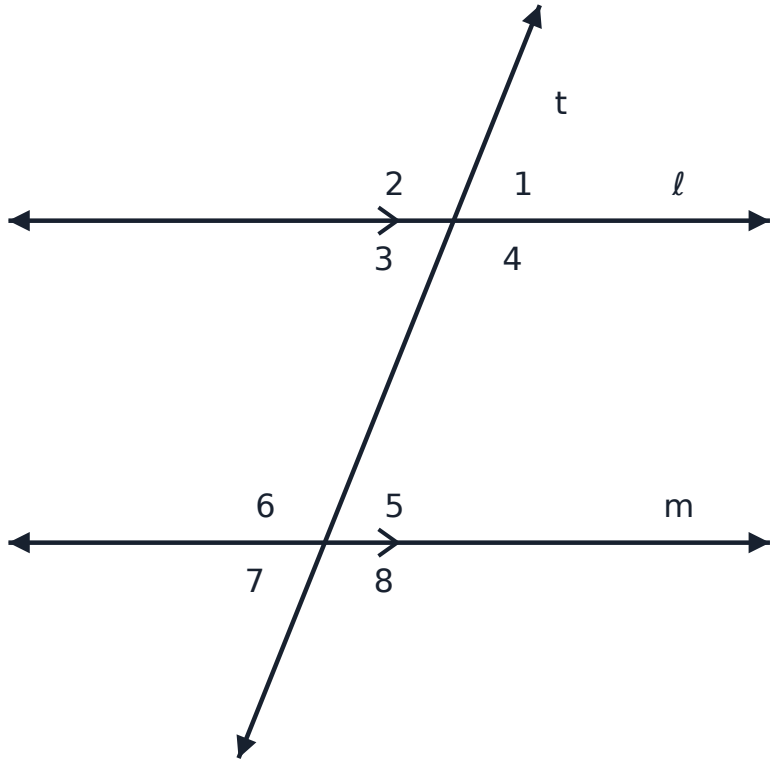
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q3/5

Lines ℓ and m are parallel and are cut by transversal t. Prove that same-side interior angles $\angle 4$ and $\angle 5$ are supplementary.

Work space



Conclusion: _____

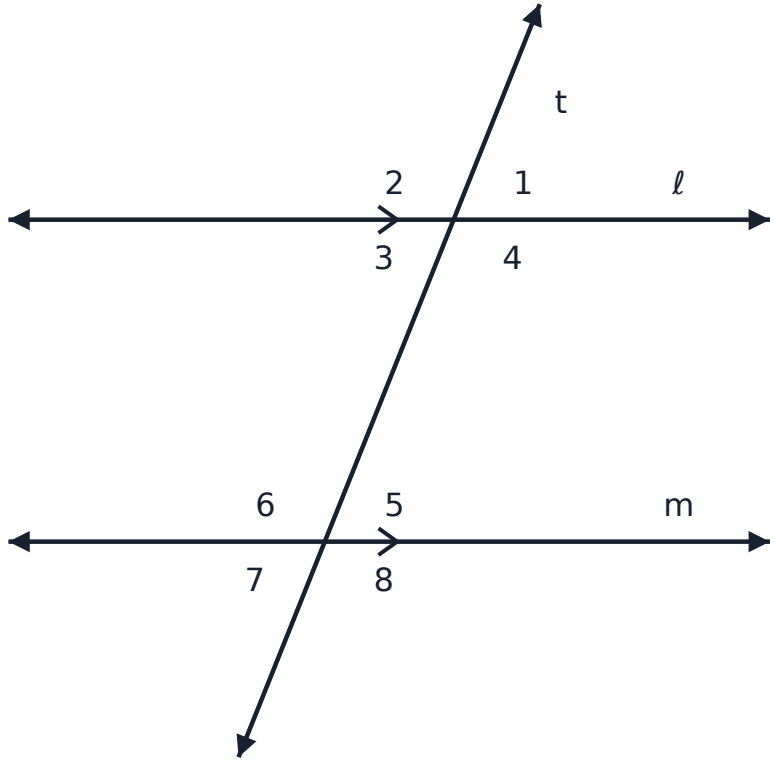
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q4/5

Lines ℓ and m are parallel and are cut by transversal t at P (on ℓ) and Q (on m). Let M be the midpoint of PQ. Explain, using the 180° rotation about M, why alternate interior angles $\angle 3$ and $\angle 5$ are congruent.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

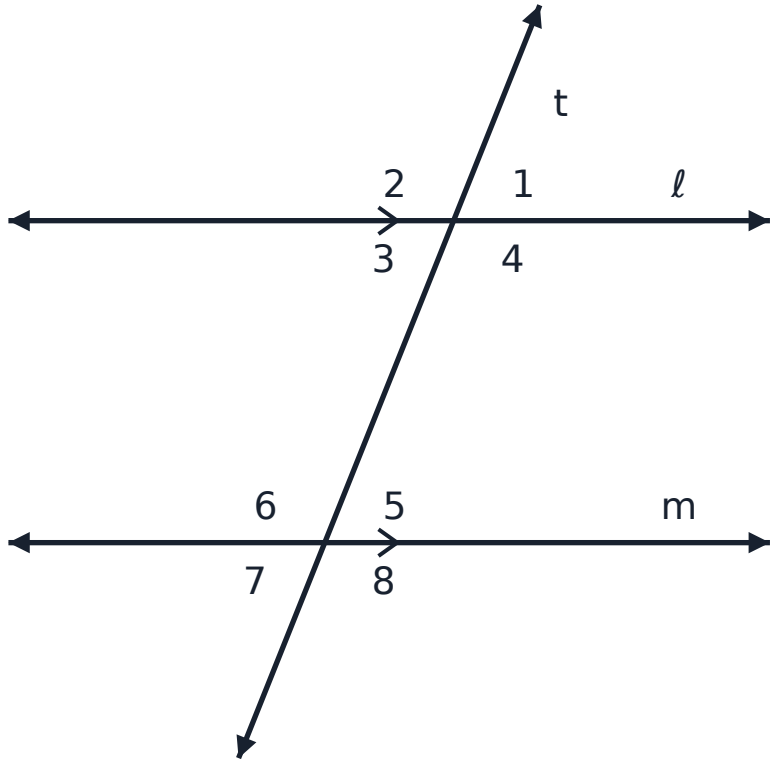
High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 1

Name and use angles formed by parallel lines and a transversal • Grade 8 • Geometry • 8.G.A.5

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ measures $70°$. Find the measure of $\angle 6$.



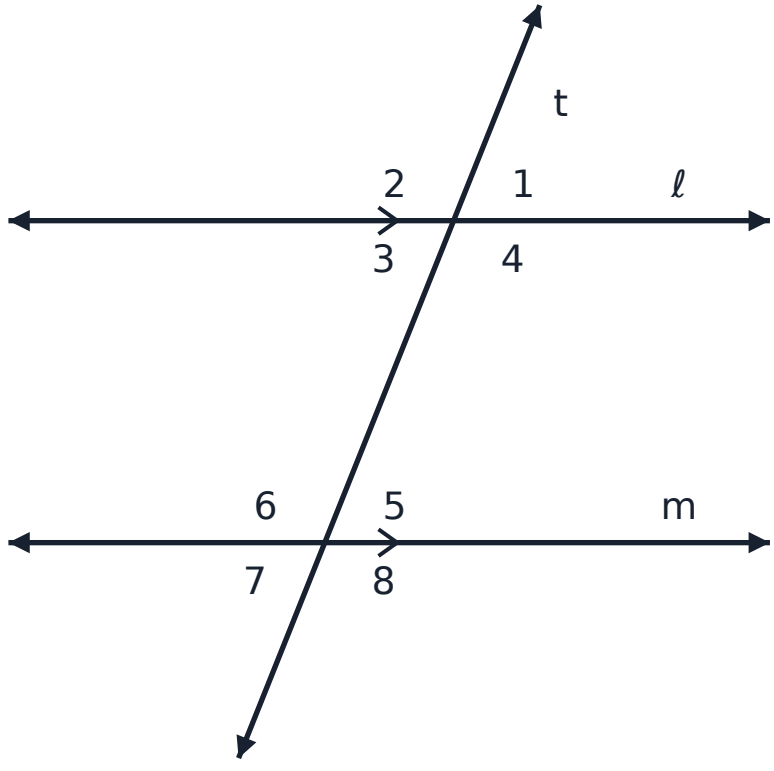
Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q1/5

Lines $\ell \parallel m$ are cut by transversal t. Which angle corresponds to $\angle 2$?



A. $\angle 7$

B. $\angle 8$

C. $\angle 6$

D. $\angle 5$

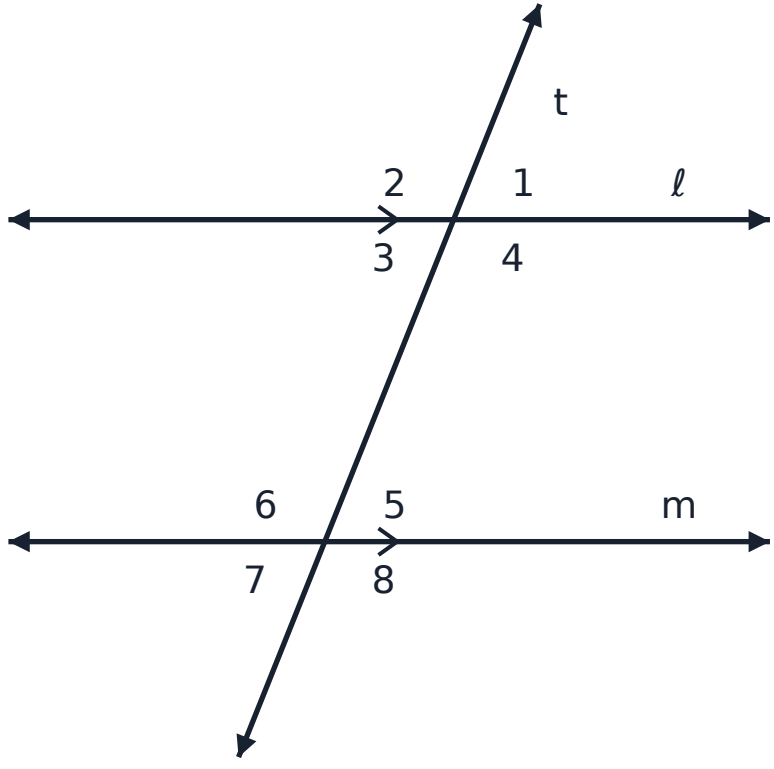
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q2/5

True or false?

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ and $\angle 5$ are same-side interior angles, so they add to 180°.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q3/5

True or false?

When two lines are cut by a transversal, corresponding angles are congruent even if the lines are not parallel.

A. True

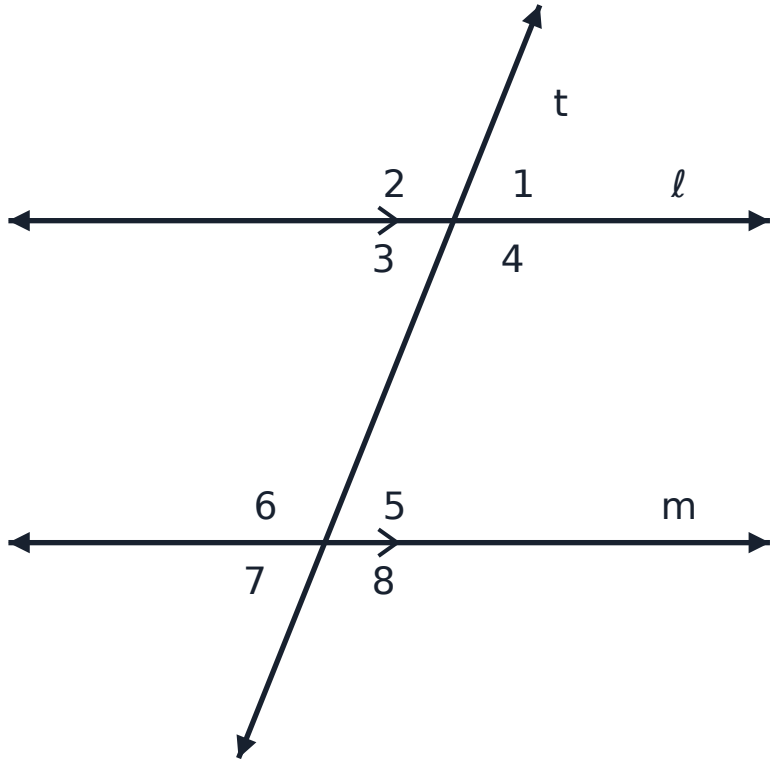
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q4/5

Lines $\ell \parallel m$ are cut by transversal t. $\angle 8$ measures 112°. Find the measure of $\angle 2$.



Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 2

Vertical angles and linear pairs as reasons • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Two lines cross. Name the reason two opposite angles are congruent.

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q1/5

Name the reason two adjacent angles on a line add to 180°.

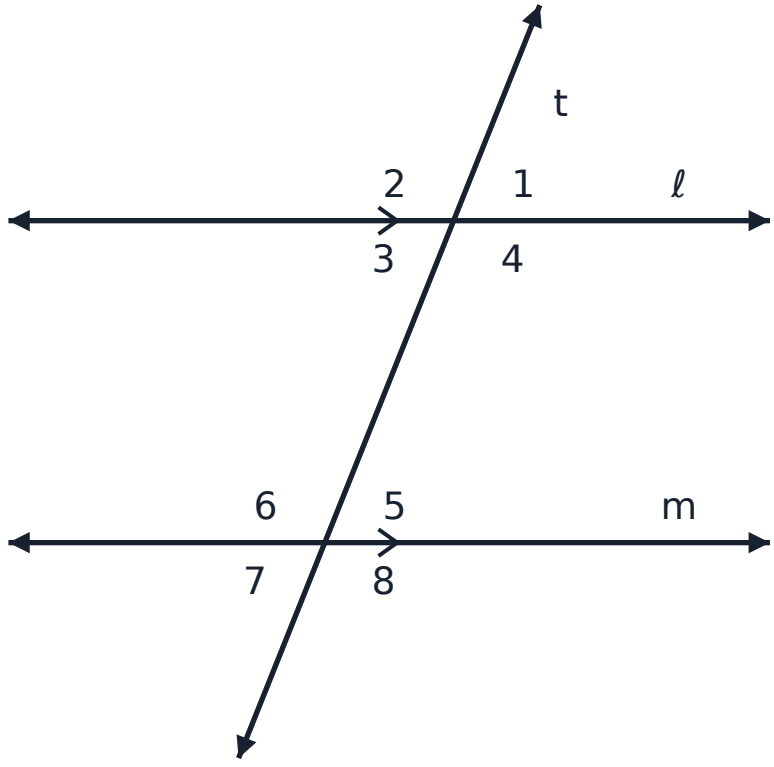
Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q2/5

At the top crossing, $\angle 1 = 110°$. What is $\angle 2$ and why?



A. 110°, linear pair

B. 70°, linear pair

C. 110°, vertical angles

D. 70°, vertical angles

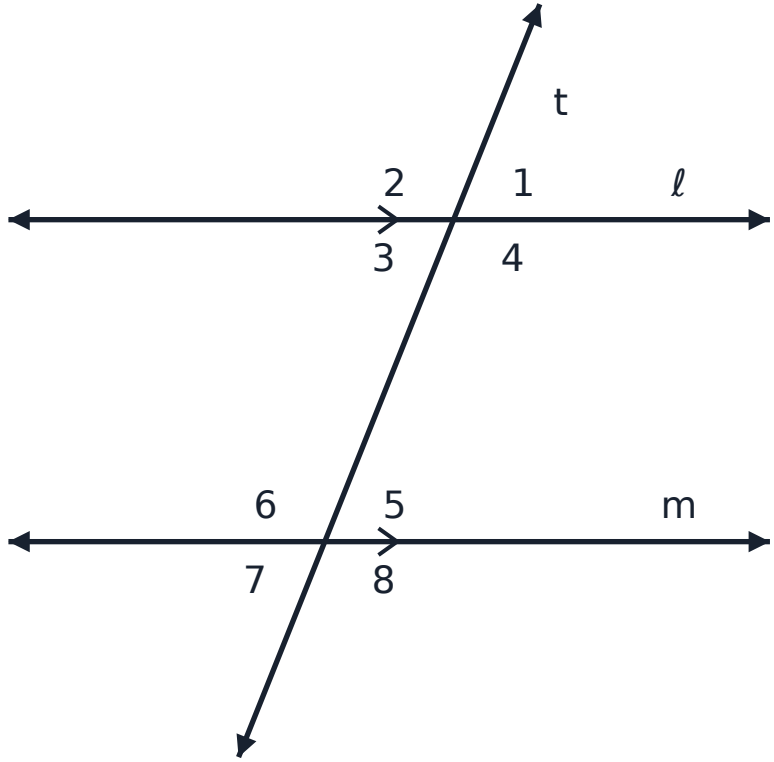
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q3/5

True or false?

Angles 1 and 3 are vertical angles in the figure.



A. True

B. False

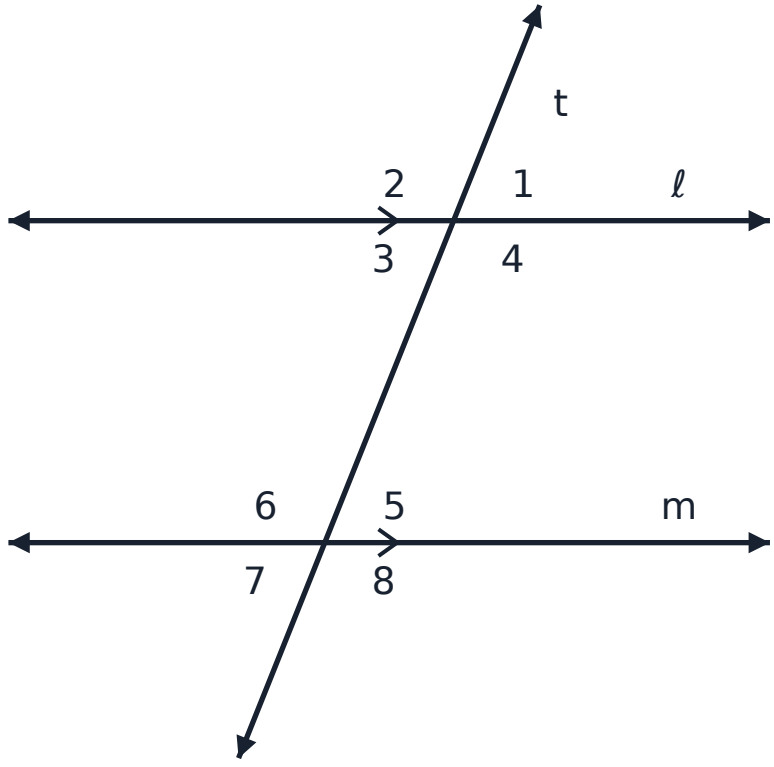
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q4/5

True or false?

Angles 1 and 4 are vertical angles in the figure.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 3

The parallel postulate and the definition of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Through a point not on a line, there is exactly one line parallel to it (in a plane).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q1/5

Line ℓ is parallel to line m. Line n passes through a point of ℓ and is also parallel to m. What can you conclude about n?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q2/5

Which pair of lines in a plane are parallel?

A. $x = 2$ and $y = 2$

B. $y = x$ and $y = -x$

C. $y = 2x + 1$ and $y = -2x + 1$

D. $y = 2x + 1$ and $y = 2x - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q3/5

True or false?

Two lines in a plane that never meet are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q4/5

Line $\ell \parallel$ line m. A translation maps m to a line through a point of ℓ. Must that image line be parallel to m or the same as m?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q5/5

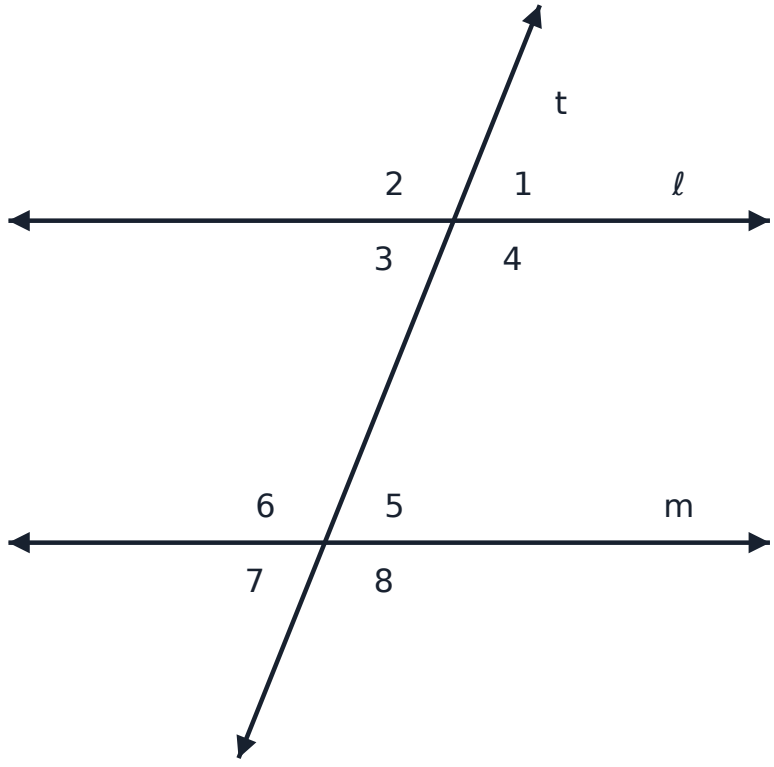
SET 18 • HSG-CO.C.9

FORWARD 1

Prove the converse: congruent alternate interior angles make lines parallel • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by transversal t, and alternate interior angles $\angle 3$ and $\angle 5$ are congruent. Prove that $\ell \parallel$ m.

Work space



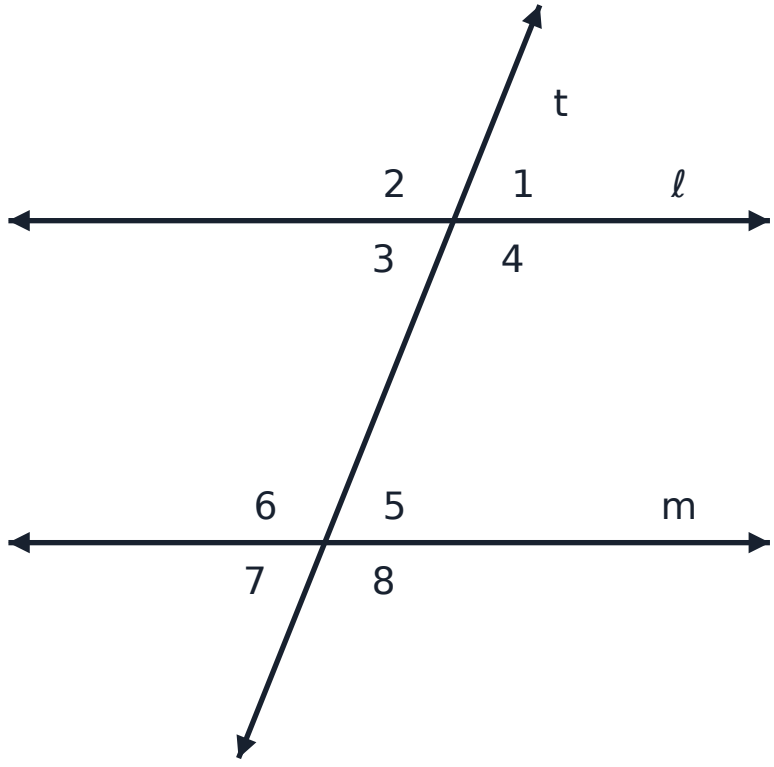
Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q1/5

Lines ℓ and m are cut by transversal t with $m\angle 4 = (5x - 20)°$ and $m\angle 6 = (3x + 30)°$. For what value of x is $\ell \parallel m$? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q2/5

True or false?

If same-side interior angles are supplementary, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q3/5

Which fact proves that lines ℓ and m are parallel?

A. A pair of corresponding angles is congruent

B. A pair of vertical angles is congruent

C. A linear pair adds to 180°

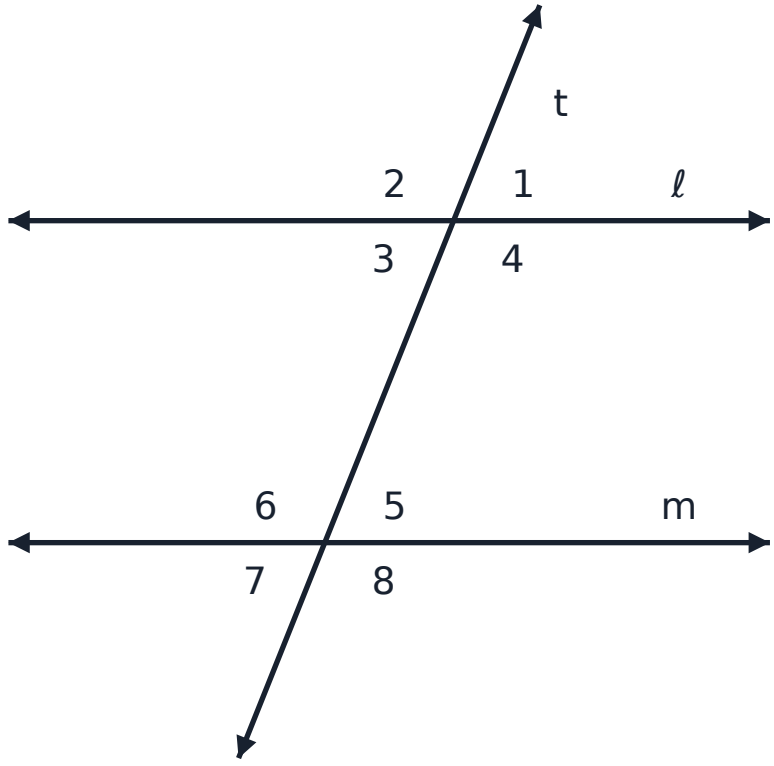
D. Two angles at the same crossing are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q4/5

Lines ℓ and m are cut by a transversal. $m\angle 1 = 72°$ and $m\angle 7 = 108°$. Are ℓ and m parallel? Explain.



Answer: _____

SET 18 • HSG-CO.C.9

FORWARD 2

Use parallel-line theorems to prove the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that the angles of triangle ABC add to 180°. Begin by drawing the line through B parallel to AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q1/5

In triangle ABC, a line through B parallel to AC makes a 50° angle with BA (on A's side) and a 65° angle with BC (on C's side). Find all three angles of the triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q2/5

True or false?

The proof of the triangle angle sum works without any fact about parallel lines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q3/5

Which theorem is used to show the new angles at B equal ∠A and ∠C?

A. Alternate interior angles theorem

B. Isosceles triangle theorem

C. Triangle inequality

D. Vertical angles theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q4/5

Explain why the exterior angle at C of triangle ABC equals m∠A + m∠B.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q5/5

SET 19 • HSG-CO.C.9 • CONGRUENCE: PROVING THEOREMS

Prove the perpendicular bisector theorem and its converse

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Use the definitions of midpoint and perpendicular bisector	**HSG-CO.A.1**
BACKWARD 2	Use SAS and SSS to prove triangles congruent	**HSG-CO.B.8**
BACKWARD 3	Corresponding parts of congruent triangles and right angles from…	**HSG-CO.B.7**
FORWARD 1	Use the perpendicular bisector to find points equidistant from two…	**HSG-GPE.B.5**
FORWARD 2	Prove that the perpendicular bisectors of a triangle meet at one p…	**HSG-C.A.3**

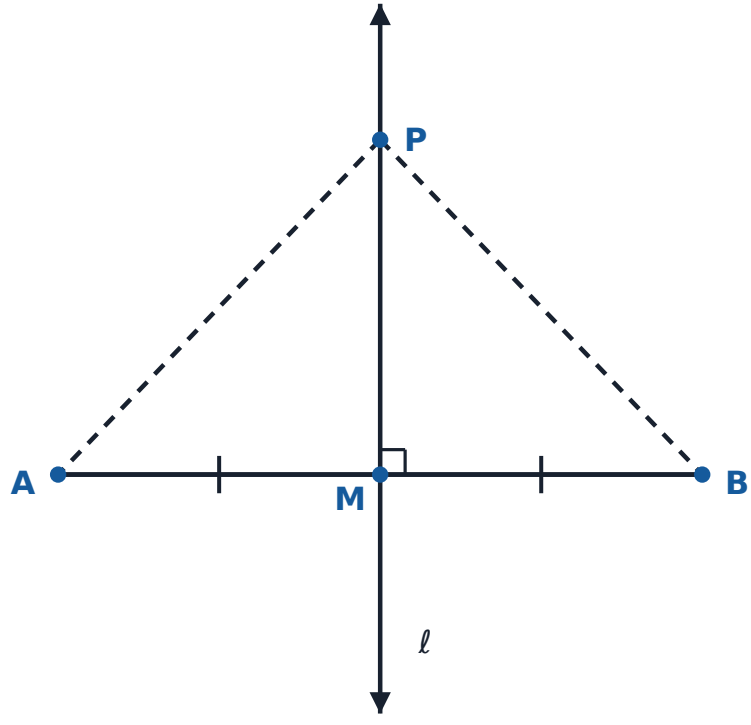
SET 19 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Line ℓ is the perpendicular bisector of segment AB, meeting AB at M. Point P is on ℓ. Prove that PA = PB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q1/5

Point P is not on line AB, and PA = PB. Let M be the midpoint of AB. Prove that line PM is perpendicular to AB, so P is on the perpendicular bisector of AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q2/5

Point P is not on line AB, and PA = PB. The bisector of ∠APB meets AB at M. Prove that line PM is the perpendicular bisector of AB (a second proof of the converse, using SAS).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q3/5

In a coordinate plane, A(−a, 0) and B(a, 0) with a > 0. Prove that every point P(0, y) on the y-axis is equidistant from A and B, and that every point P(x, y) with PA = PB has x = 0.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q4/5

A(1, 1) and B(5, 3). Prove that every point on the line $y = -2x + 8$ is equidistant from A and B, and that every point equidistant from A and B lies on $y = -2x + 8$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 1

Use the definitions of midpoint and perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

M is the midpoint of AB and AB = 14. Find AM.

AM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector of AB passes through the midpoint of AB and is perpendicular to AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q2/5

Line ℓ meets AB at M with AM = MB, but ℓ is not perpendicular to AB. What is ℓ?

A. Not related to AB

B. Parallel to AB

C. The perpendicular bisector of AB

D. A bisector of AB that is not the perpendicular bisector

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q3/5

Line ℓ is perpendicular to AB at M, and AM = 3, MB = 5. Is ℓ the perpendicular bisector of AB? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q4/5

True or false?

A segment has exactly one perpendicular bisector in a plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 2

Use SAS and SSS to prove triangles congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle PMA$ and $\triangle PMB$, PM is shared, AM = BM, and $\angle PMA \cong \angle PMB$. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q1/5

In $\triangle PMA$ and $\triangle PMB$, $PA = PB$, $AM = BM$, and PM is shared. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q2/5

Two triangles share a side and have two other pairs of congruent sides. Which criterion applies?

A. SAS

B. ASA

C. SSS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

A side shared by two triangles counts as a pair of congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

SSA is a valid congruence criterion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 3

Corresponding parts of congruent triangles and right angles from a linear pair • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

△PMA ≅ △PMB. Which angle corresponds to ∠PMA?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q1/5

Two congruent angles form a linear pair. What is each angle's measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q2/5

True or false?

If $\triangle PMA \cong \triangle PMB$, then $PA = PB$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q3/5

$\triangle XYZ \cong \triangle XWZ$. Which pair of segments must be congruent?

A. YZ and XW

B. YZ and WZ

C. XY and XZ

D. XZ and YW

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q4/5

True or false?

Two angles in a linear pair are always congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • HSG-CO.C.9

FORWARD 1

Use the perpendicular bisector to find points equidistant from two points • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Find an equation of the perpendicular bisector of the segment from A(2, 1) to B(6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q1/5

Find the point on the x-axis that is equidistant from A(2, 1) and B(6, 5).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q2/5

True or false?

The point (4, 4) is equidistant from (0, 0) and (8, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q3/5

Which point is equidistant from A(0, 0) and B(4, 0)?

A. (3, 1)

B. (4, 4)

C. (2, 7)

D. (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q4/5

Points A(−1, 2) and B(3, 2). Describe all points equidistant from A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

FORWARD 2

Prove that the perpendicular bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

Prove that the perpendicular bisectors of the sides of triangle ABC meet at one point O, and that O is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q1/5

Triangle ABC has A(0, 0), B(6, 0), C(0, 8). Find the point equidistant from all three vertices.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q2/5

True or false?

The point where the perpendicular bisectors of a triangle meet is always inside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q3/5

Three towns that are not on one straight line will share a well equally distant from all three. Where should it go?

A. Where the perpendicular bisectors of the triangle's sides meet

B. Where the angle bisectors meet

C. At the midpoint of the longest side

D. Where the medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q4/5

Explain why showing that the perpendicular bisectors of AB and BC meet at O is enough to show that all three meet at O.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q5/5

Prove that the angles of a triangle add to 180°

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Use the angle sum and parallel-line angles informally	**8.G.A.5**
BACKWARD 2	Angles that form a straight angle add to 180°	**4.MD.C.7**
BACKWARD 3	Alternate interior angles of parallel lines are congruent	**HSG-CO.C.9**
FORWARD 1	Prove the exterior angle theorem	**HSG-CO.C.10**
FORWARD 2	Generalize the angle sum to polygons	**HSG-CO.C.10**

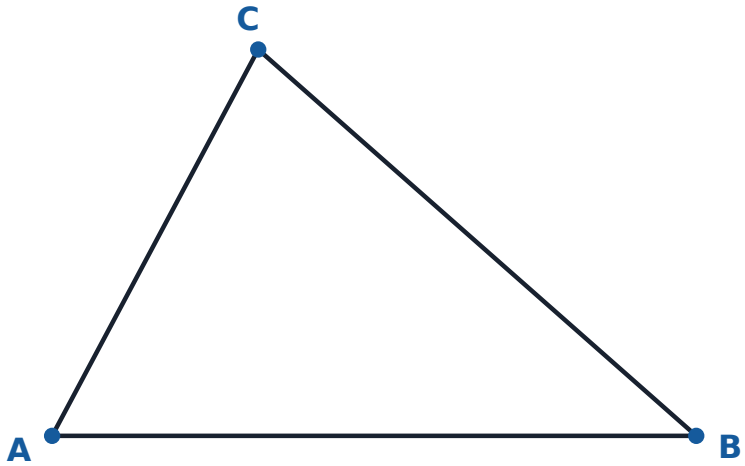
SET 20 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that
$m\angle A + m\angle B + m\angle C = 180°$ in triangle ABC. Draw the line through C parallel to AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q1/5

In triangle PQR, draw the line through P parallel to QR. Prove that the angles of triangle PQR add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q2/5

In triangle ABC, extend side BC past C to point D, and draw ray CE parallel to BA (E on the same side of line BD as A). Prove that the angles of triangle ABC add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q3/5

In triangle ABC, translate the triangle along ray AB so that A moves to B; C moves to C′ and B moves to D (on ray AB beyond B). Prove that the angles of triangle ABC add to 180°, using the three angles at B.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q4/5

In triangle PQR, extend side PQ past Q to S, and draw ray QT parallel to PR (T on the same side of line PS as R). Prove that the angles of triangle PQR add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 1

Use the angle sum and parallel-line angles informally • Grade 8 • Geometry • 8.G.A.5

Two angles of a triangle are 47° and 68°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q1/5

True or false?

Tearing off the three corners of a paper triangle and lining them up makes a straight angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q2/5

A triangle has angles x, 2x, and 3x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q3/5

Lines $\ell \parallel m$ are cut by a transversal. An angle at ℓ is 72°. Which angle at m must also be 72°?

A. Its alternate interior angle

B. Its same-side interior angle

C. Every angle at m

D. The angle next to it on the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q4/5

True or false?

A triangle can have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 2

Angles that form a straight angle add to 180° • Grade 4 • Measurement & Data • 4.MD.C.7

Three angles together form a straight angle. Two are 40° and 75°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q1/5

True or false?

The angles around a point on one side of a straight line add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q2/5

$\angle 1$, $\angle 2$, and $\angle 3$ form a straight angle, with $\angle 1 = \angle 3 = 50°$. Find $\angle 2$.

A. 50°

B. 100°

C. 80°

D. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q3/5

Angle ABC is a straight angle. Ray BD makes ∠ABD = 3x and ∠DBC = 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q4/5

True or false?

Two angles that make up a straight angle must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 3

Alternate interior angles of parallel lines are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines $\ell \parallel m$ are cut by transversal t. An alternate interior angle of a 58° angle measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q1/5

True or false?

Alternate interior angles formed by a transversal of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q2/5

Triangle ABC has A at the left end of base AB, B at the right end, and C above AB. Line ℓ through C is parallel to AB; D is on ℓ to the left of C and E is on ℓ to the right of C. Which angle is congruent to $\angle CAB$?

A. $\angle ACB$

B. $\angle ECA$

C. $\angle DCA$

D. $\angle ECB$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q3/5

Why is it allowed to draw the line through C parallel to AB?

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q4/5

True or false?

Alternate interior angles are congruent for any two lines cut by a transversal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 1

Prove the exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, side BC is extended past C to D. Prove that $m\angle ACD = m\angle A + m\angle B$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q1/5

An exterior angle of a triangle measures 125°. One remote interior angle is 48°. Find the other remote interior angle and the adjacent interior angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q2/5

True or false?

An exterior angle of a triangle is always greater than 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q3/5

The exterior angle at C is (5x + 10)° and the remote interior angles are (2x + 15)° and (2x + 20)°. Find x.

A. 5

B. 35

C. 25

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q4/5

Prove that the three exterior angles of a triangle (one at each vertex) add to 360°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 2

Generalize the angle sum to polygons • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that the interior angles of any convex polygon with n sides add to $(n - 2) \cdot 180°$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q1/5

Find the measure of each interior angle of a regular octagon.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q2/5

Each interior angle of a regular polygon is 156°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q3/5

True or false?

The interior angles of a hexagon add to 1,080°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q4/5

A pentagon has angles 100°, 110°, 120°, 95°, and x°. Find x.

A. 125

B. 115

C. 135

D. 105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q5/5

SET 21 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove that the base angles of an isosceles triangle are congruent

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Prove triangles congruent with SAS or SSS	**HSG-CO.B.8**
BACKWARD 2	Use the definitions of angle bisector and midpoint	**HSG-CO.A.1**
BACKWARD 3	Classify triangles and name legs, base, and base angles	**4.G.A.2**
FORWARD 1	Prove the converse: congruent base angles give congruent sides	**HSG-CO.C.10**
FORWARD 2	Use isosceles triangle theorems in multistep problems and proofs	**HSG-CO.C.10**

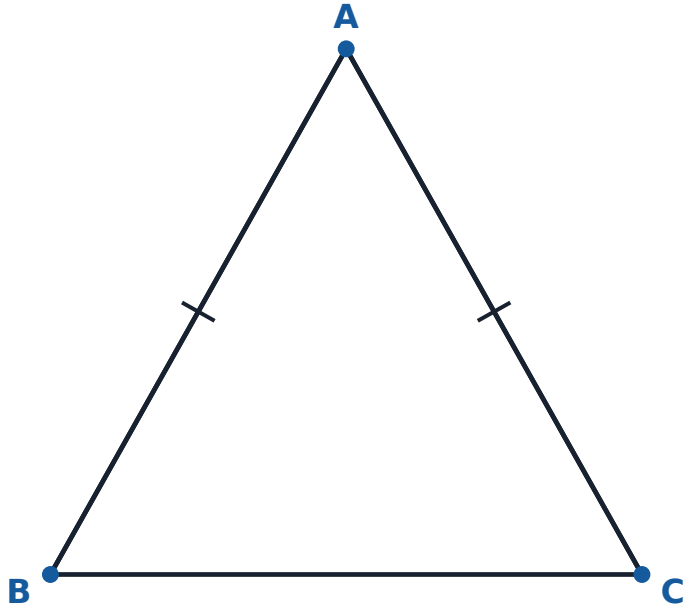
SET 21 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C. (Draw the bisector of ∠A, meeting BC at D.)

Work space



Conclusion: _____

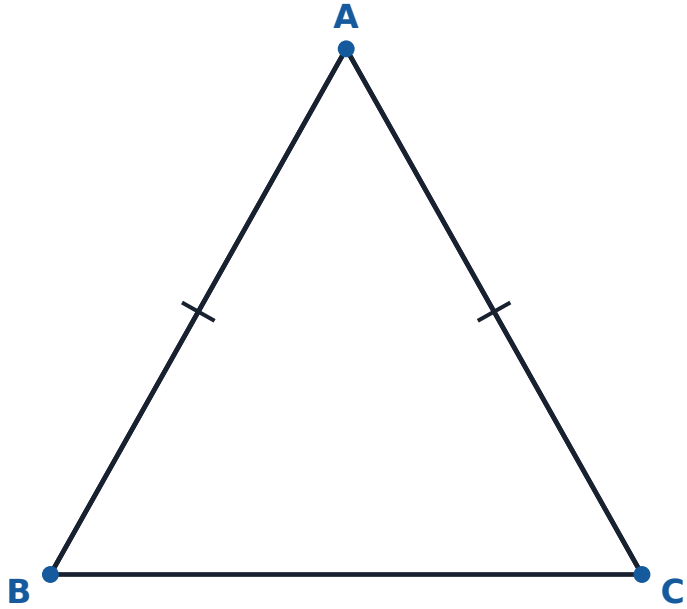
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q1/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using the midpoint M of BC.

Work space



Conclusion: _____

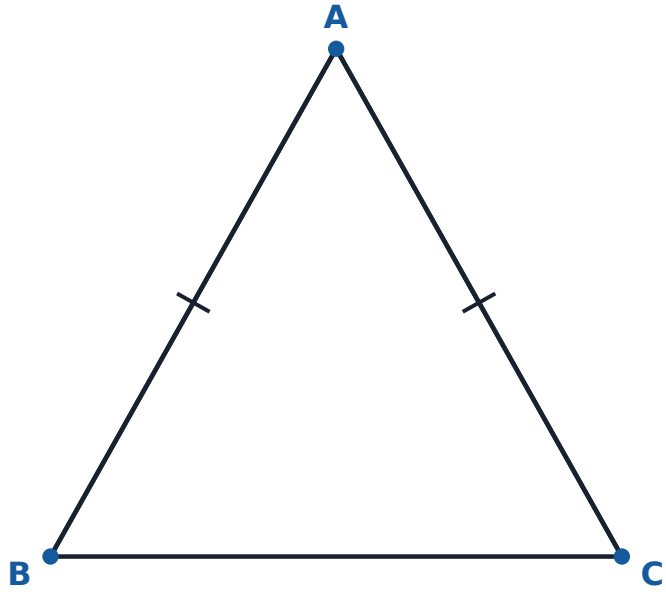
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q2/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using a rigid motion: the reflection across the bisector of ∠A.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q3/5

Prove that every equilateral triangle is equiangular (all three angles are congruent).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q4/5

Triangle ABC has A(0, h), B(−b, 0), and C(b, 0), with b > 0 and h > 0, so AB = AC. Prove that ∠B ≅ ∠C using the reflection (x, y) → (−x, y).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 1

Prove triangles congruent with SAS or SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share side AD, and AB = AC, ∠BAD ≅ ∠CAD. Which criterion shows △ABD ≅ △ACD?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q1/5

Two triangles share side AM, and AB = AC, BM = CM. Which criterion shows $\triangle ABM \cong \triangle ACM$?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q2/5

True or false?

A shared side counts as a pair of congruent sides in a congruence proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q3/5

Which information is NOT enough for congruence?

A. Three sides

B. Two sides and the included angle

C. Two angles and the included side

D. Two sides and a non-included angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q4/5

True or false?

If $\triangle ABD \cong \triangle ACD$, then $\angle ABD \cong \angle ACD$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 2

Use the definitions of angle bisector and midpoint • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Ray AD bisects ∠BAC, and m∠BAC = 64°. Find m∠BAD.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q1/5

M is the midpoint of BC, and BC = 18. Find BM.

BM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q2/5

True or false?

An angle bisector divides an angle into two congruent angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q3/5

Ray AD bisects ∠BAC. Which equation is true?

A. m∠BAD = m∠BAC

B. m∠BAD = m∠DAC

C. AD = AB

D. BD = DC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q4/5

True or false?

The bisector of $\angle A$ of triangle ABC always passes through the midpoint of BC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 3

Classify triangles and name legs, base, and base angles • Grade 4 • Geometry • 4.G.A.2

Triangle PQR has PQ = PR. Name the base and the base angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q1/5

True or false?

An isosceles triangle has at least two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q2/5

Which triangle is scalene?

A. Sides 3, 3, 3

B. Sides 5, 5, 5

C. Sides 4, 5, 6

D. Sides 4, 4, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q3/5

In isosceles triangle XYZ with XY = XZ, which angle is the vertex angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q4/5

True or false?

Every equilateral triangle is isosceles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 1

Prove the converse: congruent base angles give congruent sides • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, $\angle B \cong \angle C$. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q1/5

In triangle PQR, m∠Q = 70° and m∠R = 70°, and PQ = 3x + 2, PR = 5x − 6. Find x and PQ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q2/5

True or false?

A triangle with two congruent angles must have two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q3/5

Triangle ABC has $m\angle A = 50°$ and $m\angle B = 65°$. Which two sides are congruent?

A. No two sides

B. AB and BC

C. AB and AC

D. AC and BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q4/5

Prove that every equiangular triangle is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 2

Use isosceles triangle theorems in multistep problems and proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC and the vertex angle ∠A measures 38°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q1/5

Point D is on side BC of triangle ABC with AD = BD = CD. Find m∠BAC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q2/5

In triangle ABC, AB = AC, and D and E are points on BC with BD = CE. Prove that AD = AE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q3/5

True or false?

If the vertex angle of an isosceles triangle is 100°, each base angle is 80°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q4/5

In isosceles triangle PQR with PQ = PR, the exterior angle at Q (on the extension of RQ past Q) is 112°. Find m∠P.

A. 56°

B. 112°

C. 68°

D. 44°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q5/5

SET 22 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove the midsegment theorem

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Find midpoints of segments in the coordinate plane	**6.G.A.3**
BACKWARD 2	Find and compare slopes to decide whether lines are parallel	**8.EE.B.6**
BACKWARD 3	Find lengths with the distance formula	**8.G.B.8**
BACKWARD 4	Prove lines parallel from congruent alternate interior angles	**HSG-CO.C.9**
FORWARD 1	Prove that the midpoints of a quadrilateral's sides form a parallelo	**HSG-CO.C.11**
FORWARD 2	Midsegment triangles: similarity, scale factor, and area	**HSG-SRT.A.2**

SET 22 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Triangle ABC has A(0, 0), B(2a, 0), and C(2b, 2c), with a > 0 and c ≠ 0. D is the midpoint of AC and E is the midpoint of BC. Prove that DE ∥ AB and DE = ½AB.

Work space

Conclusion: _____

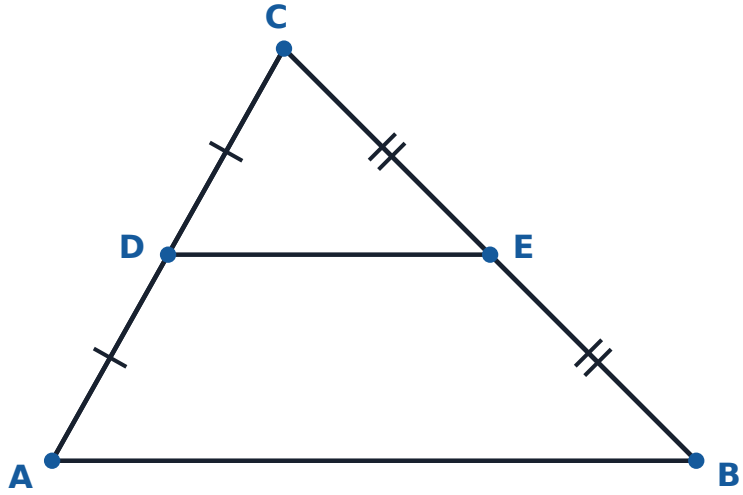
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q1/5

In triangle ABC, D is the midpoint of AC and E is the midpoint of BC. Extend DE past E to F so that EF = DE. Prove that DE ∥ AB and DE = ½AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q2/5

Triangle ABC has A(0, 0), B(2a, 0), and C(2b, 2c), with $a > 0$ and $c \neq 0$. F is the midpoint of AB and E is the midpoint of AC. Prove that FE ∥ BC and FE = ½BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q3/5

In triangle ABC, F is the midpoint of AB and E is the midpoint of AC. Extend FE past E to G so that EG = FE. Prove that FE ∥ BC and FE = ½BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q4/5

Triangle ABC has A(2p, 2q), B(2r, 2s), and C(2t, 2u), not on one line, with AB not vertical. D is the midpoint of AC and E is the midpoint of BC. Prove that DE ∥ AB and DE = ½AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 1

Find midpoints of segments in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (2, 6) to (8, 2).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from $(-3, 1)$ to $(5, -7)$.

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q2/5

Find the midpoint of the segment from (0, 0) to (2a, 2c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q3/5

True or false?

The midpoint of the segment from (1, 1) to (7, 1) is (4, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q4/5

True or false?

The midpoint of the segment from (0, 4) to (6, 0) is (6, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 2

Find and compare slopes to decide whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

Find the slope of the line through (1, 2) and (4, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

Find the slope of the line through (b, c) and (a + b, c), where $a \neq 0$.

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

True or false?

Two different nonvertical lines with the same slope are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

Which segment is parallel to the segment from (0, 0) to (4, 2)?

A. From (1, 3) to (5, 5)

B. From (2, 2) to (2, 6)

C. From (1, 3) to (3, 7)

D. From (0, 0) to (2, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

A horizontal segment and a vertical segment have the same slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 3

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

Find the distance between (b, c) and (a + b, c), with a > 0.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q1/5

Find the distance between (0, 0) and (6, 8).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q2/5

Find the length of the segment from $(-2, 5)$ to $(4, 5)$.

A. 3

B. 10

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q3/5

True or false?

The distance from (1, 2) to (4, 6) is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q4/5

True or false?

The distance from (0, 0) to (2a, 0) is a, when $a > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 4

Prove lines parallel from congruent alternate interior angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by a transversal, and alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q1/5

True or false?

If two lines are cut by a transversal and alternate interior angles are congruent, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q2/5

In $\triangle ABC$, D is the midpoint of AC and E is the midpoint of BC. DE is extended to F so that $EF = DE$, and $\triangle DEC \cong \triangle FEB$. Which fact shows $BF \parallel AC$?

A. $DE = EF$

B. $\angle DCE \cong \angle FBE$ are alternate interior angles

C. $BF = DC$

D. E is the midpoint of BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q3/5

Two lines are cut by a transversal and corresponding angles measure 70° and 70°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q4/5

True or false?

Congruent vertical angles prove that two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 1

Prove that the midpoints of a quadrilateral's sides form a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is any quadrilateral, and P, Q, R, S are the midpoints of AB, BC, CD, DA. Prove that PQRS is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q1/5

Quadrilateral ABCD has diagonals AC = 10 and BD = 14. Find the perimeter of the quadrilateral formed by the midpoints of its sides.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q2/5

True or false?

The midpoint quadrilateral of any quadrilateral is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q3/5

If the diagonals of ABCD are perpendicular, the midpoint quadrilateral is a

A. rectangle

B. trapezoid that is not a parallelogram

C. rhombus

D. kite that is not a parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q4/5

In triangle ABC, D, E, F are the midpoints of AB, BC, CA. AB = 12, BC = 16, CA = 10. Find the perimeter of triangle DEF.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 2

Midsegment triangles: similarity, scale factor, and area • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

D, E, F are the midpoints of the sides of triangle ABC. Explain why triangle DEF is similar to triangle ABC, and give the scale factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q1/5

Triangle ABC has area 48. Find the area of the triangle formed by the midpoints of its sides.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q2/5

True or false?

The four triangles formed by drawing all three midsegments of a triangle are similar to it with scale factor ⅓.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q3/5

The midsegment parallel to AB in triangle ABC is 9. A line segment connects the midpoints of that midsegment's triangle the same way. How long is the new segment parallel to AB?

A. 9

B. 18

C. 3

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q4/5

Triangle ABC has vertices A(0, 0), B(8, 0), C(2, 6). Find the vertices of its midsegment triangle and verify the ratio of areas.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q5/5

Prove that the medians of a triangle meet at one point

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Find midpoints, including with variables	**6.G.A.3**
BACKWARD 2	Use the midsegment theorem	**HSG-CO.C.10**
BACKWARD 3	Find the point a given fraction of the way along a segment	**7.RP.A.2**
BACKWARD 4	Write the equation of a line through two points	**8.F.B.4**
FORWARD 1	Locate the centroid and use the 2 : 1 ratio	**HSG-GPE.B.6**
FORWARD 2	Prove that the angle bisectors of a triangle meet at one point	**HSG-C.A.3**

SET 23 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Triangle ABC has A(0, 0), B(2b, 0), and C(2c, 2d), with b ≠ 0 and d ≠ 0. Prove that the point G((2b + 2c)/3, 2d/3) lies on all three medians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q1/5

In triangle ABC, medians BE and CF meet at G. Let H and K be the midpoints of BG and CG. Prove that BG = 2GE. Then explain why the third median also passes through G.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q2/5

Triangle ABC has A(−2a, 0), B(2a, 0), and C(0, 2b), with a > 0 and b > 0. Prove that the three medians meet at (0, 2b/3).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q3/5

Triangle PQR has P(0, 0), Q(6, 0), and R(0, 6). Find the three medians' equations and prove that they meet at one point.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q4/5

Triangle ABC has $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$, not on one line. Let $G = ((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3)$. Prove that G lies on all three medians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 1

Find midpoints, including with variables • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (2b, 0) to (2c, 2d).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from (0, 6) to (6, 0).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q2/5

A median of a triangle joins a vertex to

A. the midpoint of the opposite side

B. another vertex

C. the foot of the perpendicular from that vertex

D. the midpoint of an adjacent side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q3/5

True or false?

A triangle has exactly three medians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q4/5

True or false?

A median always bisects the angle at its vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 2

Use the midsegment theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, F and E are the midpoints of AB and AC. BC = 14. Find FE.

FE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q1/5

True or false?

The segment joining the midpoints of two sides of a triangle is parallel to the third side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q2/5

In triangle GBC, H and K are midpoints of GB and GC. Which statement is true?

A. HK ⊥ BC

B. HK ∥ GB

C. HK ∥ BC and HK = ½BC

D. HK = BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q3/5

A midsegment of a triangle is 11 cm. How long is the side parallel to it?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q4/5

True or false?

A midsegment of a triangle is always half as long as each of the other two sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 3

Find the point a given fraction of the way along a segment • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

On a number line, what point is ⅔ of the way from 0 to 9?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q1/5

On a number line, what point is ⅔ of the way from 3 to 12?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q2/5

A point is ⅔ of the way from A to D. What is the ratio of its distances from A and from D?

A. 2 : 3

B. 3 : 2

C. 1 : 2

D. 2 : 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q3/5

True or false?

The point ⅔ of the way from (0, 0) to (6, 3) is (4, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q4/5

True or false?

The point ⅔ of the way from (0, 0) to (3, 6) is (2, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 4

Write the equation of a line through two points • Grade 8 • Functions • 8.F.B.4

Write an equation of the line through (0, 0) and (3, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q1

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q1/5

Write an equation of the line through (6, 0) and (0, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q2

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q2/5

Which point is on the line $y = -2x + 6$?

A. (1, 2)

B. (3, 3)

C. (0, 2)

D. (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q3

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q3/5

True or false?

The line through (0, 6) and (3, 0) has slope −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q4

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q4/5

True or false?

The line $y = x$ passes through $(2, 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q5

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q5/5

SET 23 • HSG-CO.C.10

FORWARD 1

Locate the centroid and use the 2 : 1 ratio • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the point where the medians of the triangle with vertices (1, 2), (7, 4), and (4, 9) meet.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q1/5

Median AD of a triangle is 18 cm long. How far is the centroid from A and from D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q2/5

True or false?

The centroid divides each median so that the part from the vertex is half the part from the midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q3/5

G is the centroid of triangle ABC, and GE = 5 on median BE. Find BE.

A. 20

B. 10

C. 7.5

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q4/5

Explain why the centroid of a triangle is the average of its three vertices.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSG-CO.C.10

FORWARD 2

Prove that the angle bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

Prove that the bisectors of the three angles of triangle ABC meet at one point that is equidistant from the three sides.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q1/5

Why must the bisectors of $\angle A$ and $\angle B$ of a triangle meet inside the triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q2/5

True or false?

A point on the bisector of an angle is equidistant from the two sides of the angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q3/5

The point where the angle bisectors of a triangle meet is the center of the

A. circumscribed circle

B. nine-point circle

C. inscribed circle

D. largest square inside the triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q4/5

True or false?

The angle bisectors and the medians of every triangle meet at the same point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q5/5

Prove that opposite sides of a parallelogram are congruent

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Alternate interior angles of parallel lines	**HSG-CO.C.9**
BACKWARD 2	Prove triangles congruent with ASA	**HSG-CO.B.8**
BACKWARD 3	Know the definition of a parallelogram	**5.G.B.3**
FORWARD 1	Prove that the diagonals of a rhombus are perpendicular	**HSG-CO.C.11**
FORWARD 2	Use opposite sides with coordinates and algebra	**HSG-GPE.B.4**

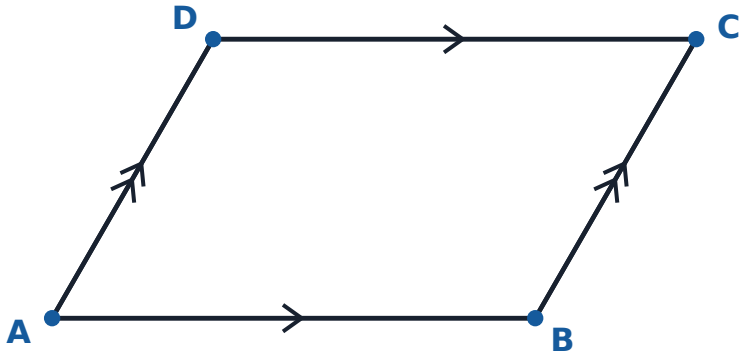
SET 24 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a parallelogram (AB ∥ DC and AD ∥ BC). Prove that AB = CD and AD = CB. (Draw diagonal AC.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q1/5

PQRS is a parallelogram (PQ ∥ SR and PS ∥ QR). Prove that PQ = RS and QR = SP, using diagonal QS.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q2/5

ABCD is a parallelogram and M is the midpoint of diagonal AC. Explain, using the 180° rotation about M, why AB = CD and AD = CB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q3/5

A parallelogram has vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c), with a > 0 and c > 0. Prove that AB = DC and AD = BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q4/5

ABCD and ABEF are parallelograms that share side AB, with C, D, E, F all different points. Prove that DC = FE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 1

Alternate interior angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

AB ∥ DC and diagonal AC is drawn. Which angle at C is congruent to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q1/5

True or false?

Alternate interior angles formed by a transversal of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q2/5

In parallelogram ABCD with diagonal BD, which angle is congruent to ∠ABD?

A. ∠ABC

B. ∠ADB

C. ∠CBD

D. ∠CDB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q3/5

In parallelogram ABCD with diagonal AC, AD ∥ BC. Which angle is congruent to ∠DAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q4/5

True or false?

Alternate interior angles of parallel lines are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 2

Prove triangles congruent with ASA • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles have two pairs of congruent angles and the congruent sides between them. Which criterion applies?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q1/5

True or false?

A diagonal shared by two triangles can be the included side in ASA.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q2/5

In $\triangle ABC$ and $\triangle CDA$, $\angle BAC \cong \angle DCA$, $\angle BCA \cong \angle DAC$, and AC is shared. Which correspondence is correct?

A. A ↔ D, B ↔ C, C ↔ A

B. A ↔ A, B ↔ D, C ↔ C

C. A ↔ C, B ↔ D, C ↔ A

D. A ↔ C, B ↔ A, C ↔ D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q3/5

After proving $\triangle ABC \cong \triangle CDA$, which side corresponds to AB?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q4/5

True or false?

ASA requires the side to be between the two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 3

Know the definition of a parallelogram • Grade 5 • Geometry • 5.G.B.3

True or false?

A parallelogram is a quadrilateral with both pairs of opposite sides parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q1/5

Which quadrilateral is always a parallelogram?

A. A trapezoid with exactly one pair of parallel sides

B. A kite that is not a rhombus

C. Any quadrilateral with one right angle

D. A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q2/5

Name the two pairs of opposite sides of parallelogram PQRS.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q3/5

True or false?

In parallelogram ABCD, sides AB and BC are opposite sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q4/5

Which sides of parallelogram WXYZ are parallel to WX and to XY?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q5/5

SET 24 • HSG-CO.C.11

FORWARD 1

Prove that the diagonals of a rhombus are perpendicular • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a rhombus (a parallelogram with all four sides congruent), with diagonals meeting at E. Prove that AC ⊥ BD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q1/5

In rhombus ABCD, diagonals AC = 16 and BD = 12. Find the side length.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q2/5

ABCD is a rhombus. Prove that diagonal AC bisects ∠BAD and ∠BCD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q3/5

Rhombus ABCD has perimeter 52 and diagonal AC = 24. Find diagonal BD.

BD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q4/5

In rhombus PQRS, m∠PQR = 110°. Find m∠PQS, where QS is a diagonal, and give the reason.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSG-CO.C.11

FORWARD 2

Use opposite sides with coordinates and algebra • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Parallelogram ABCD has A(1, 2), B(6, 2), and D(3, 5). Find C, and verify that AB = DC and AD = BC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q1/5

Explain why the vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c) describe a general parallelogram.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q2/5

In parallelogram WXYZ, $WX = 3x + 4$ and $YZ = 5x - 6$. Find x and WX.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q3/5

For the parallelogram A(0, 0), B(6, 0), C(8, 4), D(2, 4), what is AD?

A. $\sqrt{20} + 2$

B. 4

C. $2\sqrt{5}$

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q4/5

Parallelogram ABCD has A(1, 1), B(5, 1), and D(2, 4). Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q5/5

SET 101 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that opposite angles of a parallelogram are congruent

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Same-side interior angles of parallel lines	**HSG-CO.C.9**
BACKWARD 2	Name opposite and consecutive angles of a quadrilateral	**5.G.B.3**
BACKWARD 3	Prove triangles congruent with ASA and use corresponding angles	**HSG-CO.B.8**
FORWARD 1	Prove angle properties of special parallelograms	**HSG-CO.C.11**
FORWARD 2	Opposite angles and angles of rotation with coordinates	**HSG-CO.A.3**

SET 101 • HSG-CO.C.11

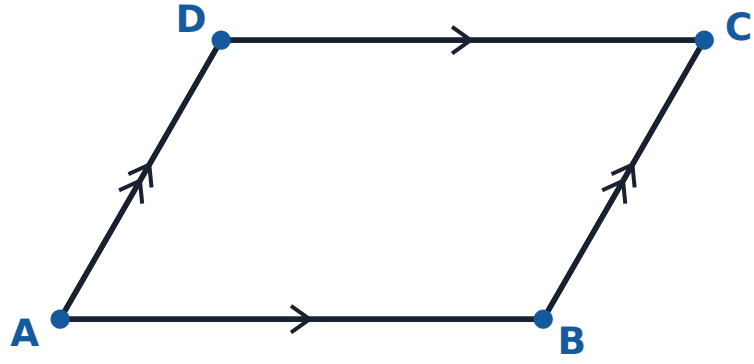
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a parallelogram.

Prove that $\angle B \cong \angle D$.

Work space



Conclusion: _____

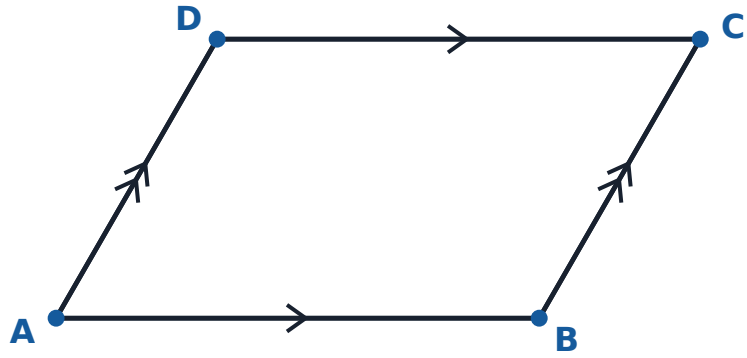
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q1/5

ABCD is a parallelogram. Prove that ∠A ≅ ∠C using same-side interior angles (no diagonal).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q2/5

PQRS is a parallelogram. Prove that $\angle P \cong \angle R$, using diagonal QS.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q3/5

ABCD is a parallelogram and M is the midpoint of diagonal AC. Explain, using the 180° rotation about M, why ∠DAB ≅ ∠BCD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q4/5

ABCD is a parallelogram with $m\angle A = x°$. Prove that $m\angle C = x°$ and $m\angle B = m\angle D = (180 - x)°$, giving a reason for each step.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q5/5

SET 101 • HSG-CO.C.11

BACKWARD 1

Same-side interior angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Same-side interior angles formed by a transversal of parallel lines are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q1/5

In parallelogram ABCD, side AB is a transversal of AD and BC. What is m∠A + m∠B?

A. It depends on the figure

B. 180°

C. 360°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q2/5

In parallelogram ABCD, m∠A = 65°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q3/5

Lines $\ell \parallel m$ are cut by a transversal; one same-side interior angle is 72°. Find the other.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q4/5

True or false?

Same-side interior angles of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q5/5

SET 101 • HSG-CO.C.11

BACKWARD 2

Name opposite and consecutive angles of a quadrilateral • Grade 5 • Geometry • 5.G.B.3

Name the two pairs of opposite angles of parallelogram WXYZ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q1/5

True or false?

In quadrilateral ABCD, $\angle A$ and $\angle B$ are consecutive angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q2/5

Which angle is opposite ∠Q in quadrilateral PQRS?

A. ∠R

B. ∠P

C. ∠Q

D. ∠S

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q3/5

Quadrilateral KLMN: name the angles consecutive to ∠K.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q4/5

True or false?

In quadrilateral ABCD, $\angle A$ and $\angle C$ share a side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q5/5

SET 101 • HSG-CO.C.11

BACKWARD 3

Prove triangles congruent with ASA and use corresponding angles • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

$\triangle ABC \cong \triangle CDA$ with A ↔ C, B ↔ D, C ↔ A. Which angle corresponds to $\angle B$?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q1/5

True or false?

After two triangles are proved congruent, their corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q2/5

Which criterion uses two angles and the side between them?

A. ASA

B. SSS

C. SAS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q3/5

△PQS ≅ △RSQ with P ↔ R. Which angle corresponds to ∠P?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q4/5

True or false?

ASA can use a shared diagonal as its side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q5/5

SET 101 • HSG-CO.C.11

FORWARD 1

Prove angle properties of special parallelograms • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Prove that a parallelogram with one right angle has four right angles (so it is a rectangle).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q1/5

In parallelogram ABCD, $m\angle A = (3x + 10)°$ and $m\angle C = (5x - 30)°$. Find x and $m\angle B$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q2/5

Rhombus ABCD has m∠A = 60°. Prove that diagonal BD divides it into two equilateral triangles.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q3/5

In parallelogram PQRS, m∠P is twice m∠Q. Find all four angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q4/5

A parallelogram has an angle of 47°. Which list gives all four angles in order?

A. 47°, 43°, 47°, 43°

B. 47°, 47°, 133°, 133°

C. 47°, 133°, 47°, 133°

D. 47°, 133°, 133°, 47°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q5/5

SET 101 • HSG-CO.C.11

FORWARD 2

Opposite angles and angles of rotation with coordinates • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Parallelogram ABCD has A(0, 0), B(4, 0), C(6, 3), D(2, 3). The 180° rotation about (3, 1.5) maps it onto itself. Which angle does ∠A map onto?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q1/5

Parallelogram ABCD has A(0, 0), B(4, 0), C(6, 3), D(2, 3). Name the image of each vertex under the 180° rotation about (3, 1.5).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q2/5

True or false?

A 180° rotation about the center of a parallelogram maps each angle onto the angle opposite it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q3/5

Which transformation shows that opposite angles of every parallelogram are congruent?

A. A 90° rotation about a vertex

B. A translation along a side

C. A reflection across a diagonal

D. A 180° rotation about the intersection of the diagonals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q4/5

Parallelogram PQRS has P(1, 1), Q(5, 1), R(7, 4), S(3, 4). Give the center and angle of a rotation that maps ∠P onto ∠R.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q5/5

SET 25 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that the diagonals of a parallelogram bisect each other

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Opposite sides of a parallelogram are congruent	**HSG-CO.C.11**
BACKWARD 2	Vertical and alternate interior angles at the crossing of the diago…	**HSG-CO.C.9**
BACKWARD 3	Find the midpoint of a segment and use the definition of bisect	**6.G.A.3**
FORWARD 1	Prove that a 180° rotation carries a parallelogram onto itself	**HSG-CO.A.3**
FORWARD 2	Use diagonal bisection with coordinates and algebra	**HSG-GPE.B.4**

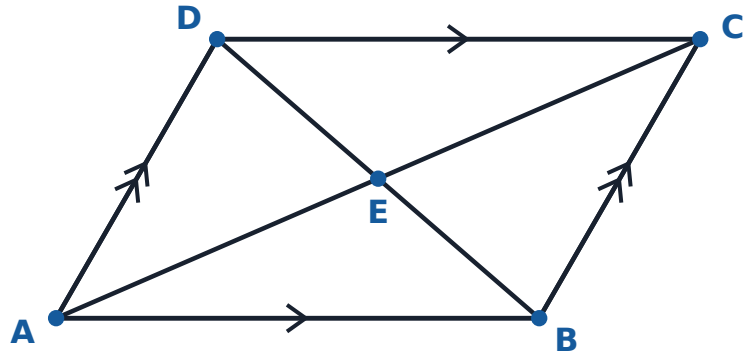
SET 25 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a parallelogram with diagonals AC and BD meeting at E. Prove that AE = CE and BE = DE.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q1/5

PQRS is a parallelogram with diagonals PR and QS meeting at T. Prove that T is the midpoint of both diagonals.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q2/5

A parallelogram has vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c). Prove that its diagonals bisect each other.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q3/5

ABCD is a parallelogram with diagonals meeting at E. Prove that AE = CE and BE = DE using triangles AED and CEB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q4/5

WXYZ is a parallelogram whose diagonals WY and XZ meet at V. Prove that V is the midpoint of WY and of XZ, using the 180° rotation about the midpoint M of WY (show that M is also the midpoint of XZ, so M = V).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 1

Opposite sides of a parallelogram are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

In parallelogram ABCD, AB = 9 and BC = 5. Find CD and DA.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q1/5

True or false?

Opposite sides of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q2/5

In parallelogram WXYZ, WX = 2x + 1 and YZ = x + 6. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q3/5

In parallelogram PQRS, which side is congruent to QR?

A. PQ

B. PR

C. SP

D. RS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q4/5

True or false?

Consecutive sides of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 2

Vertical and alternate interior angles at the crossing of the diagonals • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Diagonals of parallelogram ABCD meet at E. Which angle is vertical to ∠AEB?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q1/5

AB ∥ DC with transversal AC. Which angle is congruent to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q2/5

True or false?

Vertical angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q3/5

AB ∥ DC with transversal BD. Which angle is congruent to ∠ABD?

A. ∠CDB

B. ∠ABC

C. ∠ADB

D. ∠CBD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q4/5

True or false?

$\angle AEB$ and $\angle BEC$ are vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 3

Find the midpoint of a segment and use the definition of bisect • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (0, 0) to (a + b, c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q1/5

Find the midpoint of the segment from (a, 0) to (b, c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q2/5

True or false?

A segment bisects another segment if it passes through its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q3/5

Segment AC = 18 is bisected at E. Find AE.

AE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q4/5

True or false?

If two segments have the same midpoint, each bisects the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q5/5

SET 25 • HSG-CO.C.11

FORWARD 1

Prove that a 180° rotation carries a parallelogram onto itself • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

ABCD is a parallelogram whose diagonals meet at E. Prove that the 180° rotation about E maps ABCD onto itself.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q1/5

Parallelogram ABCD has diagonals meeting at E(2, 3), and A(−1, 1). Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q2/5

Quadrilateral ABCD has diagonals that bisect each other at E. Explain why the 180° rotation about E carries ABCD onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q3/5

The 180° rotation about the center of parallelogram PQRS maps side PQ onto

A. side SP

B. side RS

C. side QR

D. diagonal PR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q4/5

A 180° rotation about the intersection of its diagonals carries a parallelogram onto itself. Explain why this implies that opposite angles of a parallelogram are congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSG-CO.C.11

FORWARD 2

Use diagonal bisection with coordinates and algebra • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Parallelogram ABCD has A(1, 2), B(7, 2), C(9, 6). Find D and the point where the diagonals meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q1/5

In parallelogram ABCD, diagonals meet at E.
AE = 3x − 2 and CE = x + 8. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q2/5

True or false?

The diagonals of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q3/5

Three vertices of a parallelogram are (0, 0), (4, 0), (1, 3). Which could be the fourth vertex?

A. (4, 3)

B. (3, 3)

C. (5, 3)

D. (1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q4/5

In parallelogram PQRS, diagonals meet at T. PT = y + 4, TR = 2y − 1, QT = 3y. Find QS.

QS = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q5/5

SET 26 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove the converses: conditions that make a quadrilateral a parallelogram

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Converses of the parallel-line angle theorems	**HSG-CO.C.9**
BACKWARD 2	Prove triangles congruent with SSS and SAS	**HSG-CO.B.8**
BACKWARD 3	Angle sum of a quadrilateral	**8.G.A.5**
FORWARD 1	Prove that a quadrilateral is a parallelogram using coordinates	**HSG-GPE.B.4**
FORWARD 2	Decide which conditions are enough to guarantee a parallelogram	**HSG-CO.C.11**

SET 26 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

In quadrilateral ABCD, AB = CD and AD = CB. Prove that ABCD is a parallelogram. (Draw diagonal AC.)

Work space

Conclusion: _____

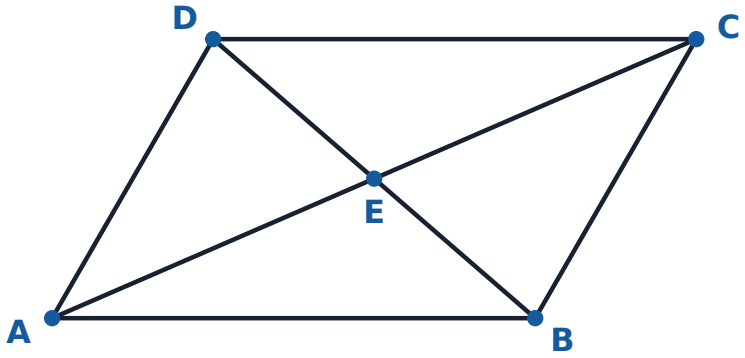
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q1/5

In quadrilateral ABCD, diagonals AC and BD bisect each other at E. Prove that ABCD is a parallelogram.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q2/5

In quadrilateral ABCD, AB ∥ DC and AB = DC. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q3/5

In quadrilateral ABCD, $\angle A \cong \angle C$ and $\angle B \cong \angle D$. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q4/5

In quadrilateral ABCD, $\angle A$ and $\angle B$ are supplementary, and $\angle B$ and $\angle C$ are supplementary. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 1

Converses of the parallel-line angle theorems • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Two lines are cut by a transversal so that alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q1/5

True or false?

If same-side interior angles are supplementary, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q2/5

Which fact proves two lines are parallel?

A. Congruent corresponding angles

B. A linear pair

C. Two acute angles

D. Congruent vertical angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q3/5

Same-side interior angles measure 104° and 76°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q4/5

True or false?

If alternate interior angles are 40° and 50°, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 2

Prove triangles congruent with SSS and SAS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

△ABC and △CDA share AC, and AB = CD, BC = DA. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q1/5

△AEB and △CED: AE = CE, BE = DE, and ∠AEB ≅ ∠CED. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q2/5

True or false?

Vertical angles can serve as the included angle in SAS.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q3/5

After $\triangle ABC \cong \triangle CDA$, which angle is congruent to $\angle BAC$?

A. $\angle BCA$

B. $\angle ABC$

C. $\angle DAC$

D. $\angle DCA$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q4/5

True or false?

SSA proves two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 3

Angle sum of a quadrilateral • Grade 8 • Geometry • 8.G.A.5

Three angles of a quadrilateral are 80°, 95°, and 110°. Find the fourth.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q1/5

True or false?

The angles of every quadrilateral add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q2/5

A quadrilateral has angles x, x, 2x, 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q3/5

Opposite angles of a quadrilateral are 70° and 70°, and another is 110°. Find the fourth.

A. 90°

B. 70°

C. 110°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q4/5

True or false?

A quadrilateral can have four acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q5/5

SET 26 • HSG-CO.C.11

FORWARD 1

Prove that a quadrilateral is a parallelogram using coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Prove or disprove that A(1, 1), B(6, 2), C(8, 6), D(3, 5) form a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q1/5

Prove or disprove that P(0, 0), Q(4, 1), R(5, 4), S(0, 3) form a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q2/5

True or false?

If the midpoints of both diagonals of a quadrilateral are the same point, the quadrilateral is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q3/5

Which test alone proves that ABCD is a parallelogram?

A. AC and BD have equal lengths

B. AB $\perp$ BC

C. AB and DC have equal slopes and equal lengths

D. AB and DC have equal lengths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q4/5

A(0, 0), B(5, 0), C(7, 3), D(x, 3) form a parallelogram ABCD. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSG-CO.C.11

FORWARD 2

Decide which conditions are enough to guarantee a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Is a quadrilateral with one pair of opposite sides congruent and the other pair parallel always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q1/5

Is a quadrilateral with both pairs of opposite angles congruent always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q2/5

True or false?

A quadrilateral with one diagonal that bisects the other must be a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q3/5

Which condition is NOT enough to prove a quadrilateral is a parallelogram?

A. One pair of opposite sides congruent

B. Both pairs of opposite sides congruent

C. Diagonals bisect each other

D. One pair of opposite sides parallel and congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q4/5

Is a quadrilateral with perpendicular diagonals always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q5/5

SET 27 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that rectangles are parallelograms with congruent diagonals

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Parallelogram properties: opposite sides and consecutive angles	**HSG-CO.C.11**
BACKWARD 2	SAS and SSS with overlapping triangles	**HSG-CO.B.8**
BACKWARD 3	Use the Pythagorean Theorem for diagonals	**8.G.B.7**
FORWARD 1	Decide whether four points form a rectangle	**HSG-GPE.B.4**
FORWARD 2	Prove diagonal properties of rhombuses and squares	**HSG-CO.C.11**

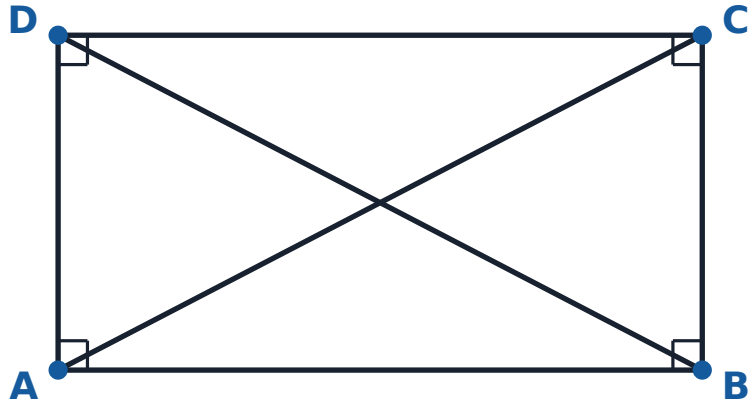
SET 27 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a rectangle. Prove that AC = BD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q1/5

ABCD is a parallelogram with AC = BD. Prove that ABCD is a rectangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q2/5

Rectangle ABCD has A(0, 0), B(a, 0), C(a, b), D(0, b), with a, b > 0. Prove that its diagonals are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q3/5

Quadrilateral ABCD has four right angles. Prove that ABCD is a parallelogram (so a rectangle is a parallelogram).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q4/5

The diagonals of quadrilateral ABCD bisect each other and are congruent. Prove that ABCD is a rectangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 1

Parallelogram properties: opposite sides and consecutive angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

In parallelogram ABCD, m∠A = 90°. Find the other three angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q1/5

True or false?

Consecutive angles of a parallelogram are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q2/5

In parallelogram PQRS, m∠P = m∠Q. Find m∠P.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q3/5

In parallelogram ABCD, which side is congruent to AD?

A. AC

B. CD

C. AB

D. BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q4/5

True or false?

In every parallelogram, all angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 2

SAS and SSS with overlapping triangles • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

△ABC and △BAD overlap and share AB. Which corresponding angle is at A in △BAD for ∠B in △ABC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q1/5

True or false?

Overlapping triangles can share a side used in a congruence proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q2/5

△ABC and △BAD: AB shared, BC = AD, AC = BD. Which criterion?

A. SSS

B. ASA

C. SAS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q3/5

△ABC and △BAD: AB shared, BC = AD, ∠ABC ≅ ∠BAD. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q4/5

True or false?

In $\triangle ABC \cong \triangle BAD$, vertex C corresponds to vertex C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 3

Use the Pythagorean Theorem for diagonals • Grade 8 • Geometry • 8.G.B.7

A rectangle is 6 by 8. Find the length of a diagonal.

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q1/5

A rectangle is 5 by 12. Find the length of a diagonal.

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q2/5

True or false?

A rectangle with sides a and b has diagonal $\sqrt{(a^2 + b^2)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q3/5

A square has side 4. Its diagonal is

A. 8

B. $4\sqrt{2}$

C. $2\sqrt{2}$

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q4/5

True or false?

The two diagonals of a 3-by-7 rectangle have different lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • HSG-CO.C.11

FORWARD 1

Decide whether four points form a rectangle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Prove or disprove that A(1, 1), B(5, 3), C(4, 5), D(0, 3) form a rectangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q1/5

Prove or disprove that P(0, 0), Q(4, 0), R(5, 3), S(1, 3) form a rectangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q2/5

True or false?

A quadrilateral whose diagonals are congruent and bisect each other is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q3/5

Which set of facts proves ABCD is a rectangle?

A. Diagonals bisect each other and are congruent

B. Diagonals are congruent

C. One angle is a right angle

D. Opposite sides are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q4/5

A(0, 0), B(6, 0), and D(0, 4) are three vertices of rectangle ABCD. Find C and verify the diagonals are congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSG-CO.C.11

FORWARD 2

Prove diagonal properties of rhombuses and squares • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Prove that a parallelogram whose diagonals are perpendicular is a rhombus.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q1/5

Which property must a quadrilateral have to be a square, using diagonals only?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q2/5

True or false?

A rectangle with congruent diagonals is a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q3/5

A parallelogram has congruent, perpendicular diagonals. It is a

A. rectangle that is not a square

B. rhombus that is not a square

C. square

D. trapezoid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q4/5

A square has diagonal 10. Find its side length (exact) and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q5/5

Construct a copy of a segment

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Compare and measure lengths	**2.MD.A.1**
BACKWARD 2	A compass draws all points at one distance from its center	**7.G.B.4**
BACKWARD 3	Know rays, segments, and their endpoints	**HSG-CO.A.1**
FORWARD 1	Construct a triangle from three given segments	**HSG-CO.D.12**
FORWARD 2	Decide whether segments can form a triangle, and construct sums	**HSG-CO.D.12**

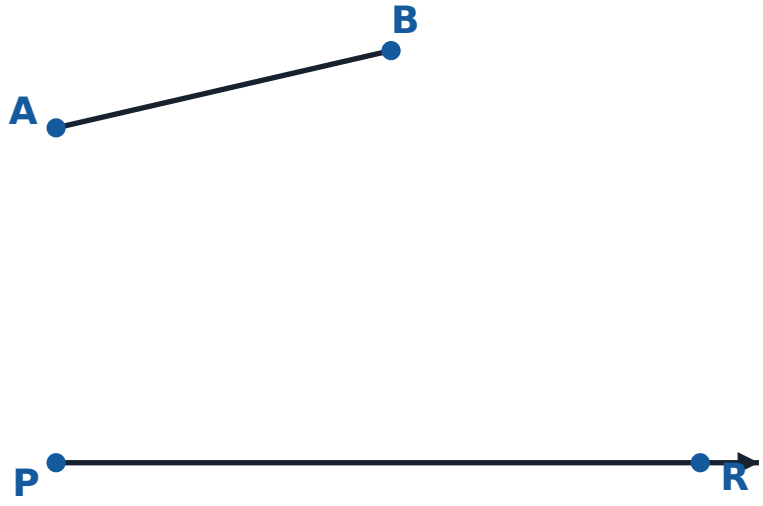
SET 28 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a point Q on ray PR so that PQ = AB.



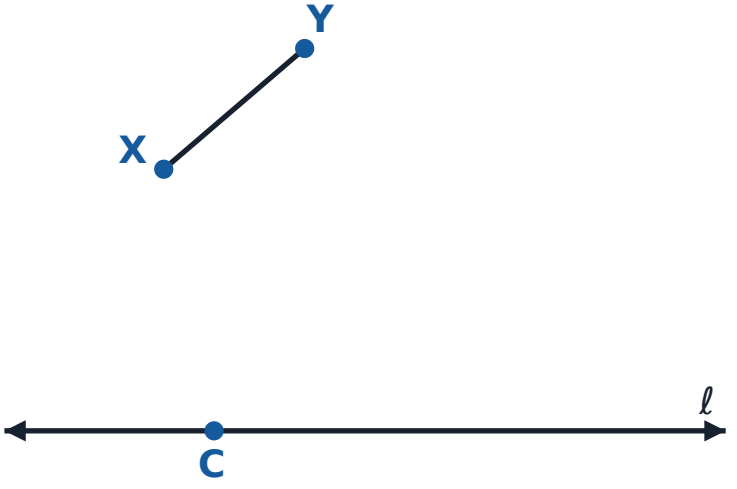
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

On line ℓ, starting at point C, construct a segment CD with CD = XY.



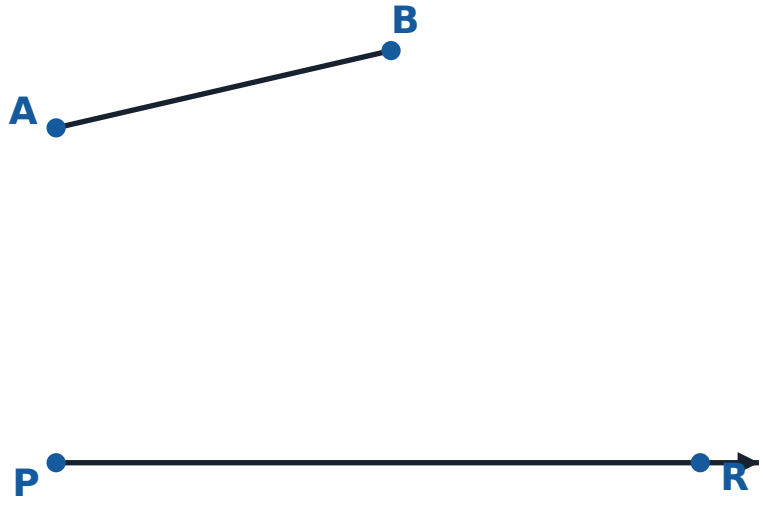
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a point Q on ray PR so that PQ = 2 · AB.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

On ray XW, construct a point Z so that XZ = AB + CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct an isosceles triangle with base PQ whose two legs are each congruent to segment AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 1

Compare and measure lengths • Grade 2 • Measurement & Data • 2.MD.A.1

Segment AB is 6 cm and segment CD is 6 cm. Are they congruent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q1/5

True or false?

Two segments are congruent if they have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q2/5

Which tool transfers a length without reading numbers?

A. A compass

B. A calculator

C. A protractor

D. A thermometer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q3/5

A segment 4.5 cm long is copied twice end to end on a line. How long is the result?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q4/5

True or false?

A segment drawn longer on the page than another is congruent to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 2

A compass draws all points at one distance from its center • Grade 7 • Geometry • 7.G.B.4

A compass is opened to 5 cm with its point on P. How far from P is every point on the arc it draws?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q1/5

True or false?

Changing the compass opening between steps of a construction keeps the copy accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q2/5

An arc centered at P crosses a ray from P at Q. What is PQ?

A. It cannot be known

B. Half the opening

C. Twice the opening

D. The compass opening

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q3/5

True or false?

Every point on a circle is the same distance from the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q4/5

An arc centered at C with radius 3 cm crosses a line through C. How many crossing points are there?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 3

Know rays, segments, and their endpoints • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Ray PR starts at P and continues through R without end.

A. True

B. False

Q is on ray PR with PQ = 5. Must Q be between P and R? Describe all the cases.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q2/5

Segment AB has endpoints

A. none

B. only B

C. only A

D. A and B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q3/5

True or false?

A point on ray PR can be on the opposite side of P from R.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q4/5

Name the endpoint of ray PR.

Endpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q5/5

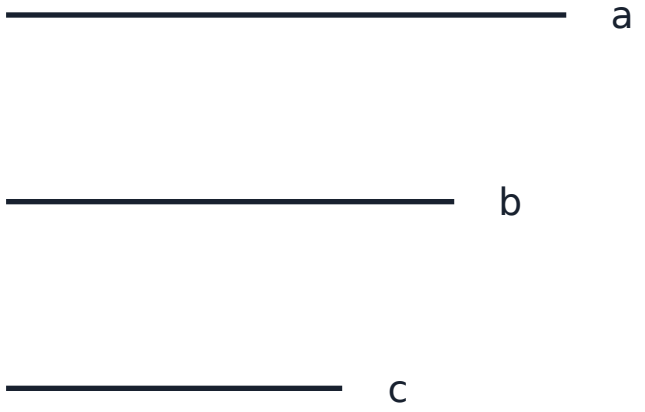
SET 28 • HSG-CO.D.12

FORWARD 1

Construct a triangle from three given segments • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a triangle whose sides are congruent to the three segments a, b, and c.



Explain why the two arcs in the triangle construction locate the third vertex.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q2/5

True or false?

Any three segments can be used to construct a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q3/5

To construct a triangle from three given segments, arcs are drawn from the two ends of the base, and they cross at two points, one on each side of the base. The two triangles formed are

A. similar but not congruent

B. different in size

C. not triangles

D. congruent (mirror images)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q4/5

Segments of length 3, 4, and 6 are given. Can a triangle be constructed? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSG-CO.D.12

FORWARD 2

Decide whether segments can form a triangle, and construct sums and differences • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Segments of length 2, 3, and 6 are given. Can a triangle be constructed? Explain what happens to the arcs.

Answer: _____

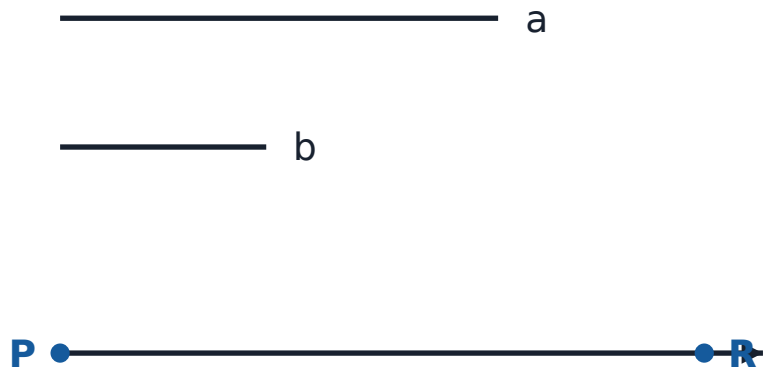
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q1/5

Compass and straightedge only; leave all arcs.

On the ray from P, construct a segment of length a − b (a is the longer segment).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q2/5

True or false?

If the arcs in a triangle construction just touch at one point on the base, the "triangle" is flat.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q3/5

Which lengths CANNOT form a triangle?

A. 1, 2, 3

B. 3, 4, 5

C. 4, 4, 7

D. 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q4/5

Two sides of a constructed triangle are 5 and 8. For what third lengths will the arcs meet in two points?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q5/5

SET 102 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Make formal constructions with a variety of tools and methods

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Congruent segments have equal lengths	**2.MD.A.1**
BACKWARD 2	A reflection maps a point to its mirror image	**8.G.A.1**
FORWARD 1	Choose and justify a method for a construction	**HSG-CO.D.12**
FORWARD 2	Locate the circumcenter by folding	**HSG-C.A.3**

SET 102 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

You have no compass, only a straightedge and a strip of paper. Describe how to copy segment AB onto line ℓ starting at point C, and explain why the copy is exact.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q1/5

Describe how to construct the perpendicular bisector of segment AB by folding paper, and explain why the crease is the perpendicular bisector.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q2/5

Describe how to bisect an angle drawn on paper by folding, and explain why the crease bisects the angle.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q3/5

You have a pencil, a pin, and a piece of string, but no compass. Describe how to construct an equilateral triangle on segment AB, and explain why it works.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q4/5

In dynamic geometry software, describe how to construct the line through P parallel to line ℓ so that it stays parallel when P or ℓ is dragged, and explain why a line drawn "by eye" is not a construction.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q5/5

SET 102 • HSG-CO.D.12

BACKWARD 1

Congruent segments have equal lengths • Grade 2 • Measurement & Data • 2.MD.A.1

Segment PQ is 4.5 cm and segment RS is 45 mm. Are they congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q1/5

True or false?

A strip of paper can carry a length from one place to another without a ruler.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q2/5

Which of these can carry a length from one segment to another without a ruler?

A. A thermometer

B. A calculator

C. A folded paper strip marked at both ends

D. A protractor

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q3/5

A string is cut to the length of segment AB. It is laid along line ℓ from C. How long is the segment it marks?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q4/5

True or false?

Two segments with different lengths can be congruent if they are drawn with the same tool.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q5/5

SET 102 • HSG-CO.D.12

BACKWARD 2

A reflection maps a point to its mirror image • Grade 8 • Geometry • 8.G.A.1

True or false?

Folding paper along a line matches each point with its reflection across the fold.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q1/5

A paper is folded so that point A lands on point B. What is the fold line, in relation to segment AB?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q2/5

A reflection keeps

A. neither

B. lengths and angle measures

C. only lengths

D. only angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q3/5

True or false?

A point on the fold line moves when the paper is folded.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q4/5

Folding makes ray BA land on ray BC. What is true of ∠ABD and ∠DBC, where BD is the crease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q5/5

SET 102 • HSG-CO.D.12

FORWARD 1

Choose and justify a method for a construction • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which tool or method would you use to find the midpoint of a segment drawn on tracing paper, without a compass? Explain why it is exact.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q1/5

A reflective device (a transparent mirror) is placed on line ℓ, and point P is seen in it. Explain how to use it to locate the reflection of P across ℓ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q2/5

Explain why measuring a 60° angle with a protractor is not a formal construction, but stepping a compass around a circle is.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q3/5

Describe a way to construct the perpendicular from P to line ℓ by folding.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q4/5

In dynamic geometry software, a student constructs a "square" by placing four points by eye. Describe a drag test that shows it is not a constructed square, and how to construct one properly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q5/5

SET 102 • HSG-CO.D.12

FORWARD 2

Locate the circumcenter by folding • High School — Geometry • Geometry: Circles • HSG-C.A.3

Describe how to find the point equidistant from the three vertices of a paper triangle by folding.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q1/5

Why do two creases suffice, without folding the third pair of vertices?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q2/5

True or false?

The point where the perpendicular-bisector creases of a triangle cross can lie outside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q3/5

Folding vertex A onto vertex B of a triangle gives a crease that is

A. the perpendicular bisector of AB

B. the bisector of $\angle C$

C. the median from C

D. parallel to AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q4/5

After finding the circumcenter O by folding, how can you draw the circumscribed circle with a pin and string?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q5/5

SET 29 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a copy of an angle

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Measure and draw angles with a protractor	**4.MD.C.6**
BACKWARD 2	Copy a segment with a compass	**HSG-CO.D.12**
BACKWARD 3	Arcs and circles in constructions	**7.G.B.4**
FORWARD 1	Justify why the angle-copy construction works	**HSG-CO.B.8**
FORWARD 2	Use angle copying to construct a parallel line	**HSG-CO.D.12**

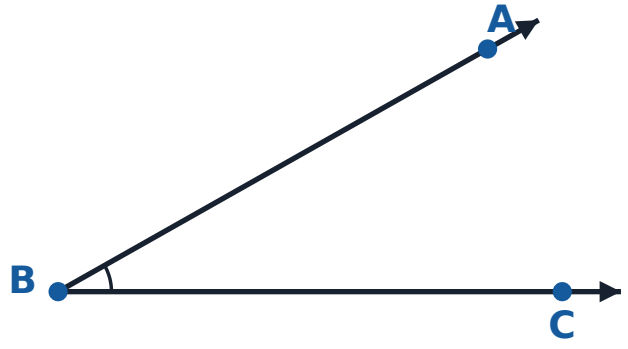
SET 29 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct an angle with vertex P and one side ray PR that is congruent to ∠ABC.



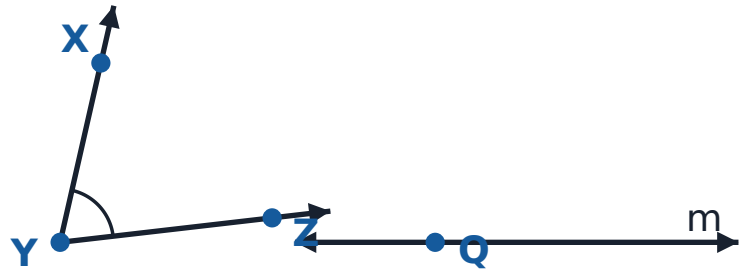
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct an angle at point Q on line m congruent to ∠XYZ, opening upward.



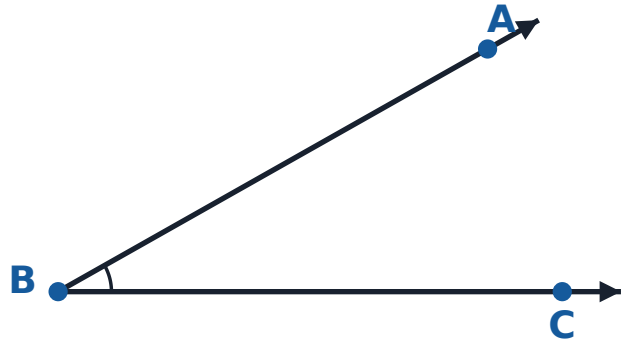
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct an angle with vertex P and one side ray PR that is congruent to ∠ABC, opening BELOW ray PR.



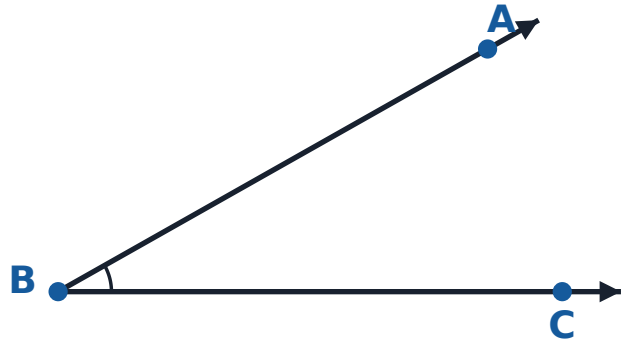
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct an angle at P on ray PR whose measure is twice m∠ABC.



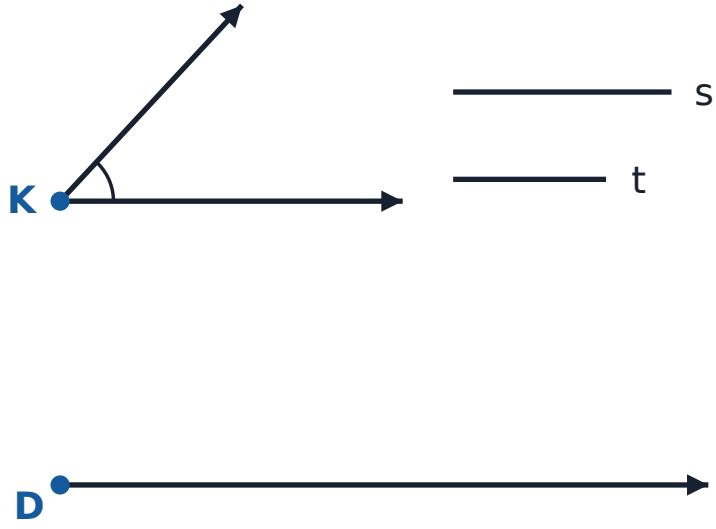
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct triangle DEF with $\angle D \cong \angle K$, DE = s, and DF = t.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 1

Measure and draw angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

A protractor shows a ray at 0° and the other ray at 65°. What is the angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q1/5

True or false?

An angle's measure does not change when its sides are drawn longer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q2/5

Which angle is obtuse?

A. 90°

B. 45°

C. 130°

D. 20°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q3/5

Two angles of 35° and 35° are placed side by side. What is the combined angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q4/5

True or false?

Two angles with sides of different lengths cannot be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 2

Copy a segment with a compass • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

To copy ∠ABC to vertex P, an arc centered at B crosses BA at D and BC at E, and the same arc centered at P crosses ray PR at F. Which length is set on the compass next, to locate the second side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q1/5

True or false?

To copy a length, set the compass to the length and do not change it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q2/5

An arc centered at F with radius DE crosses another arc at G. What is FG?

A. Half of DE

B. BD

C. DE

D. PF

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q3/5

Arcs centered at B and at P are both drawn with radius 3 cm. How far is F (on the arc at P) from P?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q4/5

True or false?

A segment copy can be checked only by measuring with a ruler.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 3

Arcs and circles in constructions • Grade 7 • Geometry • 7.G.B.4

True or false?

Two arcs centered at different points can cross at two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q1/5

Points D and E are both on an arc centered at B with radius 4. Find BD and BE.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q2/5

Which describes an arc drawn with a compass?

A. Part of a circle

B. A ray

C. A straight segment

D. A polygon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q3/5

True or false?

An arc centered at P passes through P.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q4/5

What does the point of the compass mark in a construction?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSG-CO.D.12

FORWARD 1

Justify why the angle-copy construction works • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In copying ∠ABC to ∠FPG, BD = BE = PF = PG (same arc radius) and DE = FG (same compass setting). Prove that ∠DBE ≅ ∠FPG.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q1/5

∠XYZ is copied to vertex Q, opening downward: an arc of radius r centered at Y meets the sides at U and V, the same arc centered at Q meets the given ray at U′, and an arc of radius UV centered at U′ meets it at V′. Prove that ∠U′QV′ ≅ ∠XYZ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q2/5

∠ABC is copied twice at P: first onto ray PR to give ray PG, then onto ray PG (on the side away from R) to give ray PH. Prove that m∠RPH = 2 · m∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q3/5

In copying ∠ABC to P, a student keeps the arc radius but sets the compass to BD (a radius) instead of the chord DE for the last arc. Which step of the SSS justification fails, and what goes wrong with the copy?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q4/5

Explain why the angle-copy construction works using a rigid motion instead of SSS: there is a rigid motion taking B to P and ray BC onto ray PR.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q5/5

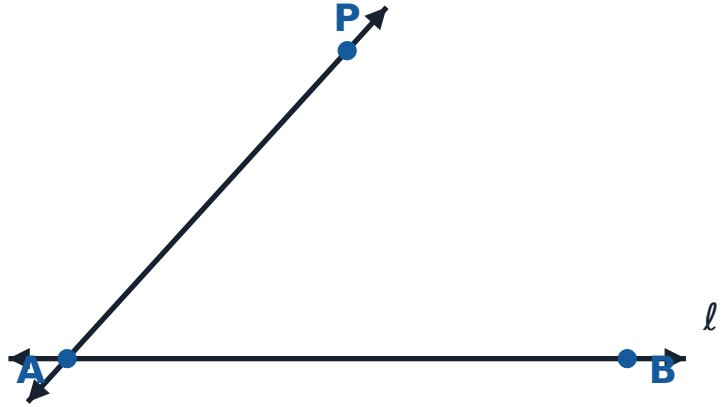
SET 29 • HSG-CO.D.12

FORWARD 2

Use angle copying to construct a parallel line • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ by copying an angle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q1/5

Why does copying a corresponding angle at P give a line parallel to ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q2/5

True or false?

Copying the angle so the two angles are congruent same-side interior angles also produces a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q3/5

To construct the line through P parallel to ℓ by copying an angle, a transversal is drawn through P meeting ℓ. Which angle is copied?

A. The angle between ℓ and itself

B. The angle the transversal makes with ℓ

C. A right angle

D. Any 45° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q4/5

Describe a different way to construct a parallel through P using two perpendicular constructions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q5/5

SET 30 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct the perpendicular bisector of a segment (bisect a segment)

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Points equidistant from A and B lie on the perpendicular bisector	**HSG-CO.C.9**
BACKWARD 2	Know midpoint and perpendicular bisector definitions	**HSG-CO.A.1**
BACKWARD 3	Arcs with one radius from two centers	**7.G.B.4**
FORWARD 1	Justify the perpendicular bisector construction	**HSG-CO.C.9**
FORWARD 2	Construct the circumcenter and circumscribed circle of a triangle	**HSG-C.A.3**

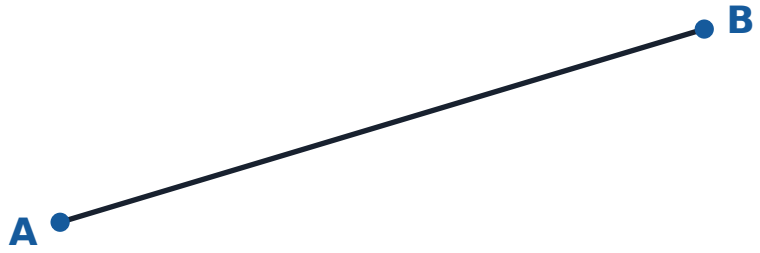
SET 30 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the perpendicular bisector of segment AB. Label the midpoint M.



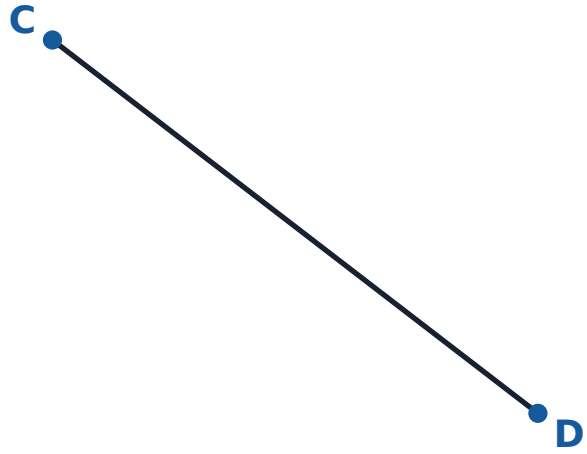
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the midpoint M of segment CD.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the point P on line ℓ that is the same distance from A as from B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the perpendicular bisector of the short vertical segment UV.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Divide segment AB into four congruent parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 1

Points equidistant from A and B lie on the perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

A point equidistant from A and B lies on the perpendicular bisector of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q1/5

Points P and Q are each equidistant from A and B. What line do P and Q determine?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q2/5

In the perpendicular bisector construction for segment AB, the two arc crossings are equidistant from A and B because

A. the crossings are on AB

B. the arcs have the same radius

C. the compass is opened to AB

D. the arcs are drawn above AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q3/5

True or false?

Every point on the perpendicular bisector of AB is equidistant from A and B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q4/5

True or false?

Two points determine more than one line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 2

Know midpoint and perpendicular bisector definitions • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

M is the midpoint of AB, and AM = 3.5. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q1/5

True or false?

The perpendicular bisector of a segment passes through its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q2/5

A line crosses AB at its midpoint at a 60° angle. It is

A. the perpendicular bisector

B. not a bisector

C. parallel to AB

D. a bisector but not the perpendicular bisector

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q3/5

Line k is perpendicular to AB and passes through its midpoint. What is k called?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q4/5

True or false?

A segment has many perpendicular bisectors in one plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 3

Arcs with one radius from two centers • Grade 7 • Geometry • 7.G.B.4

Two arcs of radius 5 are centered at A and B, and AB = 6. Do they cross?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q1/5

True or false?

If the compass opening is less than half of AB, the arcs from A and B do not meet.

A. True

B. False

Why must the opening be more than half of AB?

A. It does not matter

B. So the midpoint is on the arc

C. So the arcs from A and B cross

D. So the line is longer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q3/5

A point X is on both arcs (radius r) centered at A and B. Find XA and XB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q4/5

True or false?

The two arc crossings are on the same side of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q5/5

SET 30 • HSG-CO.D.12

FORWARD 1

Justify the perpendicular bisector construction • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

To construct the perpendicular bisector of AB, arcs of radius r (more than half of AB) centered at A and B meet at P and Q. Prove that line PQ is the perpendicular bisector of AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q1/5

Explain why the perpendicular bisector construction for AB would still work with a larger compass opening.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q2/5

True or false?

The construction works only if the arcs are drawn with radius exactly AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q3/5

Which theorem justifies that the arc crossings lie on the perpendicular bisector?

A. Midsegment theorem

B. Triangle angle sum

C. Converse of the perpendicular bisector theorem

D. Vertical angles theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q4/5

Why does folding A onto B produce the perpendicular bisector?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q5/5

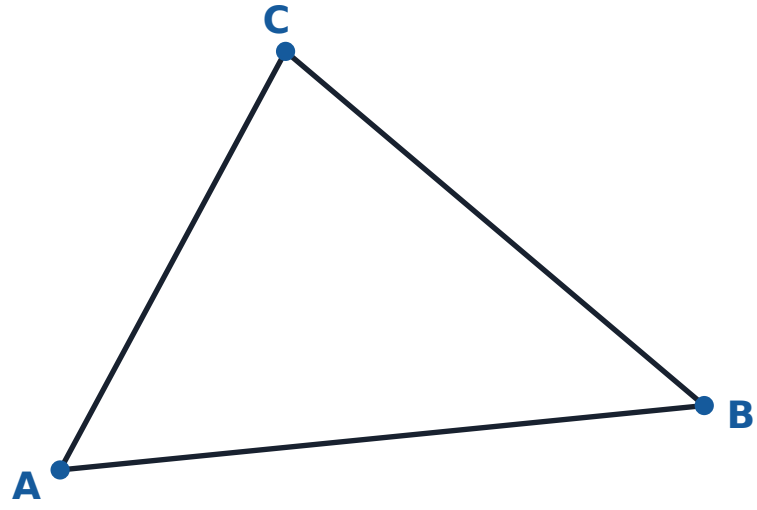
SET 30 • HSG-CO.D.12

FORWARD 2

Construct the circumcenter and circumscribed circle of a triangle • High School — Geometry • Geometry: Circles • HSG-C.A.3

Compass and straightedge only; leave all arcs.

Construct the point O equidistant from A, B, and C, and draw the circle through A, B, and C.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q1/5

Why is it enough to construct only two perpendicular bisectors to find the circumcenter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q2/5

True or false?

The circumcenter of an obtuse triangle lies on its longest side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q3/5

Where is the circumcenter of a right triangle?

A. Inside, near the centroid

B. At the midpoint of the hypotenuse

C. Outside the triangle

D. At the right-angle vertex

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q4/5

A triangle has A(0, 0), B(8, 0), C(0, 6). Where is its circumcenter, and what is the radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q5/5

SET 31 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct the bisector of an angle

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Know the definition of an angle bisector	**HSG-CO.A.1**
BACKWARD 2	Prove triangles congruent with SSS	**HSG-CO.B.8**
BACKWARD 3	Measure angles with a protractor	**4.MD.C.6**
FORWARD 1	Justify the angle bisector construction	**HSG-CO.B.8**
FORWARD 2	Construct special angles and the incenter by bisecting	**HSG-CO.D.12**

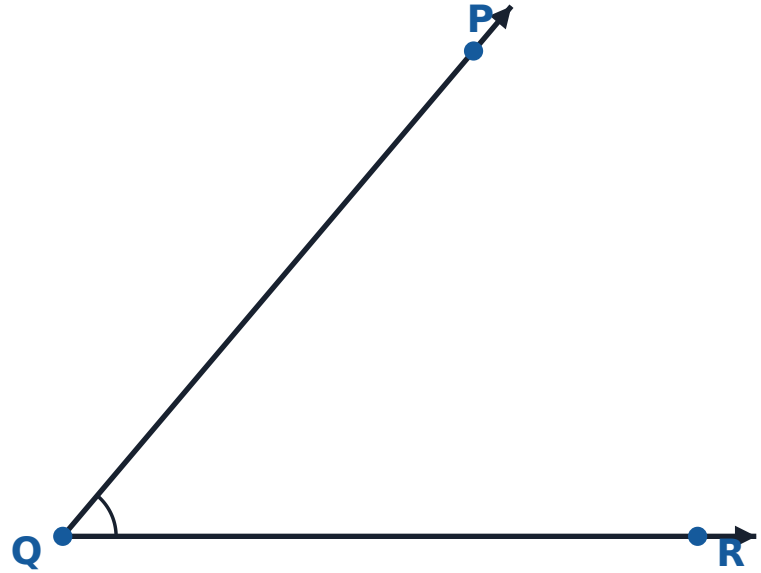
SET 31 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the bisector of $\angle PQR$.



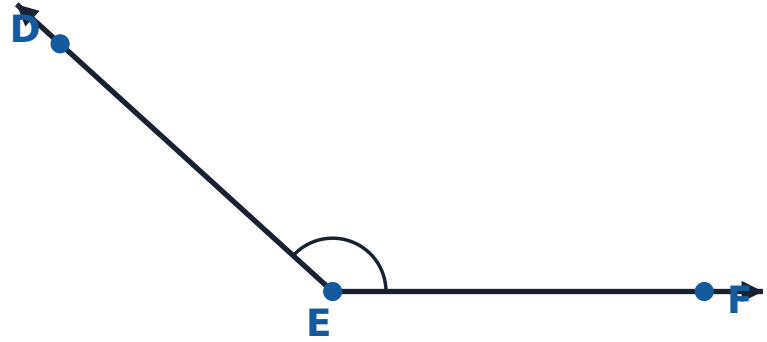
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the bisector of obtuse angle ∠DEF.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a 45° angle with vertex A on line ℓ.

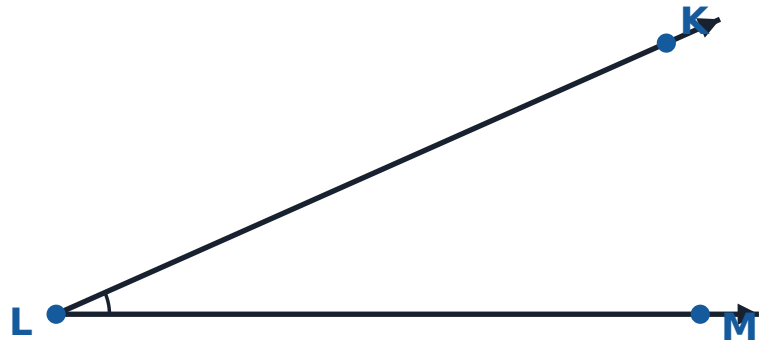
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the bisector of the narrow angle ∠KLM.



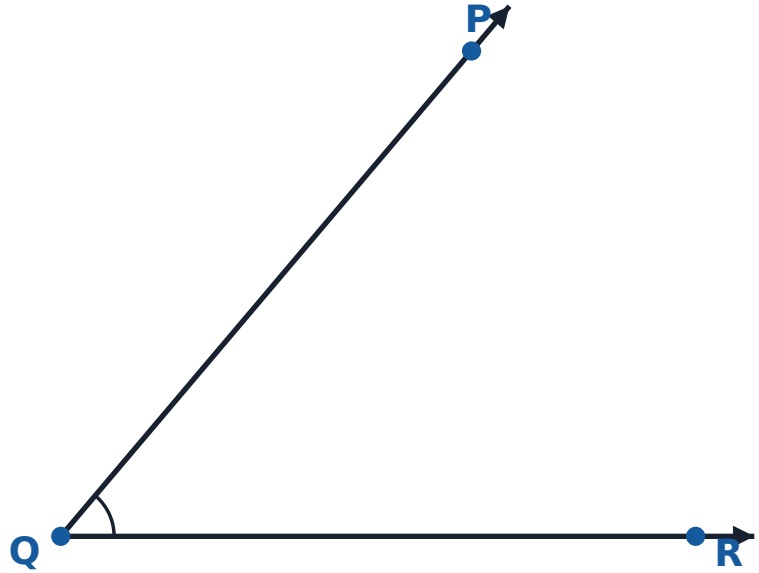
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct a ray QS inside ∠PQR so that m∠RQS = ¼ m∠PQR.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 1

Know the definition of an angle bisector • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Ray BD bisects ∠ABC, and m∠ABD = 27°. Find m∠ABC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q1/5

True or false?

An angle bisector divides the angle into two congruent angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q2/5

Ray BD bisects $\angle ABC$ with $m\angle ABC = 3x + 6$ and $m\angle ABD = 24$. Find x.

A. 6

B. 10

C. 18

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q3/5

A 90° angle is bisected. What is each part?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q4/5

True or false?

An angle bisector is a segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 2

Prove triangles congruent with SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share a side and have two more pairs of congruent sides. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q1/5

True or false?

SSS guarantees that corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q2/5

To bisect ∠ABC, an arc centered at B crosses BA at X and BC at Y, and equal arcs centered at X and Y meet at Z. Which triangles are congruent?

A. △ABC and △XYZ

B. △BXZ and △BYZ

C. △BXY and △XYZ

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q3/5

In the angle-bisector construction for ∠ABC (arc at B meeting the sides at X and Y, equal arcs from X and Y meeting at Z), △BXZ ≅ △BYZ. Which angle is congruent to ∠XBZ?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q4/5

True or false?

SSS requires knowing one angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 3

Measure angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

A protractor reads 0° on one side and 112° on the other. Half of this angle is

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q1/5

True or false?

Measuring with a protractor is a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q2/5

An angle of 150° is bisected. Each part is

A. 60°

B. 150°

C. 75°

D. 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q3/5

After bisecting, a student measures 38° and 38°. What was the original angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q4/5

True or false?

A protractor can check a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q5/5

SET 31 • HSG-CO.D.12

FORWARD 1

Justify the angle bisector construction • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In bisecting ∠ABC, an arc centered at B crosses the sides at X and Y, and equal arcs centered at X and Y meet at Z. Prove that ray BZ bisects ∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q1/5

In the angle bisector construction, an arc centered at the vertex crosses the sides at X and Y, and then arcs from X and Y cross at Z. Why must the arcs from X and Y have the same radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q2/5

True or false?

The bisector construction works for obtuse angles too.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q3/5

Which criterion justifies the angle bisector construction?

A. AAS

B. ASA

C. SSS

D. SAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q4/5

Explain why folding one side onto the other gives the bisector.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSG-CO.D.12

FORWARD 2

Construct special angles and the incenter by bisecting • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a 30° angle. (Hint: start from an equilateral triangle.)

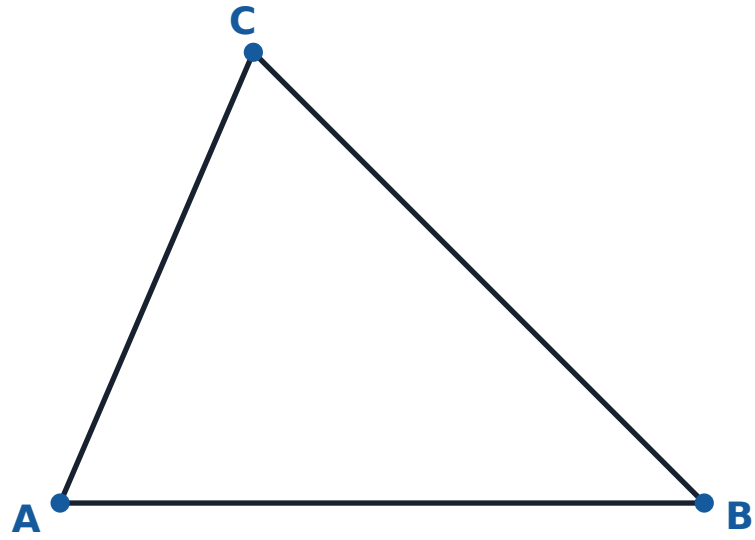
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q1/5

Compass and straightedge only; leave all arcs.

Construct the point inside triangle ABC that is equidistant from its three sides.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q2/5

True or false?

Bisecting an angle and then bisecting each half divides the angle into three equal parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q3/5

Starting from 60° and 90° angles, which angle CANNOT be constructed by bisecting angles and adding angles together?

A. 15°

B. 75°

C. 20°

D. 22.5°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q4/5

Explain how to construct a 135° angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q5/5

SET 32 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a perpendicular to a line through a point on or off the line

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Know the definition of perpendicular lines	**HSG-CO.A.1**
BACKWARD 2	Construct the perpendicular bisector of a segment	**HSG-CO.D.12**
BACKWARD 3	Recognize perpendicular lines in figures	**4.G.A.1**
FORWARD 1	Construct an altitude of a triangle and use it for area	**HSG-CO.D.12**
FORWARD 2	Construct a square on a given segment	**HSG-CO.D.12**

SET 32 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P perpendicular to line ℓ. (P is on ℓ.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the line through P perpendicular to line ℓ. (P is not on ℓ.)

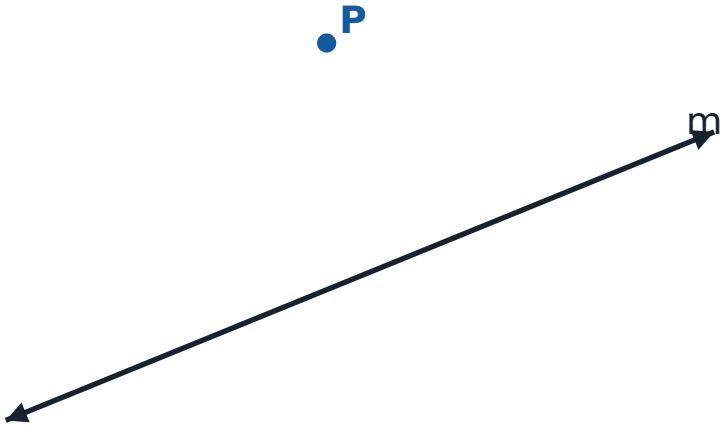
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the perpendicular from point P to the slanted line m.



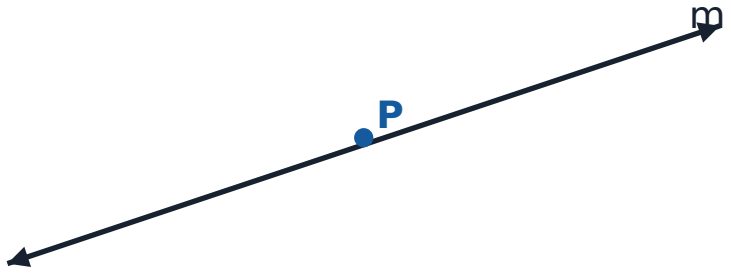
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the line perpendicular to the slanted line m at the point P on m.



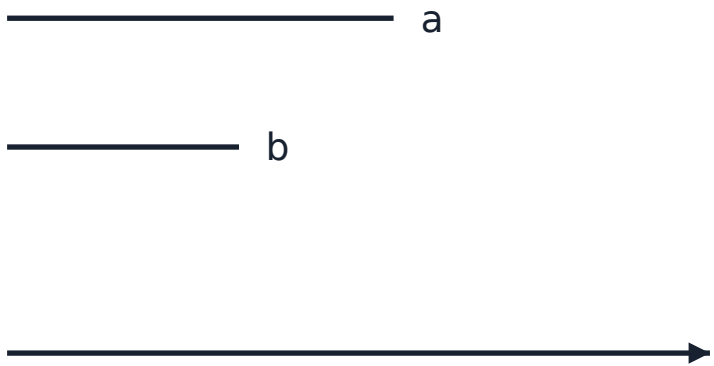
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct a right triangle whose legs are congruent to segments a and b.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 1

Know the definition of perpendicular lines • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Perpendicular lines meet at a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q1/5

Line k $\perp$ line ℓ. What is the measure of each angle they form?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q2/5

Which pair of lines is perpendicular?

A. $y = 3$ and $y = -3$

B. $y = 2x$ and $y = 2x + 1$

C. $y = x$ and $y = 2x$

D. $y = 2x$ and $y = -\frac{1}{2}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q3/5

True or false?

Through a point, there are many lines perpendicular to a given line in a plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q4/5

The shortest path from a point to a line follows which kind of line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 2

Construct the perpendicular bisector of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

After marking A and B on ℓ, which construction finishes the perpendicular?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q1/5

True or false?

The perpendicular bisector construction needs arcs of equal radius from both endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q2/5

The arc crossings of the perpendicular bisector construction are

A. equidistant from A and B

B. on segment AB

C. on line ℓ

D. at A and B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q3/5

P is on ℓ, and A and B are marked on ℓ with PA = PB. Where does the perpendicular bisector of AB cross ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q4/5

True or false?

The perpendicular bisector of AB passes through A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 3

Recognize perpendicular lines in figures • Grade 4 • Geometry • 4.G.A.1

True or false?

A square's adjacent sides are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q1/5

Which letter is made of perpendicular segments?

A. V

B. T

C. O

D. S

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q2/5

How many right angles do two perpendicular lines make?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q3/5

True or false?

Two lines that look almost perpendicular must be perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q4/5

Name a pair of perpendicular edges of a rectangular door.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q5/5

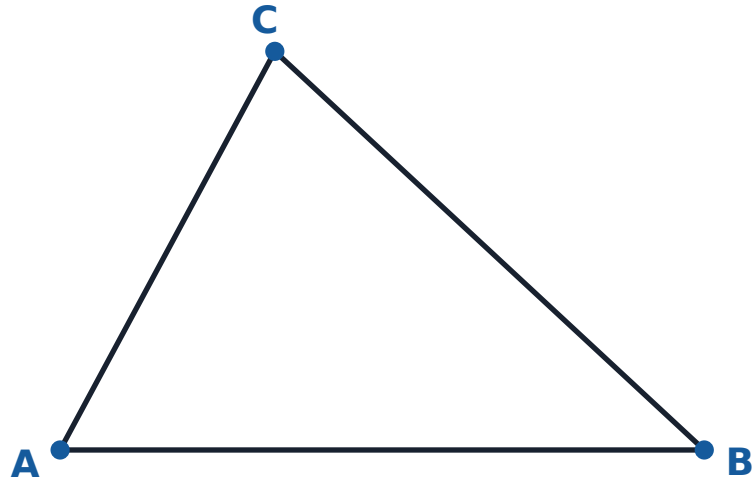
SET 32 • HSG-CO.D.12

FORWARD 1

Construct an altitude of a triangle and use it for area • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the altitude from C to side AB of triangle ABC.



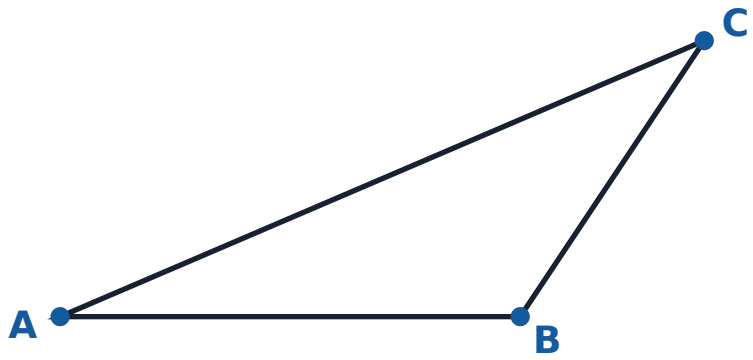
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q1/5

Compass and straightedge only; leave all arcs.

Construct the altitude from C in obtuse triangle ABC (the foot is outside AB). Extend AB as needed.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q2/5

A triangle has base 10 and its constructed altitude measures 6. Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q3/5

True or false?

Every altitude of a triangle lies inside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q4/5

In a right triangle, the altitude from the right-angle vertex to the hypotenuse is

A. the hypotenuse

B. inside the triangle

C. outside the triangle

D. a leg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSG-CO.D.12

FORWARD 2

Construct a square on a given segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a square with side AB, above AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q1/5

A square on segment AB is constructed by drawing perpendiculars to AB at A and at B and copying AB onto each, on the same side of line AB, to get D (from A) and C (from B). Explain why ABCD is a square, and why the two copies must be on the same side of AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q2/5

True or false?

A quadrilateral with four congruent sides is always a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q3/5

To construct a rectangle 2AB long and AB wide, which additional step is needed?

A. Bisect AB

B. Construct a 60° angle

C. Draw a circle through A and B

D. Copy AB twice along the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q4/5

A square is constructed on a segment of length 5. What is the length of its diagonal?

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q5/5

SET 33 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a line parallel to a given line through a point not on it

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Copy an angle with compass and straightedge	**HSG-CO.D.12**
BACKWARD 2	Converse: congruent corresponding angles give parallel lines	**HSG-CO.C.9**
BACKWARD 3	Recognize parallel lines	**4.G.A.1**
FORWARD 1	Justify the parallel-line constructions	**HSG-CO.C.9**
FORWARD 2	Divide a segment into equal parts with parallel lines	**HSG-SRT.B.4**

SET 33 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ.

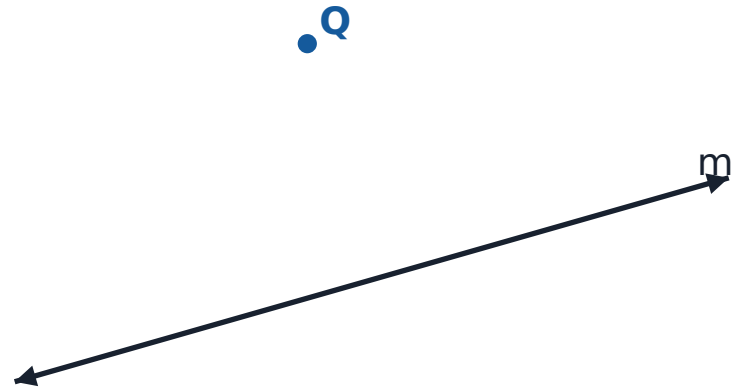
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the line through Q parallel to the slanted line m.



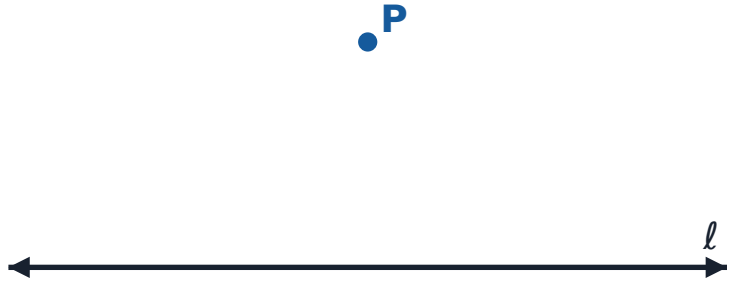
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to ℓ using two perpendiculars.



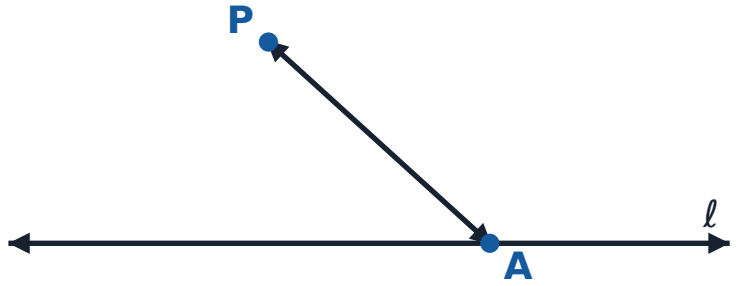
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ by copying an ALTERNATE INTERIOR angle.



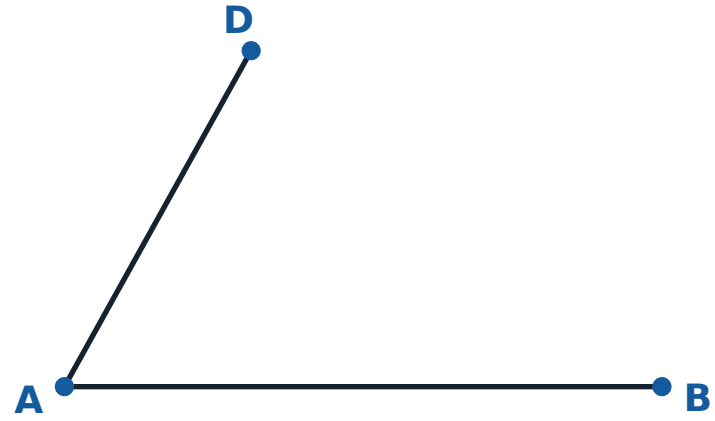
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct parallelogram ABCD with sides AB and AD.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 1

Copy an angle with compass and straightedge • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

In copying an angle, which three lengths are transferred?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q1/5

True or false?

An angle copy uses arcs of the same radius at both vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q2/5

Which step completes an angle copy?

A. Draw a circle

B. Measure with a protractor

C. Draw the ray from the new vertex through the last arc crossing

D. Bisect the angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q3/5

Why does the angle copy work?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q4/5

True or false?

An angle copy needs a protractor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 2

Converse: congruent corresponding angles give parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Congruent corresponding angles mean the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q1/5

A transversal makes 62° angles with two lines in corresponding positions. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q2/5

Which also proves the lines parallel?

A. Congruent vertical angles

B. Supplementary linear pairs

C. Congruent alternate interior angles

D. Two acute angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q3/5

True or false?

Same-side interior angles of 70° and 70° prove the lines parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q4/5

Corresponding angles are (2x + 10)° and 50°. For which x are the lines parallel?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 3

Recognize parallel lines • Grade 4 • Geometry • 4.G.A.1

True or false?

The opposite edges of a ruler are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q1/5

Which letter has parallel segments?

A. V

B. A

C. H

D. X

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q2/5

Two railroad rails stay the same distance apart. What are they modeling?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q3/5

True or false?

Lines that cross are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q4/5

In a plane, how many lines through a point NOT on a given line are parallel to that line?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q5/5

SET 33 • HSG-CO.D.12

FORWARD 1

Justify the parallel-line constructions • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

To construct a line through P parallel to ℓ, a transversal through P meets ℓ at A, and the angle at A is copied at P in the corresponding position. Prove that the new line is parallel to ℓ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q1/5

In the two-perpendicular construction, k ⊥ ℓ and n ⊥ k at P. Prove that n ∥ ℓ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q2/5

True or false?

Copying the alternate interior angle at P also gives a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q3/5

Which construction gives a line through P parallel to ℓ?

A. Copy, at P, the angle that a transversal through P makes with ℓ, in the corresponding position

B. Bisect the angle at P

C. Draw a circle centered at P

D. Draw the perpendicular bisector of the transversal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q4/5

By the parallel postulate, how many lines could the construction produce?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSG-CO.D.12

FORWARD 2

Divide a segment into equal parts with parallel lines • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

Compass and straightedge only; leave all arcs.

Divide segment AB into three congruent parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q1/5

To divide AB into three equal parts, three equal steps are marked on a ray from A, the last mark is joined to B, and parallels to that segment are drawn through the other marks. Why do the parallel lines divide AB into equal parts?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q2/5

True or false?

The same method can divide a segment into 5 equal parts only if its length is a multiple of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q3/5

To divide AB in the ratio 2 : 3, how many equal steps are marked on the auxiliary ray?

A. 2

B. 5

C. 3

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q4/5

A segment of length 12 is divided into 4 parts this way. How long is each part?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q5/5

SET 34 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct an equilateral triangle inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Central angles of a regular polygon	**HSG-CO.A.3**
BACKWARD 2	Copy a length with a compass	**HSG-CO.D.12**
BACKWARD 3	Circle vocabulary: center, radius, chord, diameter	**7.G.B.4**
FORWARD 1	Justify that the inscribed triangle is equilateral	**HSG-C.A.2**
FORWARD 2	Construct a regular dodecagon by bisecting arcs	**HSG-CO.D.13**

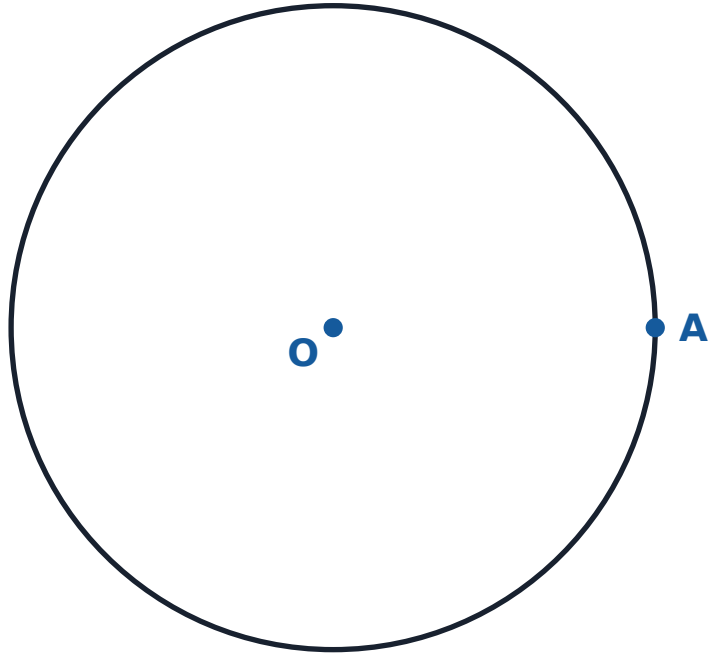
SET 34 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle, with one vertex at A.



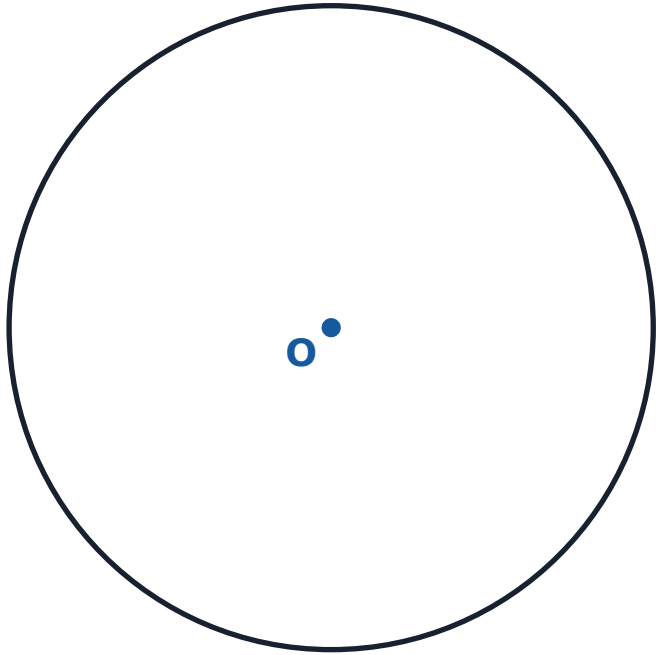
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle (choose any starting vertex).



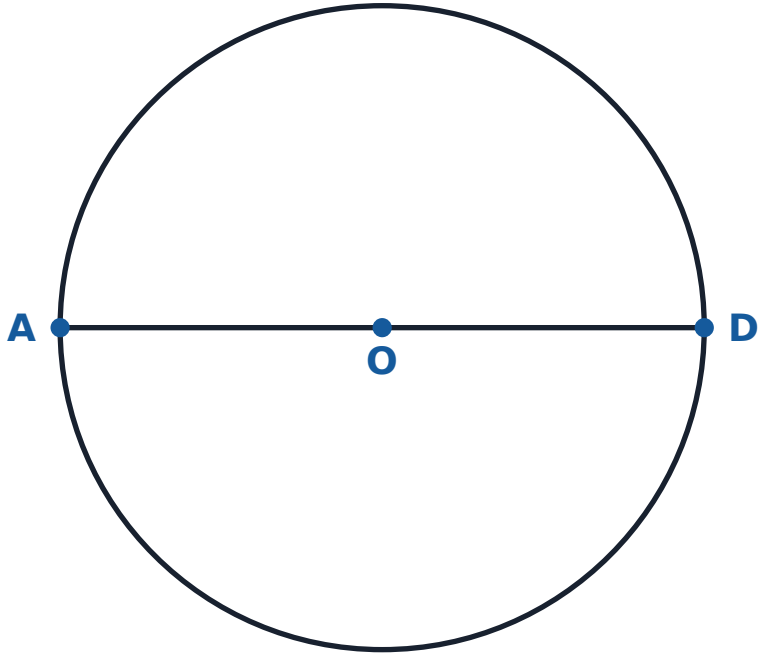
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle using a diameter: draw diameter AD, then an arc centered at D with radius DO.



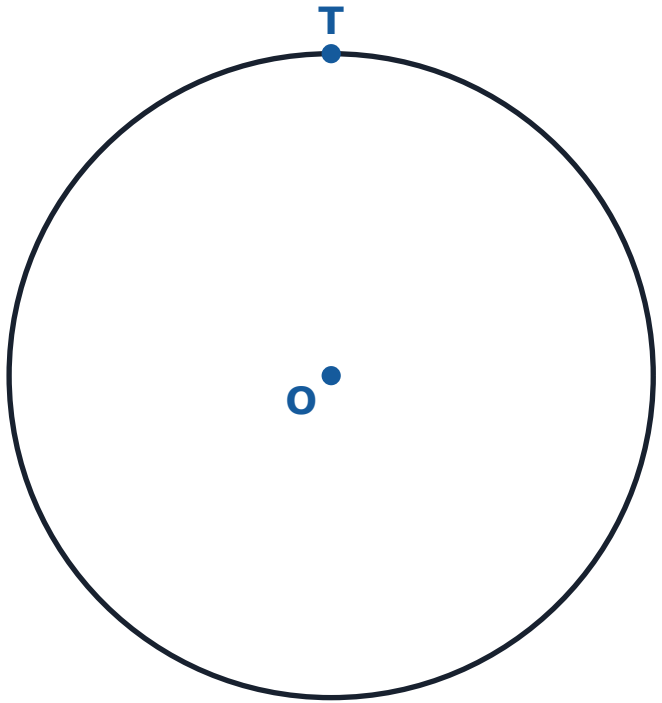
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle with one vertex at the top point T.



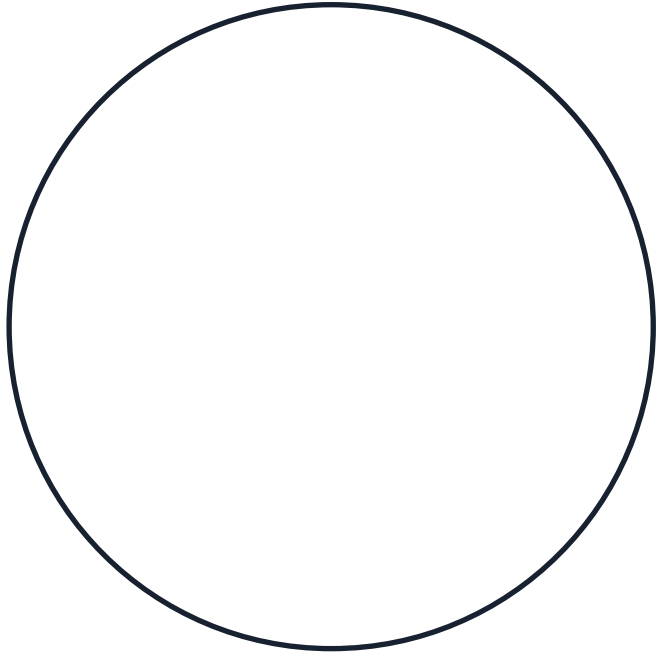
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

The center of this circle is not marked. Construct an equilateral triangle inscribed in it.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 1

Central angles of a regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

What rotation carries an equilateral triangle onto itself, about its center (smallest positive)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q1/5

Into how many equal central angles does a regular hexagon divide 360°?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q2/5

True or false?

The vertices of an inscribed equilateral triangle are 120° apart around the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q3/5

Joining every other vertex of a regular hexagon gives

A. a square

B. a rectangle

C. an equilateral triangle

D. a pentagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q4/5

True or false?

The central angles of an inscribed equilateral triangle are 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 2

Copy a length with a compass • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

The compass is set to the radius of a circle. What length does each step around the circle mark?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q1/5

True or false?

Keeping the compass opening fixed is required when stepping around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q2/5

Each radius-length chord and the two radii to its ends form

A. a square

B. an equilateral triangle

C. a right triangle

D. an isosceles triangle with a 90° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q3/5

A circle has radius 4 cm. How long is each chord marked by stepping the radius?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q4/5

True or false?

Stepping a compass set to the diameter marks six points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 3

Circle vocabulary: center, radius, chord, diameter • Grade 7 • Geometry • 7.G.B.4

What is a chord of a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q1/5

True or false?

A diameter is a chord through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q2/5

A circle has radius 6. Its diameter is

A. 3

B. 36

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q3/5

What does "inscribed in a circle" mean for a polygon?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q4/5

True or false?

An inscribed triangle can have a vertex inside the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q5/5

SET 34 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed triangle is equilateral • High School — Geometry • Geometry: Circles • HSG-C.A.2

Points A, B, C, D, E, F are marked around a circle with center O by stepping the radius. Prove that triangle ACE is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q1/5

A circle has center O and diameter AD. An arc centered at D with radius DO meets the circle at B and C. Prove that triangle ABC is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q2/5

Explain why stepping a compass set to the radius around a circle returns exactly to the starting point after six steps.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q3/5

In the stepped construction, points A, B, C, D, E, F lie on a circle with center O, each 60° from the next. Prove that ∠CAE = 60°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q4/5

Ana inscribes a triangle by marking points every 120° around the center with a protractor. Explain why her triangle is equilateral, and why her method is not a compass-and-straightedge construction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q5/5

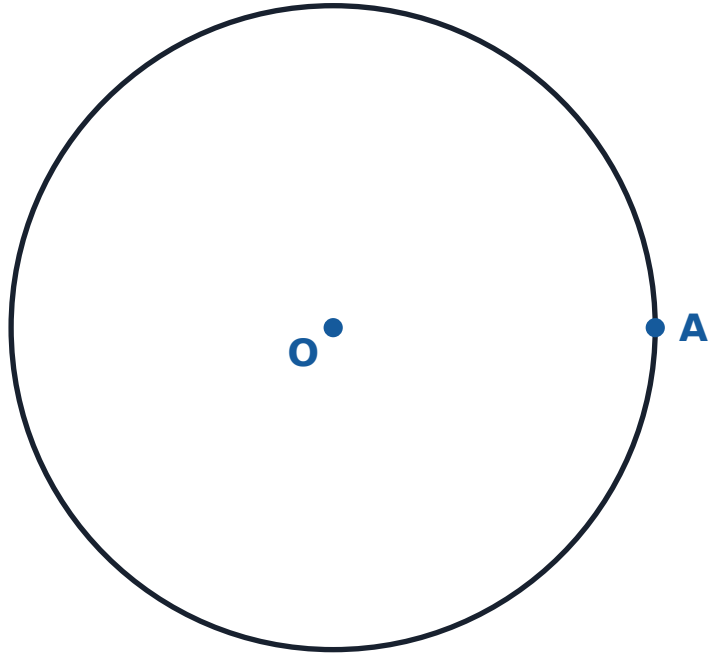
SET 34 • HSG-CO.D.13

FORWARD 2

Construct a regular dodecagon by bisecting arcs • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a regular 12-sided polygon inscribed in the circle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q1/5

Each side of the inscribed regular hexagon subtends 60° at the center. What does each side of the dodecagon subtend?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q2/5

True or false?

Bisecting the arcs of an inscribed regular hexagon gives the vertices of a regular nonagon.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q3/5

Starting from an inscribed square, bisecting arcs gives a regular

A. pentagon

B. hexagon

C. decagon

D. octagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q4/5

How many sides does the polygon have after bisecting the arcs of an inscribed equilateral triangle twice?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q5/5

SET 35 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct a square inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Construct a perpendicular bisector	**HSG-CO.D.12**
BACKWARD 2	Diagonals of a square	**HSG-CO.C.11**
BACKWARD 3	Diameters and the center of a circle	**7.G.B.4**
FORWARD 1	Justify that the inscribed quadrilateral is a square	**HSG-CO.C.11**
FORWARD 2	Construct a regular octagon inscribed in a circle	**HSG-CO.D.13**

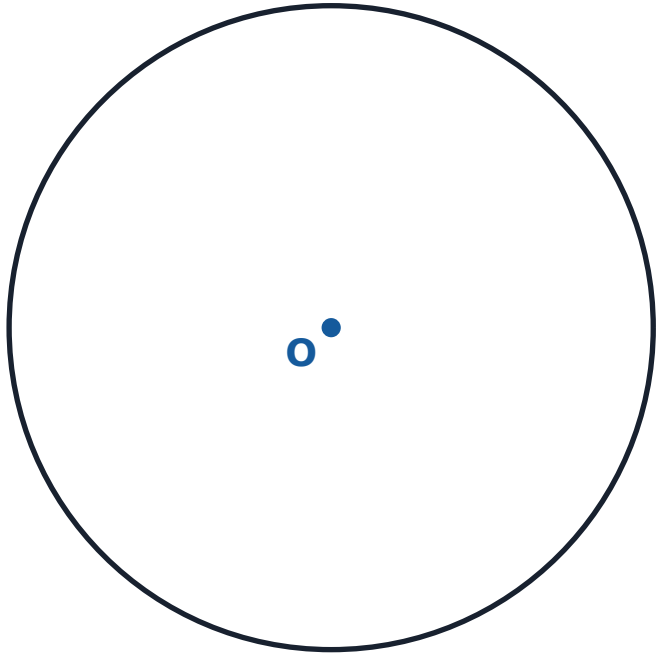
SET 35 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle.



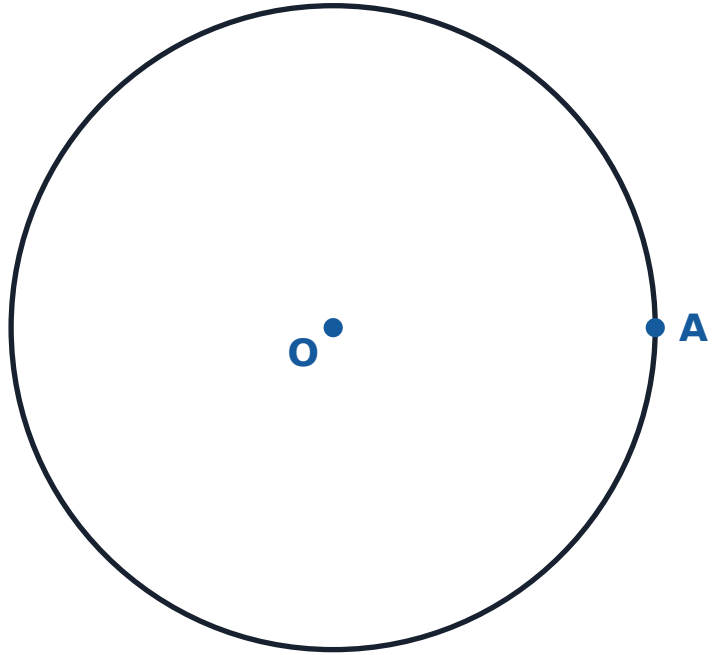
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle with one vertex at A.



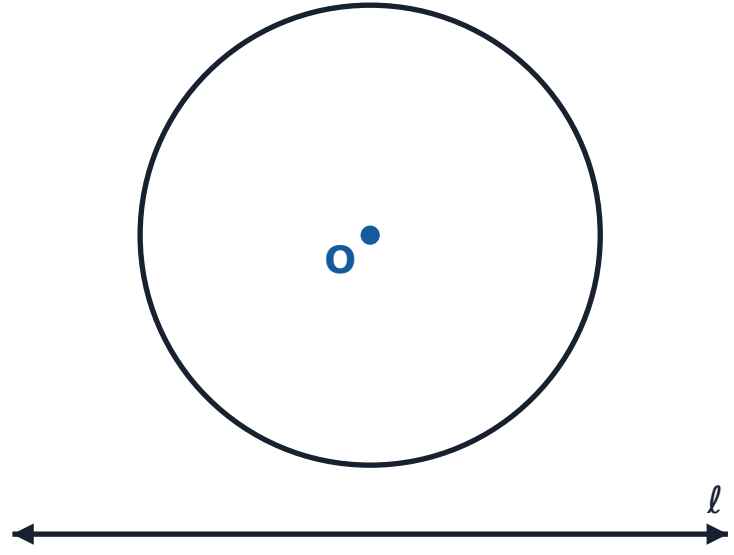
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle with one side parallel to the given line ℓ.



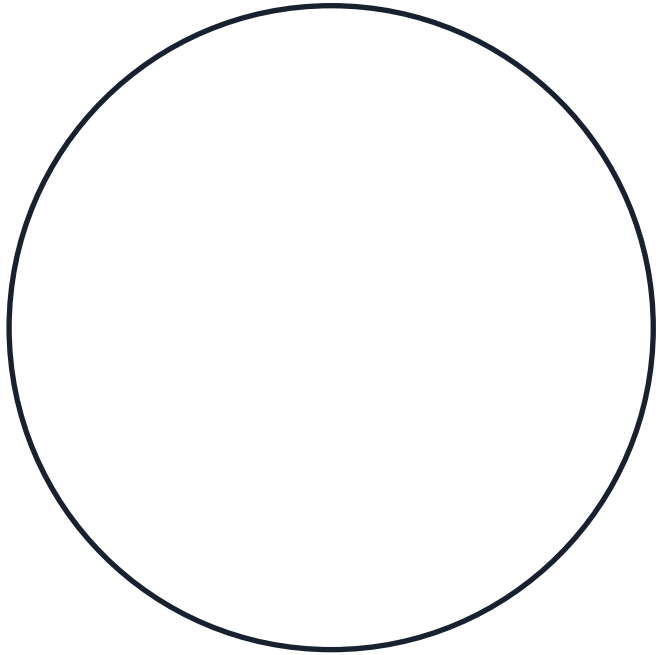
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

The center of this circle is not marked. Construct a square inscribed in it.



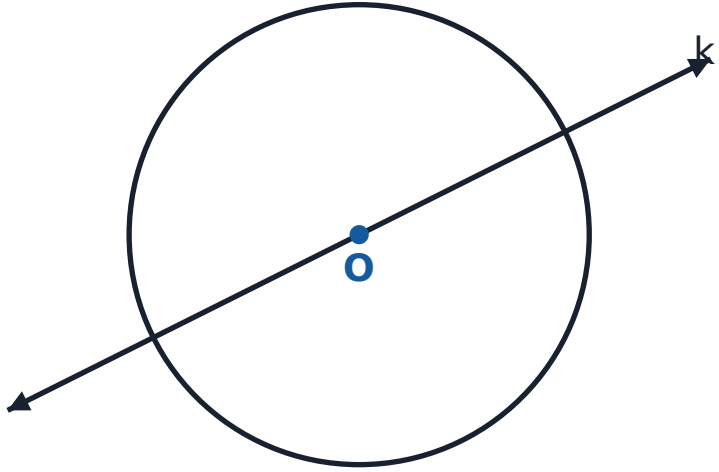
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle so that one of its diagonals lies along line k.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 1

Construct a perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction turns one diameter into two perpendicular diameters?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector of a diameter passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q2/5

Two perpendicular diameters cut the circle into

A. six 60° arcs

B. four 45° arcs

C. four 90° arcs

D. two 90° arcs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q3/5

A diameter is 10 cm. Where does its perpendicular bisector cross it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q4/5

True or false?

A perpendicular bisector can be drawn by measuring with a ruler only.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 2

Diagonals of a square • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

The diagonals of a square are congruent and perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q1/5

A quadrilateral's diagonals are congruent, perpendicular, and bisect each other. What is its most specific name?

A. rhombus that is not a square

B. square

C. trapezoid

D. kite that is not a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q2/5

A square has diagonal 8. Find its side.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q3/5

True or false?

A rectangle always has perpendicular diagonals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q4/5

The diagonals of a square meet at O. Name the angle between them.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 3

Diameters and the center of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has radius 3. How long is any diameter?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q1/5

True or false?

Every diameter passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q2/5

Where do two different diameters of a circle meet?

A. At the midpoint of a radius

B. On the circle

C. At the center

D. Nowhere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q3/5

What is the longest chord of a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q4/5

True or false?

All diameters of a circle have different lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q5/5

SET 35 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed quadrilateral is a square • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Diameters AC and BD of a circle with center O are perpendicular. Prove that ABCD is a square.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q1/5

A square is inscribed in a circle of radius 5. Find its side length and area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q2/5

True or false?

The side of a square inscribed in a circle of radius r is 2r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q3/5

The area of a square inscribed in a circle of radius r is

A. $2r^2$

B. πr^2

C. $4r^2$

D. r^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q4/5

What fraction of the circle's area does the inscribed square cover? Give the exact value.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q5/5

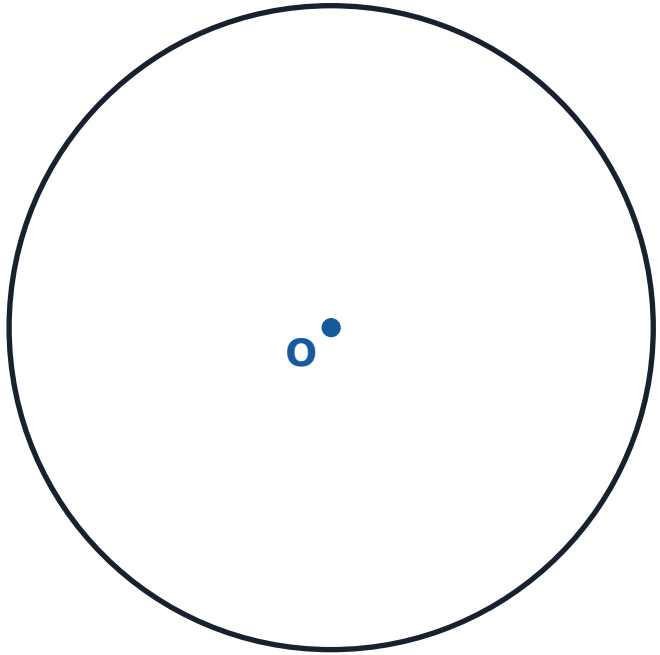
SET 35 • HSG-CO.D.13

FORWARD 2

Construct a regular octagon inscribed in a circle • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a regular octagon inscribed in the circle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q1/5

What is the central angle for each side of a regular octagon?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q2/5

True or false?

A regular octagon has interior angles of 120°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q3/5

Bisecting the arcs of an inscribed regular octagon gives a regular

A. hexagon

B. 16-gon

C. 12-gon

D. square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q4/5

A regular octagon is constructed in a circle by bisecting the four 90° central angles of an inscribed square. Explain why the eight points are equally spaced.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct a regular hexagon inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Construct an inscribed equilateral triangle	**HSG-CO.D.13**
BACKWARD 2	An equilateral triangle has equal sides and 60° angles	**4.G.A.2**
BACKWARD 3	Circle facts: radius and chord	**7.G.B.4**
FORWARD 1	Justify that the inscribed hexagon is regular	**HSG-C.A.2**
FORWARD 2	Use inscribed hexagons to estimate π	**HSG-GMD.A.1**

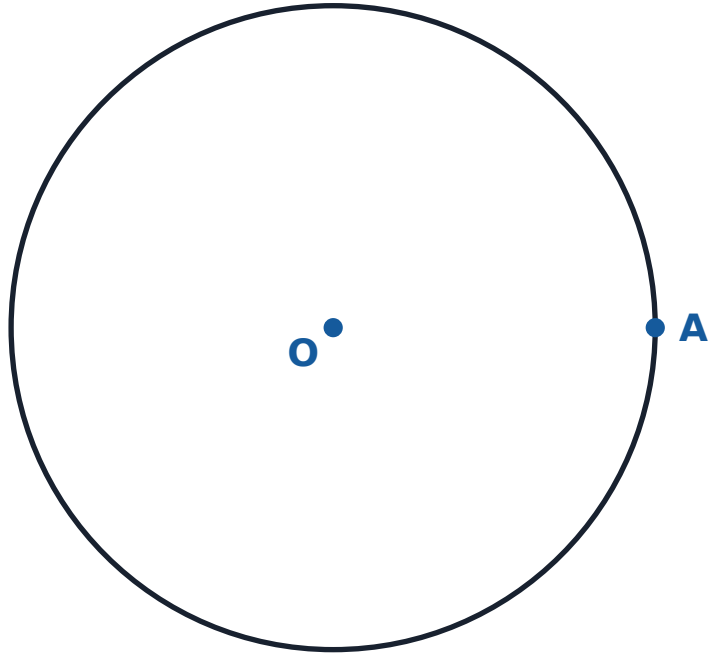
SET 36 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle, with one vertex at A.



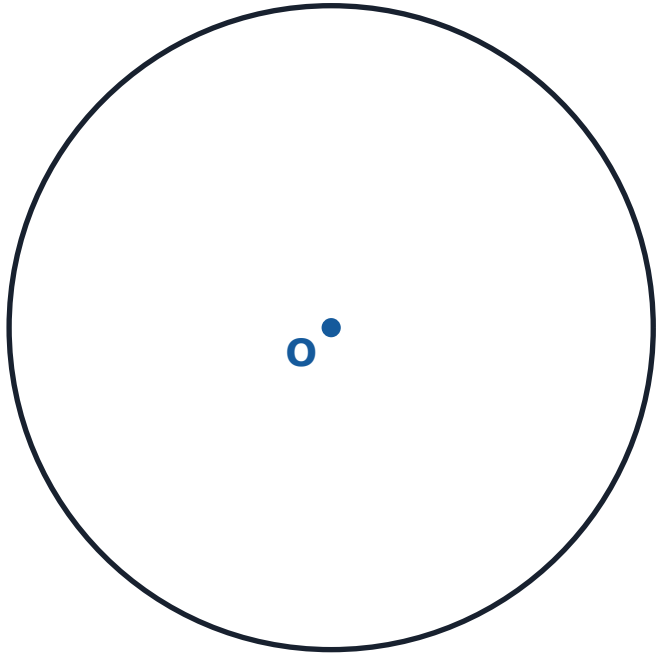
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle (choose any starting point).



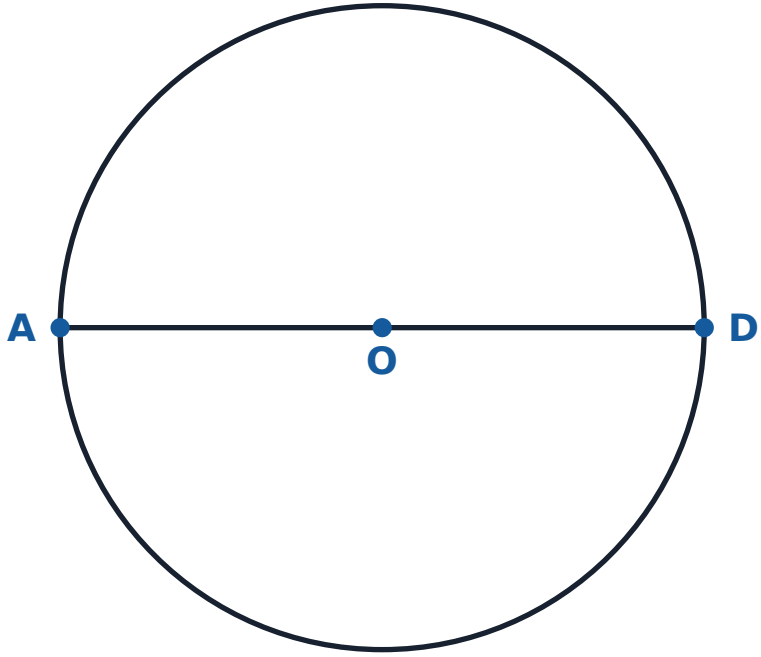
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle using diameter AD and arcs centered at A and D with radius equal to the circle's radius.



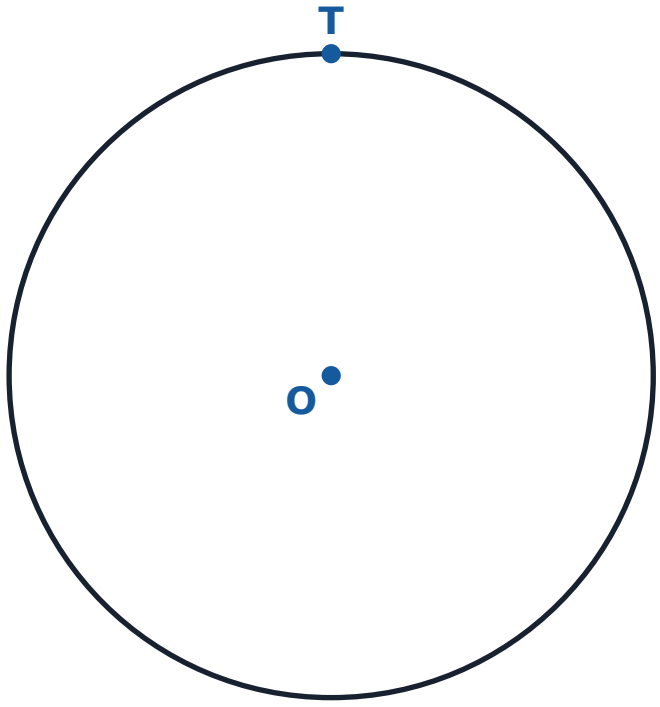
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle with one vertex at the top point T.



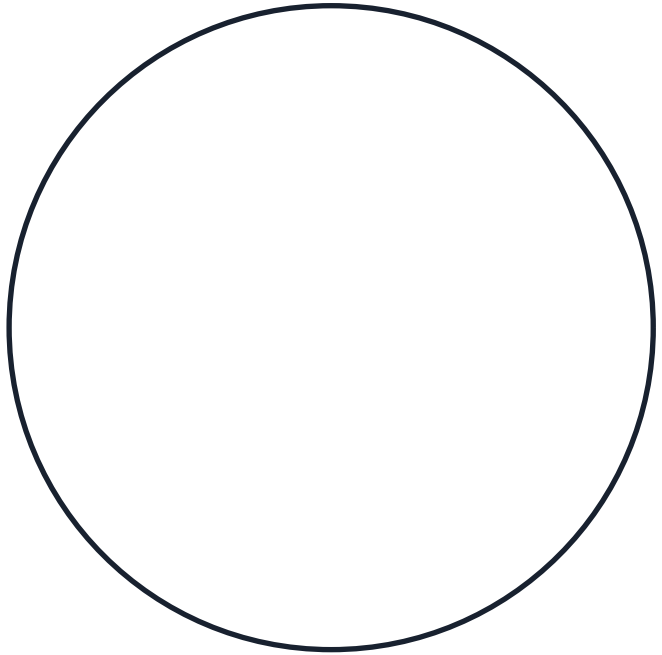
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

The center of this circle is not marked. Construct a regular hexagon inscribed in it.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 1

Construct an inscribed equilateral triangle • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

How are the inscribed equilateral triangle and the inscribed regular hexagon related?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q1/5

True or false?

Both constructions start by stepping the radius around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q2/5

How many points does stepping the radius mark?

A. 12

B. 6

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q3/5

An equilateral triangle is inscribed. How many more points must be added to make a regular hexagon?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q4/5

True or false?

The inscribed equilateral triangle and hexagon have the same side length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 2

An equilateral triangle has equal sides and 60° angles • Grade 4 • Geometry • 4.G.A.2

True or false?

Each angle of an equilateral triangle measures 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q1/5

A triangle has three sides equal to the radius of a circle. What kind of triangle is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q2/5

Six equilateral triangles meeting at a point fill an angle of

A. 300°

B. 360°

C. 60°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q3/5

A regular hexagon has side 5. How long are the segments from its center to its vertices?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q4/5

True or false?

A regular hexagon splits into four equilateral triangles from its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 3

Circle facts: radius and chord • Grade 7 • Geometry • 7.G.B.4

A chord equals the radius. What central angle does it subtend?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q1/5

True or false?

A chord equal to the radius forms an equilateral triangle with the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q2/5

A circle has radius 7. A regular inscribed hexagon has perimeter

A. 21

B. 14

C. 42

D. 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q3/5

A circle has circumference 31.4 (π ≈ 3.14). Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q4/5

True or false?

Every chord of a circle is as long as the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q5/5

SET 36 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed hexagon is regular • High School — Geometry • Geometry: Circles • HSG-C.A.2

Six points are marked around a circle with center O by stepping the radius. Prove that the hexagon they form is regular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q1/5

A compass set to the radius of a circle is stepped around the circle from a starting point. Why does the sixth step land exactly back on the starting point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q2/5

True or false?

Each interior angle of a regular hexagon is 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q3/5

A regular hexagon inscribed in a circle of radius 4 has perimeter

A. 12

B. 48

C. 24

D. 8π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q4/5

Find the area of a regular hexagon inscribed in a circle of radius 2.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSG-CO.D.13

FORWARD 2

Use inscribed hexagons to estimate π • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A regular hexagon is inscribed in a circle of radius r. Compare its perimeter with the circumference. What estimate of π does this give?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q1/5

A regular 12-gon inscribed in a circle of radius 1 has perimeter about 6.212. What estimate of π does it give, to the nearest hundredth?

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q2/5

True or false?

Doubling the number of sides of an inscribed regular polygon makes its perimeter longer than the circumference.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q3/5

Which polygon's perimeter is closest to the circumference of its circle?

A. Inscribed regular 96-gon

B. Inscribed regular hexagon

C. Inscribed square

D. Inscribed triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q4/5

Explain why an inscribed polygon's perimeter is always less than the circumference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q5/5

Verify that a dilation maps a line to a parallel line (or to itself through the center)

MAIN	Geometry main question	**HSG-SRT.A.1.a**
BACKWARD 1	Dilate points with coordinates	**8.G.A.3**
BACKWARD 2	Compare slopes to test whether lines are parallel	**8.EE.B.6**
BACKWARD 3	Write the equation of a line through two points	**8.F.B.4**
FORWARD 1	Dilate lines about a center other than the origin	**HSG-SRT.A.1.a**
FORWARD 2	Prove algebraically that a dilation maps a line to a parallel line	**HSG-SRT.A.1.a**

SET 37 • HSG-SRT.A.1.a

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

Line ℓ is $y = 2x + 1$ (it does not pass through the origin). Dilate it about the origin with scale factor 3: find the images of (0, 1) and (1, 3), write an equation of the image line, and state what this example shows about dilations and lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q1/5

Line m is $y = -x$, which passes through the origin. Dilate it about the origin with scale factor 2: find the images of $(1, -1)$ and $(3, -3)$. What is the image line, and what does this example suggest about lines through the center?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q2/5

The horizontal line y = 4 is dilated about the center C(0, 2) with scale factor ½. Find the images of (0, 4) and (6, 4), write an equation of the image line, and compare it with the original.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q3/5

Line n is y = −½x + 4. Dilate it about the center C(2, 1) with scale factor 3: find the images of (0, 4) and (4, 2), write an equation of the image line, and compare it with n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q4/5

Line p is y = 2x, which passes through C(1, 2). Dilate it about C with scale factor 4: find the images of (0, 0) and (2, 4) and the image line. What do you observe?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 1

Dilate points with coordinates • Grade 8 • Geometry • 8.G.A.3

Dilate $(2, -3)$ about the origin with scale factor 4.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

Dilate (6, 10) about the origin with scale factor ½.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

A dilation about the origin sends (3, 1) to (9, 3). What is the scale factor?

A. 6

B. 9

C. 3

D. ⅓

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

True or false?

A dilation about the origin with scale factor 1 leaves every point where it is.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

True or false?

A dilation with scale factor 2 moves the center to a new point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 2

Compare slopes to test whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

Find the slope of the line through (0, 3) and (3, 9).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q1/5

True or false?

$y = 2x + 1$ and $y = 2x + 3$ are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q2/5

Which line is parallel to $y = -3x + 2$?

A. $y = \frac{1}{3}x$

B. $y = -3$

C. $y = -3x - 5$

D. $y = 3x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q3/5

Are y = 5 and y = 2 parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q4/5

True or false?

Lines with slopes 2 and −2 are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 3

Write the equation of a line through two points • Grade 8 • Functions • 8.F.B.4

Write an equation of the line through (0, 3) and (3, 9).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q1

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q1/5

Write an equation of the line through (2, −2) and (6, −6).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q2

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q2/5

Which point is on y = 2x + 3?

A. (5, 1)

B. (2, 3)

C. (0, 2)

D. (1, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q3

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q3/5

True or false?

The line through (0, 0) and (4, 8) is $y = 2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q4

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q4/5

True or false?

The line through (1, 1) and (2, 4) has slope 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q5

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q5/5

SET 37 • HSG-SRT.A.1.a

FORWARD 1

Dilate lines about a center other than the origin • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

Dilate the line $y = x + 2$ about the center $(2, 0)$ with scale factor 2. Find the images of $(0, 2)$ and $(2, 4)$ and the image line.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q1/5

Does the line $y = x - 2$ pass through the center $(2, 0)$? What happens to it under any dilation about $(2, 0)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q2/5

True or false?

Under a dilation about (1, 1) with scale factor 3, the point (2, 1) goes to (4, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q3/5

A dilation about (0, 2) with scale factor ½ maps (4, 6) to

A. (4, 4)

B. (8, 10)

C. (2, 3)

D. (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q4/5

Line ℓ: y = 3x. A dilation about (1, 0) with scale factor 2 maps ℓ to ℓ'. Find ℓ'.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSG-SRT.A.1.a

FORWARD 2

Prove algebraically that a dilation maps a line to a parallel line • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

Prove that the dilation about the origin with scale factor k ($k > 0, k \neq 1$) maps the line $y = mx + b$ to the line $y = mx + kb$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q1/5

From the general result, when does the image of $y = mx + b$ coincide with the original line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q2/5

True or false?

The dilation $(x, y) \to (kx, ky)$ maps the vertical line $x = c$ to the line $x = c + k$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q3/5

Under the dilation about the origin with scale factor 4, the line $y = -2x + 5$ maps to

A. $y = -8x + 5$

B. $y = -2x + 20$

C. $y = -2x + 5$

D. $y = -8x + 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q4/5

Explain why measuring a few dilated lines is not a proof of the property.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q5/5

Verify that a dilation multiplies lengths by the scale factor

MAIN	Geometry main question	**HSG-SRT.A.1.b**
BACKWARD 1	Find lengths with the distance formula	**8.G.B.8**
BACKWARD 2	Scale factor in scale drawings	**7.G.A.1**
BACKWARD 3	Dilate points about the origin	**8.G.A.3**
FORWARD 1	Find the center or scale factor from a segment and its image	**HSG-SRT.A.1.b**
FORWARD 2	Perimeter and area of dilated figures	**HSG-SRT.A.2**

SET 38 • HSG-SRT.A.1.b

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

Segment AB has A(1, 2) and B(4, 6). Dilate it about the origin with scale factor 3. Find A′B′ and AB, and compare.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q1/5

Segment PQ has P(−4, 2) and Q(8, −6). Dilate it about the origin with scale factor ½. Find PQ and P′Q′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q2/5

Segment CD has C(2, 1) and D(5, 5). Dilate it about the center (2, 1) with scale factor 2. Find C′D′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q3/5

Triangle ABC has A(0, 0), B(4, 0), C(0, 3). Dilate it about (1, 1) with scale factor 2. Find all three image side lengths, compare each with the original, and say what the comparison supports.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q4/5

Segment EF has E(−3, −1) and F(3, 7). Dilate it about the origin with scale factor 0.4. Find EF and E′F′, and compare.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 1

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

Find the distance between (3, 6) and (12, 18).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q1/5

Find the distance between $(-2, 1)$ and $(4, -3)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q2/5

Find the distance between (0, 0) and (5, 12).

A. $\sqrt{17}$

B. 7

C. 17

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q3/5

True or false?

The distance between (1, 2) and (4, 6) is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q4/5

True or false?

The distance between (0, 0) and (3, 3) is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 2

Scale factor in scale drawings • Grade 7 • Geometry • 7.G.A.1

A drawing uses scale factor 4. A 3 cm segment in the original is how long in the drawing?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q1/5

A 15 m wall is drawn 5 cm long. What scale factor (drawing ÷ actual, in the same units) was used?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q2/5

True or false?

A scale factor less than 1 makes a figure smaller.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q3/5

A rectangle 4 by 6 is enlarged by a scale factor of 1.5. Its new dimensions are

A. 5.5 by 7.5

B. 4 by 9

C. 8 by 12

D. 6 by 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q4/5

True or false?

Enlarging by a scale factor of 2 doubles the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 3

Dilate points about the origin • Grade 8 • Geometry • 8.G.A.3

Dilate (4, 6) about the origin with scale factor 3.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q1/5

Dilate (−4, 2) about the origin with scale factor ½.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q2/5

Dilation about the origin sends (2, 5) to (5, 12.5). Scale factor?

A. 2.5

B. 3

C. 0.4

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q3/5

True or false?

Dilating about the origin multiplies both coordinates by the scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q4/5

True or false?

Dilating (3, 0) by 2 about the origin gives (5, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • HSG-SRT.A.1.b

FORWARD 1

Find the center or scale factor from a segment and its image • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

A dilation about the origin maps a segment of length 6 to a segment of length 15. What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q1/5

A dilation maps A(1, 1) to A′(3, 3) and B(2, 1) to B′(5, 3). Find the scale factor and the center.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q2/5

True or false?

Under a dilation with scale factor $k \neq 1$, the line through any point P other than the center and its image P′ passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q3/5

A dilation maps P(4, 0) to P′(1, 0) and its center is the origin. Scale factor?

A. 4

B. −3

C. ¼

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q4/5

A segment of length 9 is dilated by an unknown factor k, and the image is 3 units shorter than the original. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSG-SRT.A.1.b

FORWARD 2

Perimeter and area of dilated figures • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

A triangle with perimeter 18 and area 12 is dilated with scale factor 3. Find the new perimeter and area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q1/5

A square of area 50 is dilated so its area becomes 8. Find the scale factor.

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q2/5

True or false?

If lengths are multiplied by k, areas are multiplied by 2k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q3/5

A figure is dilated with scale factor ½. Its area is multiplied by

A. ½

B. 4

C. 2

D. ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q4/5

Explain why a dilation with factor k multiplies the perimeter by k.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q5/5

Decide whether two figures are similar using similarity transformations

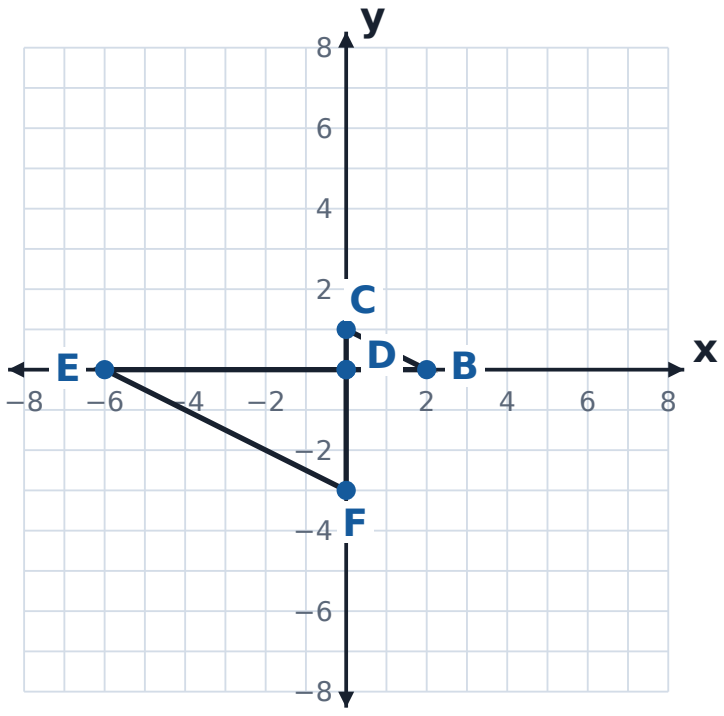
MAIN	Geometry main question	**HSG-SRT.A.2**
BACKWARD 1	Similar figures as a sequence of transformations including a dilati…	**8.G.A.4**
BACKWARD 2	Specify a sequence of rigid motions	**HSG-CO.A.5**
BACKWARD 3	Dilate figures about the origin	**8.G.A.3**
FORWARD 1	Find the scale factor and correspondence when the image is turne	**HSG-SRT.A.2**
FORWARD 2	Prove that all circles are similar	**HSG-C.A.1**

SET 39 • HSG-SRT.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Is triangle ABC similar to triangle DEF? If so, describe a similarity transformation that maps ABC onto DEF; if not, explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q1/5

Is rectangle R (4 by 6) similar to rectangle S (6 by 8)? Explain using similarity transformations.

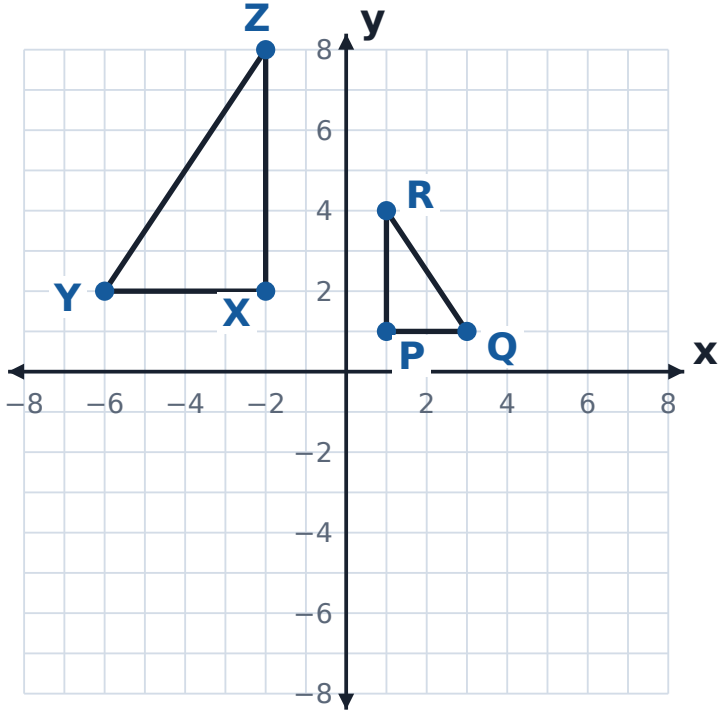
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q2/5

Is triangle PQR similar to triangle XYZ? If so, describe a similarity transformation; if not, explain.



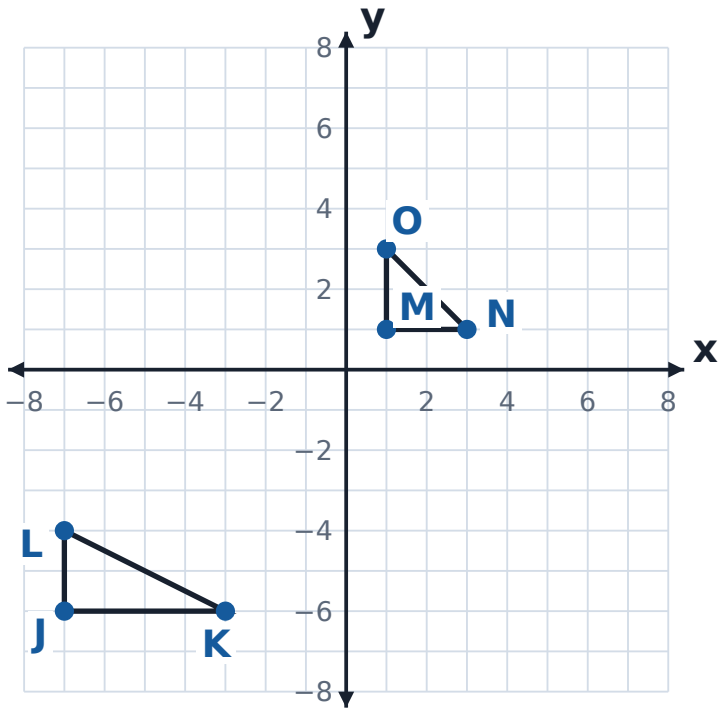
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q3/5

Is triangle JKL similar to triangle MNO? If so, describe a similarity transformation; if not, explain.



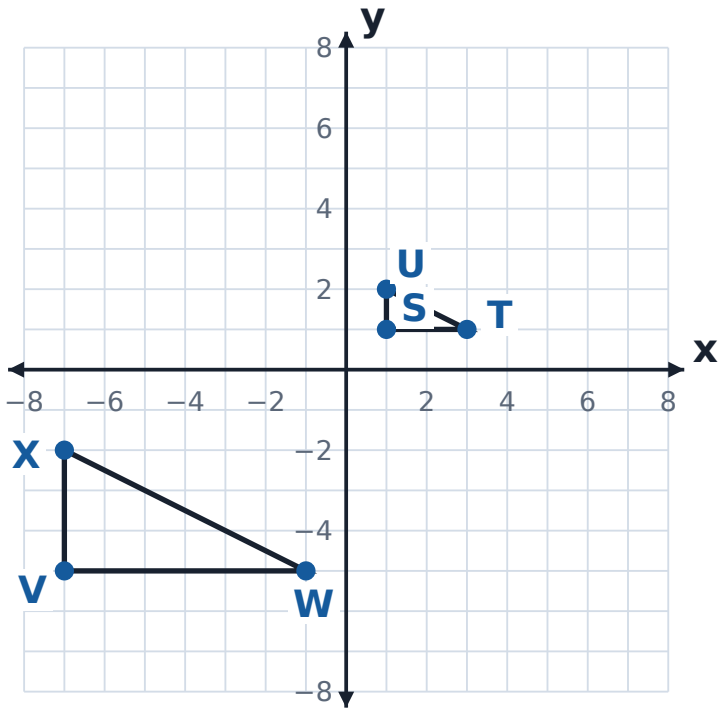
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q4/5

Is triangle STU similar to triangle VWX? If so, describe a similarity transformation; if not, explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q5/5

SET 39 • HSG-SRT.A.2

BACKWARD 1

Similar figures as a sequence of transformations including a dilation • Grade 8 • Geometry • 8.G.A.4

True or false?

A figure and its image after a dilation and a translation are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q1/5

Two figures are similar. What may be different?

A. Their shape

B. Their size

C. Their angle measures

D. The ratios of their sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q2/5

A triangle is dilated by 2 and then rotated. Its sides were 3, 4, 5. What are the image's sides?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q3/5

True or false?

Two figures with the same angles but different side ratios are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q4/5

Name the transformation that changes size.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSG-SRT.A.2

BACKWARD 2

Specify a sequence of rigid motions • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Which rigid motion maps (x, y) to (−x, y)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q1/5

Which rigid motion maps (x, y) to (−x, −y)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q2/5

Which rigid motion maps (2, 3) to (−3, 2)?

A. Translation 5 left

B. Rotation 90° counterclockwise about the origin

C. Rotation 180°

D. Reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q3/5

True or false?

A translation then a reflection is a valid sequence of rigid motions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q4/5

True or false?

A dilation is a rigid motion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSG-SRT.A.2

BACKWARD 3

Dilate figures about the origin • Grade 8 • Geometry • 8.G.A.3

Dilate triangle (0, 0), (2, 0), (0, 1) by 3 about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q1/5

True or false?

A dilation about the origin with factor 3 triples each coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q2/5

(4, −2) dilated by ½ about the origin is

A. (8, −4)

B. (2, −1)

C. (4.5, −1.5)

D. (2, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q3/5

A dilation about the origin maps (5, 5) to (2, 2). What is the factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q4/5

True or false?

A dilation can change the angles of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • BACKWARD 3 • Q5/5

SET 39 • HSG-SRT.A.2

FORWARD 1

Find the scale factor and correspondence when the image is turned or flipped • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has sides AB = 4, BC = 6, CA = 8. Triangle PQR has PQ = 9, QR = 12, RP = 6. Are they similar? If so, give the correspondence and the scale factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q1/5

A 3-4-5 triangle and a triangle with sides 7.5, 10, 12.5: similar? Scale factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q2/5

True or false?

Matching the shortest side with the shortest side, middle with middle, and longest with longest gives the correspondence for similar triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q3/5

Triangles with sides 5, 6, 7 and 10, 12, 15: are they similar?

A. Cannot tell

B. Yes, factor 2.5

C. Yes, factor 2

D. No, 15/7 ≠ 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q4/5

Triangle ABC ~ triangle DEF with a factor of 4 from ABC to DEF. DE = 10. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSG-SRT.A.2

FORWARD 2

Prove that all circles are similar • High School — Geometry • Geometry: Circles • HSG-C.A.1

Circle C_1 has center A and radius r; circle C_2 has center B and radius s. Prove that C_1 and C_2 are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q1/5

Circle P has radius 4 and circle Q has radius 10. What scale factor maps P onto Q (after a translation)?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q2/5

True or false?

Any two rectangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q3/5

Which pair of figures is NOT always similar?

A. Two equilateral triangles

B. Two squares

C. Two circles

D. Two ellipses

Explain why all equilateral triangles are similar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 39 • FORWARD 2 • Q5/5

Explain similarity of triangles as equal angles and proportional sides

MAIN	Geometry main question	**HSG-SRT.A.2**
BACKWARD 1	Read correspondence from a similarity or congruence statement	**HSG-CO.B.7**
BACKWARD 2	Solve proportions	**7.RP.A.2**
BACKWARD 3	Scale factor between figures	**7.G.A.1**
FORWARD 1	Use the AA criterion to recognize similar triangles	**HSG-SRT.A.3**
FORWARD 2	Use similarity to find unknown lengths in problems	**HSG-SRT.B.5**

SET 40 • HSG-SRT.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

A similarity transformation (a dilation with scale factor k followed by rigid motions) maps $\triangle ABC$ onto $\triangle DEF$ with $A \to D$, $B \to E$, $C \to F$. Explain why $\angle A \cong \angle D$, $\angle B \cong \angle E$, $\angle C \cong \angle F$ and $DE/AB = EF/BC = FD/CA = k$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q1/5

$\triangle ABC \sim \triangle DEF$ with $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$. $AB = 6$, $BC = 9$, $DE = 10$. Find EF and the scale factor from ABC to DEF.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q2/5

$\triangle PQR \sim \triangle STU$ with $P \leftrightarrow S$, $Q \leftrightarrow T$, $R \leftrightarrow U$. $m\angle P = 42°$ and $m\angle Q = 71°$. Find $m\angle U$.

$m\angle U =$ _____

△ABC has AB = 4, BC = 6, CA = 8, and △XYZ has XY = 6, YZ = 9, ZX = 10. Using the meaning of similarity, decide whether △ABC ~ △XYZ with A ↔ X, B ↔ Y, C ↔ Z, and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q4/5

A dilation with scale factor 2.5 followed by a rotation maps $\triangle JKL$ onto $\triangle MNO$ (J → M, K → N, L → O). $m\angle J = 50°$, $m\angle K = 60°$, JK = 4, KL = 6. Find $m\angle O$, MN and NO, and explain each answer using what the transformations do.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • MAIN • Q5/5

SET 40 • HSG-SRT.A.2

BACKWARD 1

Read correspondence from a similarity or congruence statement • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

△ABC ~ △XYZ. Which angle corresponds to ∠C?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q1/5

△ABC ~ △XYZ. Which side corresponds to BC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q2/5

△KLM ~ △RST. Which ratio equals KL/RS?

A. LM/RT

B. RS/LM

C. LM/ST

D. KM/ST

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q3/5

True or false?

$\triangle DEF \sim \triangle GHI$ means $\angle E \cong \angle H$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q4/5

True or false?

$\triangle DEF \sim \triangle GHI$ means DF corresponds to GH.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSG-SRT.A.2

BACKWARD 2

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

Solve 6/9 = 10/x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q1/5

Solve x/4 = 15/6.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q2/5

Which ratio is equal to 3 : 5?

A. 9 : 12

B. 6 : 8

C. 12 : 20

D. 5 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 40 • BACKWARD 2 • Q3/5

True or false?

$2/3 = 8/12$

A. True

B. False

True or false?

$4/6 = 6/9 = 10/14$

A. True

B. False

SET 40 • HSG-SRT.A.2

BACKWARD 3

Scale factor between figures • Grade 7 • Geometry • 7.G.A.1

A figure is enlarged so that a 4 cm side becomes 10 cm. Scale factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q1/5

True or false?

All lengths of a scale drawing are multiplied by the same number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q2/5

With scale factor 3, a 7 cm side becomes

A. 7 cm

B. 10 cm

C. 73 cm

D. 21 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q3/5

A scale drawing's 12 cm side represents 60 cm. Scale factor (drawing to actual)?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q4/5

True or false?

A scale drawing changes the angles of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 3 • Q5/5

SET 40 • HSG-SRT.A.2

FORWARD 1

Use the AA criterion to recognize similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

Two triangles have angles of 35° and 80°. Are they similar? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q1/5

Triangle ABC has m∠A = 50°, m∠B = 60°. Triangle DEF has m∠D = 60°, m∠F = 70°. Are they similar? Give the correspondence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q2/5

True or false?

Two right triangles with one pair of congruent acute angles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q3/5

Which pair is NOT necessarily similar?

A. Two triangles with angles 30° and 60°

B. Two right isosceles triangles

C. Two equilateral triangles

D. Two isosceles triangles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q4/5

In a figure, DE ∥ BC with D on AB and E on AC. Explain why △ADE ~ △ABC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSG-SRT.A.2

FORWARD 2

Use similarity to find unknown lengths in problems • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 1.8 m person casts a 2.4 m shadow when a tree casts a 16 m shadow. How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q1/5

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$. AD = 4, DB = 6, and DE = 5. Find BC.

BC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q2/5

True or false?

In similar triangles with scale factor 3, a 7 cm side corresponds to a 10 cm side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q3/5

A 6 ft pole casts a 4 ft shadow. At the same time, a building casts a 30 ft shadow. How tall is the building?

A. 36 ft

B. 45 ft

C. 20 ft

D. 24 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q4/5

Two similar triangles have perimeters 24 and 36. A side of the smaller is 10. Find the corresponding side of the larger.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 40 • FORWARD 2 • Q5/5

Establish the AA criterion with similarity transformations

MAIN	Geometry main question	**HSG-SRT.A.3**
BACKWARD 1	Prove triangles congruent with ASA	**HSG-CO.B.8**
BACKWARD 2	Dilations multiply lengths and keep angles	**HSG-SRT.A.1.b**
BACKWARD 3	Use the angle sum to find a third angle	**8.G.A.5**
FORWARD 1	Establish the SAS and SSS similarity criteria	**HSG-SRT.A.3**
FORWARD 2	Use AA to set up and solve problems	**HSG-SRT.B.5**

SET 41 • HSG-SRT.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

In △ABC and △DEF, ∠A ≅ ∠D and ∠B ≅ ∠E. Prove that △ABC ~ △DEF using similarity transformations.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q1/5

In $\triangle PQR$ and $\triangle STU$, $m\angle P = m\angle S = 40°$, $m\angle Q = m\angle T = 75°$, $PQ = 4$ and $ST = 10$. Prove that $\triangle PQR \sim \triangle STU$ by describing a similarity transformation and justifying each step.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q2/5

Prove that two right triangles with one pair of congruent acute angles are similar.

Work space

Conclusion: _____

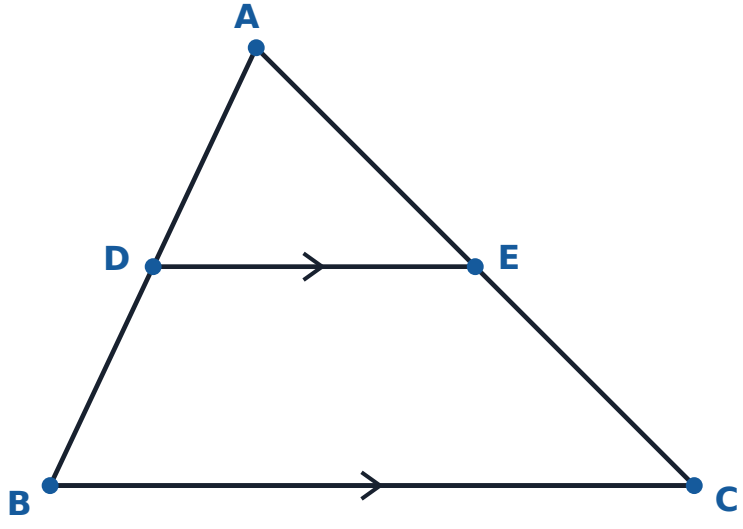
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q3/5

In △ABC, D is on AB and E is on AC with DE ∥ BC. Prove that △ADE ~ △ABC.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q4/5

Prove that two isosceles triangles with congruent vertex angles are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • MAIN • Q5/5

SET 41 • HSG-SRT.A.3

BACKWARD 1

Prove triangles congruent with ASA • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles have $\angle A \cong \angle D$, $AB = DE$, $\angle B \cong \angle E$. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q1/5

True or false?

ASA needs the side between the two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q2/5

Which set of parts gives ASA?

A. Three angles

B. Three sides

C. Two sides and the angle between them

D. Two angles and the side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q3/5

After proving $\triangle PQR \cong \triangle XYZ$, are their corresponding sides equal?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q4/5

True or false?

ASA works only for right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSG-SRT.A.3

BACKWARD 2

Dilations multiply lengths and keep angles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

A 5 cm segment is dilated by 1.6. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q1/5

True or false?

A dilation preserves angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q2/5

A dilation maps AB = 4 to A′B′ = 10. Factor?

A. 2.5

B. 6

C. 0.4

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-B2-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q3/5

What scale factor makes a 6 cm segment 6 cm long?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 41 • BACKWARD 2 • Q4/5

True or false?

A dilation by 2 doubles the angles of a triangle.

A. True

B. False

SET 41 • HSG-SRT.A.3

BACKWARD 3

Use the angle sum to find a third angle • Grade 8 • Geometry • 8.G.A.5

A triangle has angles 47° and 88°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q1/5

True or false?

If two angles of one triangle equal two angles of another, the third angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q2/5

Triangles with angles 30°, 70°, x and 70°, 80°, y. Which is true?

A. x = 80, y = 30; the triangles are similar

B. x = 80, y = 30; not similar

C. x = 70, y = 30

D. x = 30, y = 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q3/5

Two angles of a triangle are equal and the third is 40°. Find the equal angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q4/5

True or false?

Knowing one angle of each triangle is enough for similarity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • BACKWARD 3 • Q5/5

SET 41 • HSG-SRT.A.3

FORWARD 1

Establish the SAS and SSS similarity criteria • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

In $\triangle ABC$ and $\triangle DEF$, $DE/AB = DF/AC = k$ and $\angle A \cong \angle D$. Prove that $\triangle ABC \sim \triangle DEF$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q1/5

Triangles with sides 4, 6, 8 and 6, 9, 12. Similar? Which criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q2/5

True or false?

Two triangles with two pairs of proportional sides and congruent included angles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q3/5

Which is NOT a similarity criterion?

A. SAS

B. SSA

C. SSS

D. AA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q4/5

△PQR: PQ = 5, PR = 7, ∠P = 40°. △XYZ: XY = 10, XZ = 14, ∠X = 40°. Similar? Factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSG-SRT.A.3

FORWARD 2

Use AA to set up and solve problems • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Two vertical poles are 3 m and 7 m tall. A wire from the top of each to the bottom of the other crosses at a point P. How high above the ground is P? (Use similar triangles.)

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q1/5

In $\triangle ABC$, D is on AB and E is on AC with $\angle ADE \cong \angle ACB$. AD = 4, AE = 6, AB = 12. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q2/5

True or false?

A mirror on the ground can be used to find a building's height with similar triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q3/5

Diagonals of trapezoid ABCD (AB ∥ DC) meet at E. AB = 12, DC = 4, AE = 9. Find EC.

A. 6

B. 27

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q4/5

A 2 m stick casts a 1.5 m shadow. A flagpole casts a 9 m shadow. How tall is the flagpole?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 41 • FORWARD 2 • Q5/5

Prove the triangle proportionality theorem

MAIN	Geometry main question	**HSG-SRT.B.4**
BACKWARD 1	Use AA to show triangles are similar	**HSG-SRT.A.3**
BACKWARD 2	Corresponding angles of parallel lines	**HSG-CO.C.9**
BACKWARD 3	Rewrite and solve proportions algebraically	**7.RP.A.2**
FORWARD 1	Prove that parallel lines cut transversals proportionally	**HSG-SRT.B.4**
FORWARD 2	Prove the angle bisector theorem	**HSG-SRT.B.5**

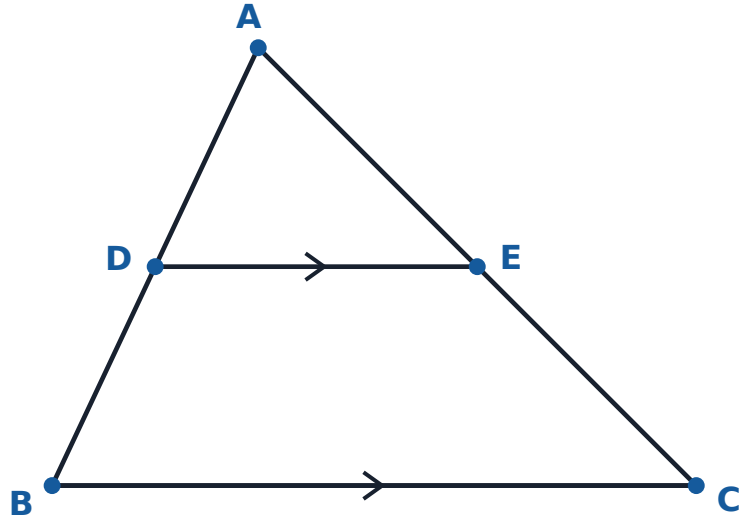
SET 42 • HSG-SRT.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In △ABC, D is on AB and E is on AC with DE ∥ BC. Prove that AD/DB = AE/EC.

Work space



Conclusion: _____

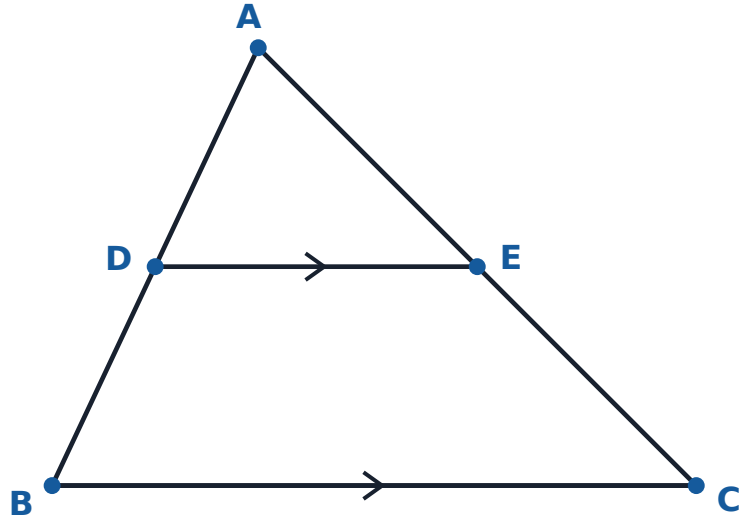
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q1/5

In △ABC, D is on AB and E is on AC with DE ∥ BC. Prove that AD/AB = AE/AC = DE/BC.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q2/5

In △PQR, a line parallel to QR meets PQ at S and PR at T. Prove that PS/SQ = PT/TR.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q3/5

Coordinate proof. Place $\triangle ABC$ with $A(0, 0)$, $B(p, 0)$ and $C(q, r)$, where $p > 0$ and $r > 0$. $D(tp, 0)$ is on AB with $0 < t < 1$, and E is on AC with $DE \parallel BC$. Prove that $AD/DB = AE/EC$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q4/5

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$. Prove that $AD/DB = AE/EC$ using the dilation about A with scale factor $k = AB/AD$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • MAIN • Q5/5

SET 42 • HSG-SRT.B.4

BACKWARD 1

Use AA to show triangles are similar • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Two pairs of congruent angles make two triangles similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q1/5

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$. Name two pairs of congruent angles in $\triangle ADE$ and $\triangle ABC$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q2/5

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. Which ratio equals AD/AB?

A. DB/AB

B. AE/AC

C. DE/AB

D. AE/EC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B1-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q3/5

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. AD = 3, AB = 9, AE = 4. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q4/5

True or false?

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$, so $\triangle ADE \sim \triangle ABC$. Then $AD/DB = DE/BC$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSG-SRT.B.4

BACKWARD 2

Corresponding angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

When a transversal crosses parallel lines, corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q1/5

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$, so AB is a transversal of DE and BC. Which angle corresponds to $\angle ADE$?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q2/5

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$. With AC as the transversal, $\angle AED$ corresponds to

A. $\angle BAC$

B. $\angle ABC$

C. $\angle DEC$

D. $\angle ACB$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q3/5

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$. $m\angle ABC = 65°$. Find $m\angle ADE$.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q4/5

True or false?

Corresponding angles are always supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSG-SRT.B.4

BACKWARD 3

Rewrite and solve proportions algebraically • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

If a/b = c/d, where a, b, c and d are all nonzero, write the equation you get by taking reciprocals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q1/5

Solve $4/(4 + x) = 6/15$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q2/5

If $(a + b)/a = (c + d)/c$, then

A. $b/a = d/c$

B. $b = d$

C. $a = c$

D. $a/b = c/d$ is false

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q3/5

True or false?

$3/5 = 6/10$ implies $5/3 = 10/6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q4/5

True or false?

$(2 + 3)/2 = 5/2$ implies $3/2 = 5/2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 42 • BACKWARD 3 • Q5/5

SET 42 • HSG-SRT.B.4

FORWARD 1

Prove that parallel lines cut transversals proportionally • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

Three parallel lines cut transversal t at A, B, C and transversal u at D, E, F (in order). Prove that AB/BC = DE/EF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q1/5

Three parallel lines cut one transversal into segments 4 and 6 and another into segments 5 and x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q2/5

True or false?

Parallel lines that cut equal segments on one transversal cut equal segments on every transversal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q3/5

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$. $AD = 5$, $DB = 3$, $AE = 7.5$. Find EC.

A. 12.5

B. 4.5

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q4/5

Lots between parallel streets have frontages 30 m and 45 m on one avenue. On another avenue the first frontage is 40 m. Find the second.

Frontage: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSG-SRT.B.4

FORWARD 2

Prove the angle bisector theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In $\triangle ABC$, the bisector of $\angle A$ meets BC at D. Prove that $BD/DC = AB/AC$. (Draw the line through C parallel to AD, meeting line AB at E.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q1/5

In $\triangle ABC$, the bisector of $\angle A$ meets BC at D. AB = 9, AC = 6, BC = 10. Find BD and DC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q2/5

True or false?

The bisector of an angle of a triangle always meets the opposite side at its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q3/5

The bisector of $\angle P$ in $\triangle PQR$ divides QR into 3 and 5. PQ = 6. Find PR.

A. 10

B. 15

C. 3.6

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q4/5

Explain why the angle bisector theorem is a relationship that similarity or proportionality, not congruence, proves.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 42 • FORWARD 2 • Q5/5

Prove the converse of the triangle proportionality theorem

MAIN	Geometry main question	**HSG-SRT.B.4**
BACKWARD 1	Use the triangle proportionality theorem	**HSG-SRT.B.4**
BACKWARD 2	SAS similarity	**HSG-SRT.A.3**
BACKWARD 3	Converse of the corresponding angles theorem	**HSG-CO.C.9**
FORWARD 1	Use the converse to decide parallelism in figures	**HSG-SRT.B.4**
FORWARD 2	Prove the trapezoid midsegment theorem	**HSG-SRT.B.5**

SET 43 • HSG-SRT.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In △ABC, D is on AB and E is on AC with AD/DB = AE/EC. Prove that DE ∥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q1/5

In $\triangle PQR$, S is on PQ and T is on PR with PS = 4, SQ = 6, PT = 6, TR = 9. Prove that $ST \parallel QR$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q2/5

In △ABC, D is on AB and E is on AC with AD/AB = AE/AC. Prove that DE ∥ BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q3/5

In $\triangle ABC$, M is the midpoint of AB and N is the midpoint of AC. Use the converse of the triangle proportionality theorem to prove that $MN \parallel BC$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q4/5

In $\triangle ABC$, D is on AB and E is on AC with AD = 3, DB = 5, AE = 4.5, EC = 7.5, and BC = 16. Prove that DE $\parallel$ BC, then find DE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • MAIN • Q5/5

SET 43 • HSG-SRT.B.4

BACKWARD 1

Use the triangle proportionality theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$. $AD = 6$, $DB = 4$, $AE = 9$. Find EC.

EC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q1/5

True or false?

In $\triangle ABC$, D is on AB and E is on AC. If $DE \parallel BC$, then $AD/DB = AE/EC$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q2/5

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$. AD = 2, AB = 8, AC = 12. Find AE.

A. 9

B. 4

C. 6

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q3/5

In $\triangle ABC$, D is on AB and E is on AC. State the converse of "If $DE \parallel BC$, then $AD/DB = AE/EC$."

Converse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q4/5

True or false?

A theorem and its converse are always both true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSG-SRT.B.4

BACKWARD 2

SAS similarity • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Two pairs of proportional sides with congruent included angles make triangles similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q1/5

In △ABC, D is on AB and E is on AC, so △ADE and △ABC share ∠A. AD/AB = AE/AC = 0.4. Are they similar? Criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q2/5

After $\triangle ADE \sim \triangle ABC$, which angle is congruent to $\angle ADE$?

A. $\angle ACB$

B. $\angle BAC$

C. $\angle ABC$

D. $\angle DEA$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-B2-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q3/5

Ratios 3/6 and 4/8 with a shared included angle: similar? Factor?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q4/5

True or false?

SAS similarity needs the angle to be between the two pairs of sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSG-SRT.B.4

BACKWARD 3

Converse of the corresponding angles theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Congruent corresponding angles make two lines parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q1/5

In $\triangle ABC$, D is on AB and E is on AC. $\angle ADE \cong \angle ABC$, with AB as the transversal. What can you conclude about DE and BC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q2/5

In $\triangle ABC$, D is on AB and E is on AC. Which angles, if congruent, prove $DE \parallel BC$?

A. $\angle A$ and $\angle B$

B. $\angle B$ and $\angle C$

C. $\angle AED$ and $\angle ACB$

D. $\angle ADE$ and $\angle AED$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q3/5

Corresponding angles measure 58° and 62°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q4/5

True or false?

Vertical angles prove lines parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • HSG-SRT.B.4

FORWARD 1

Use the converse to decide parallelism in figures • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In $\triangle ABC$, D is on AB with AD = 5, DB = 10, and E is on AC with AE = 4. How long must EC be for $DE \parallel BC$?

EC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q1/5

In $\triangle PQR$, S on PQ, T on PR: PS = x, SQ = 6, PT = 4, TR = 8. For which x is ST ∥ QR?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q2/5

True or false?

The midsegment theorem's parallel part is a special case of the converse (ratio 1 : 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q3/5

Points divide two sides in ratios 2 : 3 and 4 : 6. The segment joining them is

A. perpendicular to the third side

B. half the third side

C. parallel to the third side

D. not parallel to the third side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q4/5

In $\triangle ABC$, D is on AB with AD : DB = 1 : 2 and DE $\parallel$ BC with E on AC. Find DE : BC.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSG-SRT.B.4

FORWARD 2

Prove the trapezoid midsegment theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Trapezoid ABCD has AB ∥ DC. M and N are the midpoints of legs AD and BC. Prove that MN ∥ AB and MN = ½(AB + DC).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

A trapezoid has bases 10 and 16. Find the length of the segment joining the midpoints of the legs.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

True or false?

The midsegment of a trapezoid is half the length of the longer base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

A trapezoid's midsegment is 12 and one base is 9. Find the other base.

A. 3

B. 15

C. 6

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

Trapezoid ABCD has AB ∥ DC; M and N are the midpoints of legs AD and BC, and P is the midpoint of diagonal AC. Explain why M, P and N lie on one line.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

Prove the Pythagorean Theorem using similar triangles

MAIN	Geometry main question	**HSG-SRT.B.4**
BACKWARD 1	Show triangles similar with AA	**HSG-SRT.A.3**
BACKWARD 2	Cross-multiply proportions	**7.RP.A.2**
BACKWARD 3	Explain a proof of the Pythagorean Theorem informally	**8.G.B.6**
FORWARD 1	Use the geometric mean relationships in a right triangle	**HSG-SRT.B.5**
FORWARD 2	The Law of Cosines as a generalization of the Pythagorean Theor	**HSG-SRT.D.10**

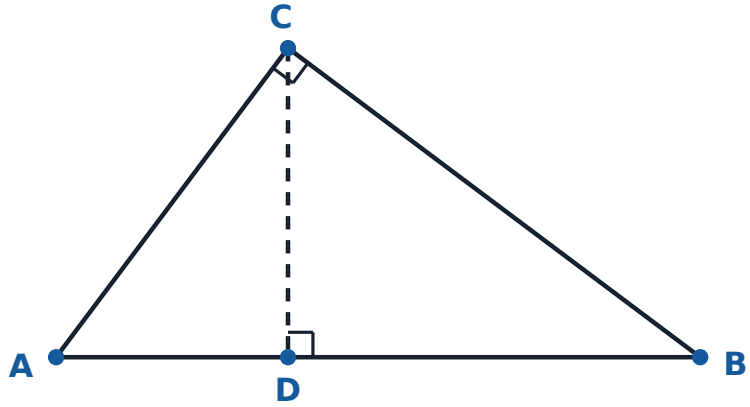
SET 44 • HSG-SRT.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

Right triangle ABC has the right angle at C, legs a = BC, b = AC, and hypotenuse c = AB. The altitude from C meets AB at D. Prove that $a^2 + b^2 = c^2$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q1/5

Right triangle PQR has the right angle at R, legs p = QR and q = PR, and hypotenuse r = PQ. The altitude from R meets PQ at S. Prove that $p^2 + q^2 = r^2$.

Work space

Conclusion: _____

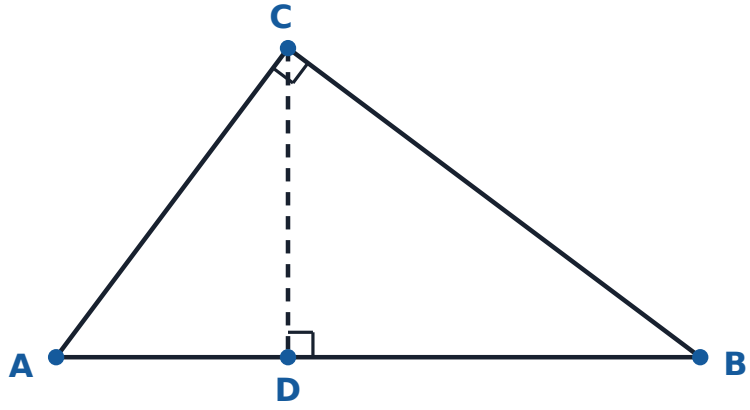
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q2/5

Isosceles right triangle ABC has the right angle at C and legs AC = BC = a. The altitude from C meets AB at D. Using similar triangles (not the Pythagorean Theorem), prove that $AB^2 = 2a^2$.

Work space



Conclusion: _____

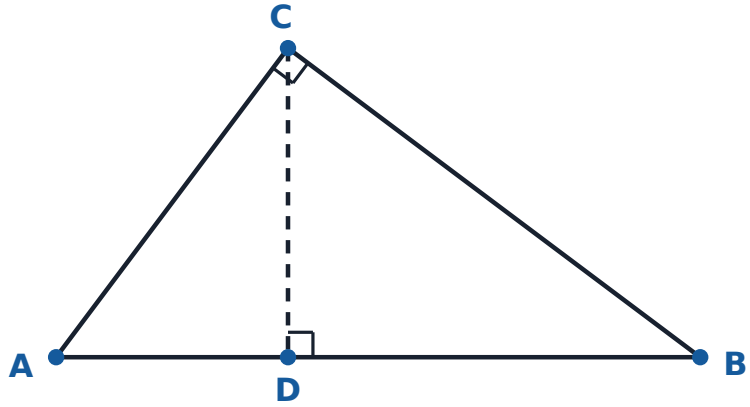
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q3/5

Right triangle ABC has the right angle at C, AC = 6 and BC = 8. The altitude from C meets AB at D; let x = AD and y = DB. Use similar triangles (not the Pythagorean Theorem) to find x, y and AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q4/5

A right triangle has legs 5 and 12. The altitude to the hypotenuse divides it into parts x (next to the leg 5) and y. Use similar triangles (not the Pythagorean Theorem) to find the hypotenuse c, x and y.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • MAIN • Q5/5

SET 44 • HSG-SRT.B.4

BACKWARD 1

Show triangles similar with AA • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Two right triangles that share an acute angle are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q1/5

Right triangle ABC has the right angle at C, and the altitude from C meets hypotenuse AB at D. Name two pairs of congruent angles in $\triangle ADC$ and $\triangle ACB$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q2/5

Right triangle ABC has the right angle at C, and the altitude from C meets hypotenuse AB at D. Then △ADC ~ △ACB. Which ratio equals AD/AC?

A. DC/AB

B. AD/AB

C. AC/AB

D. CB/AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-B1-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q3/5

Right triangle ABC has the right angle at C, and the altitude from C meets hypotenuse AB at D. $\triangle ADC \sim \triangle ACB$, AD = 4 and AB = 9. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q4/5

True or false?

Similar triangles must be the same size.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSG-SRT.B.4

BACKWARD 2

Cross-multiply proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

From AD/AC = AC/AB, write an equation without fractions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q1/5

Solve $4/x = x/9$ for $x > 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q2/5

If $p/q = q/r$, then

A. $q^2 = pr$

B. $p = r$

C. $r^2 = pq$

D. $p^2 = qr$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 44 • BACKWARD 2 • Q3/5

True or false?

From $a/b = b/c$ we get $b^2 = ac$.

A. True

B. False

True or false?

From $x/5 = 5/x$ we get $x = 5$ only.

A. True

B. False

SET 44 • HSG-SRT.B.4

BACKWARD 3

Explain a proof of the Pythagorean Theorem informally • Grade 8 • Geometry • 8.G.B.6

True or false?

The Pythagorean Theorem says $a^2 + b^2 = c^2$ for every right triangle with legs a, b and hypotenuse c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q1/5

Name a way to show the theorem by rearranging area pieces.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q2/5

Which triangle satisfies the theorem?

A. 5, 12, 13

B. 6, 7, 9

C. 5, 6, 7

D. 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q3/5

Legs 9 and 12. Hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q4/5

True or false?

Checking the theorem with three triangles proves it for all right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • HSG-SRT.B.4

FORWARD 1

Use the geometric mean relationships in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Right triangle ABC has the right angle at C, and altitude CD meets hypotenuse AB at D. AD = 4 and DB = 9. Find CD.

CD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q1/5

In right triangle ABC with the right angle at C, altitude CD meets hypotenuse AB at D. Prove that $CD^2 = AD \cdot DB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q2/5

True or false?

The altitude to the hypotenuse is the geometric mean of the two parts of the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q3/5

Right triangle ABC has its right angle at C, and the altitude from C meets hypotenuse AB at D. AD = 2 and AB = 8. Find AC.

A. 4

B. $2\sqrt{2}$

C. 6

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q4/5

A right triangle's altitude to the hypotenuse is 12 and one part of the hypotenuse is 9. Find the other part.

Part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSG-SRT.B.4

FORWARD 2

The Law of Cosines as a generalization of the Pythagorean Theorem • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

For a triangle with sides a, b, c and angle C opposite c, the Law of Cosines says $c^2 = a^2 + b^2 - 2ab\cos C$. What does it become when $C = 90°$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q1/5

A triangle has sides 5 and 8 with a 60° angle between them. Find the third side.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q2/5

True or false?

If $a^2 + b^2 > c^2$, the angle opposite c is obtuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q3/5

A triangle has sides 4, 5, 7. The largest angle is

A. right

B. impossible to tell

C. obtuse

D. acute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q4/5

Use the converse idea: is a triangle with sides 7, 24, 25 right, acute, or obtuse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 44 • FORWARD 2 • Q5/5

Use similarity criteria to solve problems in geometric figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Similarity criteria: AA, SAS, SSS	**HSG-SRT.A.3**
BACKWARD 2	Solve proportions	**7.RP.A.2**
BACKWARD 3	Identify corresponding sides in overlapping triangles	**HSG-SRT.A.2**
FORWARD 1	Solve multistep problems with two or more pairs of similar triangl…	**HSG-SRT.B.5**
FORWARD 2	Solve similarity problems with algebra (unknowns on both sides)	**HSG-SRT.B.5**

SET 45 • HSG-SRT.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In △ABC, D is on AB and E is on AC with DE ∥ BC. AD = 6, DB = 9, DE = 8. Find BC. Name the criterion you used for similar triangles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q1/5

A lamppost lights a student who is 1.6 m tall and stands 3 m from the post. The student's shadow is 2 m long, measured away from the post. How tall is the lamppost? Name the similarity criterion you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q2/5

Two triangles: $\triangle PQR$ with $PQ = 8$, $QR = 12$, $\angle Q = 50°$; $\triangle STU$ with $ST = 6$, $TU = 9$, $\angle T = 50°$. Are they similar? If so, find the ratio of their perimeters.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q3/5

To find the width w of a river, a surveyor marks points so that segments AD and BE cross at C, with ∠B and ∠E right angles. AB = 12 m, BC = 18 m and CE = 60 m; w = DE is across the river. Find w and name the similarity criterion you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q4/5

△KLM has KL = 6, LM = 8 and KM = 12, and m∠L ≈ 117°. △NOP has NO = 9, OP = 12 and NP = 18. Are the triangles similar? Name the criterion, give the scale factor, and find m∠O.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • MAIN • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 1

Similarity criteria: AA, SAS, SSS • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

Which criterion uses two pairs of congruent angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q1/5

Sides 2, 3, 4 and 4, 6, 8. Criterion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q2/5

Which information is enough for similarity?

A. Two pairs of congruent angles

B. One pair of congruent angles

C. One pair of proportional sides

D. Two sides of each triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q3/5

True or false?

SAS similarity requires the congruent angle to be between the proportional sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q4/5

True or false?

Any two right triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 2

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

Solve 6/15 = 8/x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q1/5

Solve 1.6/2 = h/5.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q2/5

Solve 12/40 = 18/x.

A. 26.7

B. 54

C. 46

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q3/5

True or false?

$3/4 = 9/12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q4/5

True or false?

$5/8 = 10/14$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSG-SRT.B.5

BACKWARD 3

Identify corresponding sides in overlapping triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

In △ABC, D is on AB and E is on AC with DE ∥ BC, so △ADE ~ △ABC. Which side of △ABC corresponds to DE?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q1/5

True or false?

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$, so $\triangle ADE \sim \triangle ABC$. Side AD corresponds to side AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q2/5

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$. In $\triangle ADE \sim \triangle ABC$, the side of the large triangle that corresponds to AD is

A. DB

B. AC

C. BC

D. AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B3-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q3/5

D is on segment AB with AD = 6 and DB = 9. What is AB?

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q4/5

True or false?

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$, so $\triangle ADE \sim \triangle ABC$. DB corresponds to DE.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 45 • BACKWARD 3 • Q5/5

SET 45 • HSG-SRT.B.5

FORWARD 1

Solve multistep problems with two or more pairs of similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Trapezoid ABCD has AB ∥ DC, AB = 15, DC = 10. The diagonals meet at E, and AC = 20. Find AE and EC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q1/5

A pinhole camera: an object 2 m tall is 8 m from the pinhole; the image is 5 cm from the pinhole on the screen. How tall is the image?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q2/5

True or false?

In a figure with two pairs of similar triangles, one ratio found in the first pair can be reused in the second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q3/5

In $\triangle ABC$, D and F are on AB and E and G are on AC, with $DE \parallel BC$ and $FG \parallel BC$. $AD = 2$, $DB = 6$, $AF = 5$ and $BC = 16$. Find FG.

A. 10

B. 12

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q4/5

Using a mirror on the ground, Ana (eyes 1.5 m high) stands 2 m from the mirror and sees the top of a tree 12 m beyond the mirror. How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSG-SRT.B.5

FORWARD 2

Solve similarity problems with algebra (unknowns on both sides) • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$. $AD = x$, $DB = 6$, $AE = 4$, $EC = x + 2$. Find x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q1/5

Similar triangles have corresponding sides $x + 1$ and $2x - 3$ with scale factor 1.5 (larger to smaller is $2x - 3$ to $x + 1$). Find x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q2/5

True or false?

A proportion with the unknown in two places can lead to a quadratic equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q3/5

In right triangle ABC with the right angle at C, the altitude CD meets hypotenuse AB at D. AD = x, DB = x + 5, and CD = 6. Find x.

A. 4

B. 3

C. 6

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q4/5

Two similar rectangles: the first is 4 by x, the second is x by 9. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 45 • FORWARD 2 • Q5/5

Use congruence criteria to prove relationships in figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Choose and apply SSS, SAS, ASA, AAS	**HSG-CO.B.8**
BACKWARD 2	Find hidden parts: shared sides, vertical angles, right angles	**HSG-CO.C.9**
BACKWARD 3	Use corresponding parts to finish a proof	**HSG-CO.B.7**
FORWARD 1	Prove relationships that need two congruences in a row	**HSG-SRT.B.5**
FORWARD 2	Decide which given data are sufficient for a proof, and supply a m…	**HSG-SRT.B.5**

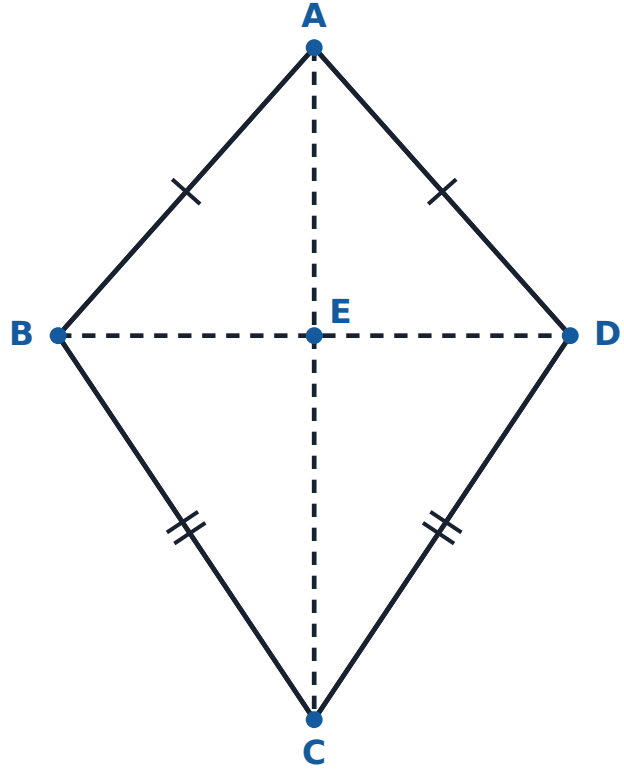
SET 46 • HSG-SRT.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Kite ABCD has AB = AD and CB = CD. Its diagonals meet at E. Prove that AC ⊥ BD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q1/5

Segments AB and CD bisect each other at M. Prove that AC ∥ DB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q2/5

Point P is on the bisector of ∠ABC. PD ⊥ BA at D and PE ⊥ BC at E. Prove that PD = PE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q3/5

In $\triangle ABC$, $AB = AC$. M is the midpoint of BC. Prove that AM bisects $\angle BAC$ and $AM \perp BC$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q4/5

In quadrilateral ABCD, AB ∥ DC and AB = DC. The diagonals AC and BD meet at E. Prove that the diagonals bisect each other.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • MAIN • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 1

Choose and apply SSS, SAS, ASA, AAS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share a side and have two more pairs of congruent sides. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q1/5

Two angles and a non-included side match. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q2/5

Which pair of facts plus a shared side gives SAS?

A. A pair of congruent sides and the congruent included angles

B. Two pairs of congruent sides not including the angle

C. Nothing more is needed

D. Two pairs of congruent angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q3/5

True or false?

A shared side is a valid pair for any criterion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q4/5

True or false?

AAA proves congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 2

Find hidden parts: shared sides, vertical angles, right angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Segments AB and CD cross at M. Which angles are vertical to ∠AMC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q1/5

True or false?

Perpendicular lines give congruent right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q2/5

Which is a hidden fact in figures where two triangles share a side?

A. All angles are equal

B. The shared side is congruent to itself

C. The triangles are right

D. Nothing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q3/5

A bisector divides ∠ABC into which two congruent angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q4/5

True or false?

Vertical angles are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSG-SRT.B.5

BACKWARD 3

Use corresponding parts to finish a proof • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

$\triangle ABC \cong \triangle ADC$. Which angle corresponds to $\angle BAC$?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q1/5

True or false?

Once triangles are congruent, all corresponding parts are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q2/5

$\triangle PBD \cong \triangle PBE$. Which conclusion follows?

A. $PD = PE$

B. $PB = PD$

C. $BD = PE$

D. $\angle D \cong \angle B$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q3/5

Two congruent angles form a linear pair. Each measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q4/5

True or false?

Corresponding parts can be used before the triangles are proved congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 46 • BACKWARD 3 • Q5/5

SET 46 • HSG-SRT.B.5

FORWARD 1

Prove relationships that need two congruences in a row • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB = CD and AD = CB. Diagonals AC and BD meet at E. Prove that E is the midpoint of AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q1/5

In $\triangle ABC$, $AB = AC$. D is on AB and E is on AC with $BD = CE$. Prove that $BE = CD$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q2/5

True or false?

Some proofs need one congruence to find a part used in a second congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q3/5

In a two-step proof, the first congruence gives $\angle 1 \cong \angle 2$. This is used to

A. measure

B. end the proof

C. supply a pair of parts for the second congruence

D. restart

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q4/5

Explain why you cannot use corresponding parts of the second pair of triangles before proving them congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSG-SRT.B.5

FORWARD 2

Decide which given data are sufficient for a proof, and supply a missing condition • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In △PQR and △PSR (sharing PR), PQ = PS. What one more fact would prove ∠QPR ≅ ∠SPR? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q1/5

A student claims a kite's diagonals bisect each other. Prove or disprove.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q2/5

True or false?

Given AB = DE, BC = EF, and $\angle C \cong \angle F$, the triangles must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q3/5

Given $\angle A \cong \angle D$ and $AB = DE$ in $\triangle ABC$ and $\triangle DEF$, which extra fact gives ASA?

A. $\angle C \cong \angle F$

B. $BC = EF$

C. $AC = DF$

D. $\angle B \cong \angle E$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q4/5

Given AB = DE and AC = DF, name an angle condition that guarantees congruence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 46 • FORWARD 2 • Q5/5

Use similarity criteria to prove relationships in figures

MAIN	Geometry main question	**HSG-SRT.B.5**
BACKWARD 1	Prove triangles similar with AA	**HSG-SRT.A.3**
BACKWARD 2	Turn a proportion into a product	**7.RP.A.2**
BACKWARD 3	Match corresponding sides of similar triangles	**HSG-SRT.A.2**
FORWARD 1	Use proved product relationships to compute	**HSG-SRT.B.5**
FORWARD 2	Prove relationships in circles with similar triangles	**HSG-C.A.2**

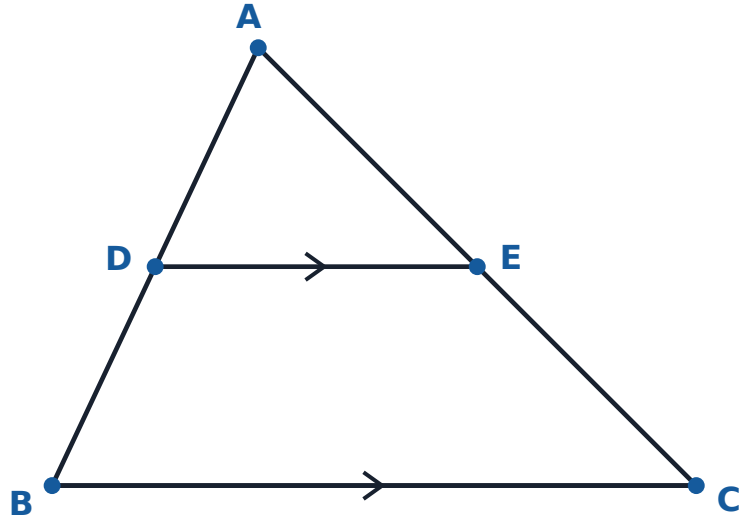
SET 47 • HSG-SRT.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. Prove that $AD \cdot AC = AE \cdot AB$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q1/5

Chords AB and CD of a circle meet at P, and ∠BAD ≅ ∠BCD (given). Prove that PA · PB = PC · PD.

Work space

Conclusion: _____

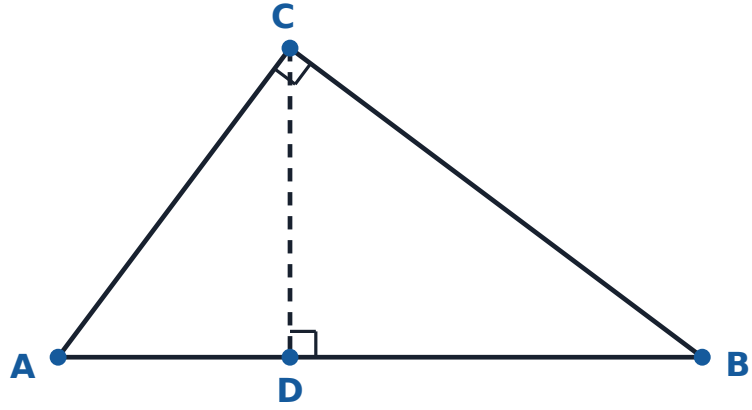
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q2/5

In right triangle ABC (right angle at C), CD is the altitude to AB. Prove that AC · BC = AB · CD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q3/5

In trapezoid ABCD with AB ∥ DC, the diagonals AC and BD meet at E. Prove that AE/EC = AB/DC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q4/5

In acute triangle ABC, the altitudes from A and B meet BC at D and AC at E. Prove that CD · CB = CE · CA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • MAIN • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 1

Prove triangles similar with AA • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

True or false?

Vertical angles can be one pair of congruent angles in an AA proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q1/5

Two triangles share ∠A and have another pair of congruent angles. Criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q2/5

Parallel lines in a figure usually supply

A. equal side lengths

B. a scale factor

C. right angles

D. congruent corresponding or alternate interior angles

△PAD ~ △PCB. Which side of △PCB corresponds to PA?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q4/5

True or false?

Similarity requires all three pairs of sides to be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 2

Turn a proportion into a product • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

From a/b = c/d, write the product relationship.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q1/5

From PA/PC = PD/PB, write the product relationship.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q2/5

If $x/3 = 4/6$, then

A. $6x = 12$

B. $x = 8$

C. $3x = 24$

D. $4x = 18$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q3/5

True or false?

$a/b = c/d$ implies $ab = cd$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q4/5

True or false?

$2/5 = 6/15$ implies $2 \cdot 15 = 5 \cdot 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSG-SRT.B.5

BACKWARD 3

Match corresponding sides of similar triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

In △ABC, D is on AB and E is on AC, and △ADE ~ △ABC. Write the proportion with AD, AB, AE, AC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q1/5

True or false?

In $\triangle ABE \sim \triangle CDE$, AB corresponds to CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q2/5

△ACB ~ △ADC. Which side corresponds to CB?

A. DC

B. AD

C. AB

D. AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q3/5

△XYZ ~ △UVW. Write all three equal ratios.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q4/5

True or false?

Any two sides of similar triangles are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSG-SRT.B.5

FORWARD 1

Use proved product relationships to compute • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Chords AB and CD meet at P with PA = 4, PB = 9, PC = 6. Find PD.

PD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q1/5

In right triangle ABC (right angle at C), AC = 6, BC = 8, AB = 10. Find the altitude CD to AB.

CD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q2/5

True or false?

If two chords meet inside a circle, the products of their segments are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q3/5

Trapezoid diagonals: AB = 18, DC = 12, AE = 9. Find EC.

A. 8

B. 4

C. 13.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q4/5

In $\triangle ABC$, D is on AB and E is on AC with $DE \parallel BC$. AD = 3, AB = 9, AC = 12. Find AE.

AE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSG-SRT.B.5

FORWARD 2

Prove relationships in circles with similar triangles • High School — Geometry • Geometry: Circles • HSG-C.A.2

From point P outside a circle, a tangent touches the circle at T and a secant meets it at A and B (A between P and B). Given $\angle PTA \cong \angle PBT$, prove that $PT^2 = PA \cdot PB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q1/5

A tangent segment from P is 8 and the secant from P has outer part 4. Find the whole secant length.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q2/5

True or false?

Two secants from an outside point P satisfy $PA + PB = PC + PD$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q3/5

From P, secants meet a circle at A, B and at C, D with PA = 3, PB = 12, PC = 4. Find PD.

A. 7

B. 1

C. 16

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q4/5

Explain why the angle facts used in these circle proofs depend on inscribed angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 47 • FORWARD 2 • Q5/5

Understand trigonometric ratios as properties of an angle, by similarity

MAIN	Geometry main question	**HSG-SRT.C.6**
BACKWARD 1	Similar triangles have equal side ratios within each triangle	**HSG-SRT.A.2**
BACKWARD 2	Name the opposite side, adjacent side, and hypotenuse for an ang	**HSG-SRT.A.3**
BACKWARD 3	Write and simplify ratios	**6.RP.A.1**
FORWARD 1	Find trig ratios from one ratio, using a similar right triangle	**HSG-SRT.C.6**
FORWARD 2	The unit circle extends the ratios beyond acute angles	**HSF-TF.A.2**

SET 48 • HSG-SRT.C.6

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

Right triangles ABC and DEF have right angles at C and F and $\angle A \cong \angle D$. Explain why $BC/AB = EF/DE$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q1/5

Two right triangles each have a 30° angle. In one, the side opposite 30° is 4 and the hypotenuse is 8. In the other, the hypotenuse is 15. Find the side opposite 30°, and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q2/5

In right triangle PQR, the right angle is at R, PR = 5, QR = 12, PQ = 13. Write sin P, cos P, and tan P.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q3/5

Right triangle ABC has the right angle at C, AC = 4 and BC = 3. Right triangle DEF has the right angle at F, ∠D ≅ ∠A and DE = 20. Find EF and DF, and explain why tan A = tan D.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q4/5

A student says cos 50° depends on how big the right triangle is. Use two right triangles with a 50° angle, one with hypotenuse 1 and one with hypotenuse 10, to explain why the student is wrong, and give cos 50° to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • MAIN • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 1

Similar triangles have equal side ratios within each triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangles with sides 3, 4, 5 and 6, 8, 10 are similar. Compare 3/5 and 6/10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q1/5

True or false?

If $\triangle ABC \sim \triangle DEF$, then $AB/AC = DE/DF$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q2/5

△ABC ~ △DEF with BC/AB = 0.8. Find EF/DE.

A. Cannot tell

B. 0.8

C. 1.25

D. 0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q3/5

A triangle has legs 2 and 3. A similar one has shorter leg 8. Find the longer leg.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q4/5

True or false?

Similar triangles have equal side lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 2

Name the opposite side, adjacent side, and hypotenuse for an angle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

In right triangle ABC with right angle at C, which side is opposite ∠A?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q1/5

In right triangle ABC with right angle at C, which side is adjacent to ∠A (not the hypotenuse)?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q2/5

In a right triangle, the hypotenuse is always

A. opposite the right angle

B. next to the right angle

C. the shortest side

D. opposite the smallest angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q3/5

True or false?

In right triangle ABC with the right angle at C, the side opposite ∠A is adjacent to ∠B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q4/5

True or false?

The opposite side of an angle depends on the triangle's size.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSG-SRT.C.6

BACKWARD 3

Write and simplify ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

Write 12 : 13 as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q1/5

Simplify 8/10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q2/5

Which equals 0.75?

A. 7/5

B. 3/7

C. 3/4

D. 4/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q3/5

True or false?

$5/13 \approx 0.385$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q4/5

True or false?

A ratio changes when both quantities are tripled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • HSG-SRT.C.6

FORWARD 1

Find trig ratios from one ratio, using a similar right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

For an acute angle θ, $\sin\theta = 3/5$. Find $\cos\theta$ and $\tan\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q1/5

$\tan \theta = 5/12$ for an acute angle θ. Find $\sin \theta$.

$\sin \theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q2/5

True or false?

For any acute angle, $(\sin \theta)^2 + (\cos \theta)^2 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q3/5

$\cos \theta = 0.6$. Which is $\sin \theta$ (θ acute)?

A. 0.8

B. 1.6

C. 0.4

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q4/5

Explain why $\sin\theta < 1$ for every acute angle θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSG-SRT.C.6

FORWARD 2

The unit circle extends the ratios beyond acute angles • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

On a unit circle, the point for angle θ is (cos θ, sin θ). Find the point for 90°.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q1/5

Find the coordinates of the point on the unit circle at 30°.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q2/5

True or false?

For acute angles, the unit-circle definition of sine gives a different value from the right-triangle definition.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q3/5

sin 150° on the unit circle equals

A. $-\frac{1}{2}$

B. $-\sqrt{3}/2$

C. $\sqrt{3}/2$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q4/5

Explain why the unit circle lets us define sine for 120°, when no right triangle has a 120° acute angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 48 • FORWARD 2 • Q5/5

Explain and use the relationship between the sine and cosine of complementary angles

MAIN	Geometry main question	**HSG-SRT.C.7**
BACKWARD 1	Complementary angles add to 90°	**7.G.B.5**
BACKWARD 2	Opposite and adjacent sides depend on the chosen angle	**HSG-SRT.C.6**
BACKWARD 3	Angle sum in a right triangle	**8.G.A.5**
FORWARD 1	Relationships between tangents of complementary angles	**HSG-SRT.C.7**
FORWARD 2	The Pythagorean identity	**HSF-TF.C.8**

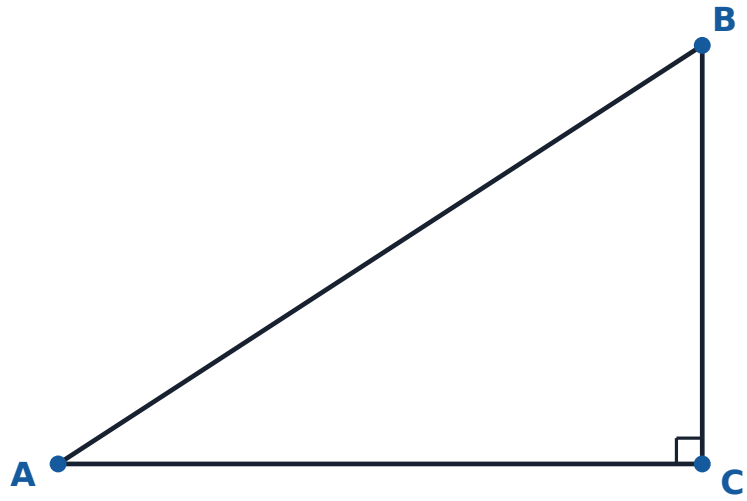
SET 49 • HSG-SRT.C.7

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7

Right triangle ABC has its right angle at C. Explain why sin A = cos B.

Work space



Not drawn to scale.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q1/5

$\sin 25° = \cos x°$, where $0 < x < 90$. Find x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q2/5

θ is acute and $\cos\theta = 0.6$. Find $\sin(90° - \theta)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q3/5

Both angles are acute and $\sin(2x + 10)° = \cos(3x)°$. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q4/5

A ladder leans against a vertical wall on level ground and makes an acute angle θ with the ground, with $\sin\theta = 0.8$. Find $\cos(90° - \theta)$, say which angle in the picture measures $90° - \theta$, and explain why the two values are equal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • MAIN • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 1

Complementary angles add to 90° • Grade 7 • Geometry • 7.G.B.5

Find the complement of 37°.

Complement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q1/5

True or false?

The two acute angles of a right triangle are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q2/5

Angles x and 2x are complementary. x =

A. 90°

B. 45°

C. 60°

D. 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q3/5

The complement of θ is $90° - \theta$. If $\theta = 64°$, find it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q4/5

True or false?

Complementary angles must be adjacent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 2

Opposite and adjacent sides depend on the chosen angle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

In right triangle ABC (right angle at C), name the side adjacent to ∠B.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q1/5

True or false?

In right triangle ABC with the right angle at C, the side opposite $\angle A$ is the side adjacent to $\angle B$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q2/5

In right triangle ABC with the right angle at C, cos B equals

A. AC/AB

B. BC/AB

C. AC/BC

D. BC/AC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q3/5

Write sin A using the sides of right triangle ABC (right angle at C).

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q4/5

True or false?

sin A and cos A are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSG-SRT.C.7

BACKWARD 3

Angle sum in a right triangle • Grade 8 • Geometry • 8.G.A.5

A right triangle has an acute angle of 28°. Find the other acute angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q1/5

True or false?

A right triangle can have two right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q2/5

In a right triangle, the acute angles are x and x + 20. x =

A. 70°

B. 45°

C. 55°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q3/5

Why do the acute angles of a right triangle add to 90°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q4/5

True or false?

The two acute angles of an isosceles right triangle are 45° each.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 49 • BACKWARD 3 • Q5/5

SET 49 • HSG-SRT.C.7

FORWARD 1

Relationships between tangents of complementary angles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7

Explain why $\tan\theta \cdot \tan(90° - \theta) = 1$ for an acute angle θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q1/5

tan 20° ≈ 0.364. Use the complementary relationship to estimate tan 70° to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q2/5

True or false?

$\sin\theta = \cos\theta$ exactly when $\theta = 45°$ (for acute θ).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q3/5

Which equals 1 for every acute θ?

A. $\tan\theta + \tan(90° - \theta)$

B. $\tan\theta \cdot \tan(90° - \theta)$

C. $\sin\theta \cdot \cos\theta$

D. $\sin\theta + \cos\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q4/5

Show that $\sin\theta / \cos(90° - \theta) = 1$ for acute θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.7 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSG-SRT.C.7

FORWARD 2

The Pythagorean identity • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8

Prove that $(\sin \theta)^2 + (\cos \theta)^2 = 1$ for every acute angle θ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q1/5

$\sin \theta = 0.28$ for acute θ. Find $\cos \theta$.

$\cos \theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q2/5

True or false?

$\sin 35° + \cos 35° = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q3/5

$\cos\theta = 12/13$ (θ acute). $\sin\theta =$

A. 1/13

B. 5/13

C. 12/5

D. 13/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q4/5

Use complementary angles to show $(\sin 20°)^2 + (\sin 70°)^2 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 49 • FORWARD 2 • Q5/5

SET 50 • HSG-SRT.C.8 • SIMILARITY, RIGHT TRIANGLES, AND TRIGONOMETRY

Find unknown sides of right triangles with trigonometric ratios in applied problems

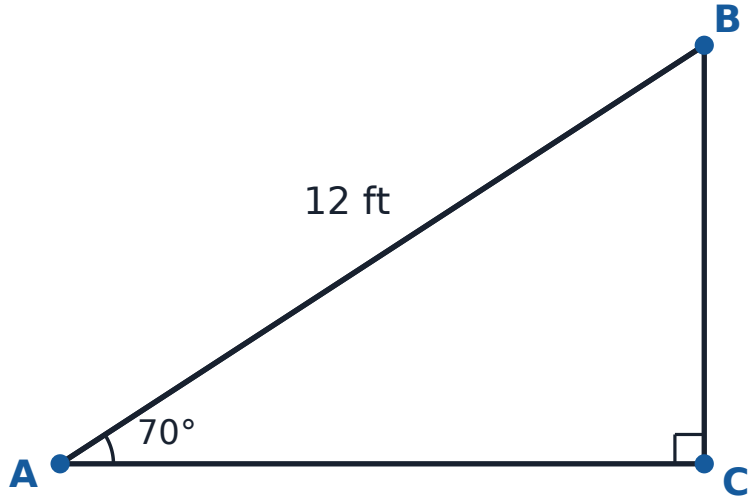
MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Choose the trigonometric ratio that links the known and unknown.	**HSG-SRT.C.6**
BACKWARD 2	Solve one-variable equations with a multiplication or division	**HSA-REI.B.3**
BACKWARD 3	Round to a stated precision and keep track of units	**HSN-Q.A.3**
FORWARD 1	Find a height from two angle-of-elevation observations	**HSG-SRT.C.8**
FORWARD 2	Find a feasible location under a clearance constraint	**HSG-MG.A.3**

SET 50 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A 12 ft ladder leans against a wall, making a 70° angle with the level ground. How high up the wall does it reach, to the nearest tenth of a foot?



Not drawn to scale.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q1/5

A wheelchair ramp rises 2.5 ft and makes an 8° angle with the ground. How long is the ramp surface, to the nearest tenth of a foot?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q2/5

From a point 25 m from the base of a tree, on level ground, the angle of elevation to the top is 38°. How tall is the tree, to the nearest tenth of a meter?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q3/5

A kite string 60 m long makes a 50° angle with the level ground. About how high is the kite above the point where the string is held (nearest meter)?

A. 78 m

B. 46 m

C. 39 m

D. 72 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q4/5

A right triangle has hypotenuse 20 and a 32° angle. Find the leg adjacent to the 32° angle, to the nearest tenth.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • MAIN • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 1

Choose the trigonometric ratio that links the known and unknown sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

You know the hypotenuse and an angle and want the opposite side. Which ratio?

A. none

B. sine

C. cosine

D. tangent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B1-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q1/5

You know the adjacent side and an angle and want the opposite side. Which ratio?

A. none

B. tangent

C. cosine

D. sine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q2/5

Which ratio uses the adjacent side and the hypotenuse?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q3/5

True or false?

To find the hypotenuse from the opposite side and the angle, use sine.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q4/5

True or false?

Tangent uses the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 2

Solve one-variable equations with a multiplication or division • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $x/12 = 0.94$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q1/5

Solve 0.14 = 2.5/x. Round to the nearest hundredth.

x ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q2/5

If $\sin 30° = h/10$, then $h =$

A. $10 \sin 30°$

B. $10 \div \sin 30°$

C. $\sin 30° \div 10$

D. $30 \div 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q3/5

True or false?

From $0.5 = 4/x$ we get $x = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q4/5

True or false?

From $\tan 40° = 20/x$ we get $x = 20 \tan 40°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B2-Q5**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSG-SRT.C.8

BACKWARD 3

Round to a stated precision and keep track of units • High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

Round 11.2763 to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q1/5

A calculation gives 19.532 m. Report it to the nearest tenth of a meter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q2/5

True or false?

Rounding in the middle of a calculation can change the final answer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q3/5

A height is measured in feet and the angle in degrees. The answer's unit is

A. degrees

B. feet per degree

C. feet

D. no unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-B3-Q4**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q4/5

True or false?

17.96 rounded to the nearest tenth is 17.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 50 • BACKWARD 3 • Q5/5

SET 50 • HSG-SRT.C.8

FORWARD 1

Find a height from two angle-of-elevation observations • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

From point A on level ground the angle of elevation to the top of a tower is 25°. From point B, 40 m closer, it is 40°. Find the tower's height to the nearest tenth of a meter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q1/5

From point A on level ground the angle of elevation to the top of a tower is 25°. From point B, 40 m closer, it is 40°. How far is point B from the base of the tower, to the nearest tenth of a meter?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q2/5

True or false?

With only one angle of elevation and no distance, the height cannot be found.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q3/5

Which equations describe two observations 40 m apart (angles 25° and 40°, distance x from the closer point)?

A. $h = 40 \tan 25°$

B. $h = x \tan 25°$ and $h = (x + 40) \tan 40°$

C. $h = x \tan 40°$ and $h = (x + 40) \tan 25°$

D. $h = x \sin 40°$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q4/5

Explain why the closer observation point has the larger angle of elevation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSG-SRT.C.8

FORWARD 2

Find a feasible location under a clearance constraint • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A ramp must rise 3 ft, and its angle with the ground may be at most 4.8°. What is the shortest horizontal run, in whole feet, that meets the requirement?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q1/5

A camera on a 6 m pole must see a point on the ground at an angle of depression of at least 20°. What is the greatest allowed distance from the pole, in tenths of a meter, that still meets the requirement?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q2/5

True or false?

Rounding UP the run of a ramp with a maximum angle can make the ramp too steep.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q3/5

A ladder must reach 10 ft up and make an angle of at most 75° with the ground. To the tenth of a foot, rounded so that the ladder still works, the shortest ladder is

A. 10.0 ft

B. 9.7 ft

C. 2.7 ft

D. 10.4 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q4/5

A building code allows stairs with an angle of at most 37°. A staircase must rise 9 ft. What is the shortest horizontal run, in tenths of a foot, that meets the code?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 50 • FORWARD 2 • Q5/5

Find unknown angles with inverse trigonometric ratios in applied problems

MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Inverse trig undoes a trig ratio	**HSG-SRT.C.6**
BACKWARD 2	Choose the ratio from the two known sides	**HSG-SRT.C.6**
BACKWARD 3	Use degree mode and round angles	**HSN-Q.A.3**
FORWARD 1	Solve a right triangle completely from two sides	**HSG-SRT.C.8**
FORWARD 2	Find directions (bearings) with inverse trigonometry	**HSG-SRT.C.8**

SET 51 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A ramp rises 1 m over a horizontal distance of 12 m. What angle does it make with the ground, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q1/5

A 6 m ladder reaches 5.5 m up a vertical wall. What angle does it make with the level ground, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q2/5

A 10 m flagpole casts a 14 m shadow on level ground. Find the angle of elevation of the sun, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q3/5

A right triangle has legs 7 and 24. What is its smaller acute angle, to the nearest tenth of a degree?

A. 73.7°

B. 16.3°

C. 32.5°

D. 26.3°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q4/5

A road rises 6 m for every 100 m of horizontal distance (a 6% grade). What angle does it make with the horizontal, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • MAIN • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 1

Inverse trig undoes a trig ratio • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

$\sin \theta = 0.5$ for an acute angle θ. Find θ.

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q1/5

$\tan \theta = 1$ for an acute angle θ. Find θ.

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q2/5

Which gives an angle from a ratio?

A. cos

B. tan

C. sin

D. $\sin^{-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q3/5

True or false?

$\cos^{-1}(0.5) = 60°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q4/5

True or false?

$\sin^{-1}(x)$ means $1 \div \sin(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 2

Choose the ratio from the two known sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

You know the opposite side and the hypotenuse. Which inverse function finds the angle?

A. none

B. $\cos^{-1}$

C. $\sin^{-1}$

D. $\tan^{-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q1/5

You know both legs. Which inverse function finds an acute angle?

A. $\sin^{-1}$

B. $\tan^{-1}$

C. $\cos^{-1}$

D. none

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q2/5

Rise 5 and run 12. Write the ratio for the angle with the ground.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q3/5

True or false?

Adjacent side and hypotenuse give the angle by $\cos^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q4/5

True or false?

Opposite side and hypotenuse give the angle by $\tan^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSG-SRT.C.8

BACKWARD 3

Use degree mode and round angles • High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

Round 4.7636° to the nearest tenth of a degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q1/5

True or false?

A calculator in radian mode gives $\sin^{-1}(0.5) \approx 0.524$, not 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q2/5

An acute angle of a right triangle must be

A. more than 90°

B. any angle

C. exactly 45°

D. between 0° and 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B3-Q3**

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q3/5

A computed angle is 112°. Can it be an acute angle of a right triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q4/5

True or false?

35.537° to the nearest tenth is 35.6°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 51 • BACKWARD 3 • Q5/5

SET 51 • HSG-SRT.C.8

FORWARD 1

Solve a right triangle completely from two sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A right triangle has legs 9 and 14. Find the hypotenuse and both acute angles (nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q1/5

A right triangle has hypotenuse 17 and one leg 8. Find the other leg and both acute angles (nearest tenth of a degree).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q2/5

True or false?

Once one acute angle is known, the other is 90° minus it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q3/5

Legs 5 and 5. Angles?

A. 45° and 45°

B. 50° and 40°

C. 30° and 60°

D. 5° and 85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q4/5

Explain how to check a solved right triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSG-SRT.C.8

FORWARD 2

Find directions (bearings) with inverse trigonometry • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A boat travels 8 km east and then 5 km north. Find the angle north of east of its straight-line direction from the start, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q1/5

A hiker walks 3 km north and 4 km west. What is the bearing angle west of north from the start, to the nearest tenth of a degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q2/5

True or false?

Bearing problems become right-triangle problems when the motions are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q3/5

A drone flies 120 m horizontally and rises 50 m. Its angle of climb is about

A. 22.6°

B. 24.6°

C. 67.4°

D. 50°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q4/5

A ship is 10 km east and 10 km south of port. Find the straight-line distance (to the nearest tenth of a km) and the angle south of east.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 51 • FORWARD 2 • Q5/5

Solve right triangles in applied problems with trigonometry and the Pythagorean Theorem

MAIN	Geometry main question	**HSG-SRT.C.8**
BACKWARD 1	Use the Pythagorean Theorem to find a side	**8.G.B.7**
BACKWARD 2	Interpret angles of elevation and depression	**HSG-SRT.C.6**
BACKWARD 3	Choose sine, cosine, or tangent	**HSG-SRT.C.6**
FORWARD 1	Solve problems that need two right triangles	**HSG-SRT.C.8**
FORWARD 2	Solve non-right triangles with the Law of Sines	**HSG-SRT.D.11**

SET 52 • HSG-SRT.C.8

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

From the top of an 80 m cliff, the angle of depression to a boat is 25°. How far is the boat from the base of the cliff, to the nearest tenth of a meter? How far is it from the observer?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q1/5

A plane flying at an altitude of 2,000 m is seen at an angle of elevation of 14°. Find the horizontal ground distance and the line-of-sight distance to the plane, to the nearest meter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q2/5

A ramp is 15 ft long and rises 2 ft. Find its horizontal run (nearest tenth of a foot) and its angle with the ground (nearest tenth of a degree).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q3/5

A 25 ft guy wire from the top of a pole meets the level ground 9 ft from the pole. How tall is the pole and what angle does the wire make with the ground (nearest tenth)?

A. 26.6 ft; 68.9°

B. 16.0 ft; 21.1°

C. 23.3 ft; 21.1°

D. 23.3 ft; 68.9°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q4/5

A right triangle has hypotenuse 25 and a 40° angle. Find both legs and the other acute angle (legs to the nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • MAIN • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 1

Use the Pythagorean Theorem to find a side • Grade 8 • Geometry • 8.G.B.7

A right triangle has hypotenuse 25 and a leg 9. Find the other leg, to the nearest tenth.

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q1/5

Legs 8 and 15. Find the hypotenuse.

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q2/5

True or false?

In a right triangle, the hypotenuse is the longest side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q3/5

Hypotenuse 13, leg 5. Other leg?

A. √194

B. 12

C. 8

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 52 • BACKWARD 1 • Q4/5

True or false?

$a^2 + b^2 = c^2$ works for every triangle.

A. True

B. False

SET 52 • HSG-SRT.C.8

BACKWARD 2

Interpret angles of elevation and depression • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

True or false?

An angle of depression is measured down from the horizontal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q1/5

The angle of depression from a cliff top to a boat is 25°. What is the angle of elevation from the boat to the cliff top?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q2/5

Which side is horizontal in an angle-of-elevation triangle?

A. The hypotenuse

B. The leg along the ground

C. The vertical leg

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q3/5

An observer looks up at 30° to a bird. Is that an angle of elevation or depression?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q4/5

True or false?

The angle of depression is measured from the vertical line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSG-SRT.C.8

BACKWARD 3

Choose sine, cosine, or tangent • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

Known: opposite side and angle. Wanted: hypotenuse. Ratio?

A. none

B. tangent

C. sine

D. cosine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q1/5

Known: adjacent side and angle. Wanted: opposite side. Ratio?

A. cosine

B. tangent

C. none

D. sine

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q2/5

Known: hypotenuse and angle. Wanted: adjacent side. Ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q3/5

True or false?

Two different ratios can lead to the same correct side if used correctly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q4/5

True or false?

Tangent relates the hypotenuse and a leg.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 52 • BACKWARD 3 • Q5/5

SET 52 • HSG-SRT.C.8

FORWARD 1

Solve problems that need two right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

From a window 20 m up, the angle of elevation to the top of a building across the street is 30° and the angle of depression to its base is 40°. How tall is the building, to the nearest tenth of a meter?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q1/5

Two observers 100 m apart on opposite sides of a tower see its top at 35° and 50°. Find the tower's height, to the nearest tenth of a meter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q2/5

True or false?

A shared side links two right triangles in such problems.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q3/5

Two observers 100 m apart on opposite sides of a tower of height h see its top at angles of elevation 35° and 50°. Which equation is correct?

A. $h = 100 \tan 85°$

B. $h/\sin 35° + h/\sin 50° = 100$

C. $h \tan 35° + h \tan 50° = 100$

D. $h/\tan 35° + h/\tan 50° = 100$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q4/5

Two observers on opposite sides of a tower each measure an angle of elevation to its top, and the distance between them is known. Explain why the tower's height cannot be found from one of the two right triangles alone.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSG-SRT.C.8

FORWARD 2

Solve non-right triangles with the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

In $\triangle ABC$, $A = 40°$, $B = 65°$, and $a = 12$. Find b to the nearest tenth.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q1/5

In $\triangle ABC$, $A = 52°$, $C = 70°$, and $c = 20$. Find a to the nearest tenth.

$a =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q2/5

True or false?

SOH-CAH-TOA applies directly to a triangle with no right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q3/5

Which data allow the Law of Sines directly?

A. Two sides and the included angle

B. Two angles and any side

C. Three sides

D. Three angles only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q4/5

Explain how dropping an altitude turns a non-right triangle into two right triangles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 52 • FORWARD 2 • Q5/5

SET 53 • HSG-C.A.1 • CIRCLES

Prove that all circles are similar

MAIN	Geometry main question	**HSG-C.A.1**
BACKWARD 1	A dilation multiplies distances by the scale factor	**HSG-SRT.A.1.b**
BACKWARD 2	Translate a point to another point	**8.G.A.3**
BACKWARD 3	A circle is all points at one distance from its center	**HSG-CO.A.1**
FORWARD 1	Find a single dilation that maps one circle onto another	**HSG-C.A.1**
FORWARD 2	Arcs of similar circles scale with the radius	**HSG-C.B.5**

SET 53 • HSG-C.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.1

Circle C has center A and radius r, and circle D has center B and radius s. Prove that the two circles are similar.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q1/5

Circle C_1 has center (1, 2) and radius 3. Circle C_2 has center (−4, 0) and radius 6. Describe a similarity transformation that maps C_1 onto C_2 and explain why it works.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q2/5

Circle C_1 has center $(0, 0)$ and radius 2. Circle C_2 has center $(3, -1)$ and radius 5. Prove that C_1 and C_2 are similar by giving a similarity transformation, and show where it sends the point $(2, 0)$ of C_1.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q3/5

Explain why two circles with the same center but different radii, r and s, are similar using only ONE transformation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q4/5

Circle C has center A and radius r, and circle D has center B and radius s. Prove that C and D are similar using a dilation FIRST and a translation SECOND.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • MAIN • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 1

A dilation multiplies distances by the scale factor • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

A dilation with factor 2.5 is applied to a segment of length 4. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q1/5

True or false?

A dilation about point O keeps O fixed and multiplies each distance from O by the factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q2/5

Which factor maps a circle of radius 4 about its center to one of radius 10?

A. 2.5

B. 0.4

C. 6

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-B1-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q3/5

A circle of radius 9 is dilated about its center by ⅓. New radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q4/5

True or false?

A dilation can turn a circle into a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 2

Translate a point to another point • Grade 8 • Geometry • 8.G.A.3

Which translation moves (1, 2) onto (−4, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q1/5

True or false?

A translation keeps the radius of a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q2/5

A circle centered at (3, 3) is translated by (x, y) → (x − 3, y + 1). New center?

A. (6, 2)

B. (0, 4)

C. (0, 3)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q3/5

A translation moves (2, −1) to (7, 4). Describe it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q4/5

True or false?

A translation changes the size of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSG-C.A.1

BACKWARD 3

A circle is all points at one distance from its center • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Every point of a circle is the same distance from its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q1/5

A circle has center (0, 0) and passes through (3, 4). Its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q2/5

A circle is completely determined by

A. its color

B. one point on it

C. its center only

D. its center and radius

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q3/5

Two circles have the same center and the same radius. What can you say?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q4/5

True or false?

A circle's radius can be different in different directions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • HSG-C.A.1

FORWARD 1

Find a single dilation that maps one circle onto another • High School — Geometry • Geometry: Circles • HSG-C.A.1

Circle P: center (0, 0), radius 2. Circle Q: center (6, 0), radius 4. Find the center of the dilation with a positive factor that maps P onto Q.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q1/5

Circle P has center (0, 0) and radius 2. Circle Q has center (6, 0) and radius 4. What is the scale factor of a dilation that maps P onto Q?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q2/5

True or false?

Two circles with different radii can always be mapped onto each other by one dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q3/5

Circles with equal radii and different centers are related by

A. nothing

B. a translation (no dilation needed)

C. a dilation with factor 2

D. a reflection only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q4/5

Circle A has radius 3 and circle B has radius 12, both centered at the origin. Which dilation maps A onto B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSG-C.A.1

FORWARD 2

Arcs of similar circles scale with the radius • High School — Geometry • Geometry: Circles • HSG-C.B.5

Two circles have radii 3 and 7. A 60° central angle cuts off an arc in each. What is the ratio of the arc lengths?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q1/5

A 90° arc of a circle with radius 4 has length 2π. How long is a 90° arc of a circle with radius 10?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q2/5

True or false?

For a fixed central angle, the arc length is the same in every circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q3/5

An arc of length 6 in a circle of radius 2 corresponds, by similarity, to what arc length in a circle of radius 5 (same angle)?

A. 2.4

B. 15

C. 30

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q4/5

Explain how the similarity of circles leads to radian measure.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 53 • FORWARD 2 • Q5/5

SET 54 • HSG-C.A.2 • CIRCLES

Relate inscribed angles and central angles

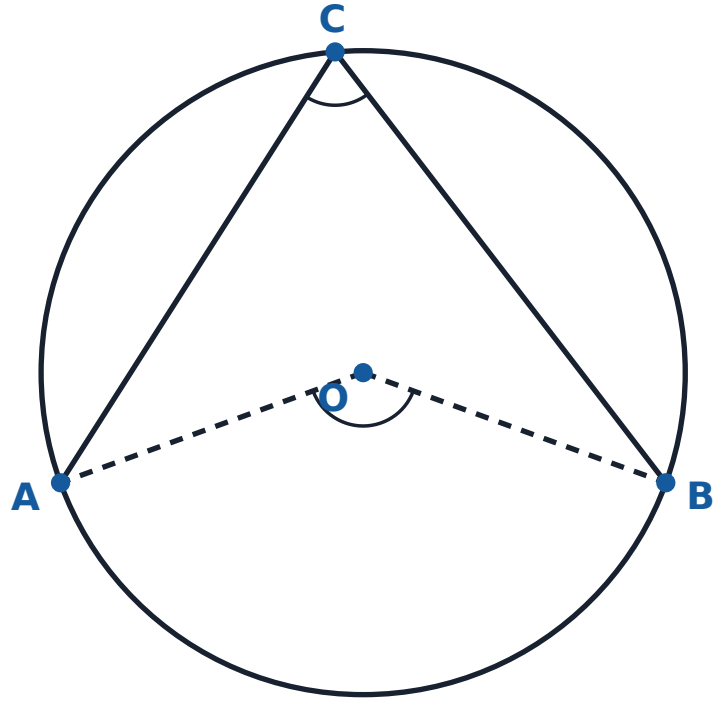
MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Base angles of an isosceles triangle are congruent	**HSG-CO.C.10**
BACKWARD 2	The exterior angle theorem	**HSG-CO.C.10**
BACKWARD 3	A central angle and its arc have the same measure	**4.MD.C.5.a**
FORWARD 1	Prove the inscribed angle theorem when the center is inside or out…	**HSG-C.A.2**
FORWARD 2	Inscribed angles on the same arc, and arc applications	**HSG-C.A.2**

SET 54 • HSG-C.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

In circle O, central angle AOB measures 110°. Inscribed angle ACB intercepts the same arc AB. Find m∠ACB.



m∠ACB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q1/5

In circle O, inscribed angle PQR measures 38°. Find the measure of arc PR (the arc it intercepts) and of central angle POR.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q2/5

In circle O, AC is a diameter and B is another point on the circle. Prove that m∠BAC = ½ m∠BOC (the inscribed angle theorem when one side of the angle is a diameter).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q3/5

Inscribed angle XYZ and central angle XOZ intercept the same arc XZ, and m∠XYZ = 47°. What is m∠XOZ?

A. 23.5°

B. 94°

C. 47°

D. 133°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q4/5

In circle O, points C and D are on the same side of chord AB, so inscribed angles ADB and ACB intercept the same arc AB. $m\angle ADB = (3x + 4)°$ and $m\angle ACB = (5x - 12)°$. Find x, $m\angle ACB$, and the central angle AOB that intercepts arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • MAIN • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 1

Base angles of an isosceles triangle are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In $\triangle AOB$, $OA = OB$ and $m\angle OAB = 35°$. Find $m\angle OBA$ and $m\angle AOB$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q1/5

True or false?

Two radii of a circle and the chord joining their ends form an isosceles triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q2/5

Isosceles triangle with vertex angle 80°. Each base angle?

A. 100°

B. 50°

C. 80°

D. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q3/5

In $\triangle OPQ$ with $OP = OQ$, $m\angle P = 2x$ and $m\angle Q = 50°$. Find x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q4/5

True or false?

In an isosceles triangle, the vertex angle always equals a base angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 2

The exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

A triangle has remote interior angles 30° and 30°. Find the exterior angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q1/5

True or false?

An exterior angle equals the sum of the two remote interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q2/5

Exterior angle 100°, one remote interior angle 45°. The other?

A. 45°

B. 80°

C. 55°

D. 135°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q3/5

If both remote interior angles equal x, the exterior angle is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q4/5

True or false?

An exterior angle equals the adjacent interior angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSG-C.A.2

BACKWARD 3

A central angle and its arc have the same measure • Grade 4 • Measurement & Data • 4.MD.C.5.a

A central angle cuts off ¼ of a circle. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q1/5

True or false?

The measure of a minor arc equals its central angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q2/5

An arc measures 70°. Its central angle measures

A. 70°

B. 140°

C. 290°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q3/5

A central angle of 150°. What fraction of the circle is its arc?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q4/5

True or false?

All arcs of 60° have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 54 • BACKWARD 3 • Q5/5

SET 54 • HSG-C.A.2

FORWARD 1

Prove the inscribed angle theorem when the center is inside or outside the angle • High School — Geometry • Geometry: Circles • HSG-C.A.2

Inscribed angle ABC in circle O has the center O inside the angle. Prove that m∠ABC = ½ m(arc AC), where arc AC is the arc intercepted by ∠ABC (the arc not containing B). (Draw diameter BD.) If ∠ABC is obtuse, arc AC is a major arc and its central angle is a reflex angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q1/5

Inscribed angle ABC in circle O has the center O outside the angle. Prove that m∠ABC = ½ m(arc AC), where arc AC is the arc intercepted by ∠ABC (the arc not containing B). (Draw diameter BD.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q2/5

True or false?

The inscribed angle theorem needs three cases for a complete proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q3/5

In the proof of the inscribed angle theorem for the case where the center lies inside the inscribed angle, a diameter splits the angle in two, and the two diameter-case results are

A. averaged

B. multiplied

C. subtracted

D. added

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q4/5

Why can't the proof of the inscribed angle theorem stop after the case in which one side of the angle is a diameter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSG-C.A.2

FORWARD 2

Inscribed angles on the same arc, and arc applications • High School — Geometry • Geometry: Circles • HSG-C.A.2

In a circle, inscribed angles ∠ADB and ∠ACB intercept the same arc AB, and m∠ADB = 41°. Find m∠ACB.

m∠ACB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q1/5

In a circle, inscribed angles ADB and ACB intercept the same arc AB (C and D are on the same side of chord AB). Prove that ∠ADB ≅ ∠ACB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q2/5

True or false?

An inscribed angle whose intercepted arc is a 200° major arc measures 200°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q3/5

Points A, B, C, D are on a circle with arc AB = 80° and arc CD = 50° (arcs not containing the other points). Chords AC and BD meet at E inside. m∠AEB?

A. 30°

B. 65°

C. 15°

D. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q4/5

An inscribed angle measures 3x + 10 and its intercepted arc measures 110°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 54 • FORWARD 2 • Q5/5

SET 55 • HSG-C.A.2 • CIRCLES

Inscribed angles on a diameter are right angles

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Use the inscribed angle theorem	**HSG-C.A.2**
BACKWARD 2	Isosceles triangles from radii	**HSG-CO.C.10**
BACKWARD 3	Triangle angle sum	**8.G.A.5**
FORWARD 1	Use the converse to find a circle through a right triangle	**HSG-C.A.2**
FORWARD 2	Use the diameter property to construct tangents from an external…	**HSG-C.A.4**

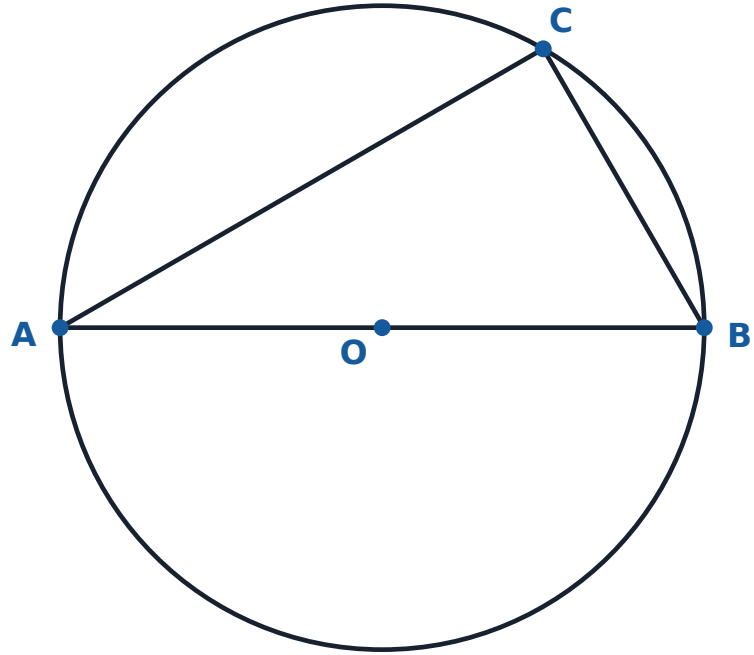
SET 55 • HSG-C.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

AB is a diameter of circle O and C is any other point on the circle. Prove that ∠ACB is a right angle.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q1/5

AB is a diameter of a circle and C is on the circle. AC = 6 and BC = 8. Find AB and the radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q2/5

AB is a diameter of circle O and C is on the circle with m∠CAB = 34°. Find m∠CBA.

m∠CBA = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q3/5

Triangle PQR is inscribed in a circle with PR a diameter. $m\angle P = (3x)°$ and $m\angle R = (2x + 15)°$. Find x and the measures of all three angles, and name the theorem you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q4/5

Points A(−5, 0), B(5, 0) and C(3, 4) are on the circle $x^2 + y^2 = 25$, and AB is a diameter. Use slopes to show that ∠ACB is a right angle, and explain how this agrees with the theorem.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • MAIN • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 1

Use the inscribed angle theorem • High School — Geometry • Geometry: Circles • HSG-C.A.2

An inscribed angle intercepts an arc of 140°. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q1/5

True or false?

An inscribed angle is half its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q2/5

A semicircle measures

A. 360°

B. 180°

C. 90°

D. 120°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q3/5

An inscribed angle measures 25°. Its arc?

Arc: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q4/5

True or false?

An inscribed angle equals its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 2

Isosceles triangles from radii • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

OA = OC (radii) and m∠OAC = 40°. Find m∠OCA.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q1/5

True or false?

Any two radii and the chord between them form an isosceles triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q2/5

In $\triangle OBC$ with $OB = OC$, $m\angle BOC = 100°$. $m\angle OBC =$

A. 80°

B. 40°

C. 100°

D. 50°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q3/5

Why are ∠OAC and ∠OCA congruent when OA = OC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q4/5

True or false?

A radius is always longer than a chord.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSG-C.A.2

BACKWARD 3

Triangle angle sum • Grade 8 • Geometry • 8.G.A.5

Two angles of a triangle are 90° and 34°. The third?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q1/5

True or false?

The acute angles of a right triangle add to 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q2/5

$2x + 2y = 180$. Then $x + y =$

A. 45

B. 360

C. 90

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q3/5

A triangle has angles x, x, and 2x. Find 2x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q4/5

True or false?

A triangle can have angles 90°, 60°, and 40°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 55 • BACKWARD 3 • Q5/5

SET 55 • HSG-C.A.2

FORWARD 1

Use the converse to find a circle through a right triangle • High School — Geometry • Geometry: Circles • HSG-C.A.2

Right triangle ABC has its right angle at C, with A(0, 0) and B(8, 6). Where is the center of the circle through A, B, and C, and what is its radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q1/5

In $\triangle ABC$, $m\angle C = 90°$. Prove that C lies on the circle with diameter AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q2/5

True or false?

The hypotenuse of a right triangle is a diameter of its circumscribed circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q3/5

A right triangle has legs 5 and 12. Radius of its circumscribed circle?

A. 6.5

B. 8.5

C. 6

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q4/5

A carpenter's square (a right angle) is placed with its corner touching a circular board and its edges crossing the edge at P and Q. What is segment PQ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 55 • FORWARD 1 • Q5/5

SET 55 • HSG-C.A.2

FORWARD 2

Use the diameter property to construct tangents from an external point • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

Compass and straightedge only; leave all arcs.

Construct the two tangent lines from point P to circle O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q1/5

To construct tangents from P to circle O, a circle with diameter OP is drawn; it meets circle O at T_1 and T_2. Why are lines PT_1 and PT_2 tangent to circle O?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q2/5

True or false?

From a point outside a circle there is exactly one tangent line to the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q3/5

To construct a tangent from a point P outside circle O, a circle with diameter OP is drawn. That circle is used because

A. it passes through the center only

B. every angle inscribed in it on OP is a right angle

C. it has the same radius as circle O

D. it is tangent to circle O

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q4/5

O is the center, OP = 10, and the radius is 6. How long is each tangent segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 55 • FORWARD 2 • Q5/5

SET 56 • HSG-C.A.2 • CIRCLES

A radius is perpendicular to the tangent at the point of tangency

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Use the Pythagorean Theorem and its converse	**8.G.B.7**
BACKWARD 2	The perpendicular is the shortest segment from a point to a line	**7.G.A.2**
BACKWARD 3	A tangent touches a circle at exactly one point	**HSG-CO.A.1**
FORWARD 1	Tangent segments from an external point are congruent	**HSG-C.A.2**
FORWARD 2	Write the equation of a tangent line in the coordinate plane	**HSG-GPE.B.5**

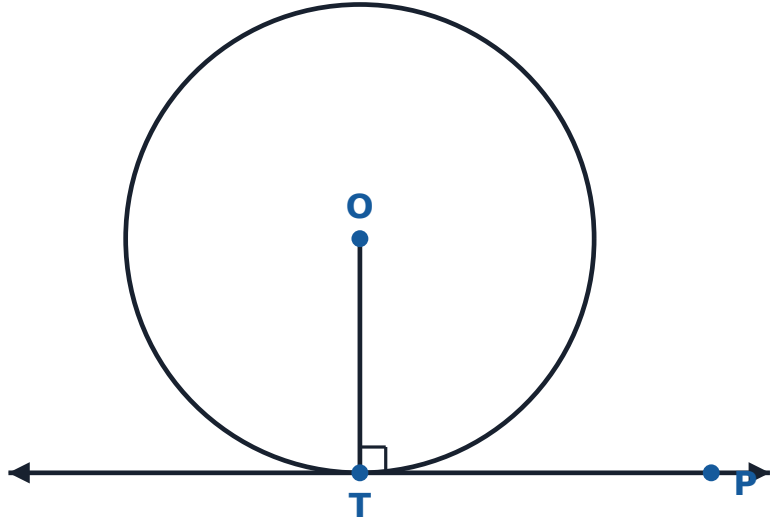
SET 56 • HSG-C.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

Line ℓ is tangent to circle O at T. Prove that OT $\perp \ell$. (Use an indirect argument.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q1/5

Line PT is tangent to circle O at T. OT = 5 and PT = 12. Find OP.

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q2/5

A tangent from P touches circle O at T. OP = 17 and the radius is 8. Find PT.

PT = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q3/5

T lies on circle O, and P is outside it, with OT = 7, PT = 24 and OP = 26. Is line PT tangent to the circle at T? Justify your answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q4/5

T is on circle O with radius 6. Line ℓ is perpendicular to OT at T, and Q is a point on ℓ with TQ = 8. Find OQ, and explain why Q is outside the circle. What does this show about line ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • MAIN • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 1

Use the Pythagorean Theorem and its converse • Grade 8 • Geometry • 8.G.B.7

Legs 5 and 12. Hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q1/5

Hypotenuse 17, leg 8. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q2/5

True or false?

A triangle with sides 7, 24, 25 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q3/5

Sides 6, 8, 11. Right triangle?

A. Yes

B. No

C. Only if 11 is a leg

D. Cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q4/5

True or false?

The Pythagorean Theorem applies to any triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 2

The perpendicular is the shortest segment from a point to a line • Grade 7 • Geometry • 7.G.A.2

True or false?

The shortest segment from a point to a line is perpendicular to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

In a right triangle, which side is longest?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

O is 4 cm from line ℓ (perpendicular distance). Another segment from O to ℓ could be

A. 4 cm only

B. 5 cm

C. 2 cm

D. 3 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

Why is the perpendicular segment shorter than any slanted one?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

True or false?

The distance from a point to a line is measured along any segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSG-C.A.2

BACKWARD 3

A tangent touches a circle at exactly one point • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

A tangent line meets the circle at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q1/5

A line meets a circle at two points. It is a

A. secant

B. radius

C. diameter

D. tangent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q2/5

Point Q on a line is inside the circle. Can the line be tangent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q3/5

What is the point where a tangent touches the circle called?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q4/5

True or false?

A tangent can pass through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 56 • BACKWARD 3 • Q5/5

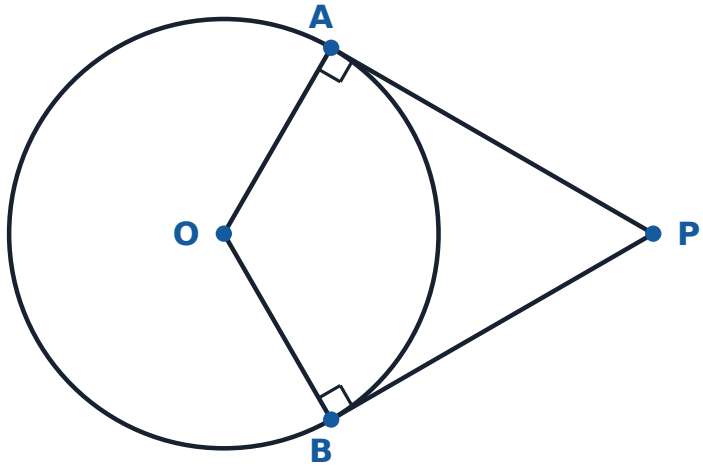
SET 56 • HSG-C.A.2

FORWARD 1

Tangent segments from an external point are congruent • High School — Geometry • Geometry: Circles • HSG-C.A.2

From point P outside circle O, tangent segments PA and PB touch the circle at A and B. Prove that PA = PB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q1/5

Tangent segments from P to a circle are 3x + 4 and 25. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q2/5

True or false?

The segment from P to the center bisects the angle between the two tangents.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q3/5

A circle is inscribed in a triangle with sides tangent to it. From one vertex, the two tangent segments are

A. equal

B. parallel

C. unequal

D. perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q4/5

A circle is tangent to all sides of a quadrilateral ABCD. AB = 9, BC = 11, CD = 13. Find DA.

DA = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSG-C.A.2

FORWARD 2

Write the equation of a tangent line in the coordinate plane • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Find an equation of the line tangent to $x^2 + y^2 = 25$ at $(3, 4)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q1/5

Find an equation of the tangent to $x^2 + y^2 = 13$ at $(2, -3)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q2/5

True or false?

A tangent line has the same slope as the radius to its point of tangency.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q3/5

The tangent to $x^2 + y^2 = 16$ at $(0, 4)$ is

A. $x = 0$

B. $y = x + 4$

C. $x = 4$

D. $y = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q4/5

Is y = 7 tangent to the circle with center (2, 3) and radius 4? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 56 • FORWARD 2 • Q5/5

SET 57 • HSG-C.A.2 • CIRCLES

Relate circumscribed angles and central angles

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	A tangent is perpendicular to the radius at the point of tangency	**HSG-C.A.2**
BACKWARD 2	The angles of a quadrilateral add to 360°	**8.G.A.5**
BACKWARD 3	Supplementary angles add to 180°	**7.G.B.5**
FORWARD 1	Use the circumscribed angle with arcs, including major arcs	**HSG-C.A.2**
FORWARD 2	Angles formed by secants and tangents outside a circle	**HSG-C.A.2**

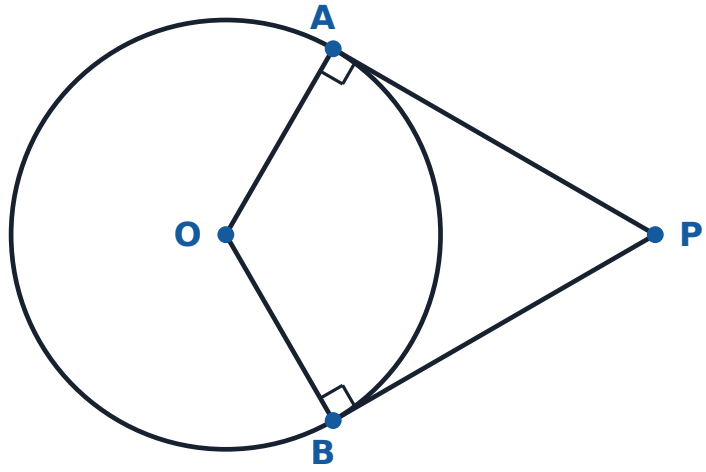
SET 57 • HSG-C.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

Tangents from P touch circle O at A and B. Prove that $m\angle APB + m\angle AOB = 180°$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q1/5

Tangents from P touch circle O at A and B. The circumscribed angle APB measures 52°. Find the central angle AOB and the minor arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q2/5

Tangents from P touch circle O at A and B. Minor arc AB measures 100°. Find the circumscribed angle APB.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q3/5

Tangents from P touch circle O at A and B. m∠APB = 90°. Find m∠AOB and classify quadrilateral OAPB as specifically as possible. Give reasons.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q4/5

Tangents from P touch circle O at A and B. $m\angle APB = (2x + 10)°$ and $m\angle AOB = (4x + 50)°$. Find x and both angle measures, and state the relationship you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • MAIN • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 1

A tangent is perpendicular to the radius at the point of tangency • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

The radius to the point of tangency is perpendicular to the tangent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q1/5

What is m∠OAP when PA is tangent at A?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q2/5

In quadrilateral OAPB with tangents PA and PB, how many right angles are certain?

A. 4

B. 0

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q3/5

Tangent PA, radius OA = 9, PA = 12. OP?

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q4/5

True or false?

The angle between a tangent and a radius can be 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 2

The angles of a quadrilateral add to 360° • Grade 8 • Geometry • 8.G.A.5

A quadrilateral has angles 90°, 90°, and 70°. The fourth?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q1/5

True or false?

A diagonal splits a quadrilateral into two triangles, so its angles add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q2/5

Two angles of a quadrilateral are right angles. The other two must

A. each be 90°

B. add to 360°

C. add to 180°

D. add to 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q3/5

Angles x, 90°, 90°, 2x in a quadrilateral. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q4/5

True or false?

The angles of a quadrilateral add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSG-C.A.2

BACKWARD 3

Supplementary angles add to 180° • Grade 7 • Geometry • 7.G.B.5

Find the supplement of 52°.

Supplement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q1/5

True or false?

Supplementary angles add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q2/5

Angles x and 3x are supplementary. x =

A. 60°

B. 90°

C. 45°

D. 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q3/5

If two angles are supplementary and one is obtuse, the other is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q4/5

True or false?

Supplementary angles must form a linear pair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • HSG-C.A.2

FORWARD 1

Use the circumscribed angle with arcs, including major arcs • High School — Geometry • Geometry: Circles • HSG-C.A.2

Tangents from P touch circle O at A and B. The circumscribed angle APB is 40°. Find the minor arc AB and the major arc AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q1/5

Tangents from P touch circle O at A and B. Show that m∠APB = ½(major arc AB − minor arc AB).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q2/5

True or false?

A circumscribed angle can measure more than 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q3/5

Tangents from P touch circle O at A and B. m∠APB = 70°. An inscribed angle intercepting the minor arc AB measures

A. 110°

B. 70°

C. 55°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q4/5

Tangents from P touch circle O at A and B. Minor arc AB is x°, and the circumscribed angle APB is (x − 20)°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSG-C.A.2

FORWARD 2

Angles formed by secants and tangents outside a circle • High School — Geometry • Geometry: Circles • HSG-C.A.2

Two secants from P meet a circle at A, B and at C, D (A between P and B, C between P and D). Prove that m∠P = ½(arc BD − arc AC).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q1/5

Two secants from P intercept arcs of 110° and 30°. Find m∠P.

m∠P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q2/5

True or false?

The angle outside a circle formed by a tangent and a secant is half the sum of the intercepted arcs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q3/5

An angle outside a circle intercepts arcs of 160° and 50°. Its measure?

A. 110°

B. 55°

C. 210°

D. 105°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q4/5

Explain why the formula for two tangents is a special case of the secant formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 57 • FORWARD 2 • Q5/5

SET 58 • HSG-C.A.2 • CIRCLES

Relationships among radii and chords

MAIN	Geometry main question	**HSG-C.A.2**
BACKWARD 1	Right triangles: Hypotenuse-Leg and the Pythagorean Theorem	**HSG-SRT.B.5**
BACKWARD 2	The perpendicular bisector theorem	**HSG-CO.C.9**
BACKWARD 3	Radii of a circle are equal	**HSG-CO.A.1**
FORWARD 1	Locate the center of a circle from chords	**HSG-C.A.2**
FORWARD 2	Find chord lengths and central angles with trigonometry	**HSG-SRT.C.8**

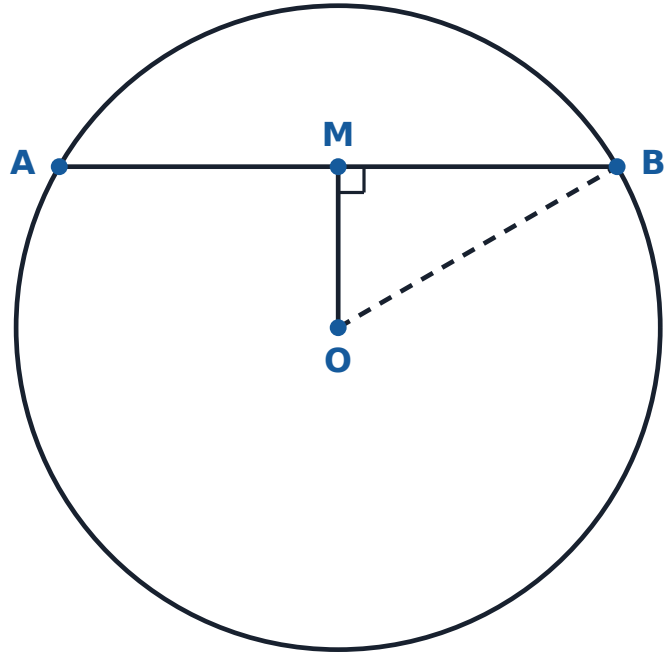
SET 58 • HSG-C.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.2

In circle O, OM ⊥ chord AB at M. Prove that M is the midpoint of AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q1/5

A circle has radius 13. A chord is 5 units from the center. How long is the chord?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q2/5

A chord of length 16 is in a circle of radius 10. How far is the chord from the center?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q3/5

In circle O, chords AB and CD are congruent. OM ⊥ AB at M and ON ⊥ CD at N. Prove that OM = ON (congruent chords are equidistant from the center).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q4/5

Circle O has radius 17. Chords AB and CD are each 30 units long. Find the distance from O to each chord, and explain why the distances are equal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • MAIN • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 1

Right triangles: Hypotenuse-Leg and the Pythagorean Theorem • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Right triangles with equal hypotenuses and one pair of equal legs are congruent by

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q1/5

Hypotenuse 10, leg 6. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q2/5

True or false?

In circle O, OM is perpendicular to chord AB at M. The radii OA and OB are congruent hypotenuses of right triangles OMA and OMB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q3/5

OM $\perp$ AB in circle O. $\triangle$OMA and $\triangle$OMB share

A. nothing

B. OM

C. AB

D. OA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q4/5

True or false?

HL works for any triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 2

The perpendicular bisector theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

A point equidistant from A and B lies on the perpendicular bisector of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q1/5

The center O of a circle is equidistant from the endpoints of chord AB. Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q2/5

The center of a circle is equidistant from the endpoints of each of its chords. So the center lies on

A. the perpendicular bisector of every chord

B. a tangent

C. the circle

D. every chord

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q3/5

Two chords of a circle are not parallel, so their perpendicular bisectors are different lines; the bisectors meet at P. What is P?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q4/5

True or false?

The perpendicular bisector of a chord can miss the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • HSG-C.A.2

BACKWARD 3

Radii of a circle are equal • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

A circle has center O. OA = 7. Find OB if B is on the circle.

OB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q1/5

True or false?

All radii of one circle have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q2/5

AB is a chord of circle O, and OA = 5. OB =

A. unknown

B. 10

C. 2.5

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q3/5

What is the longest chord in a circle of radius 6?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q4/5

True or false?

A chord always passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 58 • BACKWARD 3 • Q5/5

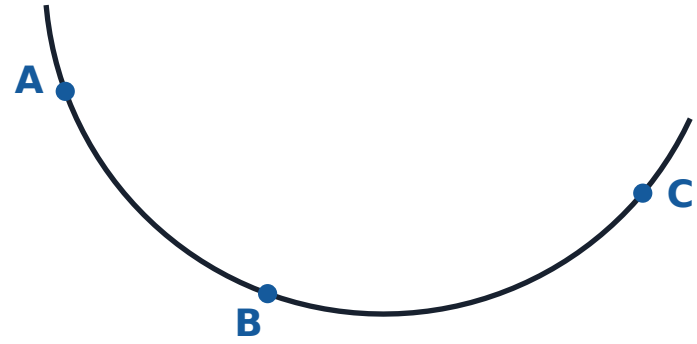
SET 58 • HSG-C.A.2

FORWARD 1

Locate the center of a circle from chords • High School — Geometry • Geometry: Circles • HSG-C.A.2

Compass and straightedge only; leave all arcs.

The arc below is part of a circle. Construct its center.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q1/5

A circle passes through (0, 0), (8, 0), and (0, 6). Find its center.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q2/5

True or false?

Three points not on one line determine exactly one circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q3/5

Why can't three points on one line lie on a circle?

A. Three points on a line always lie on a circle

B. Their perpendicular bisectors are parallel, so no center exists

C. Only two points can lie on a circle

D. They are too close

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q4/5

A circular plate is broken. Describe how to find its original radius from the piece.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSG-C.A.2

FORWARD 2

Find chord lengths and central angles with trigonometry • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A chord subtends a central angle of 80° in a circle of radius 10. Find its length to the nearest tenth.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q1/5

A chord of length 12 is in a circle of radius 10. Find the central angle to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q2/5

True or false?

A chord's length is $2r \sin \theta$ for central angle θ.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q3/5

Central angle 60°, radius 9. Chord length?

A. 4.5

B. 18

C. 9

D. $9\sqrt{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q4/5

A chord equals the radius. Find its central angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 58 • FORWARD 2 • Q5/5

SET 59 • HSG-C.A.3 • CIRCLES

Construct the inscribed circle of a triangle

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	Construct an angle bisector	**HSG-CO.D.12**
BACKWARD 2	Construct a perpendicular from a point to a line	**HSG-CO.D.12**
BACKWARD 3	Points on an angle bisector are equidistant from the sides	**HSG-SRT.B.5**
FORWARD 1	Prove that the incenter is equidistant from the sides	**HSG-C.A.3**
FORWARD 2	Locate a point equidistant from three lines in a design	**HSG-MG.A.3**

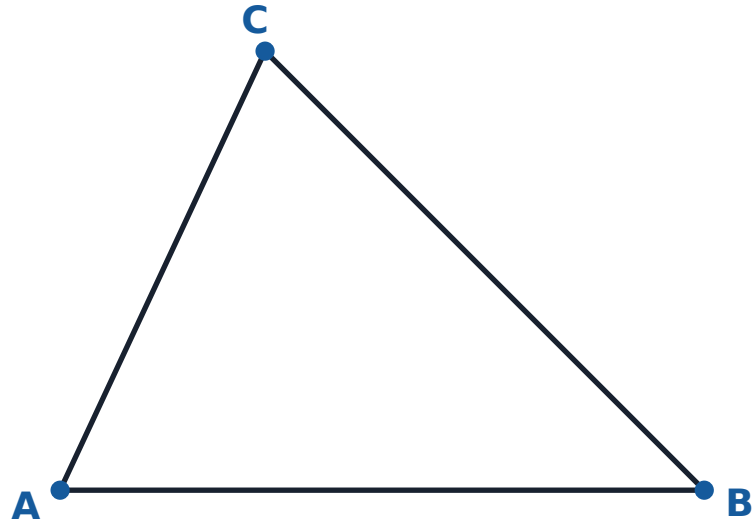
SET 59 • HSG-C.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of triangle ABC.



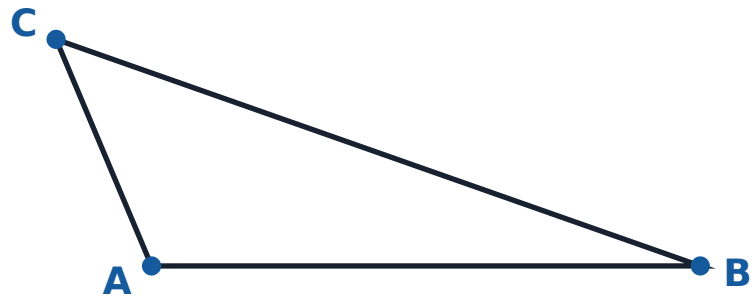
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of obtuse triangle ABC.



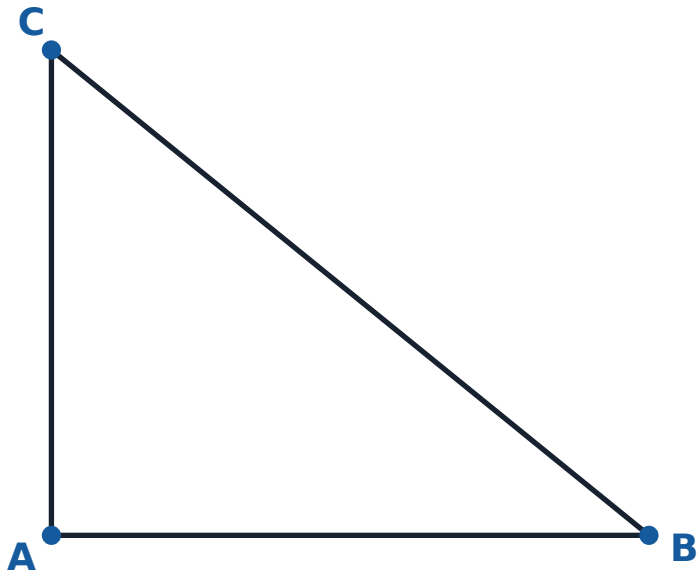
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of right triangle ABC.



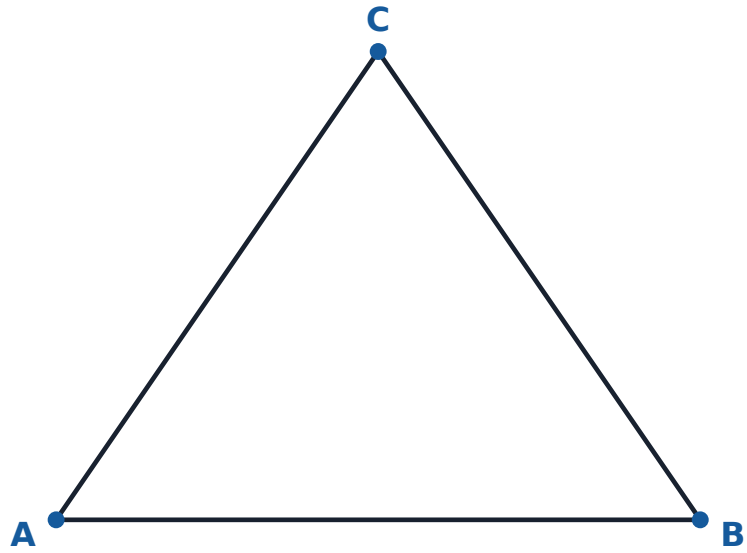
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of isosceles triangle ABC (AC = BC). Then, as evidence, construct the perpendiculars from the incenter to ALL THREE sides and mark the three points where the circle touches the sides.



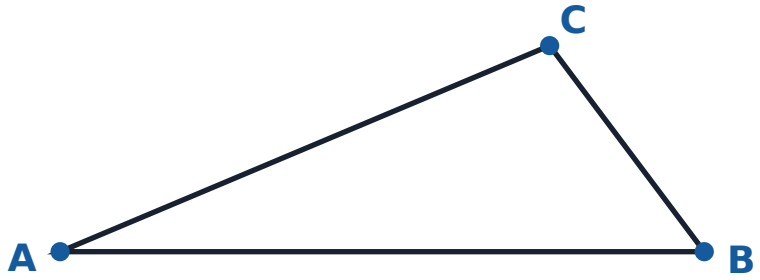
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct the inscribed circle of triangle ABC by bisecting ∠B and ∠C (not ∠A). Then construct the bisector of ∠A as a check.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • MAIN • Q5/5

SET 59 • HSG-C.A.3

BACKWARD 1

Construct an angle bisector • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

How many arcs are needed to bisect an angle with compass and straightedge?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q1/5

True or false?

An angle bisector construction ends with a ray from the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q2/5

The incenter is where the

A. medians meet

B. perpendicular bisectors meet

C. angle bisectors meet

D. altitudes meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q3/5

Why construct only two angle bisectors to find the incenter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 1 • Q4/5

True or false?

Bisecting the sides locates the incenter.

A. True

B. False

SET 59 • HSG-C.A.3

BACKWARD 2

Construct a perpendicular from a point to a line • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction finds the point where the incircle touches a side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q1/5

True or false?

The perpendicular from a point to a line is the shortest segment to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q2/5

From point I, an arc meets side AB at two points. Next step?

A. Measure AB

B. Draw a circle at A

C. Bisect ∠I

D. Arcs of equal radius from those two points, crossing on the other side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q3/5

Why not measure from I to a vertex for the radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q4/5

True or false?

The incircle passes through the vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSG-C.A.3

BACKWARD 3

Points on an angle bisector are equidistant from the sides • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

True or false?

A point on the bisector of $\angle A$ is equidistant from the two sides of $\angle A$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q1/5

The incenter is 3 cm from side AB. How far is it from side BC?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q2/5

Distance from a point to a side is measured

A. to the nearest vertex

B. along the perpendicular

C. along the bisector

D. along any segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q3/5

Why is the incenter the same distance from all three sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q4/5

True or false?

The incenter is equidistant from the three vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 59 • BACKWARD 3 • Q5/5

SET 59 • HSG-C.A.3

FORWARD 1

Prove that the incenter is equidistant from the sides • High School — Geometry • Geometry: Circles • HSG-C.A.3

The bisectors of ∠A and ∠B of △ABC meet at I. Prove that I is equidistant from all three sides and lies on the bisector of ∠C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q1/5

A triangle has area 30 and perimeter 30. Find the radius of its inscribed circle.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q2/5

True or false?

The incenter of an obtuse triangle lies outside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q3/5

A 3-4-5 right triangle has inradius

A. 2.5

B. 1

C. 1.5

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q4/5

Explain why area = r · s, where s is half the perimeter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSG-C.A.3

FORWARD 2

Locate a point equidistant from three lines in a design • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A circular fountain must touch three straight paths that form a triangle with sides 13, 14, and 15 m (area 84 m²). What is the largest possible radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q1/5

Explain why the incircle gives the largest circle that fits inside a triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q2/5

True or false?

A sprinkler equidistant from three straight fences forming a triangle should be at the circumcenter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q3/5

To be equidistant from three towns (points), use the

A. orthocenter

B. centroid

C. incenter

D. circumcenter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q4/5

An equilateral triangle has side 6. Find its inradius exactly.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 59 • FORWARD 2 • Q5/5

SET 60 • HSG-C.A.3 • CIRCLES

Construct the circumscribed circle of a triangle

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	Construct the perpendicular bisector of a segment	**HSG-CO.D.12**
BACKWARD 2	Points on the perpendicular bisector are equidistant from the end…	**HSG-CO.C.9**
BACKWARD 3	Triangle types: acute, right, obtuse	**4.G.A.2**
FORWARD 1	Find the circumcenter and radius with coordinates	**HSG-GPE.B.5**
FORWARD 2	Find the circumradius with trigonometry (extended Law of Sines)	**HSG-SRT.D.10**

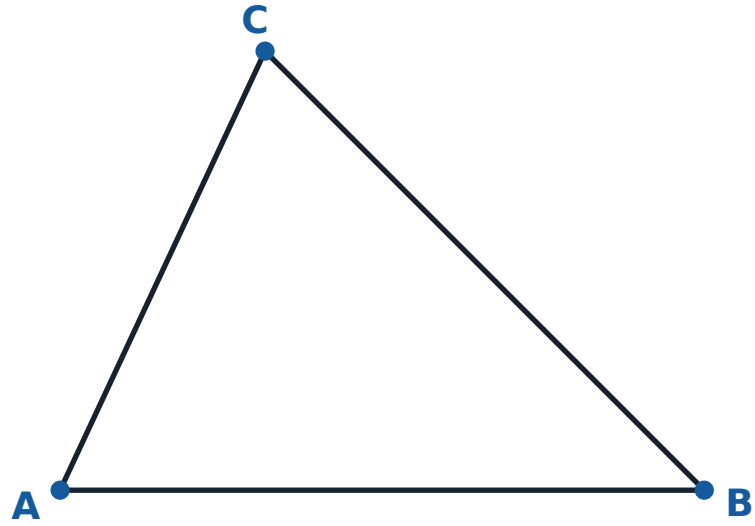
SET 60 • HSG-C.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of triangle ABC.



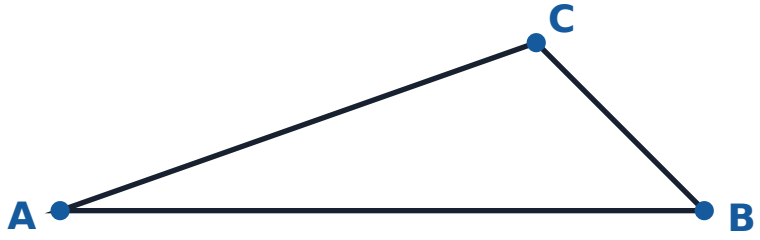
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of obtuse triangle ABC.



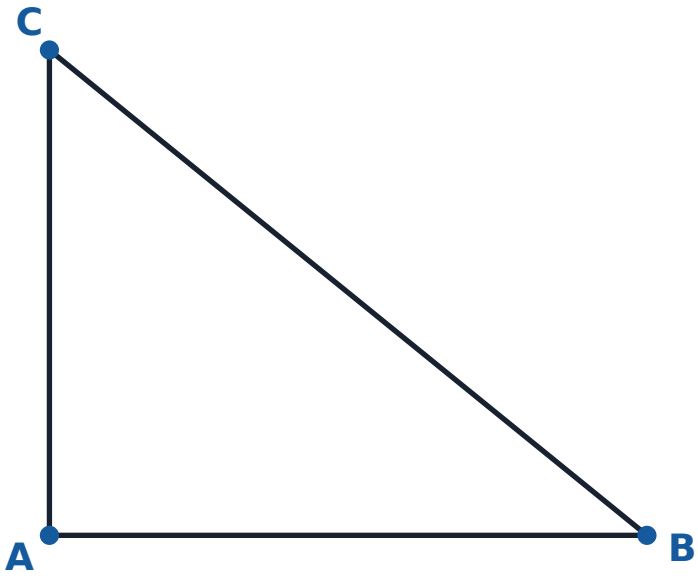
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of right triangle ABC.



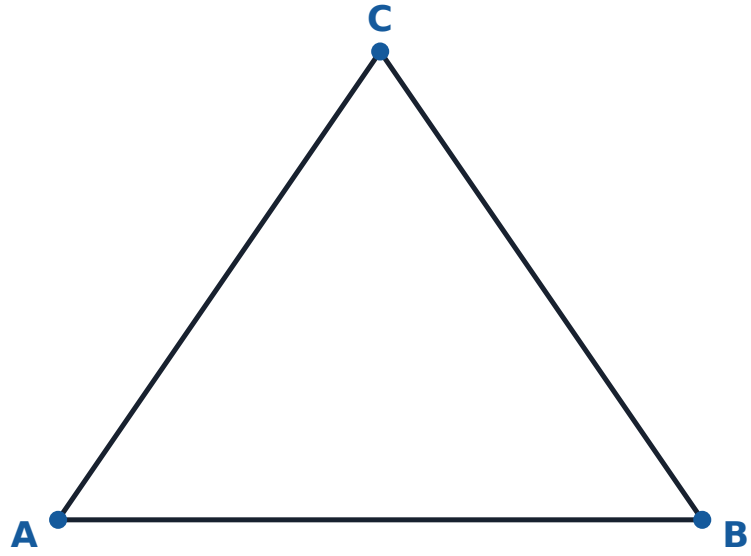
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of isosceles triangle ABC (AC = BC). Then, as evidence, construct the perpendicular bisector of the third side and show that it passes through the same center.



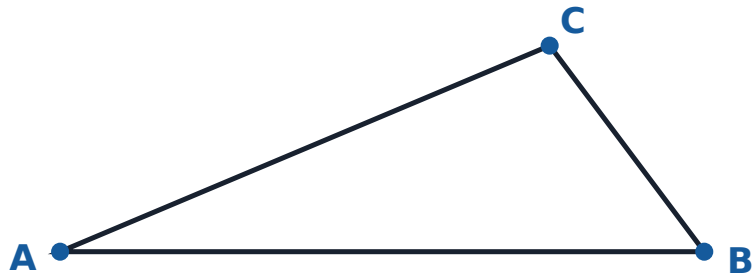
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct the circumscribed circle of triangle ABC using the perpendicular bisectors of AC and BC. Then check with the compass that the circle passes through all three vertices.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 60 • MAIN • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 1

Construct the perpendicular bisector of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction gives all points equidistant from A and B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector construction uses equal arcs from both endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q2/5

The circumcenter is where the

A. angle bisectors meet

B. sides meet

C. perpendicular bisectors of the sides meet

D. medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q3/5

After constructing two perpendicular bisectors, why is the third unnecessary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q4/5

True or false?

The arcs in a perpendicular bisector construction can have any two different radii.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 2

Points on the perpendicular bisector are equidistant from the endpoints • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Every point on the perpendicular bisector of AB is equidistant from A and B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q1/5

O is on the perpendicular bisectors of AB and BC. Compare OA, OB, OC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q2/5

The circumcircle's radius is

A. the distance from O to any vertex

B. the longest side

C. the distance from O to a side

D. half a side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q3/5

A circle passes through A, B, C. Where must its center lie relative to AB?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q4/5

True or false?

The circumcenter is equidistant from the sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSG-C.A.3

BACKWARD 3

Triangle types: acute, right, obtuse • Grade 4 • Geometry • 4.G.A.2

A triangle has angles 30°, 45°, 105°. Classify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q1/5

True or false?

A right triangle has one 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q2/5

Where is the circumcenter of a right triangle?

A. Inside

B. Outside

C. On the hypotenuse (its midpoint)

D. At the right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q3/5

A triangle has angles 50°, 60°, 70°. Classify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q4/5

True or false?

An acute triangle can have an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 60 • BACKWARD 3 • Q5/5

SET 60 • HSG-C.A.3

FORWARD 1

Find the circumcenter and radius with coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Triangle with A(0, 0), B(6, 0), C(2, 4). Find the circumcenter and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q1/5

Triangle with A(−2, 0), B(2, 0), C(0, 2). Find the circumcenter.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q2/5

True or false?

The circumcenter is the intersection of two perpendicular bisector equations.

A. True

B. False

A right triangle has vertices (0, 0), (8, 0), (0, 6). Circumradius?

A. 5

B. 10

C. 4

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q4/5

Explain how to check a computed circumcenter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSG-C.A.3

FORWARD 2

Find the circumradius with trigonometry (extended Law of Sines) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

In a triangle, $a = 10$ and $A = 30°$. Find the circumradius R using $a/\sin A = 2R$.

$R =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q1/5

Equilateral triangle with side 6. Find the circumradius exactly.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q2/5

True or false?

In any triangle, a/sin A equals the radius of the circumcircle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q3/5

A triangle with a 90° angle opposite side 14 has circumradius

A. 3.5

B. 14

C. 28

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q4/5

Explain why $a/\sin A = 2R$ using an inscribed angle on a diameter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSG-C.A.3 • CIRCLES

Prove that opposite angles of an inscribed quadrilateral are supplementary

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	The inscribed angle theorem	**HSG-C.A.2**
BACKWARD 2	Supplementary angles	**7.G.B.5**
BACKWARD 3	Arcs of a circle add to 360°	**4.MD.C.7**
FORWARD 1	Use the converse: decide whether a quadrilateral can be inscribed	**HSG-C.A.3**
FORWARD 2	Exterior angle of a cyclic quadrilateral	**HSG-C.A.3**

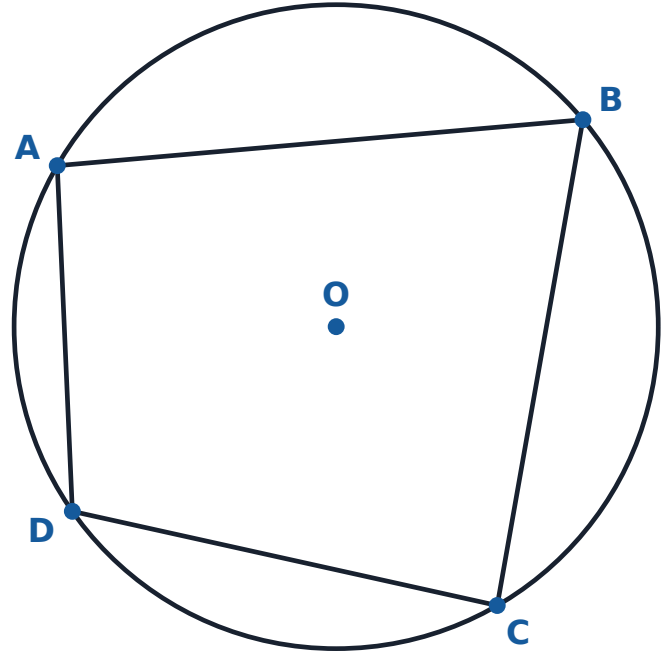
SET 61 • HSG-C.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

Quadrilateral ABCD is inscribed in circle O. Prove that $m\angle A + m\angle C = 180°$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q1/5

Quadrilateral PQRS is inscribed in a circle. Prove that $\angle Q$ and $\angle S$ are supplementary.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q2/5

Quadrilateral ABCD is inscribed in a circle with m∠A = 72° and m∠B = 105°. Find m∠C and m∠D, justifying each with the inscribed-quadrilateral theorem.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q3/5

Parallelogram ABCD is inscribed in a circle. Prove that ABCD is a rectangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q4/5

Trapezoid ABCD with AB ∥ DC is inscribed in a circle. Prove that ∠A ≅ ∠B (so the trapezoid is isosceles).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 1

The inscribed angle theorem • High School — Geometry • Geometry: Circles • HSG-C.A.2

An inscribed angle intercepts a 220° arc. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q1/5

True or false?

An inscribed angle can intercept an arc greater than 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q2/5

An inscribed angle of 70° intercepts

A. 70°

B. 290°

C. 35°

D. 140°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q3/5

Two arcs together make a whole circle. Their sum?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q4/5

True or false?

An inscribed angle equals half the central angle only when it is acute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 2

Supplementary angles • Grade 7 • Geometry • 7.G.B.5

Supplement of 72°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q1/5

True or false?

Two angles whose measures add to 180° are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q2/5

Angles 3x and 2x are supplementary. x =

A. 30°

B. 45°

C. 36°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q3/5

If $\angle A = \angle C$ and they are supplementary, find each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q4/5

True or false?

Supplementary angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 3

Arcs of a circle add to 360° • Grade 4 • Measurement & Data • 4.MD.C.7

A circle is split into arcs of 100°, 80°, and x°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q1/5

True or false?

The arcs between four points on a circle add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q2/5

Two arcs make a whole circle; one is 150°. The other is

A. 210°

B. 150°

C. 30°

D. 360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q3/5

Half of 360° is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q4/5

True or false?

The arcs of a circle add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q5/5

SET 61 • HSG-C.A.3

FORWARD 1

Use the converse: decide whether a quadrilateral can be inscribed • High School — Geometry • Geometry: Circles • HSG-C.A.3

Quadrilateral ABCD has angles A = 85°, B = 70°, C = 95°, D = 110°. Can it be inscribed in a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q1/5

An isosceles trapezoid has base angles of 64°. Can it be inscribed in a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q2/5

True or false?

Every kite can be inscribed in a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q3/5

Inscribed quadrilateral with angles (2x)°, (3x + 5)°, (2x + 20)°, and y°. If 2x and 2x + 20 are opposite, x =

A. 40

B. 50

C. 45

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q4/5

Inscribed quadrilateral: $\angle A = (5y - 10)°$, $\angle C = (3y + 30)°$, A and C opposite. Find y and $\angle A$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q5/5

SET 61 • HSG-C.A.3

FORWARD 2

Exterior angle of a cyclic quadrilateral • High School — Geometry • Geometry: Circles • HSG-C.A.3

ABCD is inscribed in a circle and side DC is extended past C to E. Prove that m∠BCE = m∠A.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q1/5

In inscribed ABCD, $\angle A = 77°$. Side DC is extended past C to E. Find $\angle BCE$.

$\angle BCE =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q2/5

True or false?

An exterior angle of a cyclic quadrilateral equals the interior angle next to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q3/5

Inscribed ABCD with $\angle B = 100°$. The exterior angle at D (extending CD past D) is

A. 50°

B. 180°

C. 100°

D. 80°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q4/5

ABCD is inscribed in a circle, and side DC is extended past C. Why is the exterior angle at C equal to ∠A, not ∠B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q5/5

SET 62 • HSG-C.B.5 • CIRCLES

Derive arc length from similarity and define radian measure

MAIN	Geometry main question	**HSG-C.B.5**
BACKWARD 1	Similar figures: lengths scale by the scale factor	**HSG-SRT.A.2**
BACKWARD 2	Circumference of a circle	**7.G.B.4**
BACKWARD 3	Find a fraction of a whole	**7.RP.A.3**
FORWARD 1	Convert between degrees and radians	**HSG-C.B.5**
FORWARD 2	Radian measure for angles of rotation and the unit circle	**HSF-TF.A.1**

SET 62 • HSG-C.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.B.5

A central angle θ cuts off arc length s_1 on a circle of radius r_1 and arc length s_2 on a circle of radius r_2 (same center). Explain, using similarity, why $s_1/r_1 = s_2/r_2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q1/5

A central angle cuts off an arc of length 12 cm on a circle of radius 4 cm. What is the angle in radians?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q2/5

Find the length of the arc cut off by a central angle of 2.5 radians in a circle of radius 6 m.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q3/5

Find the exact length of the arc cut off by a 120° central angle in a circle of radius 9.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q4/5

Use $s = r\theta$ to explain why a full turn is 2π radians and a half turn is π radians, and convert $270°$ to radians.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 1

Similar figures: lengths scale by the scale factor • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

True or false?

Under a dilation with factor k, every length (straight or curved) is multiplied by k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q1/5

Two similar circles have radii 3 and 12. Scale factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q2/5

An arc of length 5 on the small circle (radius 3) matches which arc length on the large one (radius 12)?

A. 20

B. 15

C. 8

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q3/5

Are all circles similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q4/5

True or false?

Arc lengths do not change under dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 2

Circumference of a circle • Grade 7 • Geometry • 7.G.B.4

Find the circumference of a circle with radius 5 (exact).

Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q1/5

True or false?

$C = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q2/5

A circle has diameter 8. Its circumference is

A. 64π

B. 4π

C. 16π

D. 8π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q3/5

A circle has circumference 18π. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q4/5

True or false?

$C = \pi r^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 3

Find a fraction of a whole • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

What fraction of a circle is a 90° arc?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q1/5

Find ⅓ of 18π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q2/5

An arc of 45° is what fraction of the circumference?

A. 45/100

B. ⅛

C. 1/45

D. ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q3/5

True or false?

A 120° arc is ⅓ of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q4/5

True or false?

A 60° arc is 1/60 of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q5/5

SET 62 • HSG-C.B.5

FORWARD 1

Convert between degrees and radians • High School — Geometry • Geometry: Circles • HSG-C.B.5

Convert 150° to radians (exact).

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q1/5

Convert 3π/4 radians to degrees.

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q2/5

True or false?

1 radian is about 3.14°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q3/5

Which equals π/3 radians?

A. 60°

B. 120°

C. 30°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q4/5

An arc in a circle of radius 10 has length 5π. Find the central angle in degrees.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSG-C.B.5

FORWARD 2

Radian measure for angles of rotation and the unit circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1

On the unit circle, an angle of 2 radians cuts off an arc of what length?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q1/5

A wheel of radius 0.3 m rotates about its fixed center (axle) through 10 radians. How far does a point on its rim travel along its circular path?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q2/5

True or false?

An angle of 6 radians is more than one full turn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q3/5

A point on the unit circle travels an arc of length π. The angle is

A. 1 radian

B. 2π radians

C. π°

D. π radians (180°)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q4/5

Explain why the radian measure of an angle does not depend on the radius of the circle used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSG-C.B.5 • CIRCLES

Derive and use the formula for the area of a sector

MAIN	Geometry main question	**HSG-C.B.5**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	The radius is half the diameter	**7.G.B.4**
BACKWARD 3	Fractions of a circle in degrees and radians	**HSG-C.B.5**
FORWARD 1	Solve inverse sector problems	**HSG-C.B.5**
FORWARD 2	Areas of segments and composite circular regions	**HSG-MG.A.1**

SET 63 • HSG-C.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.B.5

Derive a formula for the area of a sector with central angle θ (in degrees) in a circle of radius r. Then rewrite it for θ in radians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q1/5

Derive the formula A = ½rs for the area of a sector in terms of its radius r and arc length s. Then use it for a sector with r = 8 and s = 5.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q2/5

Derive the sector area formula for θ in radians from the proportion $A/(\pi r^2) = \theta/(2\pi)$. Then find the area of a sector with central angle 1.2 radians in a circle of radius 5.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q3/5

A circular pizza of diameter d is cut into n equal slices. Derive a formula for the area of one slice in terms of d and n, then find the area for d = 14 in. and n = 8, to the nearest tenth of a square inch.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q4/5

Solve the degree sector formula $A = (\theta/360)\pi r^2$ for θ, and use your formula to find the central angle of a sector with area 12π in a circle of radius 6.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

Find the exact area of a circle with radius 6.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q1/5

A circle has diameter 10. Exact area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q2/5

True or false?

$A = \pi r^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q3/5

A circle has area 49π. Radius?

A. 14

B. 7

C. 24.5

D. 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q4/5

True or false?

Doubling the radius doubles the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 2

The radius is half the diameter • Grade 7 • Geometry • 7.G.B.4

A circle has diameter 14 in. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q1/5

True or false?

Using the diameter in $A = \pi r^2$ gives an area four times too large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q2/5

A sector of a circle with diameter 20 uses r =

A. 10

B. 20

C. 5

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q3/5

A circle's radius is 2.5 cm. Diameter?

Diameter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q4/5

True or false?

The radius and diameter can be used interchangeably in area formulas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 3

Fractions of a circle in degrees and radians • High School — Geometry • Geometry: Circles • HSG-C.B.5

What fraction of a circle is a sector with central angle 45°?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q1/5

What fraction of a circle is a sector with central angle π/2 radians?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q2/5

A full circle measures

A. 360° or 2π radians

B. 180° or π radians

C. 360 radians

D. 2π°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q3/5

True or false?

A central angle of 1 radian is 1/(2π) of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q4/5

True or false?

A 90° sector is ⅓ of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q5/5

SET 63 • HSG-C.B.5

FORWARD 1

Solve inverse sector problems • High School — Geometry • Geometry: Circles • HSG-C.B.5

A sector with a 40° angle has area 4π. Find the radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q1/5

A sector of a circle with radius 10 has area 30. Find the central angle in radians.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q2/5

True or false?

If the radius is tripled and the angle stays the same, the sector area is tripled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q3/5

A sector has area 50 and arc length 10. Radius?

A. 20

B. 5

C. 2.5

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q4/5

Show that the area of a sector is A = ½rs, where s is the arc length.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q5/5

SET 63 • HSG-C.B.5

FORWARD 2

Areas of segments and composite circular regions • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

Find the exact area of the region between a 90° arc and its chord in a circle of radius 4 (a circular segment).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q1/5

A sprinkler waters a 120° sector of radius 9 m. Find the watered area to the nearest square meter.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q2/5

True or false?

The area of a circular segment (smaller than a semicircle) equals the sector area plus the area of the triangle formed by the two radii and the chord.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q3/5

An annulus (ring) has inner radius 3 and outer radius 5. Its area is

A. 8π

B. 4π

C. 16π

D. 25π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q4/5

A windshield wiper blade covers the part of its arm from 20 cm to 60 cm from the pivot and sweeps through 120°. Find the area it wipes to the nearest square centimeter.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q5/5

Derive the equation of a circle

MAIN	Geometry main question	**HSG-GPE.A.1**
BACKWARD 1	Distance between two points with the Pythagorean Theorem	**8.G.B.8**
BACKWARD 2	Use the Pythagorean Theorem with expressions	**8.G.B.7**
BACKWARD 3	Shifting a graph: reading h and k	**HSF-BF.B.3**
FORWARD 1	Write circle equations from other given information	**HSG-GPE.A.1**
FORWARD 2	Prove or disprove that points lie on a circle	**HSG-GPE.B.4**

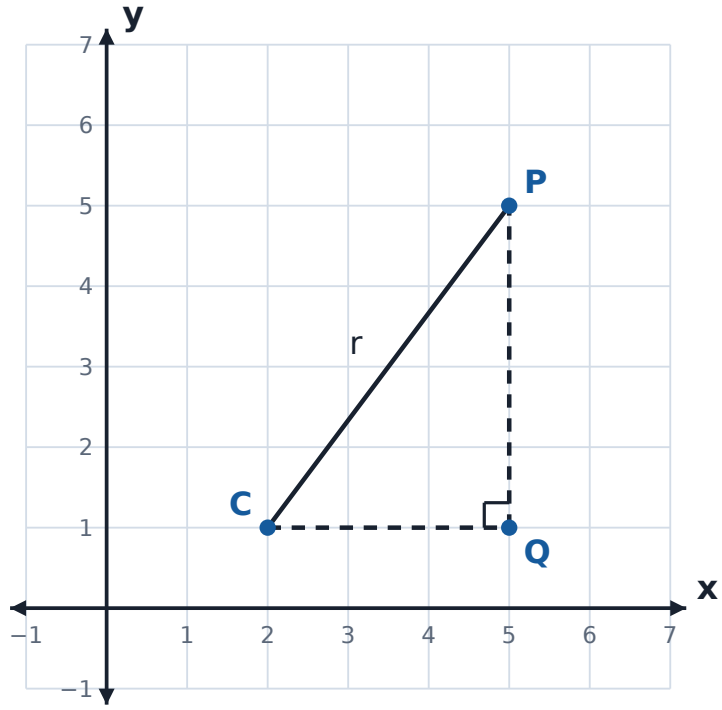
SET 64 • HSG-GPE.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

A circle has center (h, k) and radius r. Use the Pythagorean Theorem to derive the equation satisfied by every point (x, y) on the circle. The diagram shows one example, with center C(2, 1).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q1/5

Use the Pythagorean Theorem (not a memorized formula) to derive the equation of the circle with center (3, −2) and radius 5. Show the right triangle for a general point (x, y) on the circle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q2/5

A circle is centered at the origin and passes through (6, −8). Use the Pythagorean Theorem (not a memorized formula) to derive its equation, first finding the radius.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q3/5

A circle has center $(-1, 4)$ and passes through $(2, 8)$. Use the Pythagorean Theorem (not a memorized formula) to derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q4/5

Use the Pythagorean Theorem (not a memorized formula) to derive the equation of the circle with center $(-2, 5)$ and radius 3. Then use your equation to decide whether $(-2, 8)$ is on the circle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q5/5

SET 64 • HSG-GPE.A.1

BACKWARD 1

Distance between two points with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q1/5

Find the distance between (0, 0) and (5, 12).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q2/5

The legs of the right triangle joining (2, 1) and (7, 13) are

A. 2 and 1

B. 7 and 13

C. 9 and 14

D. 5 and 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q3/5

True or false?

The distance from $(-3, 0)$ to $(3, 0)$ is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q4/5

True or false?

The distance from (0, 0) to (3, 4) is 7.

A. True

B. False

SET 64 • HSG-GPE.A.1

BACKWARD 2

Use the Pythagorean Theorem with expressions • Grade 8 • Geometry • 8.G.B.7

A right triangle has legs 6 and 8. Find the hypotenuse.

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q1/5

A right triangle has legs a and b and hypotenuse c. Write the Pythagorean relationship.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q2/5

A right triangle has legs $(x - 2)$ and $(y - 1)$ and hypotenuse 4. Which equation is true?

A. $(x - 2)^2 + (y - 1)^2 = 16$

B. $x^2 + y^2 = 16$

C. $(x - 2)^2 + (y - 1)^2 = 4$

D. $(x - 2) + (y - 1) = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q3/5

True or false?

In a right triangle, the hypotenuse is opposite the right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q4/5

True or false?

$a^2 + b^2 = c^2$ holds for every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSG-GPE.A.1

BACKWARD 3

Shifting a graph: reading h and k • High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3

The graph of $y = (x - 3)^2$ is the graph of $y = x^2$ shifted in which direction and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q1/5

Write $x - (-4)$ without the double sign.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q2/5

In $(x + 2)^2$, the shift is

A. 2 down

B. 2 right

C. 2 left

D. 2 up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-B3-Q3**

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q3/5

True or false?

Replacing x with $x - 5$ shifts a graph 5 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q4/5

True or false?

Replacing y with y + 1 shifts a graph 1 unit up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q5/5

SET 64 • HSG-GPE.A.1

FORWARD 1

Write circle equations from other given information • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

A diameter of a circle has endpoints $(-2, 3)$ and $(6, -3)$. Write the equation of the circle.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q1/5

A circle has center $(4, -3)$ and is tangent to the x-axis. Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q2/5

True or false?

The circle $(x - 1)^2 + (y - 1)^2 = 2$ passes through the origin.

A. True

B. False

A circle with center (0, 5) passes through (0, −1). Its radius is

A. 4

B. 36

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q4/5

A circle is centered at (3, 3) and is tangent to both axes. Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q5/5

SET 64 • HSG-GPE.A.1

FORWARD 2

Prove or disprove that points lie on a circle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Does $(-4, 6)$ lie on the circle with center $(2, -2)$ passing through $(8, 6)$? Show the check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q1/5

Is (7, 1) inside, on, or outside the circle $(x - 3)^2 + (y + 2)^2 = 25$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q2/5

True or false?

A point is inside a circle when its distance from the center is greater than the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q3/5

The point (1, 1) is ___ the circle $x^2 + y^2 = 4$.

A. on

B. outside

C. inside

D. the center of

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q4/5

A circle has the diameter with endpoints $(-2, 3)$ and $(6, -3)$. Does $(5, 4)$ lie on it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q5/5

Complete the square to find the center and radius of a circle

MAIN	Geometry main question	**HSG-GPE.A.1**
BACKWARD 1	Complete the square in one variable	**HSA-REI.B.4.a**
BACKWARD 2	Square a binomial	**HSA-APR.A.1**
BACKWARD 3	Keep an equation balanced	**HSA-REI.A.1**
FORWARD 1	Leading coefficients and equations that are not circles	**HSG-GPE.A.1**
FORWARD 2	Complete the square for an ellipse	**HSG-GPE.A.3**

SET 65 • HSG-GPE.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Complete the square to find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$. Show the standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q1/5

Complete the square to find the center and radius of the circle $x^2 + y^2 + 8x - 10y + 5 = 0$. Show the standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q2/5

Complete the square to write $x^2 + y^2 - 2x - 14y + 25 = 0$ in standard form. Then give the center and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q3/5

Complete the square to find the center and radius of the circle $x^2 + y^2 + 10y = 0$. Show the standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q4/5

Complete the square to write $x^2 + y^2 - 4x + 6y = 3$ in standard form. Then give the center and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q5/5

SET 65 • HSG-GPE.A.1

BACKWARD 1

Complete the square in one variable • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

What number completes the square: $x^2 - 6x + ___$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q1/5

Write $x^2 + 10x + 25$ as a square.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q2/5

$x^2 - 8x + 16$ equals

A. $(x + 4)^2$

B. $(x - 8)^2$

C. $(x - 4)^2$

D. $(x - 16)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B1-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q3/5

True or false?

To complete $x^2 + bx$, add $(b/2)^2$.

A. True

B. False

True or false?

$x^2 + 6x + 6$ is a perfect square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSG-GPE.A.1

BACKWARD 2

Square a binomial • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q1/5

Expand $(y + 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q2/5

$(x + 5)^2 =$

A. $x^2 + 10x + 25$

B. $x^2 + 25$

C. $x^2 + 10x + 10$

D. $x^2 + 5x + 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q3/5

True or false?

$$(a - b)^2 = a^2 - 2ab + b^2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q4/5

True or false?

$$(x - 3)^2 = x^2 - 9$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • HSG-GPE.A.1

BACKWARD 3

Keep an equation balanced • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

If 9 is added to the left side of an equation, what must be done to the right side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q1/5

$x^2 - 6x = 7$. Add 9 to both sides and factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q2/5

$x^2 + y^2 - 4x = 5$ becomes $(x - 2)^2 + y^2 =$

A. 4

B. 5

C. 1

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B3-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q3/5

True or false?

Adding the same number to both sides keeps the solutions the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q4/5

True or false?

A number added to one side need not be added to the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q5/5

SET 65 • HSG-GPE.A.1

FORWARD 1

Leading coefficients and equations that are not circles • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Find the center and radius of
$2x^2 + 2y^2 - 8x + 12y - 6 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q1/5

Does $x^2 + y^2 + 2x + 4y + 10 = 0$ represent a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q2/5

True or false?

$x^2 + y^2 - 6x + 9 = 0$ represents a single point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q3/5

$3x^2 + 3y^2 - 12y = 0$ has radius

A. $\sqrt{12}$

B. 2

C. 12

D. 4

For what values of c is $x^2 + y^2 - 4x + c = 0$ a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q5/5

SET 65 • HSG-GPE.A.1

FORWARD 2

Complete the square for an ellipse • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Write $4x^2 + 9y^2 - 16x + 18y - 11 = 0$ in standard form and give the center.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q1/5

Find the center of $x^2 + 4y^2 + 6x - 8y + 9 = 0$.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q2/5

True or false?

After factoring 4 out of $4x^2 - 16x$, add 16 inside the parentheses to complete the square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q3/5

Which is NOT a circle?

A. $x^2 + 4y^2 = 4$

B. $2x^2 + 2y^2 = 8$

C. $x^2 + y^2 = 4$

D. $(x - 1)^2 + y^2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q4/5

Explain why $x^2 + 4y^2 = 4$ is not a circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q5/5

Derive the equation of a parabola from a focus and directrix

MAIN	Geometry main question	**HSG-GPE.A.2**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	Distance from a point to a horizontal line	**6.NS.C.8**
BACKWARD 3	Points equidistant from two objects	**HSG-CO.C.9**
BACKWARD 4	Expand and simplify squares	**HSA-APR.A.1**
FORWARD 1	Find the focus and directrix from an equation	**HSG-GPE.A.2**
FORWARD 2	Design with the reflective property of parabolas	**HSG-MG.A.3**

SET 66 • HSG-GPE.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

Derive the equation of the parabola with focus (0, 2) and directrix $y = -2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q1/5

Derive, from the definition (equal distances to the focus and the directrix), the equation of the parabola with focus $(0, -3)$ and directrix $y = 3$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q2/5

Derive, from the definition (equal distances to the focus and the directrix), the equation of the parabola with focus (2, 5) and directrix y = 1.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q3/5

Derive, from the definition (equal distances to the focus and the directrix), the equation of the parabola with focus (3, 0) and directrix $x = -3$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q4/5

Derive, from the definition (equal distances to the focus and the directrix), the equation of the parabola with focus (1, 4) and directrix y = 0 (the x-axis). Then give its vertex.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance from (3, 6) to (0, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q1/5

Write the distance from (x, y) to (0, 2) as an expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q2/5

The distance from (4, 1) to (1, 5) is

A. 7

B. 5

C. $\sqrt{7}$

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q3/5

True or false?

The distance formula comes from the Pythagorean Theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q4/5

True or false?

$\sqrt{(9 + 16)} = 3 + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 2

Distance from a point to a horizontal line • Grade 6 • The Number System • 6.NS.C.8

How far is (5, 7) from the line y = 2?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q1/5

How far is (1, −3) from the line y = 4?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q2/5

The distance from (x, y) to $y = -2$ is

A. $|x + 2|$

B. y

C. $|y - 2|$

D. $|y + 2|$

True or false?

The distance from a point to a horizontal line is measured straight up or down.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q4/5

True or false?

The distance from (6, 1) to y = 4 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 3

Points equidistant from two objects • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

The set of points equidistant from two points is the perpendicular bisector of the segment joining them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q1/5

A parabola is the set of points equidistant from what two objects?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q2/5

The vertex of a parabola is

A. at the focus

B. twice as far from the focus

C. halfway between the focus and the directrix

D. on the directrix

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q3/5

Focus (0, 6), directrix y = 0. Find the vertex.

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q4/5

True or false?

Every point of a parabola is closer to the focus than to the directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 4

Expand and simplify squares • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(y - 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B4-Q1

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q1/5

Simplify $(y + 2)^2 - (y - 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B4-Q2

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q2/5

$y^2 + 6y + 9 - (y^2 - 6y + 9) =$

A. 18

B. $2y^2$

C. 0

D. $12y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B4-Q3

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q3/5

True or false?

$(y + 3)^2 = y^2 + 6y + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B4-Q4

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q4/5

True or false?

$$(y - 2)^2 - y^2 = 4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B4-Q5

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q5/5

SET 66 • HSG-GPE.A.2

FORWARD 1

Find the focus and directrix from an equation • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

Find the focus and directrix of $y = x^2/12$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q1/5

Find the vertex, focus, and directrix of $(x - 1)^2 = -8(y + 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q2/5

True or false?

In $x^2 = 4py$ ($p \neq 0$), the focus is $|p|$ units from the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q3/5

The parabola $y^2 = -20x$ opens

A. left

B. up

C. down

D. right

Find the focus of $y = 2x^2$.

Focus: _____

SET 66 • HSG-GPE.A.2

FORWARD 2

Design with the reflective property of parabolas • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A satellite dish is 4 ft across and 1 ft deep. How far from the vertex should the receiver (at the focus) be placed?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q1/5

A headlight reflector has equation $x^2 = 12y$ (inches). Where should the bulb go?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q2/5

True or false?

Rays parallel to the axis of a parabolic mirror reflect through the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q3/5

A dish has focus 2 m above its vertex. Its equation (vertex at the origin) is

A. $x^2 = 4y$

B. $x^2 = 8y$

C. $y = 8x^2$

D. $x^2 = 2y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q4/5

A dish has equation $x^2 = 16y$ (meters) and is 8 m across. How deep is it?

Depth: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q5/5

SET 67 • HSG-GPE.B.4 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Prove or disprove the type of a quadrilateral with coordinates

MAIN	Geometry main question	**HSG-GPE.B.4**
BACKWARD 1	Find the slope between two points	**8.F.B.4**
BACKWARD 2	Find the distance between two points	**8.G.B.8**
BACKWARD 3	Properties that define special quadrilaterals	**HSG-CO.C.11**
FORWARD 1	Coordinate proofs with variables	**HSG-GPE.B.4**
FORWARD 2	Use the classification to find area	**HSG-GPE.B.7**

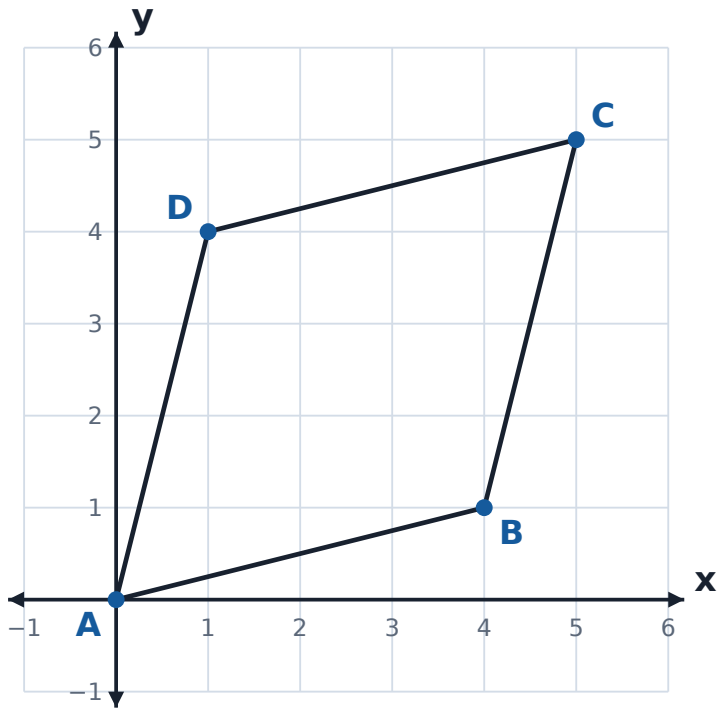
SET 67 • HSG-GPE.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Prove that ABCD with A(0, 0), B(4, 1), C(5, 5), D(1, 4) is a rhombus.

Work space



Conclusion: _____

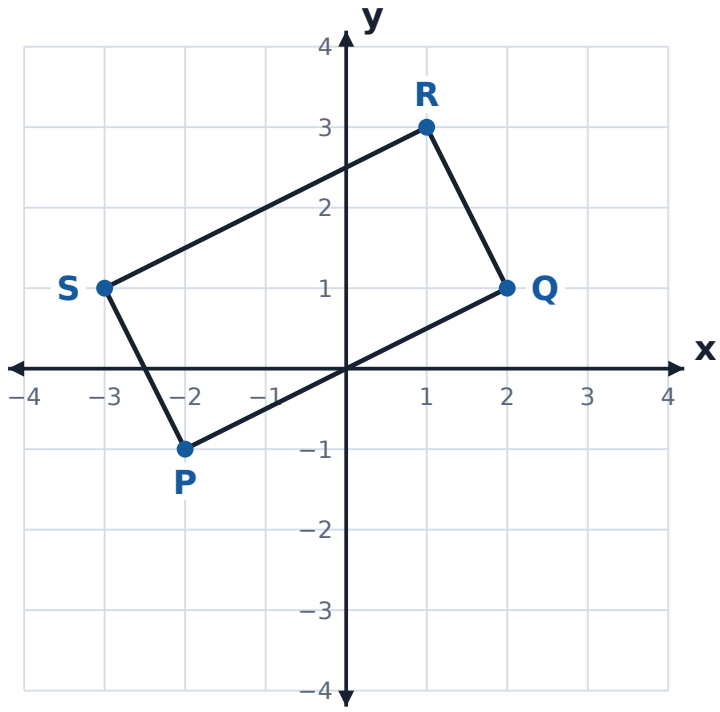
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q1/5

Prove that PQRS with P(−2, −1), Q(2, 1), R(1, 3), S(−3, 1) is a rectangle. Is it a square?

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q2/5

Prove or disprove: KLMN with K(0, 0), L(5, 0), M(6, 3), N(0, 3) is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q3/5

Prove that the triangle with vertices A(1, 1), B(4, 5), C(8, 2) is an isosceles right triangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q4/5

Prove or disprove: EFGH with E(−1, 0), F(2, −1), G(3, 2), H(0, 3) is a square.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 1

Find the slope between two points • Grade 8 • Functions • 8.F.B.4

Find the slope of the line through (1, 2) and (5, 4).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q1/5

Find the slope of the line through (0, 3) and (2, −1).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q2/5

The slope through (2, 5) and (6, 5) is

A. undefined

B. 0

C. 5

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q3/5

True or false?

A vertical segment has an undefined slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q4/5

True or false?

Slope is run divided by rise.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 2

Find the distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance between (0, 0) and (4, 1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q1/5

Find the distance between (1, 1) and (4, 5).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q2/5

Which pair of points is 10 apart?

A. (2, 3) and (8, 3)

B. (0, 0) and (6, 8)

C. (0, 0) and (10, 10)

D. (1, 1) and (5, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q3/5

True or false?

Two segments with the same length are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q4/5

True or false?

$\sqrt{10} + \sqrt{10} = \sqrt{20}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 3

Properties that define special quadrilaterals • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Which quadrilateral has four congruent sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q1/5

A parallelogram with one right angle is a ___.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q2/5

Which shows a quadrilateral is a parallelogram?

A. One pair of sides is parallel

B. Two adjacent sides are equal

C. One right angle

D. Both pairs of opposite sides are parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q3/5

True or false?

Every square is a rhombus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q4/5

True or false?

Every rhombus is a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q5/5

SET 67 • HSG-GPE.B.4

FORWARD 1

Coordinate proofs with variables • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A rectangle has vertices O(0, 0), A(a, 0), B(a, b), C(0, b). Prove that its diagonals are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q1/5

Triangle with vertices O(0, 0), A(2a, 0), B(2b, 2c), where a > 0 and c ≠ 0 (so the points are not collinear). M and N are the midpoints of OB and AB. Prove that MN ∥ OA and MN = ½OA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q2/5

True or false?

Placing a figure with a vertex at the origin and a side on the x-axis is allowed in a coordinate proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q3/5

Why do coordinates like (2a, 0) and (2b, 2c) make the midpoint proof easier?

A. The midpoints have no fractions

B. They make the triangle isosceles

C. They make a right angle

D. They make the slopes 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q4/5

Why is a proof using A(6, 0) and B(2, 4) not enough to prove the midsegment theorem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSG-GPE.B.4

FORWARD 2

Use the classification to find area • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ABCD with A(0, 0), B(4, 1), C(5, 5), D(1, 4) is a rhombus. Find its area using its diagonals.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q1/5

Rectangle PQRS has P(−2, −1), Q(2, 1), R(1, 3), S(−3, 1). Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q2/5

True or false?

Once a quadrilateral is proved a rhombus, its area is the product of two adjacent side lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q3/5

A square has vertices (0, 0), (3, 1), (2, 4), (−1, 3). Its area is

A. 16

B. √10

C. 10

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q4/5

Explain why proving the shape first matters before using a special area formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q5/5

Prove or disprove that a point lies on a circle, and other coordinate proofs

MAIN	Geometry main question	**HSG-GPE.B.4**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	The equation of a circle	**HSG-GPE.A.1**
BACKWARD 3	Substitute values into an expression	**6.EE.A.2**
FORWARD 1	Coordinate proofs about circles with variables	**HSG-GPE.B.4**
FORWARD 2	Find the tangent line to a circle at a point	**HSG-GPE.B.5**

SET 68 • HSG-GPE.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A circle is centered at (2, −1) and contains the point (2, 4). Prove or disprove that the point (5, 3) lies on the circle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q1/5

Prove or disprove that (5, 2) lies on the circle with center (1, −1) and radius 5.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q2/5

Prove or disprove that $(-3, 6)$ lies on the circle $(x - 1)^2 + (y - 2)^2 = 30$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q3/5

Right triangle ABC has A(0, 0), B(8, 0), C(0, 6). Prove that the midpoint M of hypotenuse BC is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q4/5

Prove that the point $(4, -3)$ lies outside the circle $x^2 + y^2 = 20$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q5/5

SET 68 • HSG-GPE.B.4

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

Distance from $(1, -1)$ to $(5, 2)$?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q1/5

Distance from $(2, -1)$ to $(5, 3)$?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q2/5

Distance from (0, 0) to (−3, 4) is

A. 5

B. 1

C. 7

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q3/5

True or false?

$(-3)^2 = 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q4/5

True or false?

The distance from (2, 2) to (2, 7) is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSG-GPE.B.4

BACKWARD 2

The equation of a circle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Write the equation of the circle with center $(1, -1)$ and radius 5.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q1/5

What is the radius of $(x - 1)^2 + (y - 2)^2 = 30$?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q2/5

Center of $(x + 4)^2 + y^2 = 9$?

A. $(4, 0)$

B. $(-4, 0)$

C. $(-4, 9)$

D. $(0, -4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q3/5

True or false?

A point satisfies a circle's equation exactly when it is on the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q4/5

True or false?

The radius of $x^2 + y^2 = 20$ is 20.

A. True

B. False

SET 68 • HSG-GPE.B.4

BACKWARD 3

Substitute values into an expression • Grade 6 • Expressions & Equations • 6.EE.A.2

Evaluate $(x - 1)^2 + (y - 2)^2$ when $x = -3$ and $y = 6$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q1/5

Evaluate $x^2 + y^2$ when $x = 4$, $y = -3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q2/5

$(x + 1)^2$ when $x = -4$ is

A. -9

B. 9

C. 25

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q3/5

True or false?

$(-5 - 1)^2 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q4/5

True or false?

$(2 + 1)^2 = 2^2 + 1^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q5/5

SET 68 • HSG-GPE.B.4

FORWARD 1

Coordinate proofs about circles with variables • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A(−r, 0) and B(r, 0) are endpoints of a diameter of the circle $x^2 + y^2 = r^2$, and P(x, y) is any other point on the circle. Prove that ∠APB is a right angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q1/5

Show that if (a, b) is on $x^2 + y^2 = r^2$, then (b, a) and (−a, b) are too.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q2/5

True or false?

A coordinate proof with variables covers every case at once.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q3/5

To prove four points lie on one circle, show they are

A. all the same distance from one point

B. the same distance from the origin only

C. vertices of a rectangle only

D. all on one line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q4/5

Prove or disprove: A(1, 0), B(0, 1), C(−1, 0), D(0, −2) lie on one circle centered at the origin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSG-GPE.B.4

FORWARD 2

Find the tangent line to a circle at a point • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Find the equation of the line tangent to $x^2 + y^2 = 25$ at (3, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q1/5

Find the tangent to $(x - 1)^2 + (y - 1)^2 = 25$ at $(4, 5)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q2/5

True or false?

The tangent at P is parallel to the radius drawn to P.

A. True

B. False

The radius to (0, 5) on $x^2 + y^2 = 25$ is vertical, so the tangent there is

A. $x = 0$

B. $x = 5$

C. $y = x + 5$

D. $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q4/5

Explain how the slope of the tangent at (a, b) on $x^2 + y^2 = r^2$ is found.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q5/5

Prove the slope criteria for parallel and perpendicular lines

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Similar slope triangles give the same slope	**8.EE.B.6**
BACKWARD 2	Rotate a point 90° about the origin	**HSG-CO.A.5**
BACKWARD 3	Negative reciprocals	**7.NS.A.2**
FORWARD 1	Use the criteria to classify pairs of lines	**HSG-GPE.B.5**
FORWARD 2	Use the criteria inside a general coordinate proof	**HSG-GPE.B.4**

SET 69 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Prove that two distinct nonvertical lines with the same slope m are parallel.

Work space

Conclusion: _____

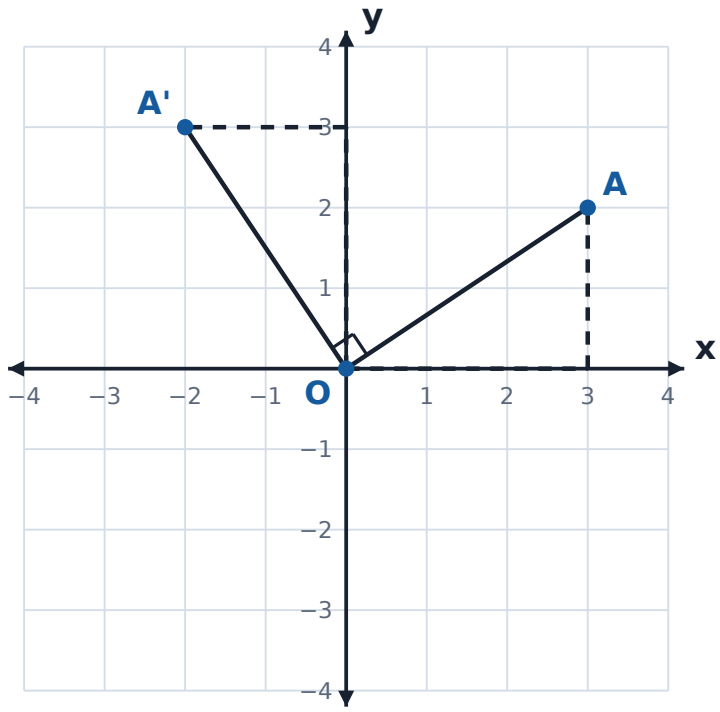
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q1/5

Line OA passes through O(0, 0) and A(a, b) with a, b ≠ 0. Rotating A 90° counterclockwise about O gives A'(−b, a). Use this to prove that perpendicular lines have slopes whose product is −1. The diagram shows the case A(3, 2).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q2/5

Prove that two distinct nonvertical lines that are parallel have equal slopes.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q3/5

Prove that if two nonvertical lines have slopes m and $-1/m$ ($m \neq 0$), they are perpendicular. (Translate both lines so they pass through the origin first.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q4/5

Lines $y = mx$ and $y = -x/m$ ($m \neq 0$) pass through the origin O. Use the points P(1, m) and Q(−m, 1) and the converse of the Pythagorean Theorem to prove that the lines are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 1

Similar slope triangles give the same slope • Grade 8 • Expressions & Equations • 8.EE.B.6

True or false?

Any two slope triangles on the same line are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q1/5

A line passes through (0, 0) and (2, 3). Find the rise for a run of 4.

Rise: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q2/5

Slope triangles with run 2, rise 3 and run 6, rise 9 give slopes

A. 6 and 18

B. 3/2 and 9/6 that differ

C. both 3/2

D. 2/3 and 6/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q3/5

Explain why the slope of a line is the same between any two of its points.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q4/5

True or false?

The slope between two points on a line depends on which points are chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 2

Rotate a point 90° about the origin • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Rotate (3, 2) 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q1/5

Rotate (1, 4) 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q2/5

The rule for a 90° counterclockwise rotation about the origin is

A. $(x, y) \to (-y, x)$

B. $(x, y) \to (-x, -y)$

C. $(x, y) \to (y, x)$

D. $(x, y) \to (y, -x)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q3/5

True or false?

A 90° rotation maps a line to a perpendicular line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q4/5

True or false?

A 90° rotation maps (2, 0) to (0, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 3

Negative reciprocals • Grade 7 • The Number System • 7.NS.A.2

Negative reciprocal of 2/3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q1/5

Multiply $(b/a)(-a/b)$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q2/5

Negative reciprocal of -4?

A. $\frac{1}{4}$

B. $-\frac{1}{4}$

C. 4

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q3/5

True or false?

A number times its negative reciprocal is -1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q4/5

True or false?

The negative reciprocal of 1 is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q5/5

SET 69 • HSG-GPE.B.5

FORWARD 1

Use the criteria to classify pairs of lines • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Are $y = 3x - 2$ and $6x - 2y = 7$ parallel, perpendicular, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q1/5

Are $2x + 5y = 1$ and $5x - 2y = 4$ parallel, perpendicular, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q2/5

True or false?

A horizontal line and a vertical line are perpendicular even though the slope product rule cannot be used.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q3/5

$y = \frac{1}{2}x + 1$ and $y = 2x - 1$ are

A. neither

B. the same line

C. perpendicular

D. parallel

Find k so that $y = kx + 1$ is perpendicular to $y = -\frac{1}{3}x$.

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q5/5

SET 69 • HSG-GPE.B.5

FORWARD 2

Use the criteria inside a general coordinate proof • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Rhombus OABC has O(0, 0), A(a, 0), B(a + b, c), C(b, c), with $b^2 + c^2 = a^2$. Prove that its diagonals are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q1/5

A coordinate proof places rhombus OABC at O(0, 0), A(a, 0), B(a + b, c), C(b, c). Why must the proof assume $b^2 + c^2 = a^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q2/5

True or false?

The slope criterion turns "perpendicular" into an algebra statement.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q3/5

In a coordinate proof that lines are parallel, you show

A. equal slopes

B. equal intercepts

C. slope product -1

D. equal lengths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q4/5

Prove or disprove with slopes: A(0, 0), B(4, 2), C(3, 4) form a right angle at B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q5/5

Write equations of parallel and perpendicular lines

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Read the slope of a line	**8.F.B.4**
BACKWARD 2	Rewrite an equation in slope-intercept form	**HSA-CED.A.4**
BACKWARD 3	Write a line from a point and a slope	**HSF-LE.A.2**
BACKWARD 4	Opposite reciprocals	**7.NS.A.2**
FORWARD 1	Write perpendicular bisectors and altitudes	**HSG-GPE.B.5**
FORWARD 2	Find the distance from a point to a line	**HSG-GPE.B.4**

SET 70 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Write the equation of the line parallel to $y = 3x - 5$ that passes through (2, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q1/5

Write the equation of the line perpendicular to $y = \frac{2}{3}x + 1$ that passes through $(4, -1)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q2/5

Write the equation of the line parallel to $2x + 5y = 10$ that passes through $(5, 3)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q3/5

Write the equation of the line perpendicular to $x = 4$ that passes through $(1, -3)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q4/5

Which line is perpendicular to $y = -4x + 1$ and passes through the origin?

A. $y = -4x$

B. $y = -\frac{1}{4}x$

C. $y = \frac{1}{4}x$

D. $y = 4x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 1

Read the slope of a line • Grade 8 • Functions • 8.F.B.4

Slope of $y = -2x + 7$?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q1/5

Slope of $y = 5$?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q2/5

Slope of $y = \frac{2}{3}x - 4$?

A. 3/2

B. $\frac{2}{3}$

C. -4

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q3/5

True or false?

In $y = mx + b$, m is the slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q4/5

True or false?

The slope of $x = 3$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 2

Rewrite an equation in slope-intercept form • High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4

Solve 2x + 5y = 10 for y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q1/5

Solve $3x - y = 4$ for y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q2/5

$6x + 3y = 9$ in slope-intercept form is

A. $y = 2x + 3$

B. $y = -6x + 9$

C. $y = -2x + 9$

D. $y = -2x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q3/5

True or false?

$x - 2y = 6$ has slope ½.

A. True

B. False

True or false?

$4x + 2y = 8$ has slope 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 3

Write a line from a point and a slope • High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the line with slope 3 through (2, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q1/5

Write the line with slope -1 through $(0, 6)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q2/5

Slope 2 through (1, 5): b =

A. 5

B. 7

C. −3

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B3-Q3**

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q3/5

True or false?

$y - 4 = 3(x - 2)$ is the line with slope 3 through (2, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q4/5

True or false?

A line through (0, 6) has y-intercept 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 4

Opposite reciprocals • Grade 7 • The Number System • 7.NS.A.2

Negative reciprocal of ⅔?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q1/5

Negative reciprocal of −5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q2/5

Negative reciprocal of $-2/5$?

A. $5/2$

B. $2/5$

C. $-2/5$

D. $-5/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q3/5

True or false?

The negative reciprocal of ¼ is −4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q4/5

True or false?

The negative reciprocal of 3 is −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q5/5

SET 70 • HSG-GPE.B.5

FORWARD 1

Write perpendicular bisectors and altitudes • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Write the equation of the perpendicular bisector of the segment from (2, 1) to (6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q1/5

Triangle ABC has A(0, 0), B(6, 0), C(2, 4). Write the equation of the altitude from A to BC.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q2/5

True or false?

A perpendicular bisector passes through the midpoint of the segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q3/5

The perpendicular bisector of the segment from (0, 0) to (0, 8) is

A. $x = 0$

B. $y = 4$

C. $x = 4$

D. $y = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q4/5

Write the perpendicular bisector of the segment from $(-3, 2)$ to $(5, 2)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSG-GPE.B.5

FORWARD 2

Find the distance from a point to a line • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Find the distance from P(5, 5) to the line $y = -x + 2$. Give the exact value.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q1/5

Find the distance from (0, 0) to the line y = 4.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q2/5

True or false?

The distance from a point to a line can be measured along any segment from the point to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q3/5

The foot of the perpendicular from (3, 0) to y = x is

A. (0, 3)

B. (0, 0)

C. (3, 3)

D. (1.5, 1.5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q4/5

Find the distance from (6, 1) to the line $y = 2x - 1$. Give the exact value.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q5/5

SET 71 • HSG-GPE.B.5 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Solve geometric problems with the slope criteria

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Slope from two points	**8.F.B.4**
BACKWARD 2	Solve a linear equation for an unknown	**HSA-REI.B.3**
BACKWARD 3	Equations of parallel and perpendicular lines	**HSG-GPE.B.5**
BACKWARD 4	Opposite sides of a parallelogram are parallel	**HSG-CO.C.11**
FORWARD 1	Find the orthocenter of a triangle	**HSG-GPE.B.5**
FORWARD 2	Shortest path to a straight line in a design	**HSG-MG.A.3**

SET 71 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

The line through (1, 2) and (4, k) is parallel to $y = 2x + 7$. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q1/5

ABCD is a parallelogram with A(0, 0), B(5, 1), C(7, 4). Find D.

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q2/5

The segment from (2, 1) to (5, k) is perpendicular to the line y = 3x. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q3/5

Triangle ABC has A(0, 0), B(8, 4), C(4, 6). Find the foot of the altitude from C to AB.

Foot: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q4/5

Triangle ABC has A(1, 1), B(4, 2), C(3, 5). At which vertex is the right angle?

A. C

B. B

C. A

D. There is no right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 1

Slope from two points • Grade 8 • Functions • 8.F.B.4

Slope through (1, 2) and (4, 8)?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q1/5

Slope through (4, 2) and (3, 5)?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q2/5

Slope through (0, 0) and (8, 4)?

A. ½

B. 8

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q3/5

True or false?

The slope from A to B equals the slope from B to A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q4/5

True or false?

The slope through (1, 1) and (1, 5) is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 2

Solve a linear equation for an unknown • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $(k - 2)/3 = 2$.

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q1/5

Solve ½x = −2x + 14.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q2/5

Solve $(k - 1)/3 = -\frac{1}{3}$.

A. -2

B. 1

C. 2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q3/5

True or false?

Multiplying both sides by 3 is a valid first step in $(k - 2)/3 = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q4/5

True or false?

The solution of $2x + 1 = 7$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 3

Equations of parallel and perpendicular lines • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Line perpendicular to y = ½x through (4, 6)?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q1/5

Line parallel to $y = -x$ through $(0, 3)$?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q2/5

Perpendicular slope to ⅓?

A. 3

B. −3

C. −⅓

D. ⅓

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q3/5

True or false?

An altitude is perpendicular to the side it meets.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q4/5

True or false?

An altitude passes through the midpoint of the opposite side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 4

Opposite sides of a parallelogram are parallel • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

In a parallelogram, both pairs of opposite sides are parallel and congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q1/5

A parallelogram has A(0, 0), B(3, 0), C(4, 2). Find D (in order ABCD).

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q2/5

In parallelogram ABCD, side AB is parallel to

A. AC

B. BC

C. AD

D. DC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q3/5

In parallelogram ABCD, going from B to C is "right 2, up 3". Going from A to D is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q4/5

True or false?

A parallelogram's adjacent sides are always perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q5/5

SET 71 • HSG-GPE.B.5

FORWARD 1

Find the orthocenter of a triangle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Triangle ABC has A(0, 0), B(6, 0), C(2, 4). Find the orthocenter (the intersection of the altitudes).

Orthocenter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q1/5

Triangle with A(0, 0), B(4, 0), C(0, 3). Where is the orthocenter?

Orthocenter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q2/5

True or false?

The orthocenter of a right triangle is at the right-angle vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q3/5

To find the orthocenter, you need at least

A. three medians

B. the circumcenter

C. one altitude

D. two altitudes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q4/5

Check that the third altitude of the triangle A(0, 0), B(6, 0), C(2, 4) also passes through (2, 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSG-GPE.B.5

FORWARD 2

Shortest path to a straight line in a design • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A well is at W(2, 9). A straight canal runs along $y = 2x - 1$ (units: 100 m). Where should a pipe from the well meet the canal so the pipe is as short as possible, and how long is it (to the nearest hundredth of a unit)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q1/5

A pipe will connect a well to a straight canal. Why is the shortest possible pipe perpendicular to the canal?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q2/5

True or false?

A pipe meeting the canal at a 60° angle is shorter than one meeting it at 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q3/5

A path from (0, 5) to the road $y = 0$ is shortest at

A. (0, 0)

B. (5, 0)

C. (−5, 0)

D. (0, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q4/5

A sensor at (3, 4) must connect to the line y = x. Where should the wire meet the line?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q5/5

Partition a directed segment in a given ratio

MAIN	Geometry main question	**HSG-GPE.B.6**
BACKWARD 1	A ratio as a fraction of the whole	**6.RP.A.1**
BACKWARD 2	Horizontal and vertical change	**6.NS.C.8**
BACKWARD 3	Find a fraction of a number	**6.NS.A.1**
FORWARD 1	Work backward: find an endpoint or the ratio	**HSG-GPE.B.6**
FORWARD 2	The centroid as a partition point	**HSG-GPE.B.4**

SET 72 • HSG-GPE.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the point P on the directed segment from A(1, 2) to B(13, 8) that partitions it in the ratio 1 : 2.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q1/5

Find the point P on the directed segment from A(−4, 6) to B(6, 1) that partitions it in the ratio 3 : 2.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q2/5

Find the point P on the directed segment from A(2, −3) to B(−6, 9) that partitions it in the ratio 1 : 3.

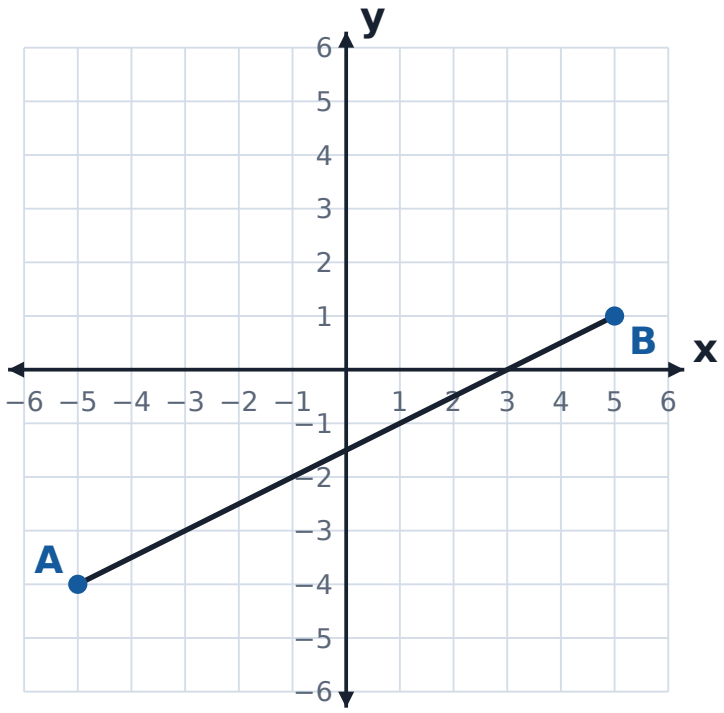
P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q3/5

Find the point P on the directed segment from A to B that partitions it in the ratio 2 : 3.



P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q4/5

Point P partitions the directed segment from B(10, 5) to A(0, 0) in the ratio 1 : 4. P is

A. (8, 4)

B. (2, 1)

C. (5, 2.5)

D. (4, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 1

A ratio as a fraction of the whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

A ratio of 2 : 3 splits a whole into how many equal parts?

Parts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q1/5

In a 1 : 2 partition, the first part is what fraction of the whole?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q2/5

In a 3 : 2 partition, the first part is

A. 3/2 of the whole

B. 2/3 of the whole

C. 3/5 of the whole

D. 2/5 of the whole

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q3/5

True or false?

A ratio of 1 : 3 means the point is ¼ of the way along.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q4/5

True or false?

A ratio of 1 : 2 means the point is halfway.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 2

Horizontal and vertical change • Grade 6 • The Number System • 6.NS.C.8

From (1, 2) to (13, 8), what is the change in x?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q1/5

From $(-4, 6)$ to $(6, 1)$, what is the change in y?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q2/5

From $(2, -3)$ to $(-6, 9)$, the change in x is

A. -8

B. -4

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q3/5

True or false?

Change in x is the ending x minus the starting x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q4/5

True or false?

The change in y from (0, 5) to (0, 1) is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 3

Find a fraction of a number • Grade 6 • The Number System • 6.NS.A.1

Find ⅓ of 12.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q1/5

Find 3/5 of −5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q2/5

2/5 of 10 is

A. 5

B. 25

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q3/5

True or false?

¼ of −8 is −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q4/5

True or false?

⅔ of 9 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q5/5

SET 72 • HSG-GPE.B.6

FORWARD 1

Work backward: find an endpoint or the ratio • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

P(3, 4) partitions the directed segment from A(1, 2) to B in the ratio 1 : 2. Find B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q1/5

P(4, 3) is on the segment from A(0, 1) to B(10, 6). In what ratio does P partition AB?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q2/5

True or false?

The midpoint partitions a segment in the ratio $1 : 1$.

A. True

B. False

P partitions AB in 3 : 1. Then P is closer to

A. B

B. neither

C. A

D. the origin

M(2, 5) is the midpoint of AB with A(−1, 1). Find B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSG-GPE.B.6

FORWARD 2

The centroid as a partition point • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Triangle with A(0, 0), B(6, 0), C(3, 9). Find the point ⅔ of the way from A to the midpoint of BC.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q1/5

Show that the centroid (the point 2 : 1 along a median from the vertex) of the triangle with vertices (x_1, y_1), (x_2, y_2), (x_3, y_3) is the average of the vertices.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q2/5

True or false?

The centroid of the triangle (0, 0), (6, 0), (3, 9) is (3, 4.5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q3/5

The centroid of (1, 2), (4, 8), (7, −1) is

A. (4, 3)

B. (4, 9)

C. (3, 3)

D. (12, 9)

Find the centroid of the triangle with vertices $(-2, 5)$, $(4, 1)$, $(7, 0)$.

Centroid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q5/5

SET 73 • HSG-GPE.B.7 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Compute perimeters of polygons with coordinates

MAIN	Geometry main question	**HSG-GPE.B.7**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	Lengths of horizontal and vertical sides	**6.G.A.3**
BACKWARD 3	Simplify square roots	**HSN-RN.A.2**
FORWARD 1	Find a missing coordinate from a perimeter	**HSG-GPE.B.7**
FORWARD 2	Fencing and cost from a coordinate plan	**HSG-MG.A.3**

SET 73 • HSG-GPE.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the perimeter of the triangle with vertices (0, 0), (3, 4), and (3, 0).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q1/5

Find the perimeter of the quadrilateral with vertices A(1, 1), B(4, 5), C(10, −3), D(7, −7).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q2/5

Find the perimeter of the triangle with vertices (0, 0), (4, 0), (2, 4). Give the exact value and a decimal to the nearest tenth.

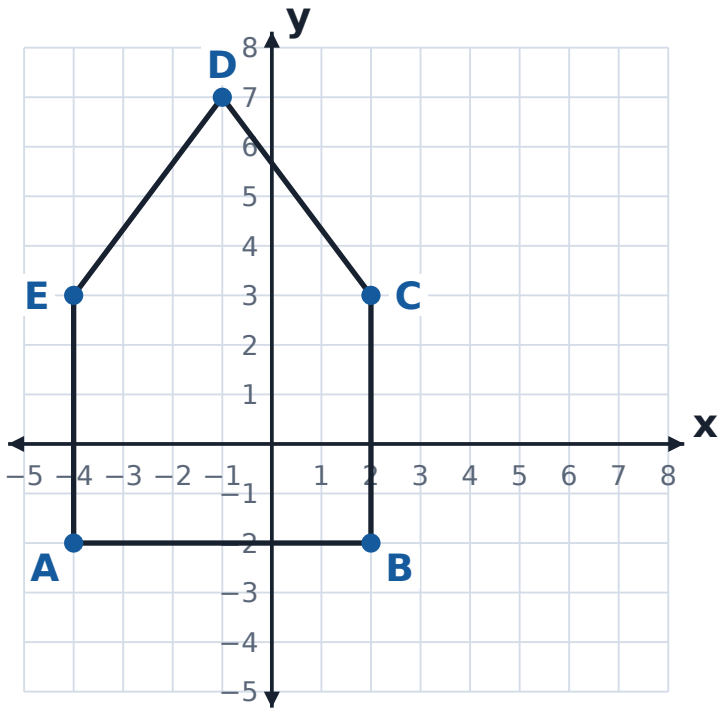
Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q3/5

Find the perimeter of pentagon ABCDE.



Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q4/5

A square has vertices (0, 0), (3, 1), (2, 4), (−1, 3). Its perimeter is

A. $4\sqrt{10}$

B. 10

C. 40

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

Distance from (2, 3) to (−1, 7)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q1/5

Distance from (4, 5) to (10, −3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q2/5

Distance from (0, 0) to (2, 4)?

A. 20

B. $2\sqrt{5}$

C. 6

D. $\sqrt{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q3/5

True or false?

The distance from (0, 0) to (3, 1) is √10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q4/5

True or false?

Distances can be negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 2

Lengths of horizontal and vertical sides • Grade 6 • Geometry • 6.G.A.3

Length of the side from $(-4, -2)$ to $(2, -2)$?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q1/5

Length of the side from (2, −2) to (2, 3)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q2/5

Length from $(-3, 4)$ to $(-3, -1)$?

A. 7

B. 3

C. 5

D. -5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q3/5

True or false?

A horizontal side's length is the absolute value of the difference of the x-coordinates of its endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q4/5

True or false?

The side from (1, 1) to (1, 9) has length 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 3

Simplify square roots • High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify $\sqrt{20}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q1/5

Simplify $\sqrt{50}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q2/5

$2\sqrt{5} + 2\sqrt{5} =$

A. $2\sqrt{10}$

B. $8\sqrt{5}$

C. $4\sqrt{10}$

D. $4\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-B3-Q3**

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q3/5

True or false?

$\sqrt{10} \approx 3.16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q4/5

True or false?

$\sqrt{9} + \sqrt{16} = \sqrt{25}$

A. True

B. False

SET 73 • HSG-GPE.B.7

FORWARD 1

Find a missing coordinate from a perimeter • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

A rectangle has vertices (0, 0), (k, 0), (k, 3), (0, 3) with $k > 0$ and perimeter 22. Find k.

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q1/5

Triangle with vertices (0, 0), (6, 0), (3, h), h > 0, has perimeter 16. Find h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q2/5

True or false?

The triangle with vertices (0, 0), (6, 0), (3, h) has the same perimeter for h = 4 and for h = −4.

A. True

B. False

A square has vertices (0, 0), (a, 0), (a, a), (0, a) with $a > 0$ and perimeter 18. Find a.

A. 4

B. 3

C. 9

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q4/5

A right triangle has vertices (0, 0), (8, 0), (8, k) with k > 0 and perimeter 24. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSG-GPE.B.7

FORWARD 2

Fencing and cost from a coordinate plan • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A garden plan has corners (0, 0), (8, 0), (8, 6), (0, 6), in meters. Fencing costs $12 per meter. Find the cost to fence it.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q1/5

A triangular lot has corners (0, 0), (30, 0), (30, 40) in feet. Edging costs $2.50 per foot. Find the total cost.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q2/5

True or false?

On a map with 1 unit = 5 m, an area of 12 square units is 60 m².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q3/5

A path has corners (0, 0), (3, 4), (9, 12) (units: 10 m). Its length is

A. 150 m

B. 210 m

C. 15 m

D. 21 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q4/5

A run around a lot with corners (0, 0), (5, 12), (5, 0) in units of 100 m: how many meters?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q5/5

SET 74 • HSG-GPE.B.7 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Compute areas of triangles and rectangles with coordinates

MAIN	Geometry main question	**HSG-GPE.B.7**
BACKWARD 1	Area of a triangle from base and height	**6.G.A.1**
BACKWARD 2	Lengths on the coordinate plane	**6.G.A.3**
BACKWARD 3	Distance between two points	**8.G.B.8**
FORWARD 1	Find a missing vertex from an area	**HSG-GPE.B.7**
FORWARD 2	Use area from coordinates with density	**HSG-MG.A.2**

SET 74 • HSG-GPE.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the area of the triangle with vertices (0, 0), (8, 0), and (3, 5).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q1/5

Rectangle ABCD has A(1, 1), B(5, 3), C(4, 5), D(0, 3). Find its area.

Area: _____

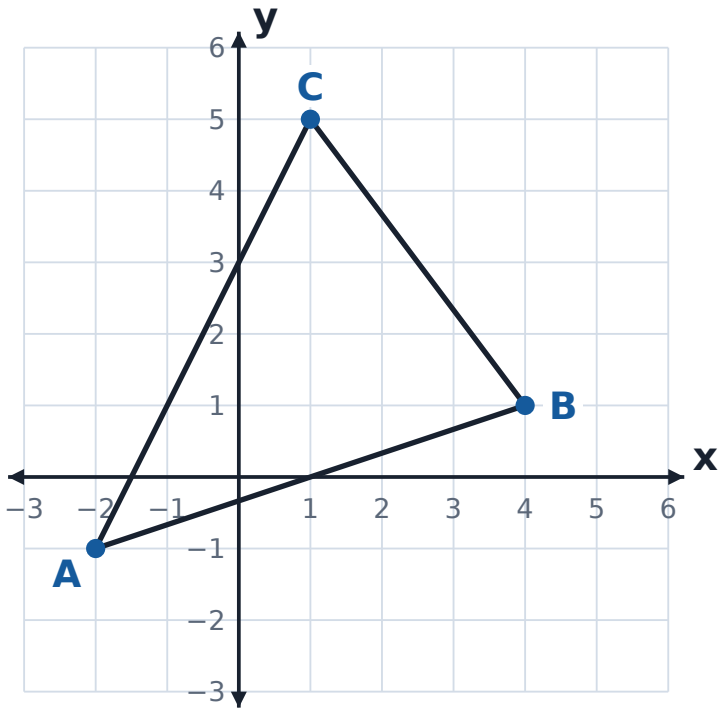
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q2/5

Find the area of triangle ABC.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q3/5

Find the area of the triangle with vertices A(0, 0), B(6, 2), C(2, 6).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q4/5

A triangle has vertices (0, 0), (10, 0), and (x, 4). What is its area?

A. 20, for any x

B. It depends on x

C. 40

D. 20 only if x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q5/5

SET 74 • HSG-GPE.B.7

BACKWARD 1

Area of a triangle from base and height • Grade 6 • Geometry • 6.G.A.1

Area of a triangle with base 8 and height 5?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q1/5

Area of a right triangle with legs 6 and 2?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q2/5

A =

A. ½bh

B. b + h

C. 2bh

D. bh

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q3/5

True or false?

The height of a triangle is perpendicular to the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q4/5

True or false?

The height must be one of the sides.

A. True

B. False

SET 74 • HSG-GPE.B.7

BACKWARD 2

Lengths on the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Length from $(-2, -1)$ to $(4, -1)$?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q1/5

Height of the point (3, 5) above the x-axis?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q2/5

Width of the rectangle enclosing x from −2 to 4?

A. 2

B. 8

C. 4

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q3/5

True or false?

A point with y-coordinate −3 is 3 units from the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q4/5

True or false?

The length from (0, 2) to (0, 9) is 11.

A. True

B. False

SET 74 • HSG-GPE.B.7

BACKWARD 3

Distance between two points • Grade 8 • Geometry • 8.G.B.8

Distance from (1, 1) to (5, 3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q1/5

Distance from (5, 3) to (4, 5)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q2/5

$\sqrt{20} \cdot \sqrt{5} =$

A. 10

B. 25

C. $\sqrt{25}$

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q3/5

True or false?

Slanted rectangle sides need the distance formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q4/5

True or false?

$\sqrt{20} \cdot \sqrt{5} = 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q5/5

SET 74 • HSG-GPE.B.7

FORWARD 1

Find a missing vertex from an area • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Triangle with vertices (0, 0), (6, 0), (2, k) has area 18. Find all possible k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q1/5

A rectangle has vertices (0, 0), (a, 0), (a, 5), (0, 5), $a > 0$, and area 35. Find a.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q2/5

True or false?

Moving the top vertex of a triangle parallel to its base does not change its area.

A. True

B. False

Triangle (0, 0), (4, 0), (x, 6). For which x is the area 12?

A. Any x

B. No x

C. x = 4 only

D. x = 2 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q4/5

Triangle (1, 1), (7, 1), (4, k) has area 21 and k > 1. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q5/5

SET 74 • HSG-GPE.B.7

FORWARD 2

Use area from coordinates with density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A park on a map has corners (0, 0), (4, 0), (4, 3), (0, 3), with 1 unit = 1 km. 1,800 people live there. Find the population density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q1/5

A triangular field has vertices (0, 0), (200, 0), (0, 150) in meters. Seeds are planted at 4 per m². How many seeds?

Seeds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q2/5

True or false?

Density based on area is an area divided by a quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q3/5

A lot with corners (0, 0), (6, 0), (6, 5), (0, 5) (meters) holds 60 plants. Density?

A. 2 plants per m²

B. 30 plants per m²

C. 11 plants per m²

D. 0.5 plants per m²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q4/5

A rectangle with corners (0, 0), (3, 0), (3, 2), (0, 2) (miles) has 4,200 residents. Find the density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q5/5

SET 75 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the circumference of a circle

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Circumference formulas	**7.G.B.4**
BACKWARD 4	Perimeters of regular polygons	**3.MD.D.8**
BACKWARD 3	All circles are similar	**HSG-C.A.1**
FORWARD 1	Trap π between inscribed and circumscribed polygons	**HSG-GMD.A.1**
FORWARD 2	Extend the circumference argument to arcs and curved boundari…	**HSG-GMD.A.1**

SET 75 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Regular hexagons are inscribed in and circumscribed about a circle of radius 1. Their perimeters are 6 and $4\sqrt{3} \approx 6.93$. Explain why this shows that $3 < \pi < 2\sqrt{3} \approx 3.46$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q1/5

Explain why the ratio of circumference to diameter is the same number (π) for every circle, so that $C = \pi d = 2\pi r$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q2/5

A wheel with diameter 70 cm rolls one full turn along level ground without slipping, and a student measures the distance traveled as 219.9 cm. Explain why that distance is the circumference, use the measurement to estimate π, and say why a measurement supports the formula but does not prove it.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q3/5

A regular hexagon is inscribed in a circle of radius 5. Find the hexagon's perimeter, compare it with the circle's circumference, and explain what happens as the number of sides increases.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q4/5

Regular polygons with 6, 12 and 24 sides are inscribed in a circle of radius 1. Their perimeters are 6, 6.212 and 6.265. Explain what these values suggest about the circumference and how the formula $C = 2\pi r$ follows for a circle of any radius r.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 1

Circumference formulas • Grade 7 • Geometry • 7.G.B.4

Find the exact circumference of a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q1/5

Find the exact circumference of a circle of diameter 9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q2/5

Half the circumference of a circle of radius r is

A. πr^2

B. πr

C. $2\pi r$

D. $r/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q3/5

True or false?

$C = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q4/5

True or false?

$A = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 4

Perimeters of regular polygons • Grade 3 • Measurement & Data • 3.MD.D.8

Find the perimeter of a regular hexagon with side 5.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B4-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q1/5

Find the perimeter of a regular octagon with side 2.5 cm.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B4-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q2/5

A regular 12-gon has perimeter 36. Each side is

A. 432

B. 3

C. 12

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B4-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q3/5

True or false?

A regular polygon with n sides of length s has perimeter ns.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B4-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q4/5

True or false?

A regular pentagon with side 4 has perimeter 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B4-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 3

All circles are similar • High School — Geometry • Geometry: Circles • HSG-C.A.1

True or false?

Any circle can be mapped to any other circle by a translation and a dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q1/5

A circle of radius 2 is dilated to radius 6. By what factor is the circumference multiplied?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q2/5

Circles of radii 3 and 12 have circumferences in the ratio

A. 1 : 16

B. 3 : 4

C. 1 : 4

D. 1 : 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q3/5

Why does C/d not depend on the size of the circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q4/5

True or false?

A larger circle has a larger value of π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q5/5

SET 75 • HSG-GMD.A.1

FORWARD 1

Trap π between inscribed and circumscribed polygons • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A circle has radius 1. Find the perimeters of the inscribed square and the circumscribed square, and use them to write bounds for π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q1/5

Inscribed and circumscribed regular 12-gons of a circle of radius 1 have perimeters about 6.212 and 6.431. What bounds for π do they give (to the nearest hundredth)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q2/5

True or false?

Using polygons with more sides gives closer bounds for π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q3/5

Why is the circumscribed polygon's perimeter larger than the circumference?

A. It has fewer sides

B. The circle lies inside it, and a curve inside a convex shape is shorter than the shape's boundary

C. Its radius is smaller

D. Its sides are curved

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q4/5

Archimedes used 96-gons and found $3\ 10/71 < \pi < 3\ 1/7$. Write these bounds as decimals to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q5/5

SET 75 • HSG-GMD.A.1

FORWARD 2

Extend the circumference argument to arcs and curved boundaries • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Use the circumference formula to explain why an arc with central angle 90° in a circle of radius 6 has length 3π, and find its length to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q1/5

Explain why the length of an arc is proportional to its central angle, so that an arc of θ° has length (θ/360) · 2πr.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q2/5

True or false?

A semicircular region of radius 5 has perimeter 5π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q3/5

A running track has two straight sides of 100 m and two semicircular ends of radius 30 m. Its length is about

A. 294 m

B. 200 m

C. 388π m

D. 388 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q4/5

The circumference argument used inscribed polygons whose perimeters approach $2\pi r$. Explain why inscribed broken lines (chords) can be used the same way to measure the length of an arc.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q5/5

SET 103 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the area of a circle

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	Area of a parallelogram	**6.G.A.1**
BACKWARD 3	Scale factors and area	**7.G.A.1**
FORWARD 1	Area of a circle as a limit of regular polygons	**HSG-GMD.A.1**
FORWARD 2	From area to volume: stacking disks	**HSG-GMD.A.1**

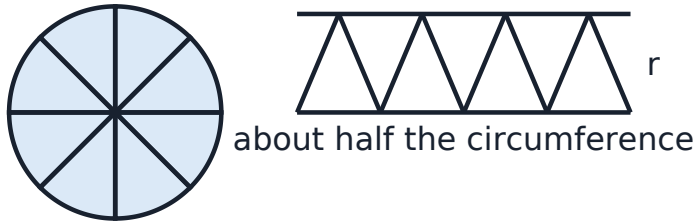
SET 103 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A circle of radius r is cut into many equal sectors, which are rearranged as shown. Explain why this shows that the area of the circle is πr^2.

Work space



Explanation: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q1/5

A circle of radius r is cut into many thin rings, like the layers of an onion. Each ring is cut open and straightened into a thin strip, and the strips are stacked from shortest (at the top) to longest (at the bottom). Explain why the stack approaches a triangle and why its area is πr^2.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q2/5

Use similarity to explain why the area of a circle of radius r equals r^2 times the area of a circle of radius 1. Then use the fact that the unit circle has area π to write the formula.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q3/5

A quarter circle of radius 10 is drawn on a grid of unit squares. Counting full squares and estimating the partial ones gives about 78.5 squares. Explain how this supports $A = \pi r^2$ and why counting is evidence rather than proof.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q4/5

Explain why the area of a circle of radius r is more than $2r^2$ and less than $4r^2$, using an inscribed square and a circumscribed square, and check that πr^2 fits these bounds.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q5/5

SET 103 • HSG-GMD.A.1

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

Find the exact area of a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q1/5

Find the exact area of a circle of diameter 12.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q2/5

A circle has area 25π. Its radius is

A. 5

B. 12.5

C. 25

D. $\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q3/5

True or false?

A circle of radius 3 has area 9π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q4/5

True or false?

Tripling the radius triples the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q5/5

SET 103 • HSG-GMD.A.1

BACKWARD 2

Area of a parallelogram • Grade 6 • Geometry • 6.G.A.1

Area of a parallelogram with base 9 and height 4?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q1/5

True or false?

A parallelogram has the same area as a rectangle with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q2/5

A parallelogram's area is

A. base × slanted side

B. ½ base × height

C. base × height

D. base + height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q3/5

A parallelogram has base πr and height r. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q4/5

True or false?

The slanted side is the height of a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q5/5

SET 103 • HSG-GMD.A.1

BACKWARD 3

Scale factors and area • Grade 7 • Geometry • 7.G.A.1

A scale drawing uses factor 3. A region of 5 cm² on the original covers how much on the drawing?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B3-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q1/5

A square of side 2 is enlarged by factor 4. New area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B3-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q2/5

Enlarging a figure by factor k multiplies its area by

A. k^3

B. k

C. $2k$

D. k^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B3-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q3/5

True or false?

Halving every length of a figure multiplies its area by ¼.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B3-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q4/5

True or false?

Enlarging by factor 3 triples the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B3-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q5/5

SET 103 • HSG-GMD.A.1

FORWARD 1

Area of a circle as a limit of regular polygons • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A regular n-gon is inscribed in a circle of radius r. Its area is ½ · perimeter · apothem. Explain why this approaches πr^2 as n increases.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q1/5

Find the area of a regular hexagon with side 2 and apothem $\sqrt{3}$. Compare with the area of its circumscribed circle (radius 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q2/5

True or false?

An inscribed polygon always has less area than its circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q3/5

As n grows, the apothem of an inscribed regular n-gon approaches

A. 0

B. 2r

C. r

D. πr

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q4/5

Why do mathematicians call this an "informal" argument?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q5/5

SET 103 • HSG-GMD.A.1

FORWARD 2

From area to volume: stacking disks • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A stack of 40 identical coins, each a disk of radius 1 cm and thickness 0.2 cm, forms a cylinder. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q1/5

True or false?

A cylinder can be thought of as a stack of congruent disks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q2/5

The volume of a stack of disks of area πr^2 and total height h is

A. $\pi r^2 h$

B. $2\pi rh$

C. πr^2

D. $\frac{1}{3}\pi r^2 h$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S103-F2-Q3**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q3/5

Explain how the area of a circle is used to find the volume of a cylinder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q4/5

Each disk in a stack has area 12 cm² and the stack is 5 cm high. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q5/5

SET 76 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal argument for the volume of a cylinder

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Volume of a rectangular prism as layers	**5.MD.C.5**
BACKWARD 2	Area of a circle	**7.G.B.4**
BACKWARD 3	Cross-sections parallel to the base	**7.G.A.3**
FORWARD 1	Cavalieri's principle with different shapes	**HSG-GMD.A.1**
FORWARD 2	Use the cylinder volume formula to solve problems	**HSG-GMD.A.3**

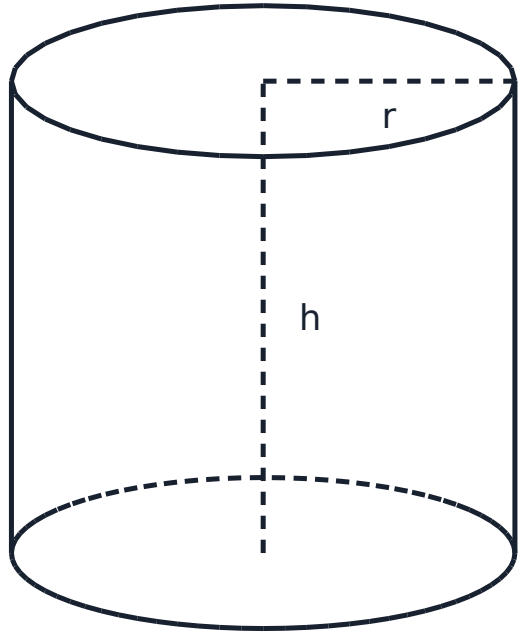
SET 76 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Explain why the volume of the cylinder shown is $V = \pi r^2 h$.

Work space



Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q1/5

A right cylinder and an oblique cylinder have the same radius r and the same height h. Explain why they have the same volume.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q2/5

A cylinder has radius r and height h. A square prism has base area πr^2 and the same height h. Use Cavalieri's principle to explain why the two solids have the same volume, and conclude the cylinder formula.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q3/5

Explain why V = Bh works for both prisms and cylinders.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q4/5

A cylinder of radius r and height h is approximated by prisms whose bases are regular polygons inscribed in its base circle. Explain why the prism volumes approach $\pi r^2 h$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 1

Volume of a rectangular prism as layers • Grade 5 • Measurement & Data • 5.MD.C.5

A prism has a 4 × 3 base and height 5. How many unit cubes are in one layer, and what is the volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q1/5

True or false?

V = Bh, where B is the area of the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q2/5

A prism has base area 20 and height 7. Volume?

A. 27

B. 140

C. 200

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q3/5

A prism has 6 layers of 10 cubes each. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q4/5

True or false?

Volume is measured in square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 2

Area of a circle • Grade 7 • Geometry • 7.G.B.4

Exact area of a circle of radius 3?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q1/5

Exact area of a circle of diameter 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q2/5

A =

A. $2\pi r$

B. πr

C. πr^2

D. πd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q3/5

True or false?

The base of a cylinder is a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q4/5

True or false?

The area of a circle of radius 2 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 3

Cross-sections parallel to the base • Grade 7 • Geometry • 7.G.A.3

True or false?

A slice of a right prism parallel to its base is congruent to the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q1/5

A cylinder is sliced parallel to its base. Describe the cross-section.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q2/5

A rectangular prism sliced parallel to its 3 × 4 base gives a

A. smaller rectangle

B. 3 × 4 rectangle

C. triangle

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q3/5

Why do all horizontal slices of a cylinder have the same area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q4/5

True or false?

Horizontal slices of a cylinder get smaller toward the top.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q5/5

SET 76 • HSG-GMD.A.1

FORWARD 1

Cavalieri's principle with different shapes • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A cylinder with base area 50 cm² and a square prism with base area 50 cm² both have height 8 cm. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q1/5

An oblique cylinder has radius 3 and vertical height 7 (its slanted side is 9). Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q2/5

True or false?

Cavalieri's principle requires equal cross-sectional areas at every height and equal heights.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q3/5

Two stacks of 30 identical cards, one straight and one slanted, have

A. different volumes

B. equal volumes

C. equal surface areas

D. no volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q4/5

State Cavalieri's principle in your own words.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSG-GMD.A.1

FORWARD 2

Use the cylinder volume formula to solve problems • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A can has radius 4 cm and height 12 cm. Find its volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q1/5

A pipe 2 m long has inner radius 0.1 m. Find the volume of water it holds, to the nearest thousandth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q2/5

True or false?

Doubling the height of a cylinder multiplies its volume by 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q3/5

Doubling the radius of a cylinder multiplies its volume by

A. 4

B. 8

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q4/5

A cylindrical tank has diameter 6 ft and height 10 ft. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q5/5

SET 77 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the volume of a pyramid

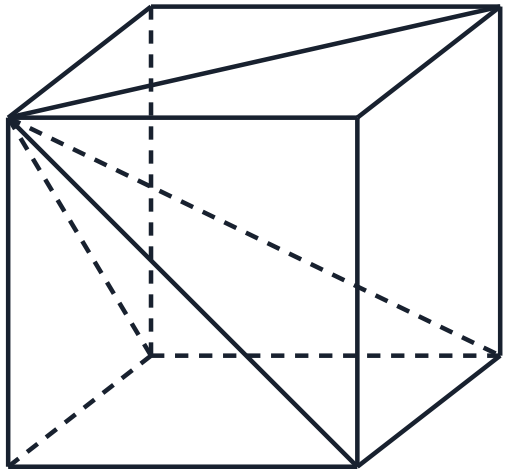
MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Volume of a prism	**7.G.B.6**
BACKWARD 2	Base area of a pyramid	**6.G.A.1**
BACKWARD 4	Split a volume into equal parts	**5.MD.C.5**
FORWARD 1	Cavalieri's principle for pyramids	**HSG-GMD.A.1**
FORWARD 2	Use the pyramid volume formula	**HSG-GMD.A.3**

SET 77 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A cube with edge s is split into three congruent pyramids that share one vertex of the cube as their apex, as shown. Use this, together with scaling and Cavalieri's principle, to explain why EVERY pyramid with base area B and height h has volume ⅓Bh.



Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q1/5

A triangular prism with base area B and height h is cut into three pyramids (tetrahedra), as Euclid did. Pairs of these pyramids have congruent bases and equal heights. Explain why each pyramid has volume ⅓Bh.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q2/5

In a class experiment, a hollow square pyramid and a hollow square prism have the same base and the same height. It takes three pyramids full of sand to fill the prism. Explain what this shows about the pyramid formula and why it is evidence, not a proof.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q3/5

A cube with edge 6 is split into three congruent pyramids that share one vertex of the cube as their apex. Find the volume of each pyramid, then show that ⅓Bh gives the same value for one pyramid, and explain why this agreement is a check, not a proof for all pyramids.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q4/5

A square pyramid has base side s and height h. Imagine it built from n stacked square slabs, each of thickness h/n, where the slab k steps from the top has side ks/n. The total slab volume is $(s^2h/n^3) \times$ (the sum of the squares 1^2, 2^2, 3^2, and so on up to n^2). For n = 10 and n = 100 this equals $0.385s^2h$ and $0.338s^2h$. Explain how this supports $V = \frac{1}{3}s^2h$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 1

Volume of a prism • Grade 7 • Geometry • 7.G.B.6

A prism has base area 36 and height 6. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q1/5

A cube has side 6. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q2/5

V =

A. Bh

B. ⅓Bh

C. 2Bh

D. B + h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q3/5

True or false?

A cube is a prism.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q4/5

True or false?

A prism's volume depends on the number of its faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 2

Base area of a pyramid • Grade 6 • Geometry • 6.G.A.1

A square base has side 10. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q1/5

A triangular base has base 6 and height 4. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q2/5

The base of a square pyramid is a

A. triangle

B. square

C. rectangle that is not a square

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q3/5

True or false?

B in $V = \frac{1}{3}Bh$ means the area of the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q4/5

True or false?

B is the length of a base edge.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 4

Split a volume into equal parts • Grade 5 • Measurement & Data • 5.MD.C.5

A box of volume 60 cm³ is split into three equal parts. Volume of each?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B4-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q1/5

Three congruent blocks fill a cube of volume 27. Volume of one block?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B4-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q2/5

A solid made of non-overlapping parts has volume

A. the largest part's volume

B. the sum of the parts' volumes

C. the product of the parts' volumes

D. the average volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B4-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q3/5

True or false?

If three equal pieces fill a prism, each piece has ⅓ of the prism's volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B4-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q4/5

True or false?

Cutting a solid into pieces changes its total volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B4-Q5

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q5/5

SET 77 • HSG-GMD.A.1

FORWARD 1

Cavalieri's principle for pyramids • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A square pyramid and a triangular pyramid both have base area 50 cm² and height 9 cm. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q1/5

An oblique pyramid has a square base with side 4 and vertical height 6. Find its volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q2/5

True or false?

Two pyramids with equal base areas and equal heights have equal volumes even if their apexes are in different positions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q3/5

Why do two pyramids with equal base areas and equal heights have equal cross-section areas at every level?

A. Their bases have the same shape

B. Both cross-sections are scaled copies of the bases by the same factor

C. They have the same slant height

D. They have the same apex

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q4/5

A cross-section of a pyramid halfway up has what fraction of the base area?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSG-GMD.A.1

FORWARD 2

Use the pyramid volume formula • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A square pyramid has base side 9 m and height 10 m. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q1/5

A rectangular pyramid has a 6 in. by 4 in. base and height 7.5 in. Find its volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q2/5

True or false?

A pyramid with the same base and height as a prism of volume 90 has volume 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q3/5

A pyramid with base area 48 and height 5 has volume

A. 120

B. 53

C. 240

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q4/5

A paperweight is a pyramid whose triangular base has base 6 cm and height 5 cm; the pyramid is 8 cm tall. Find its volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q5/5

SET 104 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the volume of a cone

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	The cylinder volume argument	**HSG-GMD.A.1**
BACKWARD 3	The pyramid volume formula	**HSG-GMD.A.1**
FORWARD 1	Volumes of frustums (cones with the top cut off)	**HSG-GMD.A.3**
FORWARD 2	Use the cone volume formula	**HSG-GMD.A.3**

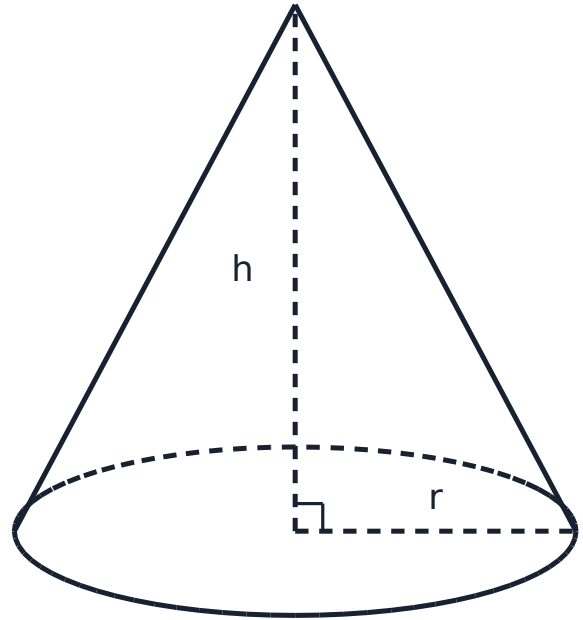
SET 104 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Explain why the volume of a cone is $\frac{1}{3}\pi r^2 h$, using pyramids.

Work space



Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q1/5

A cone has base radius r and height h. A square pyramid has base area πr^2 and the same height h. Use Cavalieri's principle to explain why the cone's volume is $\frac{1}{3}\pi r^2 h$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q2/5

In an experiment, three cones full of water exactly fill a cylinder with the same base radius and height. Explain what this shows about the cone formula and why it is evidence, not a proof.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q3/5

A cone with radius r and height h is approximated by n stacked disks of thickness h/n; the disk k steps from the top has radius kr/n. The total disk volume is $(\pi r^2 h/n^3) \times$ (the sum of the squares 1^2, 2^2, 3^2, and so on up to n^2), which equals $0.385\pi r^2 h$ for n = 10 and $0.338\pi r^2 h$ for n = 100. Explain how this supports $V = \frac{1}{3}\pi r^2 h$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q4/5

A cone has base radius r and height h. A plane parallel to the base cuts the cone at height y above the base. Explain why the cross-section is a circle of radius r(h − y)/h, and why this fact is what makes the cone's volume ⅓ of the cylinder's rather than ½.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q5/5

SET 104 • HSG-GMD.A.1

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

Exact area of a circle of radius 7?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q1/5

Exact area of a circle of diameter 8?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q2/5

The base of a cone is a

A. triangle

B. sector

C. square

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q3/5

True or false?

The base area of a cone with radius r is πr^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q4/5

True or false?

The base area of a cone with radius 2 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q5/5

SET 104 • HSG-GMD.A.1

BACKWARD 2

The cylinder volume argument • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

A cylinder's volume is $\pi r^2 h$ because it is a stack of disks of area πr^2.

A. True

B. False

Cylinder radius 3, height 5: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q2/5

A cone fits inside a cylinder with the same base and height. The cone's volume is

A. ½ of the cylinder's

B. ⅓ of the cylinder's

C. ⅔ of the cylinder's

D. equal to the cylinder's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q3/5

A cylinder holds 60 cm³. A cone with the same base and height holds

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q4/5

True or false?

A cone is a cylinder with the top cut off flat.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q5/5

SET 104 • HSG-GMD.A.1

BACKWARD 3

The pyramid volume formula • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A pyramid has base area 30 and height 8. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q1/5

True or false?

$V = \frac{1}{3}Bh$ holds for every pyramid, whatever the shape of its base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q2/5

A pyramid and a prism have the same base and height. The pyramid's volume is

A. ⅓ of the prism's

B. ½ of the prism's

C. 3 times the prism's

D. equal to the prism's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q3/5

A pyramid has volume 40 and height 6. Base area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q4/5

True or false?

A pyramid has half the volume of the prism with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q5/5

SET 104 • HSG-GMD.A.1

FORWARD 1

Volumes of frustums (cones with the top cut off) • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cone has radius 6 and height 12. The top cone of height 4 is cut off by a plane parallel to the base. Find the exact volume of the remaining frustum.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q1/5

A cup is a frustum with bottom radius 3 cm, top radius 4 cm and height 6 cm. Extend it to a full cone and find the cup's volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q2/5

True or false?

A frustum's volume is the large cone's volume minus the small cone's volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q3/5

A cone is cut halfway up its height, parallel to the base. The small top cone has what fraction of the whole cone's volume?

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. $\frac{1}{8}$

D. $\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q4/5

Explain why cutting a cone halfway up leaves 7/8 of its volume in the bottom part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q5/5

SET 104 • HSG-GMD.A.1

FORWARD 2

Use the cone volume formula • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cone has radius 3 in. and height 8 in. Find its volume to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q1/5

A cone-shaped cup has diameter 6 cm and height 9 cm. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q2/5

True or false?

A cone with the same base and height as a cylinder of volume 90 has volume 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q3/5

A cone has base area 48 and height 5. Its volume is

A. 240

B. 53

C. 80

D. 120

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q4/5

A pile of sand is a cone 2 m tall with base radius 3 m. Find its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q5/5

SET 78 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

Use volume formulas for cylinders, pyramids, and cones

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	Find a missing side with the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Evaluate a formula and round	**7.EE.B.3**
BACKWARD 4	Which solid uses ⅓?	**HSG-GMD.A.1**
FORWARD 1	Composite and hollow solids	**HSG-GMD.A.3**
FORWARD 2	Mass from volume and density	**HSG-MG.A.2**

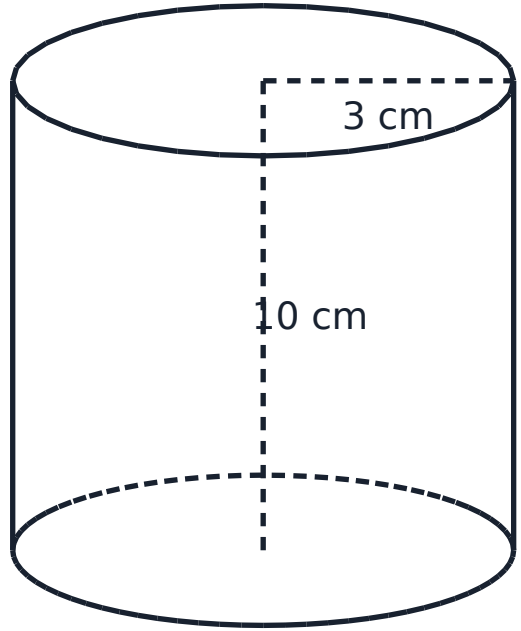
SET 78 • HSG-GMD.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cylinder has radius 3 cm and height 10 cm. Find its volume, exact and to the nearest tenth.

Volume: _____



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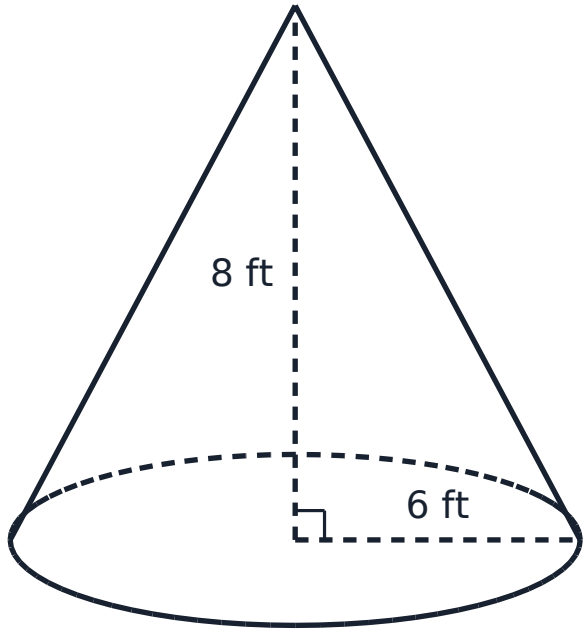
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q1/5

A cone has radius 6 ft and height 8 ft. Find its volume, exact and to the nearest tenth.

Volume: _____



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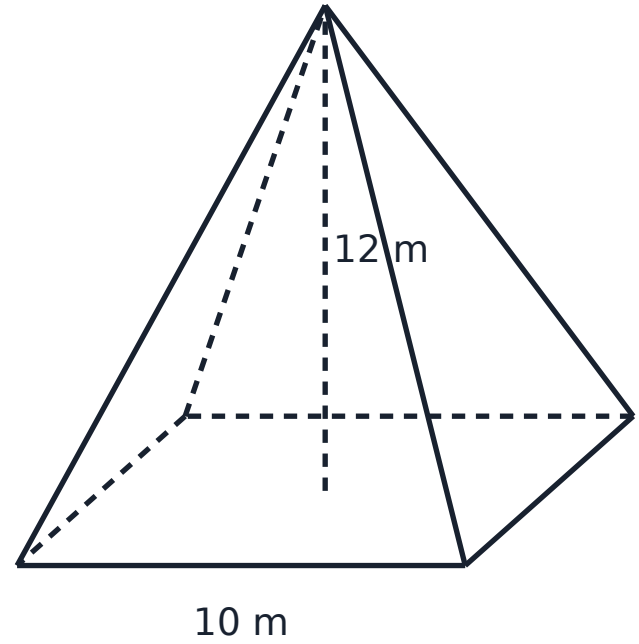
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q2/5

A square pyramid has base side 10 m and height 12 m. Find its volume.

Volume: _____



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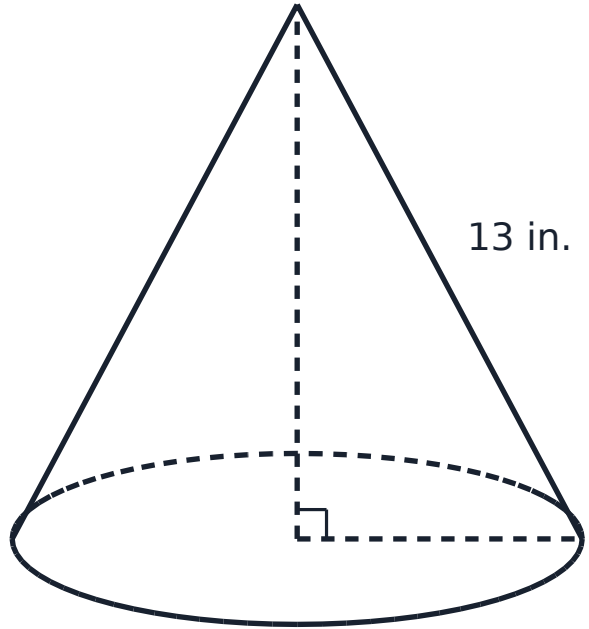
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q3/5

A cone has diameter 10 in. and slant height 13 in. Find its volume, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q4/5

A cylinder and a cone have the same radius and height. The cylinder holds 600 mL. The cone holds

A. 300 mL

B. 200 mL

C. 600 mL

D. 1,800 mL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

Exact area of a circle with radius 6?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q1/5

Exact area of a circle with diameter 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q2/5

A base of radius 3 has area

A. 9

B. 9π

C. 3π

D. 6π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q3/5

True or false?

Squaring the radius comes before multiplying by π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q4/5

True or false?

πr^2 with $r = 3$ is $(3\pi)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 2

Find a missing side with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

A right triangle has hypotenuse 13 and one leg 5. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q1/5

A cone has radius 6 and slant height 10. Height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q2/5

Radius, height, and slant height of a cone form a

A. square

B. right triangle

C. equilateral triangle

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q3/5

True or false?

The slant height is longer than the height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q4/5

True or false?

The slant height can be used as the height in $V = \frac{1}{3}\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 3

Evaluate a formula and round • Grade 7 • Expressions & Equations • 7.EE.B.3

Evaluate ⅓ · 36 · 8.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q1/5

Round 90π to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q2/5

$96\pi \approx$

A. 301.6

B. 301.5

C. 3,016

D. 96.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q3/5

True or false?

Using 3.14 for π may change the last digit of a rounded answer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q4/5

True or false?

$\frac{1}{3} \cdot 100 \cdot 12 = 1{,}200$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 4

Which solid uses ⅓? • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

Pyramids and cones use $V = \frac{1}{3}Bh$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q1/5

Which has the formula V = Bh: a cone or a cylinder?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q2/5

$V = \frac{1}{3}\pi r^2 h$ is the volume of a

A. cylinder

B. cone

C. prism

D. sphere

Why does a cone have ⅓ of the matching cylinder's volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q4/5

True or false?

A pyramid uses $V = Bh$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q5/5

SET 78 • HSG-GMD.A.3

FORWARD 1

Composite and hollow solids • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A silo is a cylinder (radius 4 m, height 10 m) topped by a cone (radius 4 m, height 3 m). Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q1/5

A pipe is a cylinder of length 20 cm with outer radius 5 cm and inner radius 4 cm. Find the volume of the material, exact.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q2/5

True or false?

The volume of a composite solid is the sum of the volumes of its non-overlapping parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q3/5

A cylinder (V = 300) has a cone with the same base and height carved out. Remaining volume?

A. 300

B. 200

C. 0

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q4/5

A pencil tip is a cone of radius 0.4 cm and height 1.5 cm on a cylinder of radius 0.4 cm and length 15 cm. Find the total volume to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q5/5

SET 78 • HSG-GMD.A.3

FORWARD 2

Mass from volume and density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A steel cylinder has radius 2 cm and height 5 cm. Steel has density 7.85 g/cm³. Find the mass to the nearest gram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q1/5

A cone of ice cream (radius 3 cm, height 10 cm) has density 0.55 g/cm³. Find the mass to the nearest gram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q2/5

True or false?

Mass = volume ÷ density.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q3/5

A pyramid of volume 400 cm³ is made of material with density 2 g/cm³. Mass?

A. 200 g

B. 402 g

C. 1,600 g

D. 800 g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q4/5

A gold cube has volume 8 cm³ and mass 154.4 g. Find the density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q5/5

SET 79 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

Use the volume formula for spheres, including composite solids

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Cubes of numbers	**6.EE.A.1**
BACKWARD 2	Radius from diameter	**7.G.B.4**
BACKWARD 3	Multiply fractions by numbers	**7.NS.A.2**
BACKWARD 4	Cylinder and cone volumes	**HSG-GMD.A.3**
FORWARD 1	Scaling spheres	**HSG-GMD.A.3**
FORWARD 2	Model real objects as spheres	**HSG-MG.A.1**

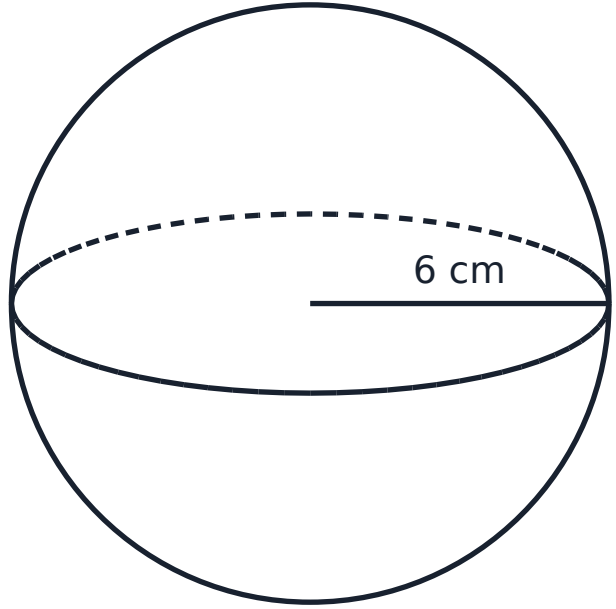
SET 79 • HSG-GMD.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

Find the volume of a sphere with radius 6 cm, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q1/5

Find the volume of a ball with diameter 10 in., exact and to the nearest tenth.

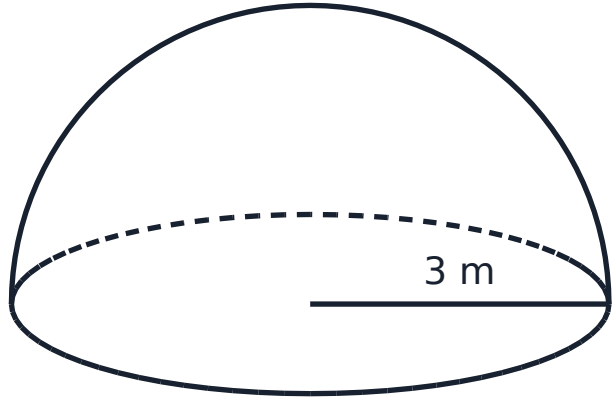
Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q2/5

Find the volume of a hemisphere with radius 3 m, exact and to the nearest tenth.



Not drawn to scale.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q3/5

A capsule is a cylinder of radius 3 cm and length 10 cm with a hemisphere on each end. Find its volume, exact and to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q4/5

An ice cream cone (radius 2, height 9) is topped by a hemisphere of radius 2. The total exact volume is

A. 52π/3

B. 28π

C. 12π

D. 44π/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 1

Cubes of numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

$6^3 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q1/5

$5^3 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q2/5

$3^3 =$

A. 9

B. 81

C. 6

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q3/5

True or false?

$2^3 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q4/5

True or false?

$4^3 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 2

Radius from diameter • Grade 7 • Geometry • 7.G.B.4

A ball has diameter 10. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q1/5

True or false?

Using the diameter in 4/3 πr³ makes the answer 8 times too large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q2/5

A sphere with diameter 12 has r^3 =

A. 216

B. 144

C. 36

D. 1,728

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q3/5

A sphere has radius 2.5. Diameter?

Diameter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q4/5

True or false?

The radius of a sphere is twice its diameter.

A. True

B. False

SET 79 • HSG-GMD.A.3

BACKWARD 3

Multiply fractions by numbers • Grade 7 • The Number System • 7.NS.A.2

$4/3 \cdot 216 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q1/5

½ · 4/3 · 27 =

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q2/5

$4/3 \cdot 125 =$

A. 375/4

B. 500/3

C. 125/3

D. 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q3/5

True or false?

$4/3 \cdot 27 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q4/5

True or false?

$4/3 \cdot 6 = 24$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 4

Cylinder and cone volumes • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

Cylinder radius 3, length 10: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q1/5

Cone radius 2, height 9: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q2/5

A cylinder and a cone with radius 3 and height 4 have volumes

A. 36π and 12π

B. 12π and 36π

C. 12π and 4π

D. 36π and 36π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q3/5

True or false?

Two hemispheres of radius r make one sphere of radius r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q4/5

True or false?

A composite volume is the product of the parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q5/5

SET 79 • HSG-GMD.A.3

FORWARD 1

Scaling spheres • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

The radius of a sphere is doubled. By what factor is the volume multiplied?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q1/5

A sphere fits exactly inside a cube with side 6. Find the exact volume of the space in the cube outside the sphere.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q2/5

True or false?

A sphere of radius 3 has 9 times the volume of a sphere of radius 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q3/5

A balloon's radius grows from 10 cm to 15 cm. Its volume is multiplied by

A. 1.5

B. 2.25

C. 5

D. 3.375

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q4/5

A sphere and a cylinder have the same radius r, and the cylinder's height is 2r. What fraction of the cylinder's volume is the sphere's?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q5/5

SET 79 • HSG-GMD.A.3

FORWARD 2

Model real objects as spheres • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

Model Earth as a sphere with radius 6,371 km. Estimate its volume in cubic kilometers, in scientific notation to 3 significant figures.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q1/5

A basketball has circumference 75 cm. Model it as a sphere and find its volume to the nearest 100 cm^3.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q2/5

True or false?

Modeling an orange as a sphere ignores its small bumps.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q3/5

A scoop of ice cream is modeled as a sphere of radius 3 cm. Its volume is about

A. 28 cm^3

B. 113 cm^3

C. 339 cm^3

D. 36 cm^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q4/5

Give one reason a sphere is a good model for a ball bearing and one limitation of the model for a pumpkin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q5/5

SET 80 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

Solve for an unknown dimension from a volume

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Solve a one-step equation with a coefficient	**HSA-REI.B.3**
BACKWARD 2	Square roots and cube roots	**8.EE.A.2**
BACKWARD 3	Rearrange a formula	**HSA-CED.A.4**
FORWARD 1	Find dimensions to meet a target volume	**HSG-GMD.A.3**
FORWARD 2	Design a container that meets constraints	**HSG-MG.A.3**

SET 80 • HSG-GMD.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cylinder has volume 360π cm³ and radius 6 cm. Find its height.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q1/5

A cone has volume 96π in^3 and height 8 in. Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q2/5

A sphere has volume 36π m³. Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q3/5

A square pyramid has volume 256 ft³ and height 12 ft. Find the side of its base.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q4/5

A cylinder holds 500 cm³ and has height 10 cm. Its radius is closest to

A. 7.1 cm

B. 4.0 cm

C. 2.0 cm

D. 15.9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q5/5

SET 80 • HSG-GMD.A.3

BACKWARD 1

Solve a one-step equation with a coefficient • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $36\pi h = 360\pi$.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q1/5

Solve ⅓x = 12.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q2/5

Solve 4x = 108.

A. 25

B. 104

C. 432

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q3/5

True or false?

To solve $36\pi h = 360\pi$, dividing both sides by π is allowed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q4/5

True or false?

To solve ⅓x = 12, divide by 3.

A. True

B. False

SET 80 • HSG-GMD.A.3

BACKWARD 2

Square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve $r^2 = 36$ for $r > 0$.

$r =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q1/5

Solve $r^3 = 27$.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q2/5

$\sqrt[3]{125} =$

A. 25

B. 5

C. 41.7

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{64} = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q4/5

True or false?

If $r^3 = 8$, then $r = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSG-GMD.A.3

BACKWARD 3

Rearrange a formula • High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4

Solve $V = \pi r^2 h$ for h.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q1/5

Solve $V = \frac{1}{3}Bh$ for B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q2

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q2/5

$V = 4/3\ \pi r^3$ solved for r^3 is

A. $4V/(3\pi)$

B. $V/(4\pi)$

C. $3V/(4\pi)$

D. $3\pi V/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B3-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q3/5

True or false?

Solving $V = \pi r^2 h$ for r needs a square root.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q4/5

True or false?

Solving $V = Bh$ for h gives $h = VB$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q5/5

SET 80 • HSG-GMD.A.3

FORWARD 1

Find dimensions to meet a target volume • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A drink can must hold at least 355 cm^3 and have radius 3.3 cm. What is the smallest height, in tenths of a centimeter, that meets the requirement?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q1/5

A spherical tank must hold at least 1,000 m^3. What is the smallest radius, in tenths of a meter, that meets the requirement?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q2/5

True or false?

To double the volume of a cylinder while keeping its radius, double its height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q3/5

A cone has the same volume as a cylinder of equal radius. The cone's height is

A. 9 times the cylinder's

B. ⅓ of the cylinder's

C. 3 times the cylinder's

D. equal to the cylinder's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q4/5

A cube has the same volume as a sphere of radius 3. Find the cube's edge to the nearest tenth.

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q5/5

SET 80 • HSG-GMD.A.3

FORWARD 2

Design a container that meets constraints • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A cylindrical vase must hold 2,000 cm^3 and be 20 cm tall. Its radius must be at least 5 cm to fit a bouquet. Does the design work? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q1/5

A conical paper cup must hold at least 150 cm^3 and have radius 3.5 cm. What is the smallest height, in tenths of a centimeter, that meets the requirement?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q2/5

True or false?

A container's volume requirement fixes each of its dimensions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q3/5

A cylinder must hold 100π cm^3. Which (r, h) works?

A. (5, 4)

B. (5, 20)

C. (4, 5)

D. (10, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q4/5

Give two different cylinders (radius and height) that each hold 64π cm^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q5/5

Identify cross-sections of three-dimensional objects

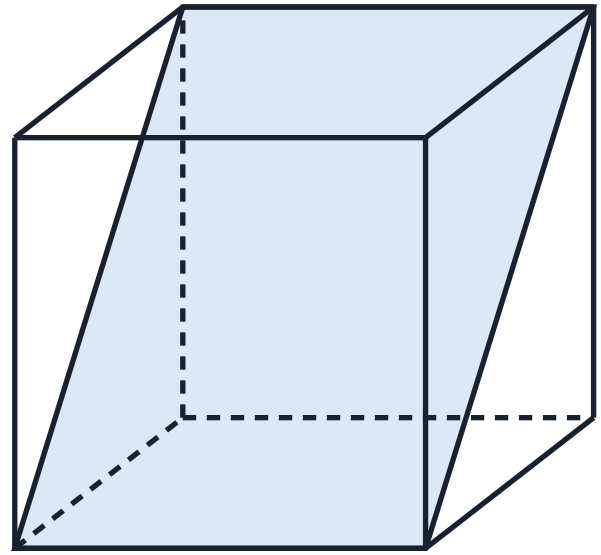
MAIN	Geometry main question	**HSG-GMD.B.4**
BACKWARD 1	Cross-sections of right rectangular prisms and pyramids	**7.G.A.3**
BACKWARD 2	Properties of polygons	**5.G.B.4**
BACKWARD 3	Parts of solids	**6.G.A.4**
FORWARD 1	Areas of cross-sections	**HSG-GMD.B.4**
FORWARD 2	Use cross-section areas to compare volumes (Cavalieri)	**HSG-GMD.A.2**

SET 81 • HSG-GMD.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

A plane cuts the cube as shown. Name the shape of the cross-section.



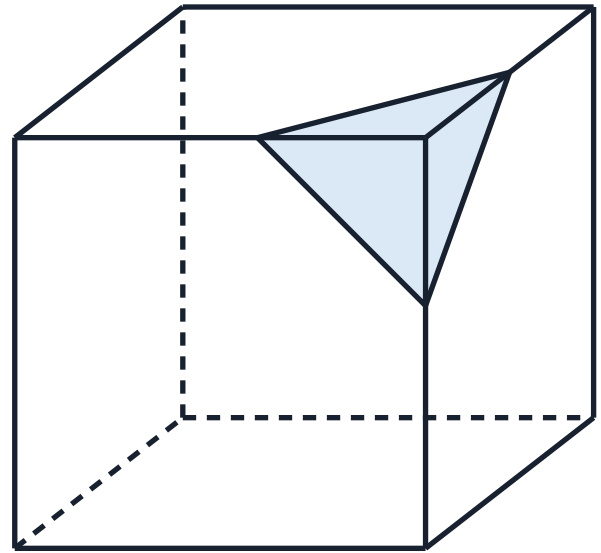
Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q1/5

A plane cuts off one corner of the cube as shown. Name the shape of the cross-section.



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q2/5

A right circular cone is cut by a vertical plane through its apex. The cross-section is

A. a circle

B. a rectangle

C. a triangle (isosceles)

D. an ellipse

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q3/5

Name the cross-section of a cylinder cut by a plane (a) parallel to its base, (b) perpendicular to its base through its axis, and (c) slanted, cutting the curved side only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q4/5

A sphere of radius 10 is cut by a plane 6 units from its center. Name the shape of the cross-section and find its radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 1

Cross-sections of right rectangular prisms and pyramids • Grade 7 • Geometry • 7.G.A.3

A rectangular prism is cut parallel to its base. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q1/5

A square pyramid is cut parallel to its base. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q2/5

A square pyramid cut vertically through its apex and two opposite base edges' midpoints gives a

A. trapezoid

B. square

C. circle

D. triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q3/5

True or false?

A vertical cut of a rectangular prism parallel to a side face gives a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q4/5

True or false?

A cut of a cube parallel to a face can be a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 2

Properties of polygons • Grade 5 • Geometry • 5.G.B.4

A quadrilateral with four right angles and unequal adjacent sides is a

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q1/5

True or false?

Every square is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q2/5

A polygon with six sides is a

A. hexagon

B. pentagon

C. octagon

D. heptagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q3/5

A triangle with two equal sides is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q4/5

True or false?

A trapezoid has no parallel sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 3

Parts of solids • Grade 6 • Geometry • 6.G.A.4

How many faces does a cube have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q1/5

What is the shape of the base of a cone?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q2/5

How many edges meet at each vertex of a cube?

A. 6

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q3/5

True or false?

A cylinder has two congruent circular bases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q4/5

True or false?

A sphere has one edge.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q5/5

SET 81 • HSG-GMD.B.4

FORWARD 1

Areas of cross-sections • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

A cube has edge 4. Find the area of the diagonal cross-section through two opposite edges.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q1/5

A cone has radius 6 and height 12. Find the area of the cross-section parallel to the base at height 4 (measured up from the base).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q2/5

True or false?

A sphere of radius 5 cut 3 units from its center gives a circle of radius 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q3/5

A cube can have a cross-section that is a regular

A. octagon

B. pentagon

C. heptagon

D. hexagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q4/5

A square pyramid has base side 10 and height 15. Find the exact area of the cross-section parallel to the base 5 units below the apex.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSG-GMD.B.4

FORWARD 2

Use cross-section areas to compare volumes (Cavalieri) • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

Two solids have height 10 and every horizontal cross-section of each has area 7. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q1/5

A hemisphere has radius 5. Find the area of its cross-section 3 units above the base.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q2/5

True or false?

Cavalieri's principle can compare two solids of different heights.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q3/5

A hemisphere of radius r rests on its flat base. At height y above the base plane (0 ≤ y ≤ r), its horizontal cross-section has area

A. $\pi(r^2 - y^2)$

B. $\pi(r - y)^2$

C. πr^2

D. πy^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q4/5

Explain why cross-section areas are useful for finding volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q5/5

SET 82 • HSG-GMD.B.4 • GEOMETRIC MEASUREMENT AND DIMENSION

Identify three-dimensional objects generated by rotations of two-dimensional objects

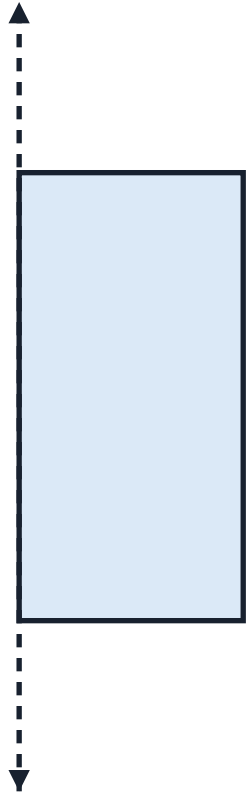
MAIN	Geometry main question	**HSG-GMD.B.4**
BACKWARD 1	Rotation of a point about a center	**HSG-CO.A.4**
BACKWARD 2	Cross-sections perpendicular to the axis	**HSG-GMD.B.4**
BACKWARD 3	Names of solids	**6.G.A.4**
FORWARD 1	Volumes of solids of rotation	**HSG-GMD.B.4**
FORWARD 2	Design an object as a solid of rotation	**HSG-MG.A.3**

SET 82 • HSG-GMD.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

The rectangle is rotated 360° about the dashed line. Name the solid formed.



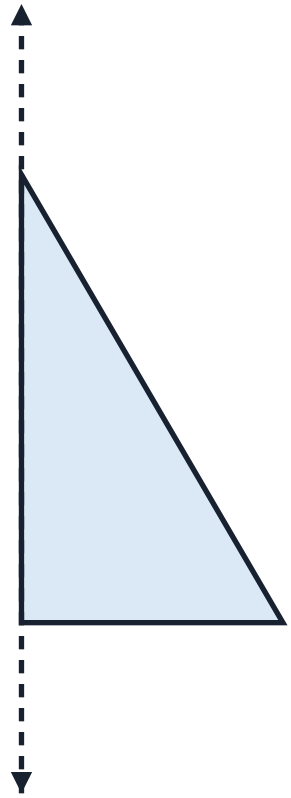
Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q1/5

The right triangle is rotated 360° about the dashed line. Name the solid formed.



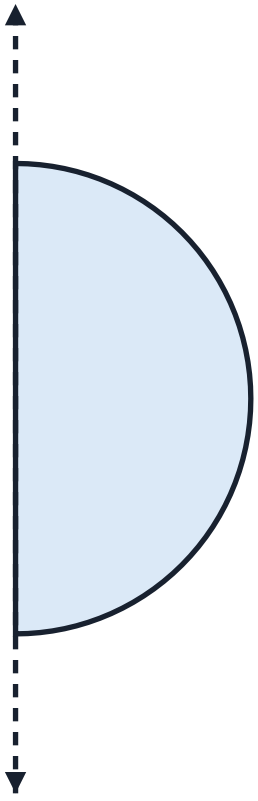
Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q2/5

The semicircle is rotated 360° about the dashed line. Name the solid formed.



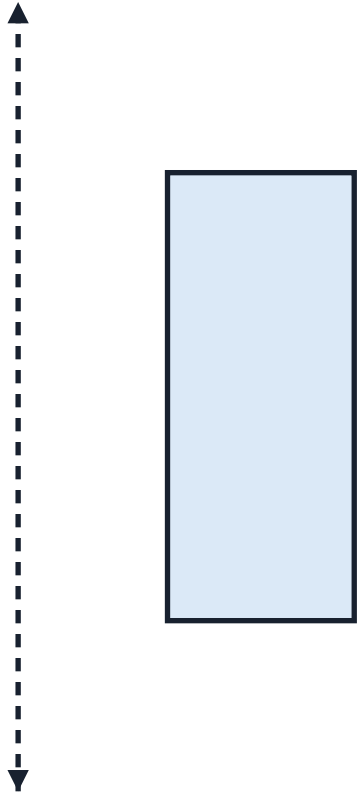
Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q3/5

The rectangle is rotated 360° about the dashed line, which does not touch it. Describe the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q4/5

A right triangle is rotated about its hypotenuse. The solid formed is

A. a sphere

B. one cone

C. two cones joined at their bases

D. a cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 1

Rotation of a point about a center • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

True or false?

In a plane, rotating a point P (P different from the center O) 360° about O traces a circle with center O and radius OP.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q1/5

A point 3 cm from an axis is rotated about the axis. What does it trace?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q2/5

A point ON the axis of rotation

A. traces a line

B. stays fixed

C. traces a circle

D. disappears

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q3/5

A segment parallel to the axis, 2 units away, is rotated about it. What surface does it trace?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q4/5

True or false?

Points farther from the axis trace smaller circles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 2

Cross-sections perpendicular to the axis • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

True or false?

A solid of rotation cut perpendicular to its axis has a circular cross-section (or a ring).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q1/5

A cone is cut perpendicular to its axis. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q2/5

A pipe cut perpendicular to its axis gives a

A. rectangle

B. ring (annulus)

C. circle

D. triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q3/5

A sphere is cut through its center. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q4/5

True or false?

A cylinder cut perpendicular to its axis gives a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 3

Names of solids • Grade 6 • Geometry • 6.G.A.4

Name the solid with two parallel circular bases and a curved side.

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q1/5

Name the solid with one circular base and an apex.

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q2/5

All points equidistant from a center in space form a

A. sphere

B. cone

C. circle

D. cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q3/5

True or false?

A cone has one flat face.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q4/5

True or false?

A sphere has a flat base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q5/5

SET 82 • HSG-GMD.B.4

FORWARD 1

Volumes of solids of rotation • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

A 3-by-5 rectangle is rotated about its 5-unit side. Find the exact volume of the solid.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q1/5

A 3-by-5 rectangle is rotated about its 3-unit side. Find the exact volume of the solid.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q2/5

True or false?

Rotating one rectangle about two different sides can give solids with different volumes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q3/5

A right triangle with legs 3 (on the axis) and 4 is rotated about the leg of length 3. Volume?

A. 48π

B. 9π

C. 12π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q4/5

A semicircle of radius 3 is rotated about its diameter. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q5/5

SET 82 • HSG-GMD.B.4

FORWARD 2

Design an object as a solid of rotation • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A lathe turns a 2 cm by 10 cm rectangle about its 10 cm side to make a wooden dowel. Find the exact volume of wood.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q1/5

A designer wants a cone-shaped lampshade by rotating a right triangle about a 30 cm leg. What other leg length gives a base of diameter 40 cm?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q2/5

True or false?

A hemispherical bowl can be modeled by rotating a right triangle about one of its legs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q3/5

To make a ring-shaped washer by rotation, rotate a rectangle about a line in the rectangle's plane, parallel to one of its sides, that

A. passes through the rectangle's center

B. contains one side

C. lies outside the rectangle

D. cuts through the rectangle off-center

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q4/5

A vase is made by rotating a rectangle 6 cm wide and 20 cm tall about a line in the rectangle's plane, parallel to its long side and 2 cm from it, outside the rectangle. Find the exact volume of material.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q5/5

SET 83 • HSG-MG.A.1 • MODELING WITH GEOMETRY

Model objects with geometric shapes

MAIN	Geometry main question	**HSG-MG.A.1**
BACKWARD 1	Volume formulas	**HSG-GMD.A.3**
BACKWARD 2	Radius from a circumference	**7.G.B.4**
BACKWARD 3	Convert cubic units	**6.RP.A.3.d**
FORWARD 1	Judge a model against a measurement	**HSG-MG.A.1**
FORWARD 2	Use the model to estimate mass with density	**HSG-MG.A.2**

SET 83 • HSG-MG.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

A tree trunk is about 6 m tall and has circumference 2.2 m. Model it as a cylinder and estimate its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q1/5

A pile of sand is shaped like a cone with base diameter 8 m and height 3 m. Estimate its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q2/5

Which shape is the best model for a soup can?

A. Cone

B. Cylinder

C. Square pyramid

D. Sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q3/5

A pool is modeled as a rectangular prism 25 m long, 10 m wide, and 1.8 m deep. How many liters of water does it hold? (1 m^3 = 1,000 L)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q4/5

A pine tree is modeled as a cone. Name one measurement you would take and one way the model differs from the real tree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 1

Volume formulas • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

Exact volume of a cylinder with r = 2 and h = 5?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q1/5

Exact volume of a cone with r = 3 and h = 4?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q2/5

Volume of a 2 × 3 × 4 prism?

A. 52

B. 9

C. 26

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q3/5

True or false?

$V = \frac{1}{3}\pi r^2 h$ is the cone formula.

A. True

B. False

True or false?

A sphere's volume is $\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 2

Radius from a circumference • Grade 7 • Geometry • 7.G.B.4

A circle has circumference 10π. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q1/5

A circle has circumference 2.2 m. Radius to the nearest hundredth?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q2/5

$r =$

A. $C/(2\pi)$

B. πC

C. $2\pi C$

D. C/π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q3/5

True or false?

A tape measure around a trunk gives its circumference.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q4/5

True or false?

The radius is C/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 3

Convert cubic units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

How many liters are in 2 m^3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q1/5

How many cubic centimeters are in 1 L?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q2/5

$450\ m^3 = ___\ L$

A. 45,000

B. 0.45

C. 450,000

D. 4,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q3/5

True or false?

$1\ \text{mL} = 1\ \text{cm}^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q4/5

True or false?

$1\ m^3 = 100\ L$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q5/5

SET 83 • HSG-MG.A.1

FORWARD 1

Judge a model against a measurement • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

A water bottle is modeled as a cylinder with radius 3.8 cm and height 17 cm. The label says 750 mL. Find the model volume to the nearest mL and the percent error.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q1/5

A silo is modeled as a cylinder (r = 3 m, h = 12 m) with a hemisphere on top. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q2/5

True or false?

A model that overestimates by 3% may still be useful for planning.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q3/5

A football (pointed ends) is best modeled by

A. two cones joined at their bases

B. a cube

C. a sphere

D. a cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q4/5

Why might a cylinder model overestimate the volume of a bottle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSG-MG.A.1

FORWARD 2

Use the model to estimate mass with density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A tree trunk modeled as a cylinder has volume about 2.3 m^3. The wood has density 600 kg/m^3. Estimate the trunk's mass to the nearest 10 kg.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q1/5

A sand pile modeled as a cone has volume 16π m³. The sand has density 1,600 kg/m³. Estimate its mass in metric tons (1 t = 1,000 kg), to the nearest ton.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q2/5

True or false?

A more accurate volume model gives a more accurate mass estimate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q3/5

A pool holds 450 m³ of water (1,000 kg/m³). Mass?

A. 450,000 kg

B. 450 kg

C. 45,000 kg

D. 4,500 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q4/5

Explain how a geometric model and a density work together to estimate the mass of a log.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q5/5

SET 84 • HSG-MG.A.2 • MODELING WITH GEOMETRY

Apply density based on area

MAIN	Geometry main question	**HSG-MG.A.2**
BACKWARD 1	Unit rates	**6.RP.A.3.b**
BACKWARD 2	Areas of rectangles and circles	**7.G.B.6**
BACKWARD 3	Compare ratios	**7.RP.A.2**
FORWARD 1	Convert area units in density problems	**HSG-MG.A.2**
FORWARD 2	Plan capacity from an area density	**HSG-MG.A.3**

SET 84 • HSG-MG.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A city has 840,000 people and an area of 120 square miles. Find its population density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q1/5

A circular forest region has radius 3 km and 4,500 trees. Find the tree density to the nearest tree per square kilometer.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q2/5

Town A has 12,000 people on 30 km². Town B has 9,000 people on 20 km². Which town is more densely populated? Show the densities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q3/5

A region has a population density of 250 people per km² and is a 4 km by 6 km rectangle. Estimate its population.

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q4/5

Which is a unit of population density based on area?

A. people per mile

B. people per square mile

C. square miles per person per year

D. people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 1

Unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

840 items in 12 boxes: items per box?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q1/5

$45 for 9 m² of tile: dollars per square meter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q2/5

Unit rate for 300 people on 4 km²?

A. 0.013 km² per person

B. 1,200 people per km²

C. 296 people per km²

D. 75 people per km²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q3/5

True or false?

A unit rate compares a quantity to one unit of another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q4/5

True or false?

600 people on 3 km² is 1,800 people per km².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 2

Areas of rectangles and circles • Grade 7 • Geometry • 7.G.B.6

Area of a 4 km by 6 km rectangle?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q1/5

Exact area of a circle with radius 3 km?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q2/5

A square region 5 mi on a side has area

A. 20 mi²

B. 10 mi²

C. 5 mi²

D. 25 mi²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q3/5

True or false?

Area units are square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q4/5

True or false?

A circle with radius 3 has area 6π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 3

Compare ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

Which is greater: 12,000/30 or 9,000/20?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q1/5

True or false?

To compare densities, compare the unit rates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q2/5

Which rate is greatest?

A. 80 per 4 km²

B. 50 per 2 km²

C. 60 per 3 km²

D. 20 per 1 km²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q3/5

A has 100 trees on 4 acres; B has 120 trees on 6 acres. Which is denser?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q4/5

True or false?

The town with more people is always denser.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q5/5

SET 84 • HSG-MG.A.2

FORWARD 1

Convert area units in density problems • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

1 square mile is about 2.59 km². Convert 7,000 people per square mile to people per km² (nearest whole number).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q1/5

A lawn of 400 m² needs grass seed at 35 g per m². How many kilograms of seed are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q2/5

True or false?

A density in people per km² is a larger number than the same density in people per mi².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q3/5

A farm plants 2 trees per 10 m². How many trees on 1 hectare (10,000 m²)?

A. 2,000

B. 200

C. 5,000

D. 20,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q4/5

A park of 3 hectares (1 ha = 10,000 m²) has 5,400 visitors. Visitors per 100 m²?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q5/5

SET 84 • HSG-MG.A.2

FORWARD 2

Plan capacity from an area density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

Safety rules allow 2 people per m² in a plaza. The plaza is a 30 m by 40 m rectangle. Maximum crowd?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q1/5

A festival expects 5,000 people and wants at most 1.5 people per m². What is the smallest whole number of square meters that meets the limit?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q2/5

True or false?

When planning for a safety limit, round the capacity up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q3/5

A circular dance floor of radius 5 m allows 3 people per m². Capacity is about

A. 235 people

B. 471 people

C. 75 people

D. 47 people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q4/5

A circular dance floor of radius 5 m allows at most 3 people per m², and 3 · 25π ≈ 235.6. Explain why the capacity is 235 people, not 236.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q5/5

SET 85 • HSG-MG.A.2 • MODELING WITH GEOMETRY

Apply density based on volume

MAIN	Geometry main question	**HSG-MG.A.2**
BACKWARD 1	Volume of prisms and cylinders	**HSG-GMD.A.3**
BACKWARD 2	Unit rates	**6.RP.A.3.b**
BACKWARD 3	Products of numbers in scientific notation	**8.EE.A.4**
FORWARD 1	Use density to identify materials and find volumes	**HSG-MG.A.2**
FORWARD 2	Design within a mass limit	**HSG-MG.A.3**

SET 85 • HSG-MG.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

An aluminum block measures 2 cm by 3 cm by 5 cm and has mass 81 g. Find its density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q1/5

A cylindrical tank (radius 0.5 m, height 2 m) is full of water (1,000 kg/m³). Find the mass of the water to the nearest kilogram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q2/5

A heater must supply 20 BTU per cubic foot of room. A room is 12 ft by 15 ft with an 8 ft ceiling. How many BTU are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q3/5

A sample has 3×10^6 bacteria per mL. How many bacteria are in 250 mL?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q4/5

Water has density 1 g/cm³. Which object floats in water?

A. A 1,000 g brick with volume 400 cm³

B. A 600 g block with volume 500 cm³

C. A 100 g ball with volume 50 cm³

D. A 400 g block with volume 500 cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 1

Volume of prisms and cylinders • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

Volume of a 2 × 3 × 5 cm block?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q1/5

Exact volume of a cylinder with r = 0.5 m, h = 2 m?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q2/5

Volume of a 12 × 15 × 8 ft room?

A. 1,440 ft³

B. 180 ft³

C. 2,880 ft³

D. 35 ft³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q3/5

True or false?

Volume is needed before a volume density can be used.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q4/5

True or false?

A 1 m cube has volume 3 m^3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 2

Unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

81 g in 30 cm³: grams per cm³?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q1/5

20 BTU per ft³ for 100 ft³: total?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q2/5

500 g in 250 cm³ is

A. 250 g/cm³

B. 2 g/cm³

C. 0.5 g/cm³

D. 750 g/cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q3/5

True or false?

Amount = rate × quantity.

A. True

B. False

True or false?

Density = volume ÷ mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 3

Products of numbers in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

$(3 \times 10^6) \times 250$ in scientific notation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q1/5

Write 2,400,000 in scientific notation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q2/5

$(2 \times 10^3)(4 \times 10^2) =$

A. 6×10^5

B. 8×10^5

C. 8×10^6

D. 8×10^1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q3/5

True or false?

$10^3 \times 10^2 = 10^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q4/5

True or false?

$5 \times 10^2 \times 3 = 15 \times 10^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q5/5

SET 85 • HSG-MG.A.2

FORWARD 1

Use density to identify materials and find volumes • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A ring has mass 9.65 g and volume 0.5 cm³. Gold has density 19.3 g/cm³. Could the ring be pure gold?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q1/5

A 1,350 g piece of aluminum (2.7 g/cm³) is melted into a cube. Find the edge of the cube, to the nearest tenth of a centimeter.

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q2/5

True or false?

Two objects with equal mass but different volumes have different densities.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q3/5

Oil (0.9 g/cm³) in a 2,000 cm³ jug has mass

A. 1,800 g

B. 2,000 g

C. 2,222 g

D. 1,100 g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q4/5

A balloon holds 0.02 m³ of helium (0.18 kg/m³). Mass of the helium in grams?

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q5/5

SET 85 • HSG-MG.A.2

FORWARD 2

Design within a mass limit • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A truck can carry 12,000 kg. Gravel has density 1,500 kg/m³. What is the largest volume of gravel it can carry?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q1/5

A shelf holds at most 40 kg. Books have density 0.8 g/cm³. How many cubic centimeters of books can it hold?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q2/5

True or false?

Mass limits become volume limits by multiplying by the density.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q3/5

A cylindrical water tank must weigh at most 3,140 kg when full (water 1,000 kg/m³, ignore the tank). With radius 1 m, its height can be at most about

A. 10 m

B. 3.14 m

C. 1 m

D. 0.32 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q4/5

A concrete block must be under 25 kg. Concrete is 2,400 kg/m³. Will a 20 cm × 20 cm × 30 cm block work?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q5/5

SET 86 • HSG-MG.A.3 • MODELING WITH GEOMETRY

Design to satisfy physical constraints

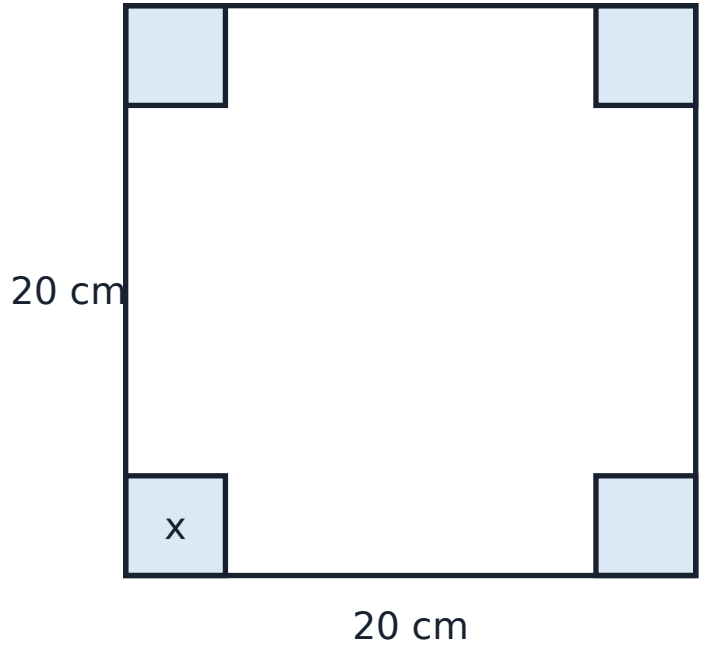
MAIN	Geometry main question	**HSG-MG.A.3**
BACKWARD 1	Volume of a rectangular prism	**7.G.B.6**
BACKWARD 2	Use the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Ratios and rates	**7.RP.A.2**
BACKWARD 4	Inequalities as limits	**7.EE.B.4**
FORWARD 1	Satisfy several constraints at once	**HSG-MG.A.3**
FORWARD 2	Find the best design with a function graph	**HSF-IF.C.7.c**

SET 86 • HSG-MG.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Compare the volumes for x = 2, 3, and 4. Which gives the largest box?



Not drawn to scale.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q1/5

Accessibility rules allow a ramp slope of at most 1 : 12 (1 unit of rise per 12 units of run). A ramp must rise 30 in. What is the shortest horizontal run allowed, in feet?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q2/5

A rectangular pen will use 40 m of fencing. Compare a 10 m by 10 m pen with a 5 m by 15 m pen. Which encloses more area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q3/5

A cylindrical tank must fit in a closet 2 m wide, 2 m deep, and 3 m tall. What is the largest possible volume, to the nearest tenth of a cubic meter?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q4/5

A 13 ft ladder leans on a wall. For safety its base must be at least 5 ft from the wall. The highest point it can reach is

A. 18 ft

B. 12 ft

C. 13 ft

D. 8 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 1

Volume of a rectangular prism • Grade 7 • Geometry • 7.G.B.6

Volume of a box 14 cm by 14 cm by 3 cm?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q1/5

A 20 cm side with 3 cm cut from each end leaves

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q2/5

Volume of a box 12 by 12 by 4?

A. 28

B. 576

C. 48

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q3/5

True or false?

Cutting larger corners makes the box taller but its base smaller.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q4/5

True or false?

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner. The box's base side is $20 - x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 2

Use the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

Hypotenuse 13, leg 5: other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q1/5

Legs 30 in. and 360 in.: hypotenuse to the nearest inch?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q2/5

A ladder, wall, and ground form a

A. right triangle

B. circle

C. square

D. equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q3/5

True or false?

A ladder leans against a vertical wall on level ground. The ladder is the hypotenuse of the right triangle formed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q4/5

True or false?

The ladder's height on the wall is 13 + 5.

A. True

B. False

SET 86 • HSG-MG.A.3

BACKWARD 3

Ratios and rates • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

A ratio of 1 : 12 for a rise of 2 in. gives what run?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q1/5

Is a slope of 3 : 30 steeper than 1 : 12?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q2/5

360 in. =

A. 30 ft

B. 36 ft

C. 12 ft

D. 3.6 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q3/5

True or false?

A ratio of 1 : 12 means 1 inch up for every 12 inches across.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q4/5

True or false?

A 1 : 8 ramp is less steep than a 1 : 12 ramp.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 4

Inequalities as limits • Grade 7 • Expressions & Equations • 7.EE.B.4

The run r must satisfy r ≥ 360. Is r = 300 allowed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B4-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q1/5

A diameter d must satisfy d ≤ 2. Largest radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B4-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q2/5

"At most 3 m" means

A. ≥ 3

B. < 3 only

C. ≤ 3

D. = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B4-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q3/5

True or false?

"At least 5 ft" allows exactly 5 ft.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B4-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q4/5

True or false?

"At most 12" excludes 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B4-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q5/5

SET 86 • HSG-MG.A.3

FORWARD 1

Satisfy several constraints at once • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Which whole-number values of x give a volume of at least 500 cm^3 AND a height of at most 3 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q1/5

A ramp must rise 2 ft and fit within 20 ft of horizontal space. Can it meet the 1 : 12 limit? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q2/5

True or false?

A design that satisfies one constraint may fail another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q3/5

A rectangular garden needs area 48 m² and width at most 6 m. Which works?

A. 6 m wide by 8 m long

B. 4 m wide by 10 m long

C. 7 m wide by 7 m long

D. 8 m wide by 6 m long

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q4/5

A cylinder 3 m tall must hold at least 10 m³ and stand on a 2 m by 2 m floor space. Is that possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • FORWARD 1 • Q5/5

SET 86 • HSG-MG.A.3

FORWARD 2

Find the best design with a function graph • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Write its volume as a function V(x) and state the domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q1/5

An open box made from a 20 cm by 20 cm sheet with x-cm corner squares cut out has volume $V(x) = x(20 - 2x)^2$. Use a graph or table to estimate the x that gives the largest volume, to the nearest tenth, and that volume to the nearest cm^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q2/5

True or false?

$V(x) = x(20 - 2x)^2$ has zeros at $x = 0$ and $x = 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q3/5

For an open box made from a 20 cm by 20 cm sheet, $V(x) = x(20 - 2x)^2$. The graph of V on $0 < x < 10$ rises and then falls. Its highest point gives

A. the cut that makes no box

B. the base area

C. the largest volume

D. the smallest volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S86-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q4/5

For an open box made from a 20 cm by 20 cm sheet with x-cm corner cuts, $V(x) = x(20 - 2x)^2$ and V is largest near x = 3.3. Why is x = 5 not the best cut even though it is halfway between 0 and 10?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 86 • FORWARD 2 • Q5/5

SET 87 • HSG-MG.A.3 • MODELING WITH GEOMETRY

Design to minimize cost

MAIN	Geometry main question	**HSG-MG.A.3**
BACKWARD 1	Surface area of prisms and cylinders	**7.G.B.6**
BACKWARD 4	Areas of surfaces with openings	**7.G.B.6**
BACKWARD 3	Unit prices and rounding up purchases	**7.EE.B.3**
FORWARD 1	Compare designs with equal volume for least material	**HSG-MG.A.3**
FORWARD 2	Decide between designs using expected cost	**HSS-MD.B.7**

SET 87 • HSG-MG.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A can with radius 4 cm and height 10 cm is made of metal costing $0.002 per cm². Find the area of metal (top, bottom, and side) and the cost, to the nearest cent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q1/5

Can A has r = 4 cm, h = 10 cm. Can B has r = 5 cm, h = 6.4 cm. Show that they hold the same volume, then decide which uses less metal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q2/5

A wall 12 m by 3 m has a window 1.5 m by 1.2 m. Paint covers 10 m² per liter and is sold only in whole liters at $8 each. Find the cost to paint the wall once.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q3/5

Two closed cardboard boxes each hold 600 cm³: box A is 10 cm by 10 cm by 6 cm and box B is 5 cm by 10 cm by 12 cm. Cardboard costs $0.01 per cm² (ignore thickness and overlaps). Find the cost of each box and choose the cheaper one.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q4/5

A 4 m by 3.5 m floor is covered with 0.5 m square tiles that cost $3.25 each and are sold individually. Plan A uses these tiles; plan B uses sheet flooring at $11 per m². Find the cost of each plan and choose the cheaper one.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • MAIN • Q5/5

SET 87 • HSG-MG.A.3

BACKWARD 1

Surface area of prisms and cylinders • Grade 7 • Geometry • 7.G.B.6

Exact area of the two circular ends of a can with radius 4?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q1/5

Exact area of the side of a can with radius 4 and height 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q2/5

Surface area of a 2 × 3 × 4 box?

A. 24

B. 48

C. 52

D. 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q3/5

True or false?

The side of a cylinder unrolls into a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 1 • Q4/5

True or false?

A cylinder's surface area is $\pi r^2 h$.

A. True

B. False

SET 87 • HSG-MG.A.3

BACKWARD 4

Areas of surfaces with openings • Grade 7 • Geometry • 7.G.B.6

A wall 12 m by 3 m has a 1.5 m by 1.2 m window. Area to paint?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B4-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q1/5

A 4 m by 3.5 m floor: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B4-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q2/5

A 5 m by 2.4 m wall has a 0.9 m by 2 m door. Paintable area?

A. 10.2 m²

B. 12 m²

C. 1.8 m²

D. 13.8 m²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B4-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q3/5

True or false?

Openings in a surface are subtracted from its area before buying material.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B4-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q4/5

True or false?

A floor of 14 m² needs 14 tiles that are 0.5 m on a side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B4-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 87 • BACKWARD 4 • Q5/5

SET 87 • HSG-MG.A.3

BACKWARD 3

Unit prices and rounding up purchases • Grade 7 • Expressions & Equations • 7.EE.B.3

Paint needed is 3.42 L, sold in whole liters. How many liters must be bought?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q1/5

Metal costs $0.002 per cm². Cost of 350 cm²?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q2/5

Fabric is $6 per yard, sold in whole yards. You need 4.2 yd. Cost?

A. $30

B. $25.20

C. $24

D. $26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q3/5

True or false?

When items are sold in whole units, round the amount needed up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q4/5

True or false?

3.1 cans of paint can be bought as 3 cans.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 87 • BACKWARD 3 • Q5/5

SET 87 • HSG-MG.A.3

FORWARD 1

Compare designs with equal volume for least material • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A closed box with a square base must hold 1,000 cm^3 (ignore the thickness of the material). Compare the surface areas for base sides 8, 10, and 12 cm. Which uses the least material?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q1/5

A closed cylinder must hold 1,000 cm^3. Compare its surface area (to the nearest cm^2) for radii 4 cm, 5.42 cm and 7 cm, and choose the radius that uses the least material.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q2/5

True or false?

Among closed boxes with a square base and a fixed volume (ignoring material thickness), the cube has the least surface area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q3/5

For a closed cylinder with a fixed volume, the least surface area occurs when

A. the height equals the radius

B. the height is as large as possible

C. the radius is as large as possible

D. the height equals the diameter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q4/5

Give one reason a manufacturer might NOT choose the least-material can shape.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 87 • FORWARD 1 • Q5/5

SET 87 • HSG-MG.A.3

FORWARD 2

Decide between designs using expected cost • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

Packaging A costs $1.20 per item and 2% of items break. Packaging B costs $0.90 and 8% break. A break costs $20. Find the expected cost per item for each and choose.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q1/5

The standard metal for a can never fails. A cheaper metal fails 5% of the time, and each failure costs $40 to replace. By how much must the cheaper metal's price per can be below the standard price for its expected cost per can to be strictly lower?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q2/5

True or false?

The cheapest material always gives the cheapest design.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q3/5

Expected cost =

A. upfront cost × probability

B. upfront cost + probability of failure × cost of failure

C. cost of failure only

D. upfront cost − probability

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q4/5

Name one cost besides material that a can designer might consider.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S87-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 87 • FORWARD 2 • Q5/5

SET 105 • HSG-MG.A.3 • MODELING WITH GEOMETRY

Design typographic grid systems

MAIN	Geometry main question	**HSG-MG.A.3**
BACKWARD 1	Proportional reasoning with ratios	**7.RP.A.2**
BACKWARD 2	Operate with decimals	**6.NS.B.3**
BACKWARD 3	Factors of a whole number	**4.OA.B.4**
FORWARD 1	Choose grids that support several layouts	**HSG-MG.A.3**
FORWARD 2	Scale a grid layout to a new page size	**HSG-SRT.A.2**

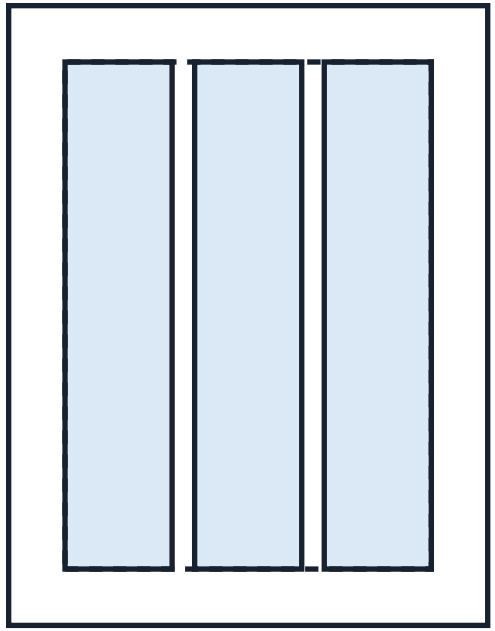
SET 105 • HSG-MG.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A page is 8.5 in. wide with 1 in. margins on each side. The text area has 3 equal columns separated by 2 gutters of 0.2 in. Find the width of each column to the nearest hundredth of an inch.

Width: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q1/5

A poster grid uses the golden ratio (about 1.618 : 1) for height : width. A poster is 60 cm wide. Its height is about

A. 120 cm

B. 97.1 cm

C. 37.1 cm

D. 61.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q2/5

A 12-column grid on a 6.5 in. text area has 11 gutters of 0.125 in. Find the width of each column to the nearest thousandth of an inch.

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q3/5

A web page 1,200 px wide uses a 12-column grid with 24 px gutters between columns and no outer margins. Find the column width and the width of an element that spans 4 columns (including the gutters inside it).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q4/5

A 6 in. by 9 in. page has 0.75 in. margins on all sides. A modular grid has 3 columns and 4 rows with 0.25 in. gutters in both directions. Find the width and height of each module to the nearest hundredth of an inch.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • MAIN • Q5/5

SET 105 • HSG-MG.A.3

BACKWARD 1

Proportional reasoning with ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

Height : width = 1.618 : 1 and width = 10. Height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q1/5

Split 6.1 in. into 3 equal columns. Width of each (nearest hundredth)?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q2/5

A 2 : 3 rectangle 30 cm wide (width = 2 parts) is

A. 60 cm tall

B. 20 cm tall

C. 33 cm tall

D. 45 cm tall

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q3/5

True or false?

Equal columns divide the available width equally.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q4/5

True or false?

A 1 : 2 ratio means the second is half the first.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 105 • BACKWARD 1 • Q5/5

SET 105 • HSG-MG.A.3

BACKWARD 2

Operate with decimals • Grade 6 • The Number System • 6.NS.B.3

$8.5 - 2 \times 1 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q1/5

$11 \times 0.125 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q2/5

6.1 ÷ 3 to the nearest hundredth is

A. 2.3

B. 2.04

C. 2.03

D. 18.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q3/5

True or false?

$5.125 \div 12 \approx 0.427$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q4/5

True or false?

$0.2 + 0.2 = 0.04$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 105 • BACKWARD 2 • Q5/5

SET 105 • HSG-MG.A.3

BACKWARD 3

Factors of a whole number • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all factors of 12.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q1/5

Which numbers from 2 to 6 are factors of 10?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q2/5

Which is a factor of 12?

A. 8

B. 4

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q3/5

True or false?

12 has more factors than 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q4/5

True or false?

5 is a factor of 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 105 • BACKWARD 3 • Q5/5

SET 105 • HSG-MG.A.3

FORWARD 1

Choose grids that support several layouts • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A 12-column grid can also be used as

A. 4 columns of 3, 3 of 4, or 2 of 6

B. 5 equal columns

C. 7 equal columns

D. only 12 columns

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q1/5

Why do designers often choose 12-column grids?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q2/5

True or false?

A 10-column grid can be split into 3 equal groups of whole columns.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q3/5

A 12-column grid has 78 px columns and 24 px gutters. How wide is an element that spans 6 columns?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q4/5

A 6.5 in. text area must hold 4 columns, each 1.5 in. wide, with equal gutters between them. Find the exact gutter width.

Gutter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 105 • FORWARD 1 • Q5/5

SET 105 • HSG-MG.A.3

FORWARD 2

Scale a grid layout to a new page size • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

A poster layout is 60 cm wide and 97.1 cm tall. It is reduced to a flyer 24 cm wide with the same shape. Find the scale factor and the flyer's height.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q1/5

In the poster above, a column is 15 cm wide and a gutter is 1.2 cm. Find their widths on the flyer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q2/5

True or false?

Scaling a layout by factor k multiplies the area of each module by k^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q3/5

A layout is enlarged by factor 1.5. A 4 cm by 6 cm module becomes

A. 5.5 cm by 7.5 cm

B. 4 cm by 9 cm

C. 9 cm by 13.5 cm

D. 6 cm by 9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q4/5

Explain why a grid keeps its proportions when every measurement is multiplied by the same factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S105-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 105 • FORWARD 2 • Q5/5

SET 88 • HSG-SRT.D.9 • ADVANCED EXTENSION (+)

Derive and use the formula A = ½ab sin C

MAIN	Geometry main question	**HSG-SRT.D.9**
BACKWARD 1	The sine ratio in a right triangle	**HSG-SRT.C.6**
BACKWARD 2	Area of a triangle from base and height	**6.G.A.1**
BACKWARD 3	The included angle of two sides	**HSG-CO.B.8**
FORWARD 1	Solve for a side or an angle from an area	**HSG-SRT.D.9**
FORWARD 2	From equal areas to the Law of Sines	**HSG-SRT.D.10**

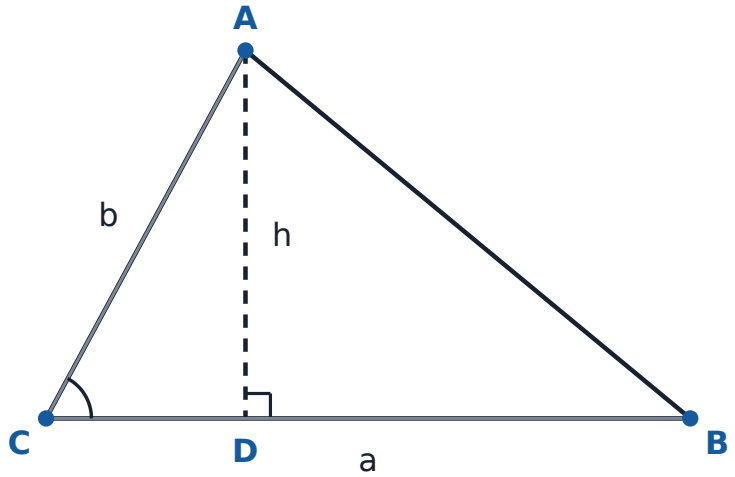
SET 88 • HSG-SRT.D.9

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

In triangle ABC, the altitude from A meets line CB at D and has length h. Derive the formula Area = ½ab sin C. Your derivation must cover both cases: ∠C acute (D on side CB, as shown) and ∠C obtuse (D on line CB extended beyond C). (If ∠C is right, h = b and sin C = 1.)

Work space



Not drawn to scale.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q1/5

A triangle has sides a = 8 and b = 11 with included angle C = 40°. Find its area to the nearest tenth.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q2/5

Two sides of a triangle are 6 and 10, and the angle between them is 150°. Find the exact area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q3/5

A parallelogram has sides 7 cm and 12 cm with an angle of 55° between them. Find its area to the nearest tenth.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q4/5

Triangle PQR has PQ = 9, QR = 14 and an obtuse angle Q = 115°. Find its area to the nearest tenth, and explain why ½ab sin C still works when the included angle is obtuse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • MAIN • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 1

The sine ratio in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

In a right triangle, sin of an acute angle = ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q1/5

Hypotenuse 10, angle 30°. Find the side opposite the angle.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q2/5

If $\sin C = h/b$, then $h =$

A. $b + \sin C$

B. $\sin C/b$

C. $b/\sin C$

D. $b \sin C$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q3/5

True or false?

$\sin 90° = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q4/5

True or false?

$\sin C = \text{adjacent} \div \text{hypotenuse}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 88 • BACKWARD 1 • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 2

Area of a triangle from base and height • Grade 6 • Geometry • 6.G.A.1

Base 8, height 5: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q1/5

True or false?

Area = ½ · base · height

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q2/5

The height must be ___ to the base.

A. equal

B. adjacent

C. parallel

D. perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q3/5

Base a and height b sin C: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q4/5

True or false?

A slanted side can be used as the height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 88 • BACKWARD 2 • Q5/5

SET 88 • HSG-SRT.D.9

BACKWARD 3

The included angle of two sides • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In triangle ABC, which angle is included between sides AB and AC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q1/5

True or false?

SAS uses the angle between the two given sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q2/5

For sides a (BC) and b (CA), the included angle is

A. C

B. none

C. A

D. B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q3/5

Why must the angle be included for ½ab sin C?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q4/5

True or false?

Any angle of the triangle works in ½ab sin C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 88 • BACKWARD 3 • Q5/5

SET 88 • HSG-SRT.D.9

FORWARD 1

Solve for a side or an angle from an area • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

A triangle has sides 10 and 12 and area 30. Find all possible measures of the included angle.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q1/5

A triangle has area 50, one side 20, and an included angle of 45°. Find the other side to the nearest tenth.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q2/5

True or false?

Two different triangles can have the same two sides and the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q3/5

The largest area of a triangle with sides 6 and 8 occurs when the included angle is

A. 60°

B. 180°

C. 90°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q4/5

Find the area of an equilateral triangle with side 6 using ½ab sin C (exact).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 88 • FORWARD 1 • Q5/5

SET 88 • HSG-SRT.D.9

FORWARD 2

From equal areas to the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

The area of triangle ABC can be written as ½bc sin A, ½ac sin B, or ½ab sin C. Use this to show that sin A/a = sin B/b = sin C/c.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q1/5

Use sin A/a = sin B/b: A = 30°, a = 5, b = 8. Find sin B.

sin B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q2/5

True or false?

The Law of Sines can be derived from the area formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q3/5

Dividing ½bc sin A = ½ac sin B by ½abc gives

A. a sin A = b sin B

B. sin A/a = sin B/b

C. sin A = sin B

D. a/b = sin A sin B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S88-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q4/5

Why is it allowed to divide by ½abc?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S88-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 88 • FORWARD 2 • Q5/5

SET 89 • HSG-SRT.D.10 • ADVANCED EXTENSION (+)

Prove the Law of Sines

MAIN	Geometry main question	**HSG-SRT.D.10**
BACKWARD 1	Use trigonometry in right triangles	**HSG-SRT.C.8**
BACKWARD 2	The area formula ½ab sin C	**HSG-SRT.D.9**
BACKWARD 3	Proportions and cross-multiplying	**7.RP.A.2**
FORWARD 1	Use the Law of Sines (AAS and ASA)	**HSG-SRT.D.10**
FORWARD 2	Surveying with the Law of Sines	**HSG-SRT.D.11**

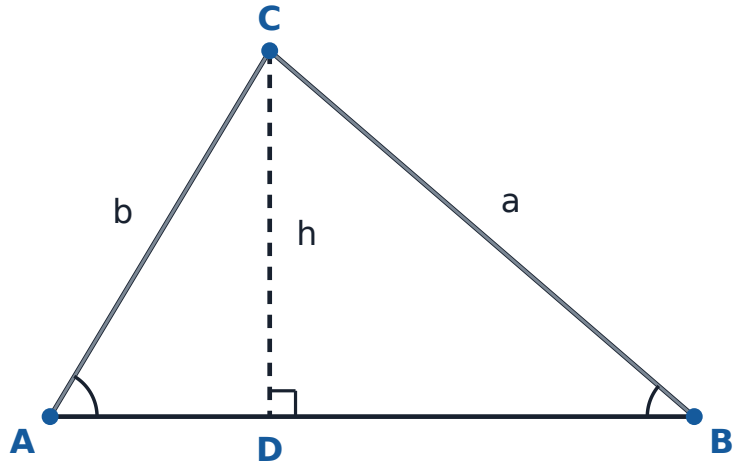
SET 89 • HSG-SRT.D.10

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

In triangle ABC, CD is the altitude to AB. Prove that a/sin A = b/sin B.

Work space



Not drawn to scale.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q1/5

Triangle ABC has an obtuse angle at A, so the altitude from C meets line AB extended past A at D. Prove that a/sin A = b/sin B in this case.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q2/5

In acute triangle ABC, AE is the altitude from A to BC. Prove that b/sin B = c/sin C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q3/5

Triangle ABC has a right angle at C. Prove that a/sin A = b/sin B = c/sin C in this case.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q4/5

Triangle ABC with acute angle A is inscribed in a circle of radius R. Draw the diameter BD. Prove that a/sin A = 2R, and explain why this gives the Law of Sines.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • MAIN • Q5/5

SET 89 • HSG-SRT.D.10

BACKWARD 1

Use trigonometry in right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

Right triangle: hypotenuse 12, angle 40°. Opposite side to the nearest tenth?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q1/5

True or false?

In an acute triangle, an altitude splits the triangle into two right triangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q2/5

If sin B = h/a, then h =

A. h sin B

B. a/sin B

C. a sin B

D. sin B/a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q3/5

Right triangle: opposite 5, hypotenuse 10. Angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q4/5

True or false?

sin is defined only for angles in right triangles, so it cannot be used after drawing an altitude.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 89 • BACKWARD 1 • Q5/5

SET 89 • HSG-SRT.D.10

BACKWARD 2

The area formula ½ab sin C • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

Triangle ABC has b = 4, c = 5, and A = 90°. What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q1/5

True or false?

½bc sin A, ½ac sin B, and ½ab sin C all give the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q2/5

Area of a triangle with sides 6, 6 and included 30°?

A. 3

B. 9

C. 36

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-B2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q3/5

Write the area of triangle ABC using sides a and c.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q4/5

True or false?

½ab sin C uses angle A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 89 • BACKWARD 2 • Q5/5

SET 89 • HSG-SRT.D.10

BACKWARD 3

Proportions and cross-multiplying • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

Solve x/6 = 10/4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q1/5

True or false?

If b sin A = a sin B, then a/sin A = b/sin B (when the sines are not 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q2/5

From 3x = 4y, x/y =

A. 3/4

B. 4/3

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q3/5

Rewrite $p \cdot q = r \cdot s$ as a proportion with p and r on top.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q4/5

True or false?

$a/b = c/d$ means $a + d = b + c$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 89 • BACKWARD 3 • Q5/5

SET 89 • HSG-SRT.D.10

FORWARD 1

Use the Law of Sines (AAS and ASA) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

In triangle ABC, A = 40°, B = 60°, and a = 10. Find b to the nearest tenth.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q1/5

In triangle ABC, A = 40°, B = 60°, and a = 10. Find C, and find c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q2/5

True or false?

Knowing two angles and any side determines the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q3/5

$A = 30°, B = 45°, a = 6.\ b =$

A. $3\sqrt{2} \approx 4.2$

B. $6\sqrt{2} \approx 8.5$

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S89-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q4/5

Why must the third angle be found before using the Law of Sines for c in an ASA problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 89 • FORWARD 1 • Q5/5

SET 89 • HSG-SRT.D.10

FORWARD 2

Surveying with the Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

Two observers 500 m apart sight a tree across a river. The angles at the observers (between the baseline and the tree) are 70° and 60°. How far is the tree from the observer at the 70° angle (nearest meter)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q1/5

A surveyor measures a baseline of 200 m and angles of 52° and 71° to a peak. Find the distance from the 52° end to the peak (nearest meter).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q2/5

True or false?

Surveyors use a measured baseline and two angles to find distances they cannot walk.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q3/5

In a surveying triangle, the side opposite the largest angle is

A. equal to the others

B. the shortest

C. the longest

D. the baseline always

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q4/5

Explain why measuring angles is easier than measuring the distance across the river directly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S89-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 89 • FORWARD 2 • Q5/5

SET 90 • HSG-SRT.D.10 • ADVANCED EXTENSION (+)

Prove the Law of Cosines

MAIN	Geometry main question	**HSG-SRT.D.10**
BACKWARD 1	The Pythagorean Theorem	**8.G.B.7**
BACKWARD 2	The distance formula	**8.G.B.8**
BACKWARD 3	Sine and cosine in a right triangle	**HSG-SRT.C.6**
BACKWARD 4	Square a binomial	**HSA-APR.A.1**
FORWARD 1	Use the Law of Cosines (SAS and SSS)	**HSG-SRT.D.10**
FORWARD 2	Distances between moving objects	**HSG-SRT.D.11**

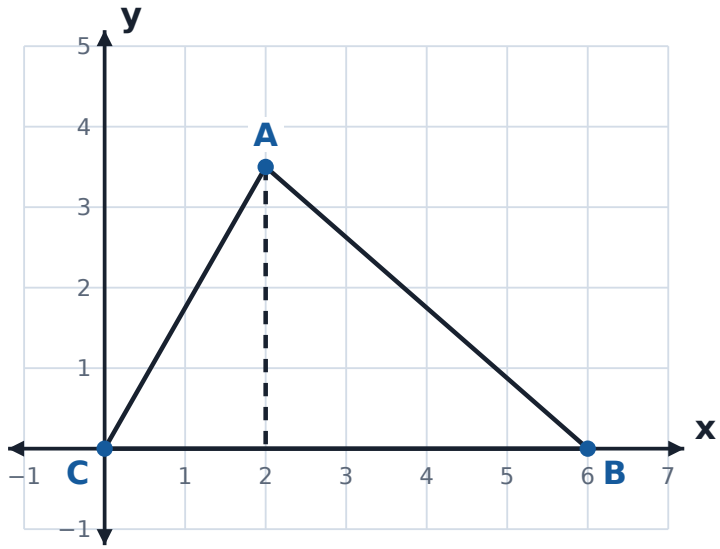
SET 90 • HSG-SRT.D.10

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

Place triangle ABC with C at the origin and B at (a, 0), so A = (b cos C, b sin C). Prove that $c^2 = a^2 + b^2 - 2ab\cos C$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q1/5

In acute triangle ABC, the altitude from A meets BC at D. Let CD = x. Use the two right triangles to prove $c^2 = a^2 + b^2 - 2ab \cos C$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q2/5

In triangle ABC, $\angle C$ is obtuse, so the altitude from A meets line BC extended beyond C at D. Let CD = x. Prove that $c^2 = a^2 + b^2 - 2ab \cos C$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q3/5

Use the Law of Cosines to prove that in triangle ABC, if $c^2 > a^2 + b^2$, then $\angle C$ is obtuse.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-M-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q4/5

Use the Law of Cosines to prove both directions: in triangle ABC, $\angle C = 90°$ if and only if $c^2 = a^2 + b^2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • MAIN • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 1

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

Legs 5 and 12: hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q1/5

Hypotenuse b, leg x: the other leg squared?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q2/5

In triangle ABC, the altitude from A to BC has length h and foot D, with CD = x and BC = a. In right triangle ABD, $h^2 + (a - x)^2 =$

A. a^2

B. c^2

C. x^2

D. b^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q3/5

True or false?

The Pythagorean Theorem applies to the right triangles made by an altitude.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q4/5

True or false?

$a^2 + b^2 = c^2$ in every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 90 • BACKWARD 1 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 2

The distance formula • Grade 8 • Geometry • 8.G.B.8

Distance from (0, 0) to (3, 4)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q1/5

Write the squared distance from (p, q) to (a, 0).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q2/5

Squared distance from (1, 2) to (4, 6)?

A. 5

B. 25

C. 49

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q3/5

True or false?

The distance formula comes from the Pythagorean Theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q4/5

True or false?

Distance from (2, 0) to (5, 0) is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 90 • BACKWARD 2 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 3

Sine and cosine in a right triangle • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

Hypotenuse b and angle C: adjacent side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B3-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q1/5

Hypotenuse b and angle C: opposite side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B3-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q2/5

For a 3-4-5 triangle, $(\sin)^2 + (\cos)^2$ of an acute angle =

A. 25

B. 1

C. 12/25

D. 7/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B3-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q3/5

True or false?

$\sin^2 C + \cos^2 C = 1$ for acute angles C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B3-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q4/5

True or false?

$\cos C = \text{opposite} \div \text{hypotenuse}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B3-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 90 • BACKWARD 3 • Q5/5

SET 90 • HSG-SRT.D.10

BACKWARD 4

Square a binomial • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(b \cos C - a)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B4-Q1

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q1/5

Expand $(a - x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B4-Q2

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q2/5

$(p - q)^2 =$

A. $p^2 - 2pq + q^2$

B. $p^2 - pq + q^2$

C. $p^2 - q^2$

D. $p^2 + q^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B4-Q3

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q3/5

True or false?

$$(a - x)^2 - x^2 = a^2 - 2ax$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B4-Q4

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q4/5

True or false?

$$(a - x)^2 = a^2 - x^2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-B4-Q5

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 90 • BACKWARD 4 • Q5/5

SET 90 • HSG-SRT.D.10

FORWARD 1

Use the Law of Cosines (SAS and SSS) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

In triangle ABC, a = 7, b = 9, and C = 50°. Find c to the nearest tenth.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q1/5

A triangle has sides 5, 7, and 8. Find the angle opposite the side of length 7.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q2/5

True or false?

The Law of Cosines can find an angle when all three sides are known.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q3/5

Sides 3, 4, 6. The angle opposite 6 is

A. obtuse

B. acute

C. right

D. impossible to tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q4/5

Find the largest angle of the triangle with sides 4, 5, and 6, to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 90 • FORWARD 1 • Q5/5

SET 90 • HSG-SRT.D.10

FORWARD 2

Distances between moving objects • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

Two ships leave port at the same time on courses 70° apart. One travels 20 km and the other 30 km. How far apart are they, to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q1/5

A golfer's ball is 150 yd from the tee; the hole is 180 yd from the tee; the angle between the two directions is 12°. How far is the ball from the hole (nearest yard)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q2/5

True or false?

The Law of Cosines can be used only when the angle between the two known sides is a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q3/5

Two hikers walk 3 km and 5 km from camp in directions 60° apart. Distance between them?

A. 8 km

B. √34 ≈ 5.8 km

C. 2 km

D. √19 ≈ 4.4 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S90-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q4/5

Two ships leave port on courses 70° apart. Why can't the Pythagorean Theorem alone give the distance between them?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S90-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 90 • FORWARD 2 • Q5/5

SET 91 • HSG-SRT.D.11 • ADVANCED EXTENSION (+)

Apply the Law of Sines, including the ambiguous case

MAIN	Geometry main question	**HSG-SRT.D.11**
BACKWARD 1	The Law of Sines	**HSG-SRT.D.10**
BACKWARD 2	Inverse sine	**HSG-SRT.C.8**
BACKWARD 3	Triangle angle sum	**HSG-CO.C.10**
BACKWARD 4	Sine of supplementary angles	**HSG-SRT.D.9**
FORWARD 1	Solve both triangles in the ambiguous case	**HSG-SRT.D.11**
FORWARD 2	Bearings and navigation with the Law of Sines	**HSG-MG.A.1**

SET 91 • HSG-SRT.D.11

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

Points A and B are 120 m apart on one bank of a river. A tree C is on the other bank. $\angle A = 72°$ and $\angle B = 63°$. Find AC to the nearest tenth of a meter.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q1/5

In triangle ABC, A = 30°, a = 6, and b = 10. How many triangles are possible? Find all possible measures of B to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q2/5

In triangle ABC, A = 40°, a = 12, and b = 10. How many triangles are possible? Find B to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q3/5

In triangle ABC, A = 50°, a = 5, and b = 10. How many triangles are possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q4/5

A is acute and $b \sin A < a < b$. How many triangles have these measurements?

A. Infinitely many

B. One

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-M-Q5**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • MAIN • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 1

The Law of Sines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

Write the Law of Sines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q1/5

A = 30°, B = 90°, a = 5. b =

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q2/5

Solving a/sin A = b/sin B for sin B gives

A. b sin A/a

B. a/(b sin A)

C. sin A

D. a sin A/b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B1-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q3/5

True or false?

Each side in the Law of Sines is paired with its opposite angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q4/5

True or false?

$a/\sin B = b/\sin A$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 91 • BACKWARD 1 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 2

Inverse sine • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

sin B = 0.5, B acute. B =

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q1/5

sin B = 0.8. Find the acute B to the nearest tenth.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q2/5

$\sin^{-1}(1) =$

A. 0°

B. 180°

C. 45°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q3/5

True or false?

The calculator's $\sin^{-1}$ always gives an angle between $-90°$ and $90°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q4/5

True or false?

$\sin^{-1}(1.2)$ is an angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 91 • BACKWARD 2 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 3

Triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Two angles are 72° and 63°. Third angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q1/5

True or false?

A triangle cannot have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q2/5

A = 40° and B = 147.6°. Is this a triangle?

A. Only if C is 0°

B. No: the sum exceeds 180°

C. Yes

D. Only if right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q3/5

A = 30°, B = 123.6°. Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q4/5

True or false?

The angles of a triangle add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 91 • BACKWARD 3 • Q5/5

SET 91 • HSG-SRT.D.11

BACKWARD 4

Sine of supplementary angles • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9

True or false?

$\sin 150° = \sin 30°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B4-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q1/5

Name an obtuse angle with the same sine as 56.4°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B4-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q2/5

sin 120° equals

A. cos 120°

B. sin 60°

C. sin 30°

D. −sin 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-B4-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q3/5

Why might sin B = 0.83 give two possible triangles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B4-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q4/5

True or false?

An obtuse angle has a negative sine.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-B4-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.9 • Set 91 • BACKWARD 4 • Q5/5

SET 91 • HSG-SRT.D.11

FORWARD 1

Solve both triangles in the ambiguous case • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

A = 30°, a = 6, b = 10. For the triangle with acute B, find C and c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q1/5

In triangle ABC, A = 30°, a = 6, and b = 10, so B ≈ 56.4° or B ≈ 123.6°. For the triangle with obtuse B, find C and c to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q2/5

True or false?

In the ambiguous case the two triangles have different third sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q3/5

$A = 30°$, $b = 10$, and $a = 5$. How many triangles?

A. Three

B. Two

C. One (a right triangle)

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S91-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q4/5

$A = 120°$, $a = 8$, $b = 10$. How many triangles? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 91 • FORWARD 1 • Q5/5

SET 91 • HSG-SRT.D.11

FORWARD 2

Bearings and navigation with the Law of Sines • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

A lighthouse is spotted from a ship at an angle of 35° to its course. After the ship sails 8 km farther, the angle is 70°. How far is the ship from the lighthouse at the second sighting (nearest tenth)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q1/5

A forest ranger at A sees a fire at an angle of 48° from the line to a second tower B, 10 km away. From B, the angle is 61°. How far is the fire from A (nearest tenth)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q2/5

True or false?

Bearings and sightings are modeled as triangles with a known side and two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q3/5

A ship sights a lighthouse at 35° to its course; after sailing 8 km farther along the course, the angle is 70°. In the triangle formed by the two ship positions and the lighthouse, the angle at the lighthouse is

A. 75°

B. 110°

C. 70°

D. 35°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q4/5

Name one reason a navigation model using a flat triangle might be inaccurate over very long distances.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S91-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 91 • FORWARD 2 • Q5/5

SET 92 • HSG-SRT.D.11 • ADVANCED EXTENSION (+)

Apply the Law of Cosines, including resultant forces

MAIN	Geometry main question	**HSG-SRT.D.11**
BACKWARD 1	The Law of Cosines	**HSG-SRT.D.10**
BACKWARD 2	Inverse cosine	**HSG-SRT.C.8**
BACKWARD 3	Angles in a parallelogram	**HSG-CO.C.11**
FORWARD 1	Solve a triangle with both laws	**HSG-SRT.D.11**
FORWARD 2	Velocity vectors: airspeed and wind	**HSN-VM.A.3**

SET 92 • HSG-SRT.D.11

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

To find the length AB across a lake, a surveyor at C measures CA = 350 m, CB = 420 m, and ∠C = 68°. Find AB to the nearest tenth of a meter.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q1/5

Two forces of 40 N and 60 N act on an object with an angle of 50° between them. Find the magnitude of the resultant force to the nearest tenth.

Resultant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q2/5

A triangle has sides 8, 10, and 13. Find its largest angle to the nearest tenth of a degree.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q3/5

A plane flies 200 km, turns 25° to the right, and flies 150 km. How far is it from its start, to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q4/5

Which law should be used first to solve a triangle given all three sides?

A. The Law of Sines

B. The Pythagorean Theorem

C. The angle sum

D. The Law of Cosines

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • MAIN • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 1

The Law of Cosines • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

Write the Law of Cosines for side c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q1/5

a = 3, b = 4, C = 90°: c?

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q2/5

Solved for cos C:

A. $(a^2 + b^2 - c^2)/(2ab)$

B. $(c^2 - a^2 - b^2)/(2ab)$

C. $2ab/(a^2 + b^2 - c^2)$

D. $(a^2 + b^2 + c^2)/(2ab)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-B1-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q3/5

True or false?

The angle in the formula is the one opposite the side being found.

A. True

B. False

True or false?

$$c^2 = a^2 + b^2 + 2ab \cos C$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 92 • BACKWARD 1 • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 2

Inverse cosine • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

$\cos C = 0.5$. $C =$

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q1/5

$\cos C = -0.03125$. C to the nearest tenth?

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q2/5

A negative cosine means the angle is

A. acute

B. right

C. obtuse

D. zero

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q3/5

True or false?

$\cos^{-1}$ gives angles from 0° to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q4/5

True or false?

$\cos^{-1}(0) = 0°$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 92 • BACKWARD 2 • Q5/5

SET 92 • HSG-SRT.D.11

BACKWARD 3

Angles in a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

Consecutive angles of a parallelogram are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q1/5

One angle of a parallelogram is 50°. The angle next to it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q2/5

Forces at 50° form a parallelogram; the angle opposite the resultant in the triangle is

A. 130°

B. 25°

C. 50°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q3/5

A plane turns 25°. The angle inside the triangle at the turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q4/5

True or false?

The angle between two forces is the angle used in the Law of Cosines for the resultant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 92 • BACKWARD 3 • Q5/5

SET 92 • HSG-SRT.D.11

FORWARD 1

Solve a triangle with both laws • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11

In triangle ABC, b = 12, c = 15, A = 40°. Find a to the nearest tenth.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q1/5

In triangle ABC, b = 12, c = 15, and A = 40°. Find B to the nearest tenth of a degree (use the Law of Sines or Cosines).

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q2/5

True or false?

After the Law of Cosines, it is safest to find the smaller remaining angle with the Law of Sines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q3/5

Given two sides and the included angle, use first

A. $\sin^{-1}$

B. the Law of Sines

C. the Law of Cosines

D. the angle sum

Use Heron's formula or ½ab sin C to find the area of the triangle with sides 8, 10, 13 (nearest tenth).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.11 • Set 92 • FORWARD 1 • Q5/5

SET 92 • HSG-SRT.D.11

FORWARD 2

Velocity vectors: airspeed and wind • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3

A plane flies due east at 300 km/h; a wind blows due north at 50 km/h. Find its ground speed (nearest tenth) and direction (degrees north of east, nearest tenth).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q1/5

A boat heads across a river at 8 km/h; the current is 3 km/h at 90° to its heading. Find its actual speed (nearest tenth).

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q2/5

True or false?

The resultant of two velocities is always found by adding their speeds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q3/5

Velocities of 10 and 10 with 120° between them have a resultant of

A. 0

B. 17.3

C. 10

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S92-F2-Q4**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q4/5

A plane's velocity is 400 km/h; a 60 km/h wind blows at 45° to its heading. Find the ground speed (nearest tenth).

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S92-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3 • Set 92 • FORWARD 2 • Q5/5

SET 93 • HSG-C.A.4 • ADVANCED EXTENSION (+)

Construct a tangent line from a point outside a circle

MAIN	Geometry main question	**HSG-C.A.4**
BACKWARD 1	Construct the midpoint of a segment	**HSG-CO.D.12**
BACKWARD 2	Inscribed angles on a diameter	**HSG-C.A.2**
BACKWARD 3	Tangent is perpendicular to the radius	**HSG-C.A.2**
FORWARD 1	Prove that the two tangent segments are congruent and compute…	**HSG-C.A.4**
FORWARD 2	Lines of sight to the horizon	**HSG-MG.A.1**

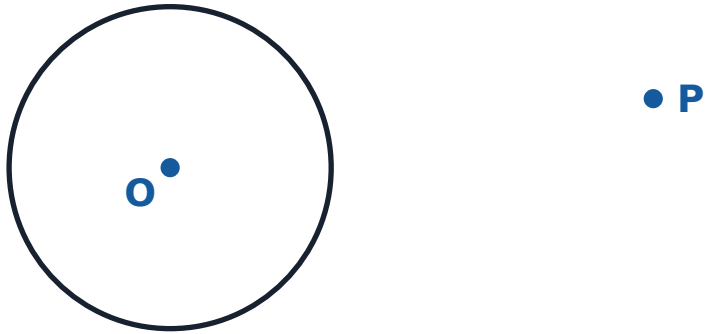
SET 93 • HSG-C.A.4

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

Compass and straightedge only; leave all arcs.

Construct a line through P tangent to the circle with center O.



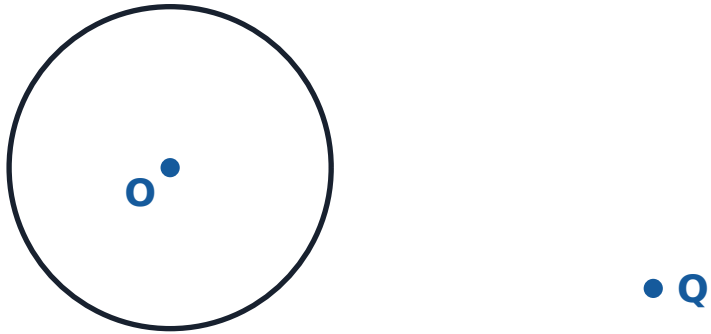
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct BOTH tangent lines from Q to the circle with center O.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a line through R tangent to the circle with center O. Then write one sentence explaining why your line is tangent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct both tangent lines from S to the circle with center O. Then, as evidence, set your compass to ST_1 and show that the same opening reaches T_2 from S.

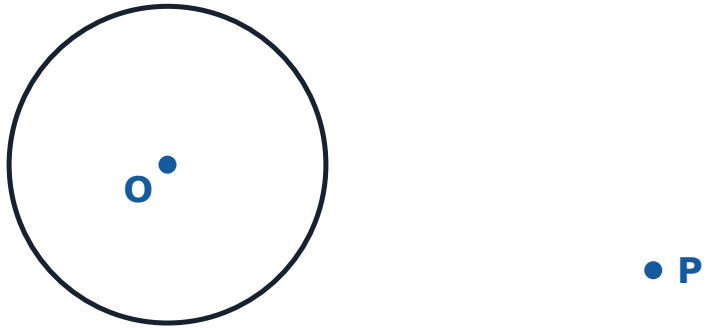
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct both tangent lines from P to the circle with center O. Then construct the bisector of the angle formed by the two tangents and show that it passes through O.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • MAIN • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 1

Construct the midpoint of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction finds the midpoint of OP?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector crosses the segment at its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q2/5

The perpendicular bisector construction uses arcs

A. of any radius

B. centered at the midpoint

C. of equal radius from both endpoints

D. from one endpoint only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q3/5

To construct a tangent from a point P outside circle O, why is the midpoint of OP needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q4/5

True or false?

The midpoint can be found by eye in a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 93 • BACKWARD 1 • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 2

Inscribed angles on a diameter • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

An angle inscribed in a semicircle is a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q1/5

A point T is on the circle with diameter OP. m∠OTP = ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q2/5

The intercepted arc of an angle inscribed in a semicircle is

A. 180°

B. 360°

C. 90°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q3/5

To construct a tangent from a point P outside circle O, why is a circle with diameter OP drawn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q4/5

True or false?

An inscribed angle on a diameter can be 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 2 • Q5/5

SET 93 • HSG-C.A.4

BACKWARD 3

Tangent is perpendicular to the radius • High School — Geometry • Geometry: Circles • HSG-C.A.2

True or false?

A line perpendicular to a radius at its endpoint on the circle is tangent to the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q1/5

A tangent meets the radius at what angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q2/5

If PT $\perp$ OT and T is on circle O, then PT is

A. a secant

B. a diameter

C. a chord

D. tangent to the circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q3/5

OT = 5 and OP = 13 with PT tangent at T. PT = ?

PT = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q4/5

True or false?

A tangent passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 93 • BACKWARD 3 • Q5/5

SET 93 • HSG-C.A.4

FORWARD 1

Prove that the two tangent segments are congruent and compute lengths • High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4

PA and PB are tangent to circle O at A and B. Prove that PA = PB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q1/5

A circle has radius 5 and P is 13 from its center. Find the length of each tangent segment from P.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q2/5

True or false?

OP bisects the angle between the two tangents.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q3/5

Tangents from P make a 60° angle. The angle AOB is

A. 30°

B. 120°

C. 60°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F1-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q4/5

Radius 6, tangent segment 8. Distance from P to the center?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Circles • HSG-C.A.4 • Set 93 • FORWARD 1 • Q5/5

SET 93 • HSG-C.A.4

FORWARD 2

Lines of sight to the horizon • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

Model Earth as a sphere of radius 6,371 km and ignore atmospheric refraction (light travels in straight lines). A person's eyes are 2 m (0.002 km) above the sea. How far is the horizon (tangent distance), to the nearest tenth of a km?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q1/5

Model Earth as a sphere of radius 6,371 km and ignore atmospheric refraction. From the top of a 100 m (0.1 km) tower at sea level, how far is the horizon, to the nearest km?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q2/5

True or false?

The line of sight to the horizon passes through Earth's center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q3/5

Raising the eye height by a factor of 4 multiplies the horizon distance by about

A. 16

B. 8

C. 2

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q4/5

In the model of the distance to the horizon from a point above Earth, why is the right angle of the triangle at the horizon point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S93-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 93 • FORWARD 2 • Q5/5

SET 94 • HSG-GPE.A.3 • ADVANCED EXTENSION (+)

Derive the equation of an ellipse from its foci

MAIN	Geometry main question	**HSG-GPE.A.3**
BACKWARD 1	The distance formula	**8.G.B.8**
BACKWARD 2	Derive a circle equation	**HSG-GPE.A.1**
BACKWARD 3	Expand squares	**HSA-APR.A.1**
FORWARD 1	Read foci and axes from an ellipse equation	**HSG-GPE.A.3**
FORWARD 2	Design with the reflective property of ellipses	**HSG-MG.A.3**

SET 94 • HSG-GPE.A.3

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

An ellipse is the set of points whose distances to the foci $(-3, 0)$ and $(3, 0)$ add to 10. Derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q1/5

Derive, from the definition (constant sum of distances to the foci), the equation of the ellipse with foci (0, −4) and (0, 4) whose distances to the foci add to 10.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q2/5

Derive, from the definition (constant sum of distances to the foci), the equation of the ellipse with foci (±5, 0) and vertices (±13, 0).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q3/5

Derive, from the definition (constant sum of distances to the foci), the equation of the ellipse with foci (1, 2) and (7, 2) whose distances to the foci add to 10. (You may translate the figure so the center is at the origin, derive, then translate back.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q4/5

Derive, from the definition (constant sum of distances to the foci), the equation of the ellipse with foci $(-c, 0)$ and $(c, 0)$ whose distances add to $2a$ ($a > c > 0$). Show that it can be written $x^2/a^2 + y^2/b^2 = 1$ with $b^2 = a^2 - c^2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • MAIN • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 1

The distance formula • Grade 8 • Geometry • 8.G.B.8

Distance from (x, y) to (3, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q1/5

Distance from (0, 4) to (3, 0)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q2/5

Distance from (5, 0) to (−3, 0)?

A. √34

B. 64

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q3/5

True or false?

The distance formula has a square root.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q4/5

True or false?

$$\sqrt{(a^2 + b^2)} = a + b$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 94 • BACKWARD 1 • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 2

Derive a circle equation • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Equation of the circle with center (0, 0) and radius 5?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q1/5

True or false?

A circle is an ellipse whose two foci coincide.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q2/5

$x^2/25 + y^2/25 = 1$ is

A. an ellipse with different axes

B. a parabola

C. a circle of radius 5

D. a hyperbola

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q3/5

A circle is the set of points at a fixed distance from

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q4/5

True or false?

An ellipse has one focus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 94 • BACKWARD 2 • Q5/5

SET 94 • HSG-GPE.A.3

BACKWARD 3

Expand squares • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(x + 3)^2 - (x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q1/5

Expand $(x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B3-Q2

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q2/5

$(3x - 25)^2 =$

A. $9x^2 + 625$

B. $9x^2 - 75x + 625$

C. $3x^2 - 150x + 625$

D. $9x^2 - 150x + 625$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-B3-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q3/5

True or false?

$$(a - b)^2 = a^2 - 2ab + b^2$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q4/5

True or false?

$(x + 3)^2 = x^2 + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 94 • BACKWARD 3 • Q5/5

SET 94 • HSG-GPE.A.3

FORWARD 1

Read foci and axes from an ellipse equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Find the foci of $x^2/25 + y^2/9 = 1$.

Foci: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q1/5

Find the vertices and foci of $x^2/16 + y^2/25 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q2/5

True or false?

For an ellipse, $c^2 = a^2 - b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q3/5

Which ellipse is closest to a circle?

A. $x^2/100 + y^2/9 = 1$

B. $x^2/25 + y^2/4 = 1$

C. $x^2/25 + y^2/24 = 1$

D. $x^2/25 + y^2/1 = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S94-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q4/5

Find the length of the major axis of $x^2/49 + y^2/36 = 1$.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 94 • FORWARD 1 • Q5/5

SET 94 • HSG-GPE.A.3

FORWARD 2

Design with the reflective property of ellipses • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A whispering gallery is an ellipse 40 m long and 16 m wide. How far from the center should two people stand (at the foci), to the nearest tenth?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q1/5

A medical device places a kidney stone at one focus of an ellipse $x^2/100 + y^2/64 = 1$ (cm). Where is the emitter placed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q2/5

True or false?

Sound from one focus of an elliptical room reflects back to the same focus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q3/5

A garden ellipse is drawn with a 10 m rope tied to stakes 6 m apart. Its width (minor axis) is

A. 8 m

B. 10 m

C. 4 m

D. 6 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q4/5

Explain how the string-and-stakes method uses the definition of an ellipse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S94-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 94 • FORWARD 2 • Q5/5

SET 95 • HSG-GPE.A.3 • ADVANCED EXTENSION (+)

Derive the equation of a hyperbola from its foci

MAIN	Geometry main question	**HSG-GPE.A.3**
BACKWARD 1	The distance formula	**8.G.B.8**
BACKWARD 2	Derive an ellipse equation	**HSG-GPE.A.3**
BACKWARD 3	Absolute value as distance	**6.NS.C.7**
FORWARD 1	Read foci and asymptotes from a hyperbola equation	**HSG-GPE.A.3**
FORWARD 2	Locate a position from a time difference	**HSG-MG.A.3**

SET 95 • HSG-GPE.A.3

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

A hyperbola is the set of points whose distances to the foci $(-5, 0)$ and $(5, 0)$ differ by 6 (in absolute value). Derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-M-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q1/5

Derive, from the definition (constant difference of distances to the foci), the equation of the hyperbola with foci $(0, \pm 13)$ whose distances to the foci differ by 10 (in absolute value).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q2/5

Derive, from the definition (constant difference of distances to the foci), the equation of the hyperbola with foci (±10, 0) and vertices (±6, 0).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q3/5

Derive, from the definition (constant difference of distances to the foci), the equation of the hyperbola with foci $(-c, 0)$ and $(c, 0)$ whose distances differ by $2a$ $(0 < a < c)$. Show that it can be written $x^2/a^2 - y^2/b^2 = 1$ with $b^2 = c^2 - a^2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q4/5

Derive, from the definition (constant difference of distances to the foci), the equation of the hyperbola with foci (±5, 0) whose distances to the foci differ by 8. Then give its asymptotes.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • MAIN • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 1

The distance formula • Grade 8 • Geometry • 8.G.B.8

Distance from (x, y) to (−5, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q1/5

Distance from (3, 0) to (5, 0) minus distance from (3, 0) to (−5, 0)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q2/5

Distance from (6, 0) to (−10, 0)?

A. 4

B. 60

C. 16

D. −16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q3/5

True or false?

Distances are never negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 95 • BACKWARD 1 • Q4/5

True or false?

Distance from (0, 13) to (0, −13) is 13.

A. True

B. False

SET 95 • HSG-GPE.A.3

BACKWARD 2

Derive an ellipse equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

True or false?

An ellipse uses a constant SUM of distances to the foci.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q1/5

Ellipse with foci (±3, 0) and sum 10: equation?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q2/5

For an ellipse, $b^2 =$

A. ac

B. $a^2 - c^2$

C. $a^2 + c^2$

D. $c^2 - a^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-B2-Q3**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q3/5

What changes in the definition to get a hyperbola?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q4/5

True or false?

An ellipse equation has a minus sign between the squared terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • BACKWARD 2 • Q5/5

SET 95 • HSG-GPE.A.3

BACKWARD 3

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7

$|4 - 10| =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q1/5

$|10 - 4| =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q2/5

$|a - b|$ means

A. the size of the difference

B. $a + b$

C. a negative number

D. always $a - b$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q3/5

True or false?

$|x| = 6$ has two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q4/5

True or false?

$|-3| = -3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 95 • BACKWARD 3 • Q5/5

SET 95 • HSG-GPE.A.3

FORWARD 1

Read foci and asymptotes from a hyperbola equation • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Find the foci of $x^2/16 - y^2/9 = 1$.

Foci: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q1/5

Find the vertices and asymptotes of $y^2/4 - x^2/9 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q2/5

True or false?

For a hyperbola, $c^2 = a^2 + b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q3/5

Which equation is a hyperbola?

A. $x^2/4 + y^2/9 = 1$

B. $y = x^2/4$

C. $x^2/4 - y^2/9 = 1$

D. $x^2 + y^2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S95-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q4/5

Explain how to tell an ellipse from a hyperbola by its equation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 95 • FORWARD 1 • Q5/5

SET 95 • HSG-GPE.A.3

FORWARD 2

Locate a position from a time difference • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

Two radio stations are 100 km apart at (±50, 0). A ship receives signals showing it is 60 km closer to one station than the other. Write the equation of the hyperbola it lies on.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q1/5

Sound travels 0.34 km/s. Two microphones 2 km apart hear a sound 3 s apart. What is the difference in distances to the source?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q2/5

True or false?

For one pair of stations, a single nonzero time difference (with the sign known, that is, which signal arrived first) places the ship on one branch of a hyperbola, not at a single point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S95-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q3/5

Radio stations are at $(-50, 0)$ and $(50, 0)$ (km). A ship on the x-axis, to the right of the origin, is 60 km closer to the right station than to the left one. Its position is

A. $(30, 0)$

B. $(40, 0)$

C. $(50, 0)$

D. $(60, 0)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q4/5

One pair of stations and one nonzero signed time difference place a ship on one branch of a hyperbola. What extra measurement is needed to fix its position, and why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S95-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 95 • FORWARD 2 • Q5/5

SET 96 • HSG-GMD.A.2 • ADVANCED EXTENSION (+)

Use Cavalieri's principle to justify the volume of a sphere

MAIN	Geometry main question	**HSG-GMD.A.2**
BACKWARD 1	Cavalieri's principle for prisms and cylinders	**HSG-GMD.A.1**
BACKWARD 2	Cross-section radius with the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Volumes of cylinders and cones	**HSG-GMD.A.3**
BACKWARD 4	Area of a ring	**7.G.B.4**
FORWARD 1	Use Cavalieri's principle for other solids	**HSG-GMD.A.2**
FORWARD 2	Model a dome as a hemispherical shell	**HSG-MG.A.2**

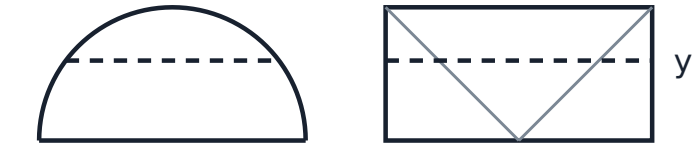
SET 96 • HSG-GMD.A.2

MAIN

Geometry main question • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

A hemisphere of radius r and a cylinder of radius r and height r with a cone removed (as shown) sit on the same plane. Use Cavalieri's principle to show that the hemisphere's volume is $\frac{2}{3}\pi r^3$, and conclude that a sphere has volume $4/3\ \pi r^3$.

Work space



hemisphere, radius r cylinder minus cone

Explanation: _____

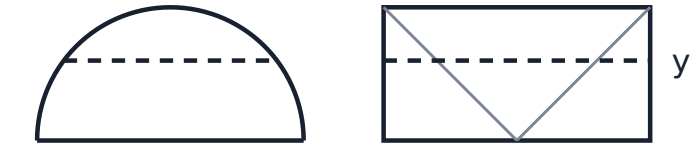
Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-M-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q1/5

Let r = 6. A hemisphere of radius 6 rests on its flat base, and a cylinder of radius 6 and height 6 has a cone removed whose vertex is at the center of the bottom base and whose base is the top of the cylinder. Show that the two cross-sections have equal area at every height y (0 ≤ y ≤ 6), and use Cavalieri's principle to find the hemisphere's and the sphere's volumes.

Work space



hemisphere, radius r cylinder minus cone

Explanation: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-M-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q2/5

A sphere of radius r fits exactly inside a cylinder of radius r and height 2r. Use the Cavalieri result for the hemisphere to explain why the sphere has ⅔ of the cylinder's volume.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-M-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q3/5

Explain, using Cavalieri's principle, why an oblique cylinder of radius r and vertical height h has the same volume as a right cylinder of radius r and height h.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-M-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q4/5

A student turns the removed cone upright (base on the bottom of the cylinder, vertex at the top) and compares the remaining solid with the hemisphere. Explain why Cavalieri's principle cannot be used for this comparison, by comparing the two cross-section areas at height y = 0.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-M-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • MAIN • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 1

Cavalieri's principle for prisms and cylinders • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

Solids with equal heights and equal cross-section areas at every level have equal volumes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q1/5

An oblique cylinder has radius 2 and height 5. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q2/5

A prism and a cylinder both have base area 30 and height 4. Volumes?

A. Cannot tell

B. 30 and 4

C. 120 and 40

D. Both 120

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q3/5

State what must be equal for Cavalieri's principle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q4/5

True or false?

Cavalieri's principle needs the cross-sections to be the same shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 96 • BACKWARD 1 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 2

Cross-section radius with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

A sphere of radius 5 is sliced 3 units from its center. Radius of the slice?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q1/5

A hemisphere of radius r rests on its flat base; y is the height above the base plane (0 ≤ y ≤ r). Find the radius of the horizontal slice at height y.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q2/5

Radius 10, height 6: slice radius

A. 4

B. 8

C. 16

D. $\sqrt{136}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q3/5

True or false?

The slice radius, the height, and the sphere radius form a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q4/5

True or false?

A hemisphere of radius r rests on its flat base; y is the height above the base plane (0 ≤ y ≤ r). Its horizontal slices get bigger as y increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 96 • BACKWARD 2 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 3

Volumes of cylinders and cones • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

Cylinder r = 6, h = 6: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q1/5

Cone r = 6, h = 6: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q2/5

Cylinder minus cone (same r and h = r) =

A. $4/3\ \pi r^3$

B. πr^3

C. $\frac{2}{3}\pi r^3$

D. $\frac{1}{3}\pi r^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q3/5

True or false?

A cone is ⅓ of the cylinder with the same base and height.

A. True

B. False

True or false?

$\pi r^3 - \frac{1}{3}\pi r^3 = \frac{1}{3}\pi r^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 96 • BACKWARD 3 • Q5/5

SET 96 • HSG-GMD.A.2

BACKWARD 4

Area of a ring • Grade 7 • Geometry • 7.G.B.4

Ring with outer radius 5 and inner radius 3: exact area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q1/5

True or false?

Ring area = $\pi(R^2 - r^2)$

A. True

B. False

Ring with outer 4 and inner 1:

A. 3π

B. 15π

C. 9π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q3/5

Ring with outer r and inner y: area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q4/5

True or false?

Ring area = $\pi(R - r)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-B4-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 96 • BACKWARD 4 • Q5/5

SET 96 • HSG-GMD.A.2

FORWARD 1

Use Cavalieri's principle for other solids • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

A twisted tower and a straight prism both have height 50 m and every horizontal slice is a 10 m by 10 m square. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F1-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q1/5

A pyramid with base area 36 and height 10 and a cone with base area 36 and height 10: compare volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F1-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q2/5

True or false?

Cavalieri's principle shows that every pyramid with base area B and height h has volume ⅓Bh, once one such pyramid is known.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F1-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q3/5

Two solids of height 8 have slice areas 5 and 5 at every level. Volumes?

A. 40 and 13.3

B. 5 each

C. Both 40

D. Cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S96-F1-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q4/5

Two solids have equal heights, but at one level their slice areas differ. Can Cavalieri's principle be used? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F1-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 96 • FORWARD 1 • Q5/5

SET 96 • HSG-GMD.A.2

FORWARD 2

Model a dome as a hemispherical shell • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A concrete dome is a hemispherical shell with outer radius 10 m and inner radius 9.5 m. Find the volume of concrete to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q1/5

A concrete dome is a hemispherical shell with outer radius 10 m and inner radius 9.5 m. Concrete has density 2.4 t/m³. Estimate the dome's mass to the nearest ton.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q2/5

True or false?

A shell's volume is the outer volume minus the inner volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q3/5

A hemisphere of radius 3 has volume

A. 9π

B. 27π

C. 36π

D. 18π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q4/5

A hemispherical shell has outer radius 10 m and inner radius 9.5 m. Why is its volume NOT $\frac{2}{3}\pi(10 - 9.5)^3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S96-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 96 • FORWARD 2 • Q5/5